Azure Administration Final Project: Case Study

The Azure Administration case study document has been structured with a common Phase 1 for all covering essentials Compute, Networking, Storage, Database services and Phase 2 with unique enterprise-level implementations, each enriched with all required sections: Problem Statement, Objectives, Expected Solutions, Tools & Technologies, Implementation Flow, Best Practices, and Common Pitfalls.

Phase 1 Common: - Azure Essentials

Problem Statement: Every enterprise begins its Azure journey by establishing a solid foundation of compute,network, and storage services. the goal is to understand and deploy these core services efficiently.

Objectives:

- Set up foundational Azure infrastructure
- Ensure security, availability, and connectivity
- Prepare base environment for advanced scenarios

Expected Solutions:

- Virtual Machines behind Load Balancer
- Storage accounts with blob and file share
- VNet with subnet, NSG, public/private IPs
- Azure SQL Database deployment
- Resource monitoring and backup configuration

Tools and Technologies:

- Azure Portal / CLI, Azure VM, VNet, NSG, Storage Account, SQL DB
- Azure Monitor, Recovery Services Vault

Implementation Flow:

1. Create Resource Group
2. Set up VNet (192.168.0.0/16) with Subnet (192.168.1.0/24)
3. Deploy 2 VMs (Windows), secure with NSG
4. Implement Load Balancer (Layer 4) for the VMs
5. Create Azure Storage (Blob, File Share, SAS key)
6. Map File Share on VMs

7. Provision Azure SQL Database (basic tier)

8. Set up Recovery Services Vault for VM backups

9. Configure CPU alert (80%) via Azure Monitor

10. Create Azure AD Group (e.g., ITAdmins), assign RBAC

Best Practices:

• Use tags and consistent naming

• Apply least privilege for access controls

• Backup policies and test restores regularly

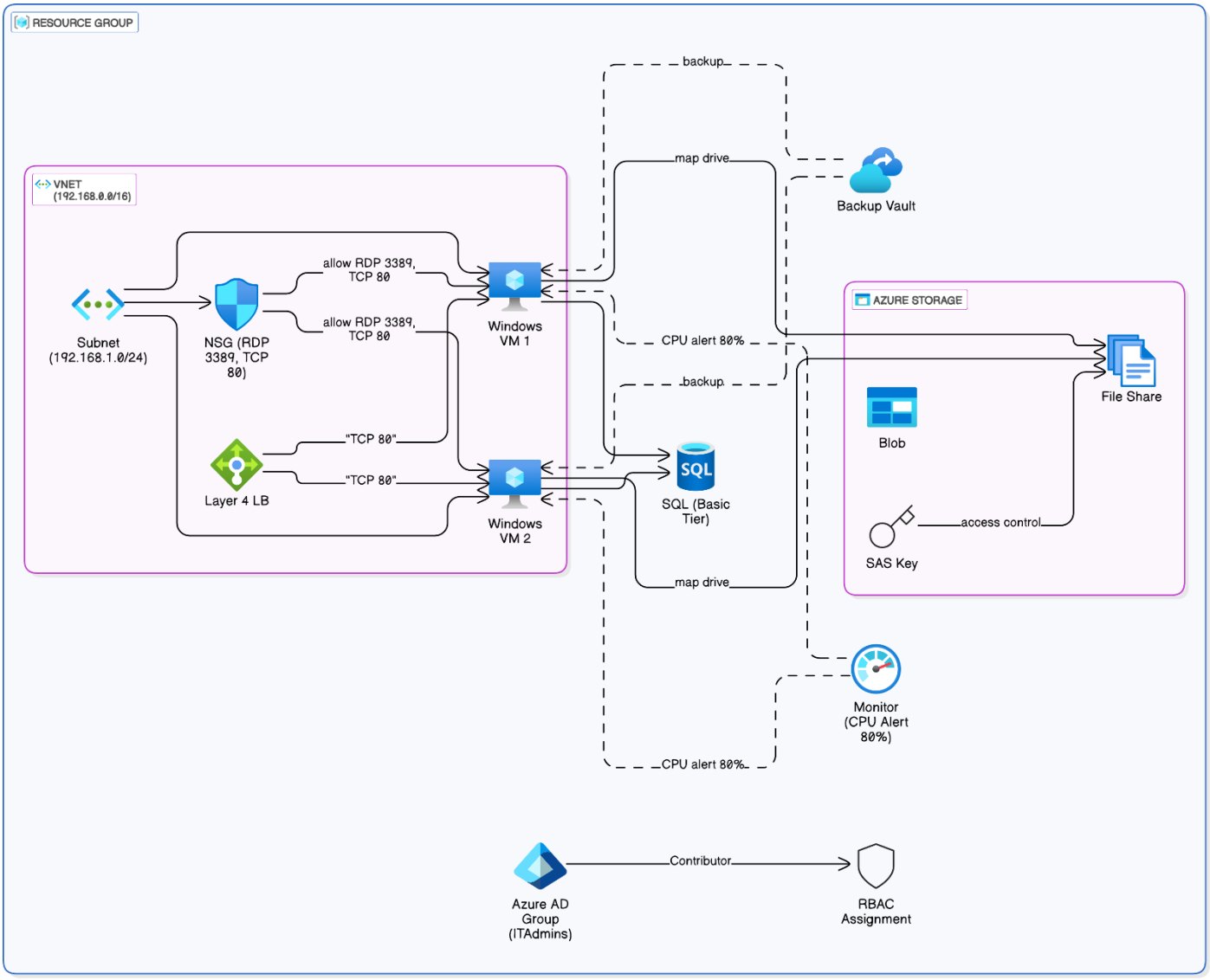
Common Pitfalls to Avoid:

• Misconfigured NSGs blocking required ports

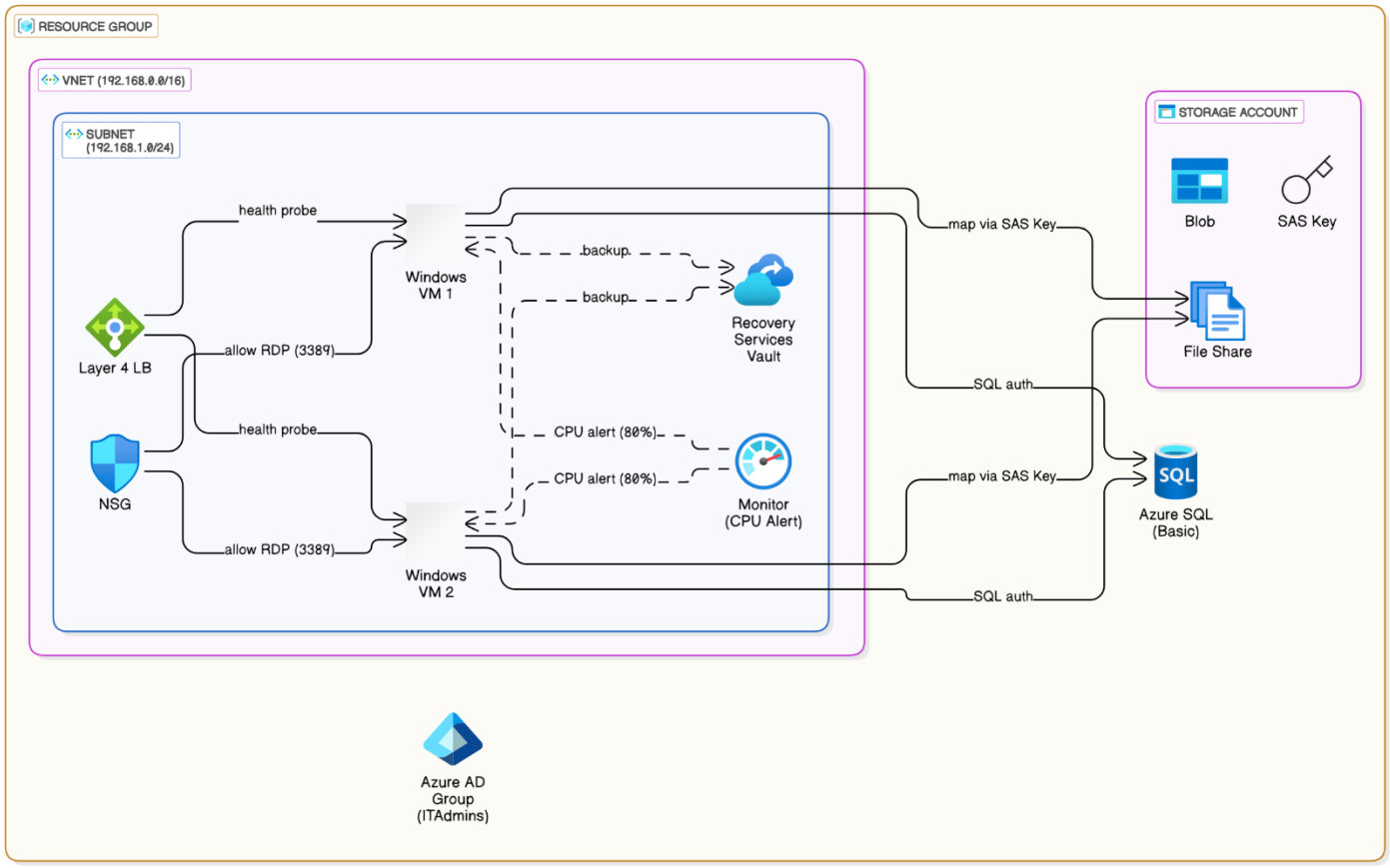
• Overlooking SAS key expiration

• Improper subnet IP planning

Network Diagram



Architecture Diagram



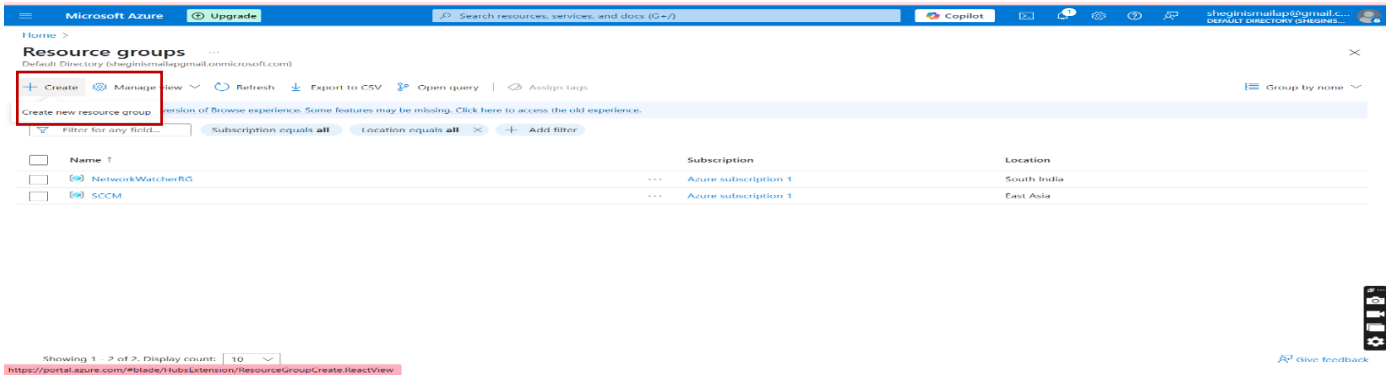
Implementation demonstration

Steps to Create Resource Group in Azure

Method 1: Using Azure Portal

1. **Sign in** to the Azure Portal.
2. In the left-hand menu, select **Resource Groups**.
3. Click **+ Create**.
4. Under **Project Details**, select your **Subscription**.
5. Under **Resource Group Name**, here it's enter BaseInfra-RG.
6. For **Region**,here it's selected Central India**,add tags**.
7. Click **Review + Create**.

Click **Create**—Azure will provision the resource group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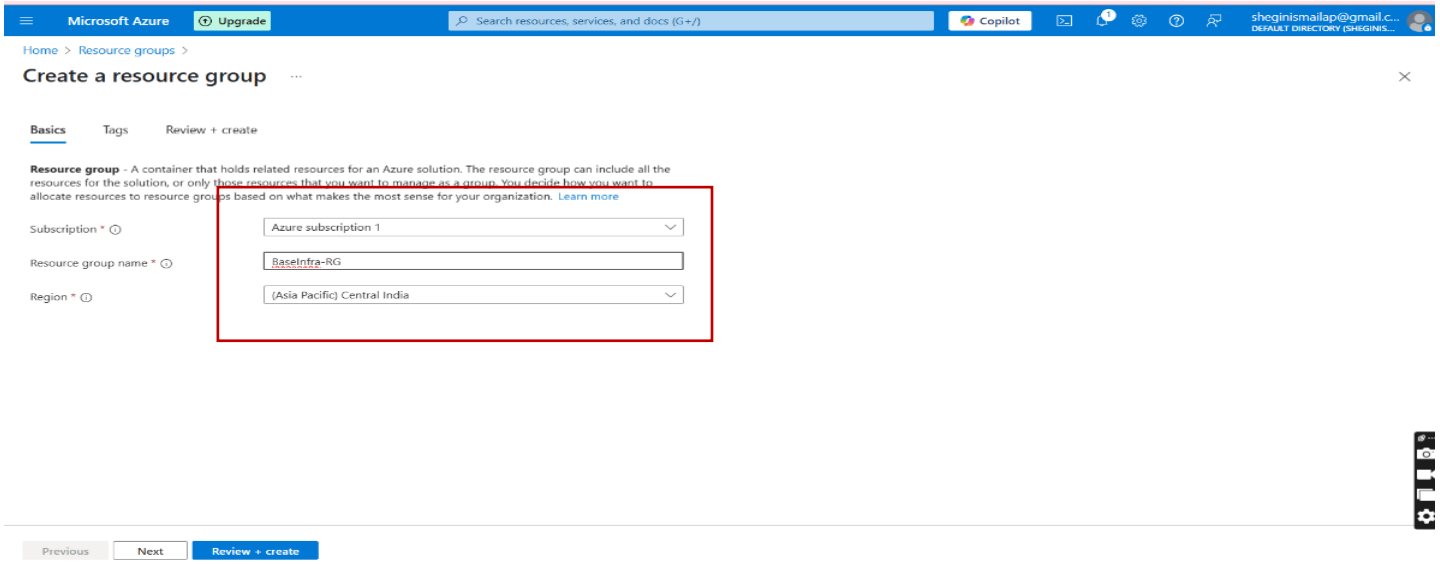
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY (SHEGINIS...
Home >
Resource groups
Default Directory (sheginismailapgmail.onmicrosoft.com)
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Group by none
Create new resource group
Filter for any field...
Subscription equals all
Location equals all
Add filter
Name
Subscription
Location
NetworkWatcherRG
Azure subscription 1
South India
SCCM
Azure subscription 1
East Asia
Showing 1 - 2 of 2. Display count: 10
https://portal.azure.com/#blade/HubsExtension/ResourceGroupCreate.ReactView
Give feedback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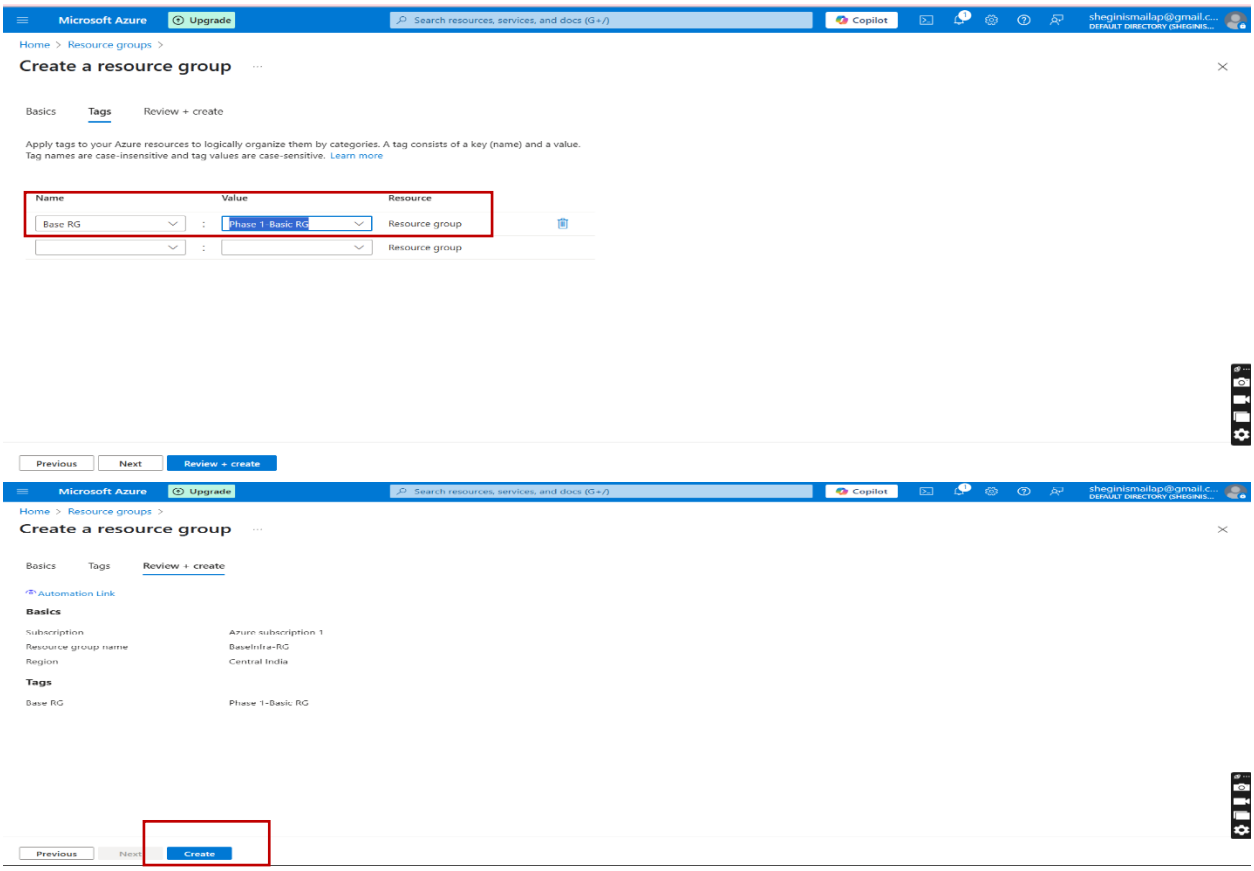
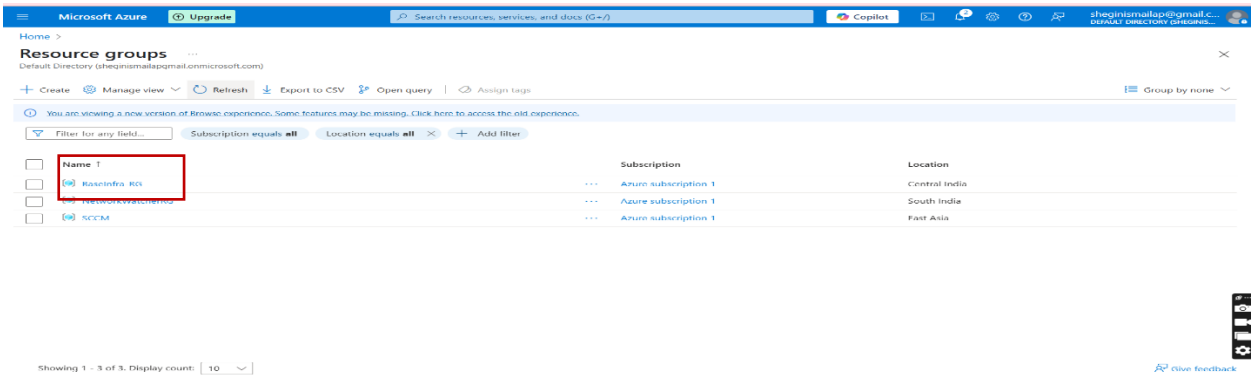
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY (SHEGINIS...
Home > Resource groups >
Create a resource group
Basics
Tags
Review + create
Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. Learn more
Subscription *
Azure subscription 1
Resource group name *
BaseInfra-RG
Region *
(Asia Pacific) Central India
Previous
Next
Review + create



RESULT



Steps for Virtual Network (VNet) with a Subnet in Azure using the **Portal** (GUI):

Step 1: Create a Virtual Network

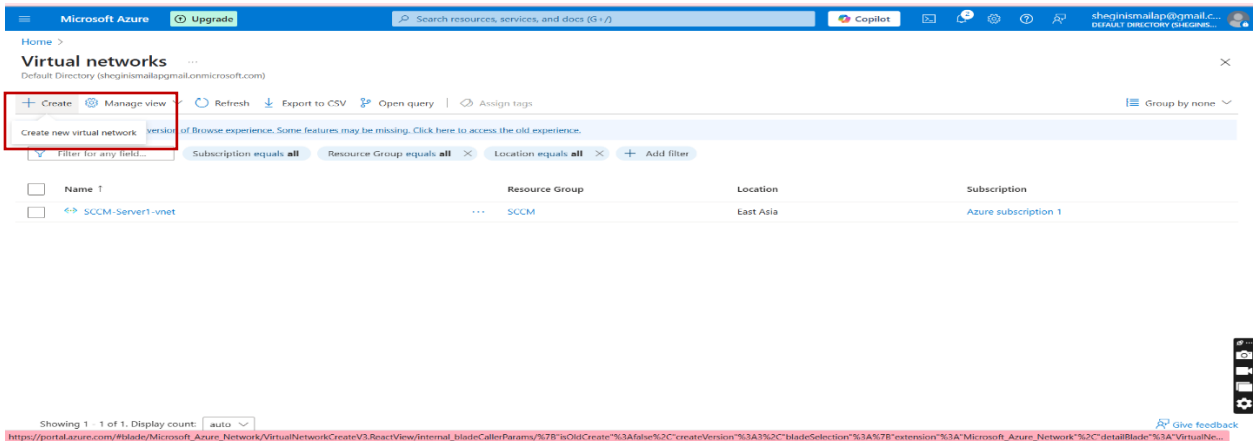
1. Navigate to **Virtual Networks**.
2. Click **Create**.
3. Select the existing **Resource Group**.
4. Choose a **Region** for the network.
5. In the **IP Address space**, enter 192.168.0.0/16.
6. Click **Next: Security**, then **Next: Tags**, then **Review + Create** → **Create**.

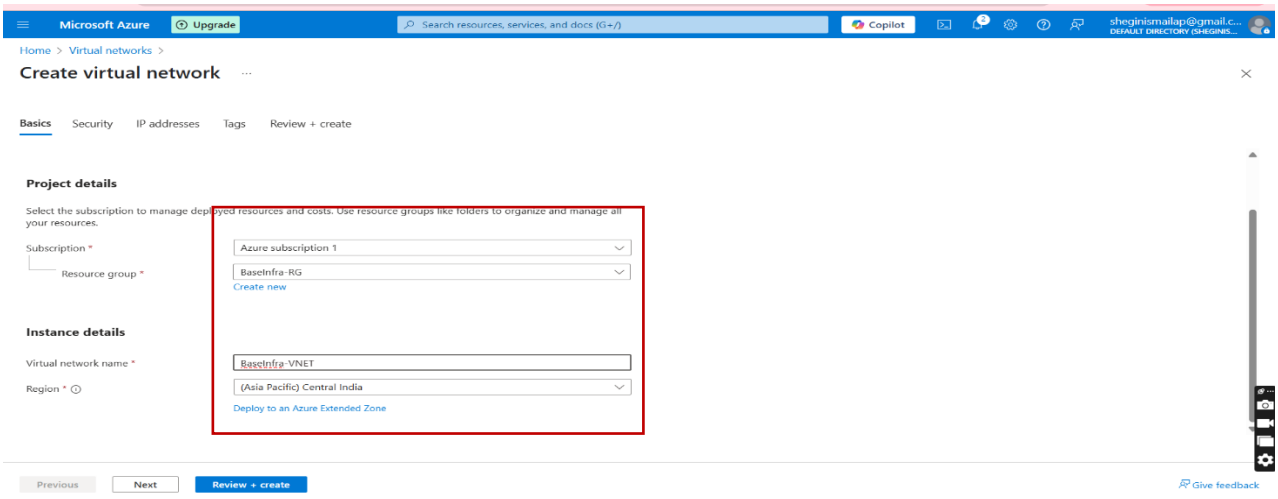
Step 2: Create a Subnet

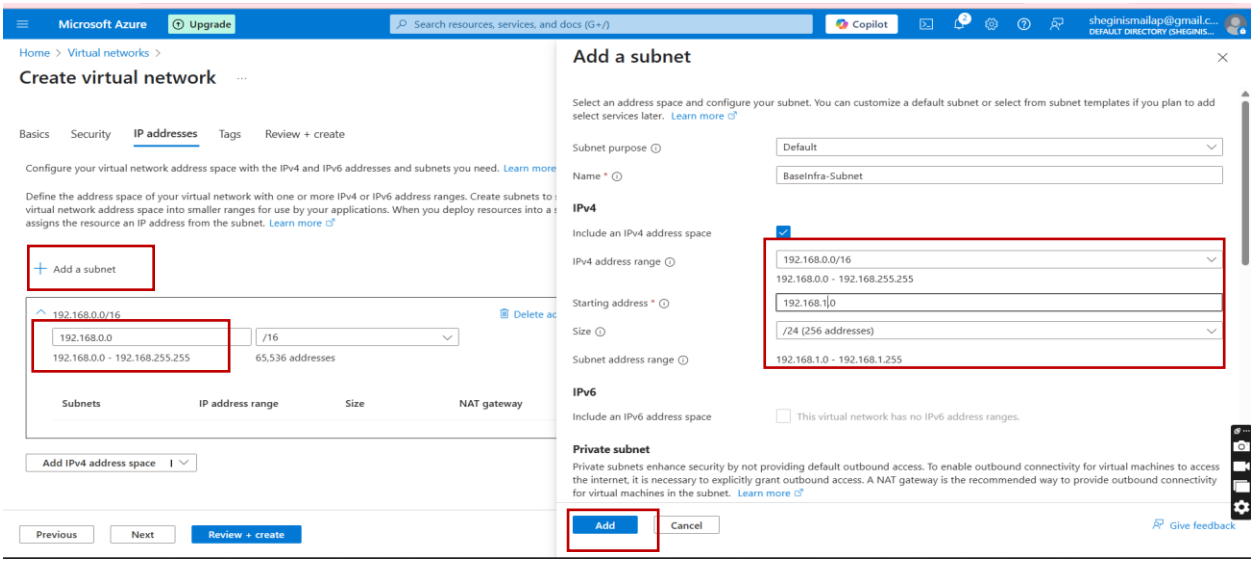
1. Open the **Virtual Network**.
2. Go to the **Subnets** tab.
3. Click **Add Subnet**.
4. Enter 192.168.1.0/24 in the **Subnet Address Range**.
5. (Optional) Associate a **Network Security Group (NSG)** if needed.
6. Click **Save**.

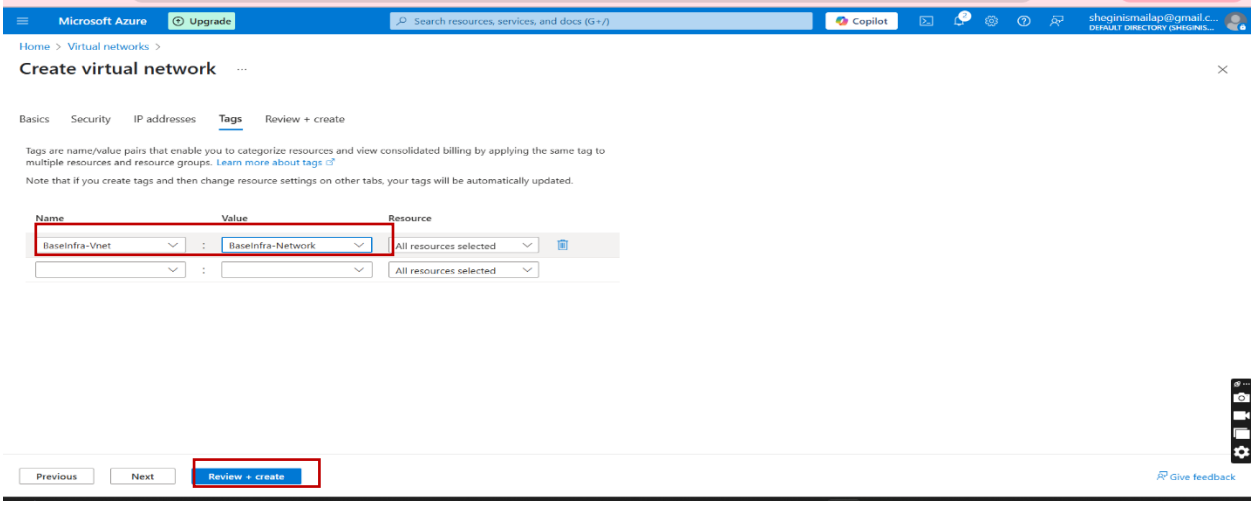
Step 3: Verify Configuration

1. Go to **Virtual Networks**.
2. Open the newly created network and check the **Subnets** tab.
3. Confirm that the address space (192.168.1.0/24) is correctly set up.






Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Virtual networks >
Create virtual network
Basics Security IP addresses Tags Review + create
Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. Learn more
Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to
virtual network address space into smaller ranges for use by your applications. When you deploy resources into a
assigns the resource an IP address from the subnet. Learn more
Add a subnet
192.168.0.0/16
192.168.0.0
192.168.0.0 - 192.168.255.255
/16
65,536 addresses
Subnets
IP address range
Size
NAT gateway
Add IPv4 address space
Previous
Next
Review + create
Add a subnet
Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add
select services later. Learn more
Subnet purpose
Default
Name *
BaseInfra-Subnet
IPv4
Include an IPv4 address space
IPv4 address range
192.168.0.0/16
192.168.0.0 - 192.168.255.255
Starting address *
192.168.1.0
Size
/24 (256 addresses)
Subnet address range
192.168.1.0 - 192.168.1.255
IPv6
Include an IPv6 address space
This virtual network has no IPv6 address ranges.
Private subnet
Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access
the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity
for virtual machines in the subnet. Learn more
Add
Cancel
Give feedback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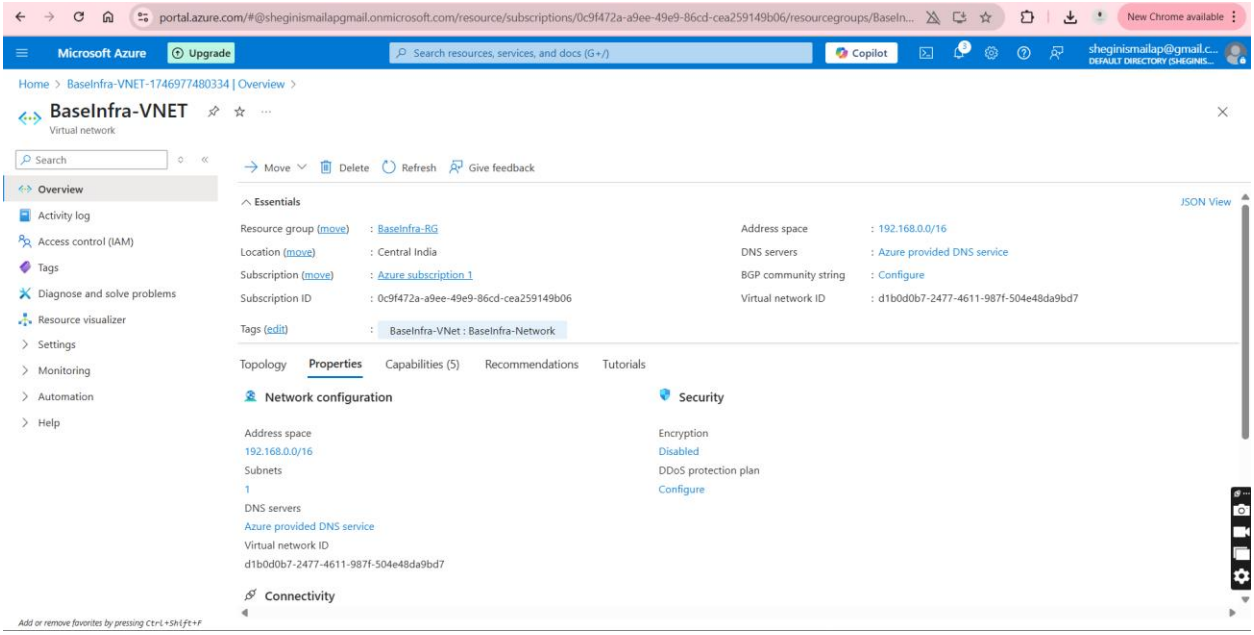
Copilot
Home > Virtual networks >
Create virtual network
Basics Security IP addresses Tags Review + create
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to
multiple resources and resource groups. Learn more about tags
Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.
Name
Value
Resource
BaseInfra-Vnet
BaseInfra-Network
All resources selected
All resources selected
Previous
Next
Review + create
Give feedback

RESULT


portal.azure.com/#@sheginismailapgmail.onmicrosoft.com/resource/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourcegroups/BaseIn...
New Chrome available
Microsoft Azure
Upgrade
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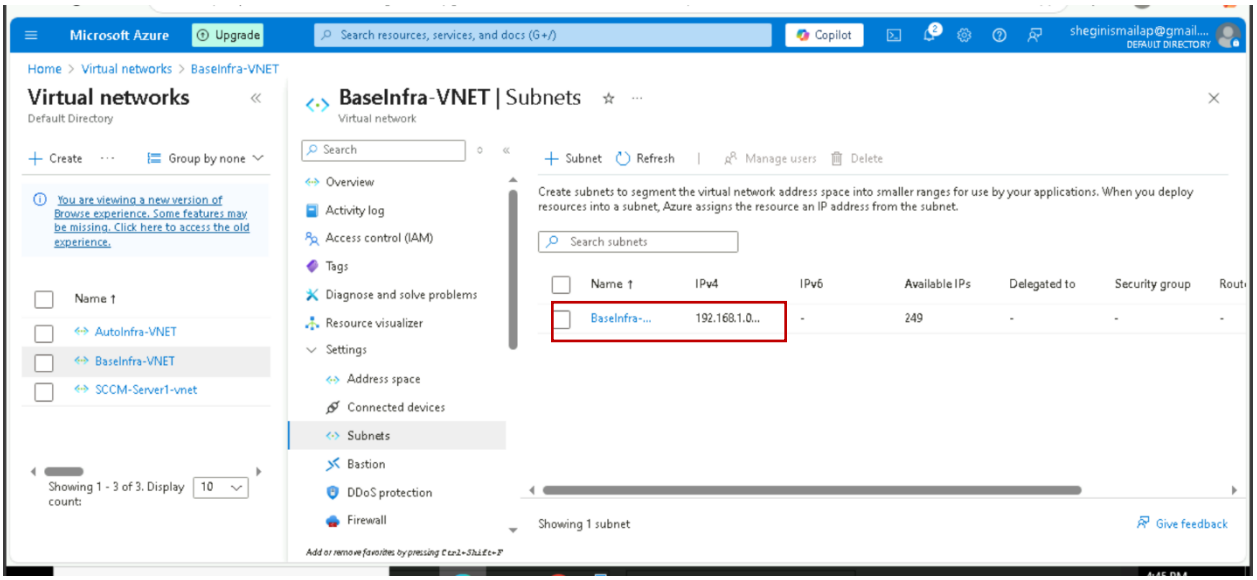
Copilot
Home > BaseInfra-VNET-1746977480334 | Overview >
BaseInfra-VNET
Virtual network
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
Move
Delete
Refresh
Give feedback
Essentials
JSON View
Resource group (move) : BaseInfra-RG
Location (move) : Central India
Subscription (move) : Azure subscription 1
Subscription ID : 0c9f472a-a9ee-49e9-86cd-cea259149b06
Tags (edit) : BaseInfra-VNet : BaseInfra-Network
Address space : 192.168.0.0/16
DNS servers : Azure provided DNS service
BGP community string : Configure
Virtual network ID : d1b0d0b7-2477-4611-987f-504e48da9bd7
Topology
Properties
Capabilities (5)
Recommendations
Tutorials
Network configuration
Address space
192.168.0.0/16
Subnets
1
DNS servers
Azure provided DNS service
Virtual network ID
d1b0d0b7-2477-4611-987f-504e48da9bd7
Security
Encryption
Disabled
DDoS protection plan
Configure
Connectivity
Add or remove favorites by pressing Ctrl+Shift+F



Steps for Deploy Two Windows VMs with Built-in NSG

Step 1:

1. Go to **Virtual Machines** > Click **Create**.
2. Select the **Resource Group**.
3. Enter **VM Name** (e.g., VM1).
4. Choose **Region**, then select **Windows Server** as the image.
5. Configure **Size, Username & Password**.
6. Under **Networking**, select:
    - **VNet & Subnet**.
    - **NSG: Create New or Use Existing** (select the built-in option).
7. Click **Review + Create** → **Create**.
8. Repeat the process for the **second VM (VM2)**.

Step 2: Configure NSG Rules for Security

Since the **NSG was assigned during VM creation**, modify rules as needed:

1. Open **Azure Portal** and navigate to **Network Security Groups**.
2. Select the **NSG associated with the VM**.
3. Go to **Inbound Security Rules**:
    - **Allow RDP (Port 3389)** for remote access (if required).

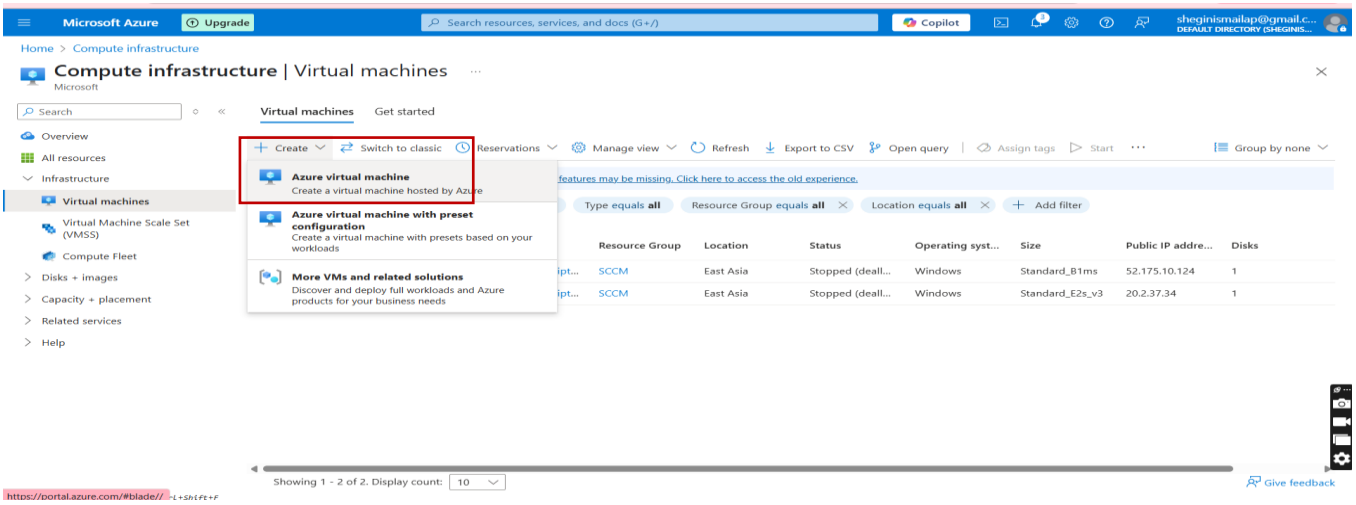
- **Allow Web Traffic (Port 80/443)** if hosting applications.
- **Block all other inbound traffic** for security.

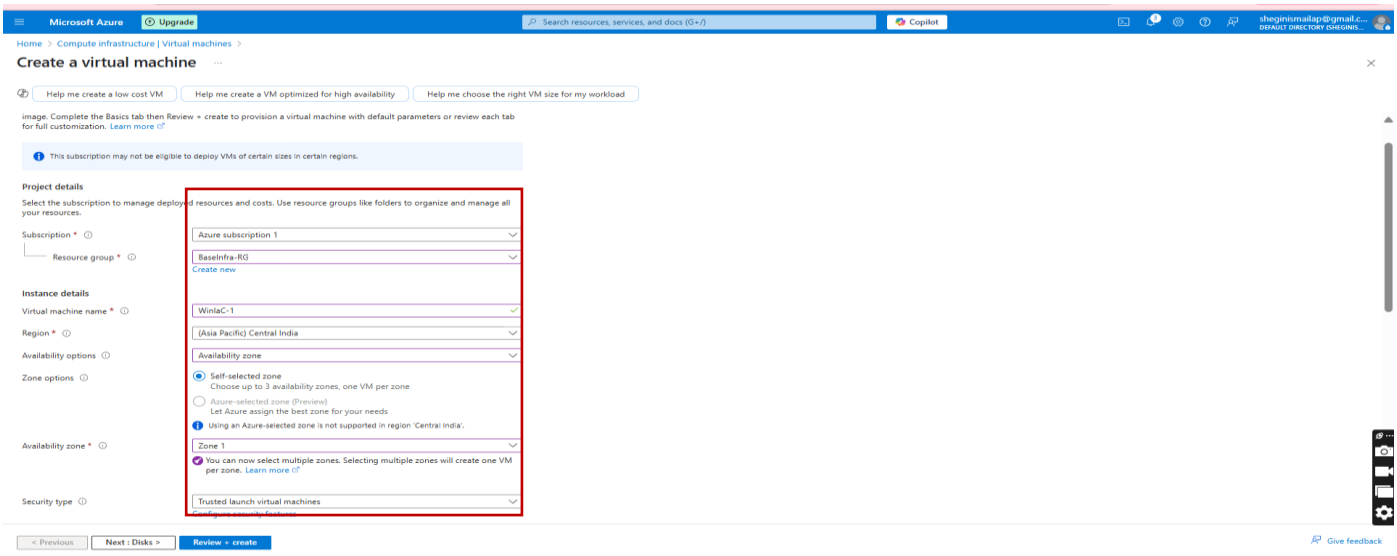
4. Click **Save**.

Step 4: Validate Deployment

1. Navigate to **Virtual Machines** and check their status.
2. Try **connecting via RDP** (if enabled).
3. Verify **NSG rules** to confirm security.

VM1:

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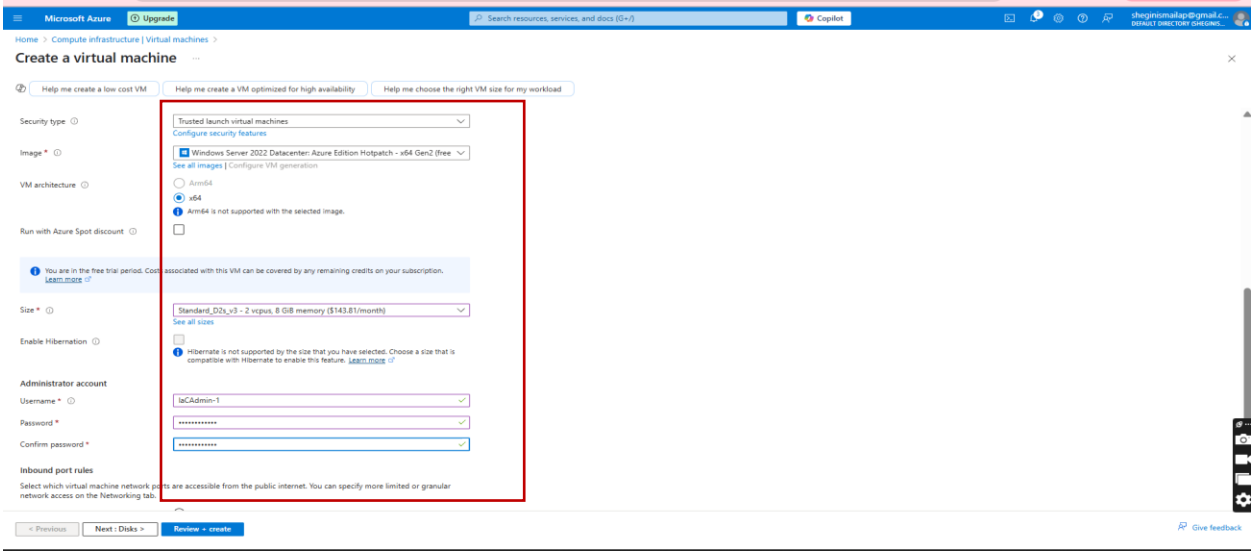
Copilot
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Project details
Subscription *
Azure subscription 1
Resource group *
BaseInfra-RG
Create new
Instance details
Virtual machine name *
WinIaC-1
Region *
(Asia Pacific) Central India
Availability options
Availability zone
Zone options
Self-selected zone
Choose up to 3 availability zones, one VM per zone
Azure-selected zone (Preview)
Let Azure assign the best zone for your needs
Using an Azure-selected zone is not supported in region 'Central India'.
Availability zone *
Zone 1
You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more
Security type
Trusted launch virtual machines
Configure security features
< Previous
Next : Disks >
Review + create
Give feedback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Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Security type
Trusted launch virtual machines
Configure security features
Image *
Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2 (free
See all images | Configure VM generation
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Run with Azure Spot discount
You are in the free trial period. Cost associated with this VM can be covered by any remaining credits on your subscription.
Learn more
Size *
Standard_D2s_v3 - 2 vcpus, 8 GiB memory ($143.81/month)
See all sizes
Enable Hibernation
Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernate to enable this feature. Learn more
Administrator account
Username *
IaCAdmin-1
Password *
Confirm password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
< Previous
Next : Disks >
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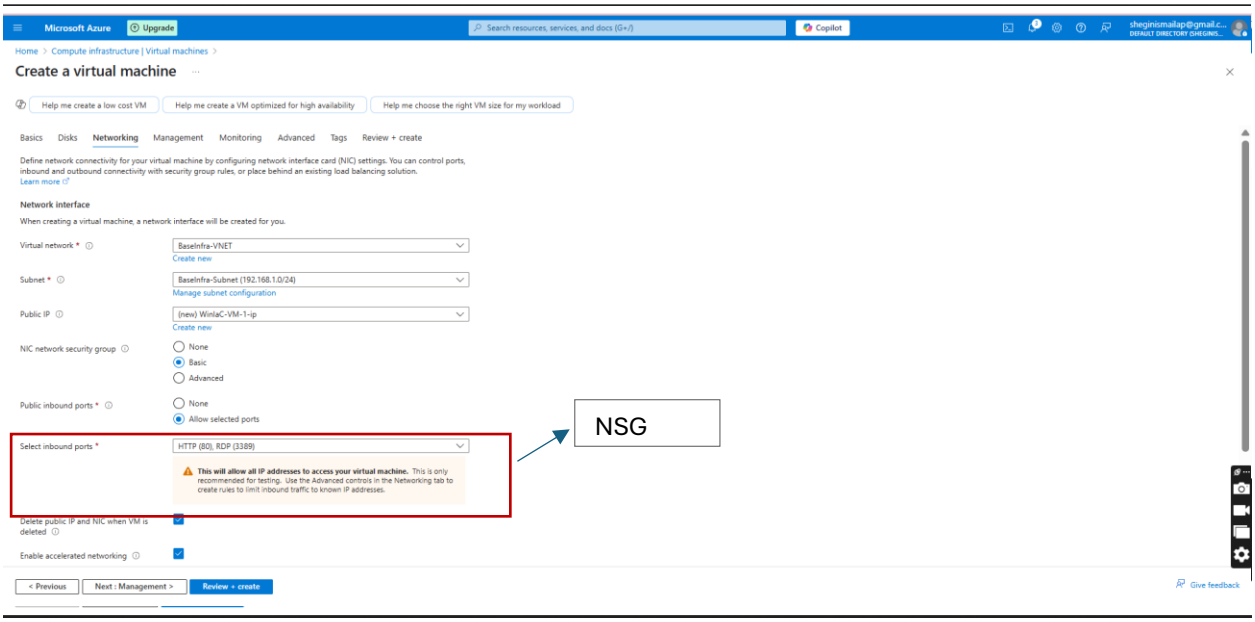
Give feedback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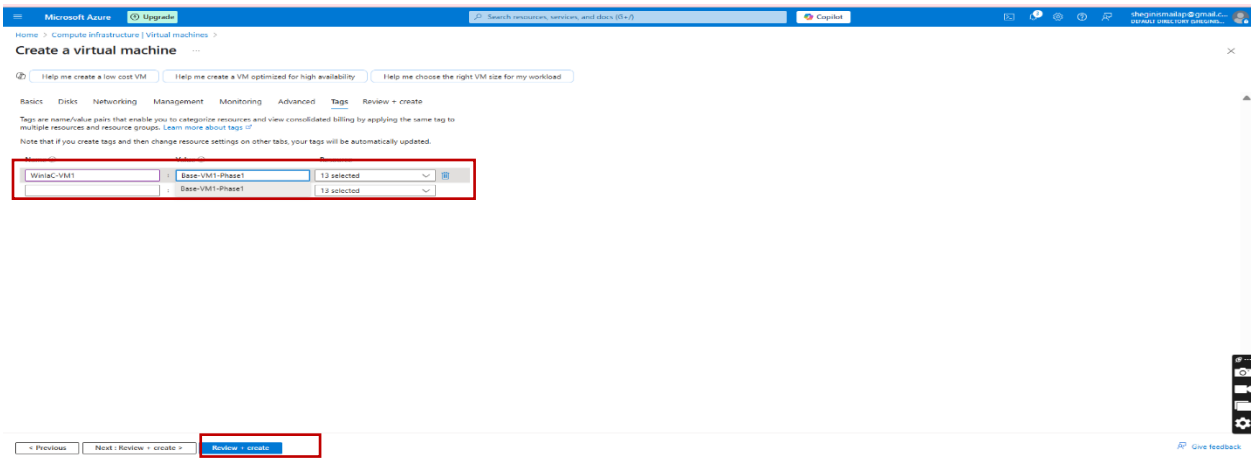
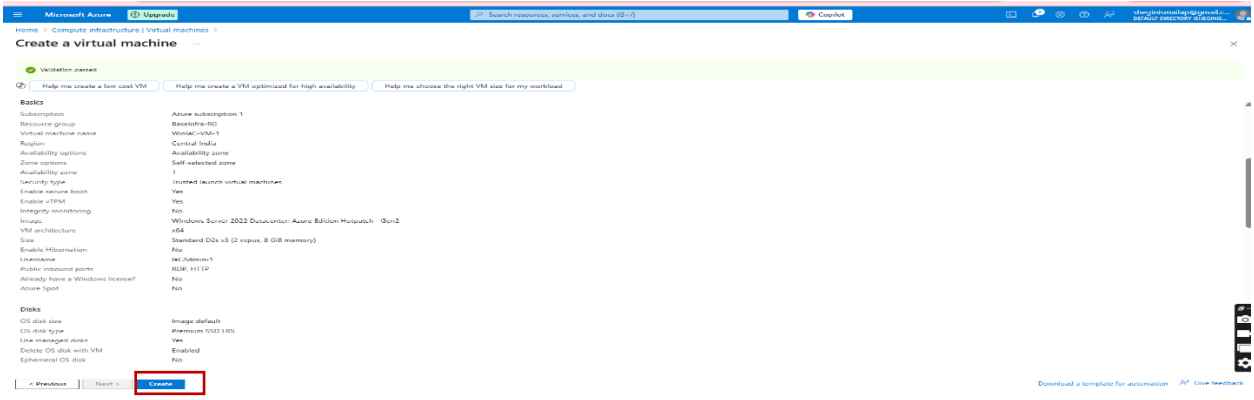
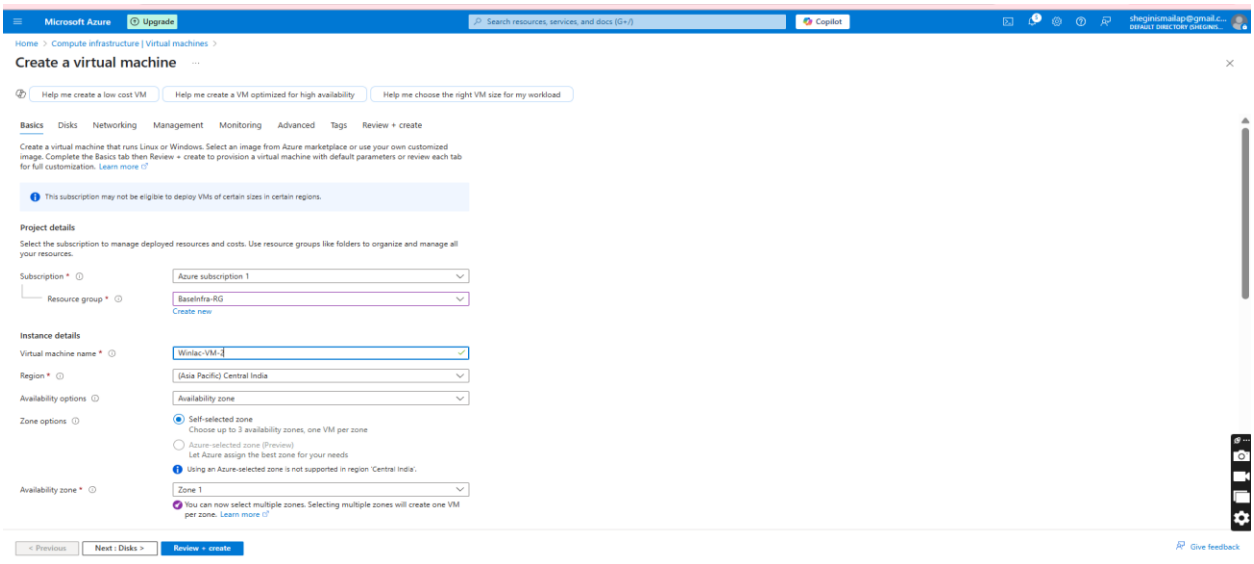
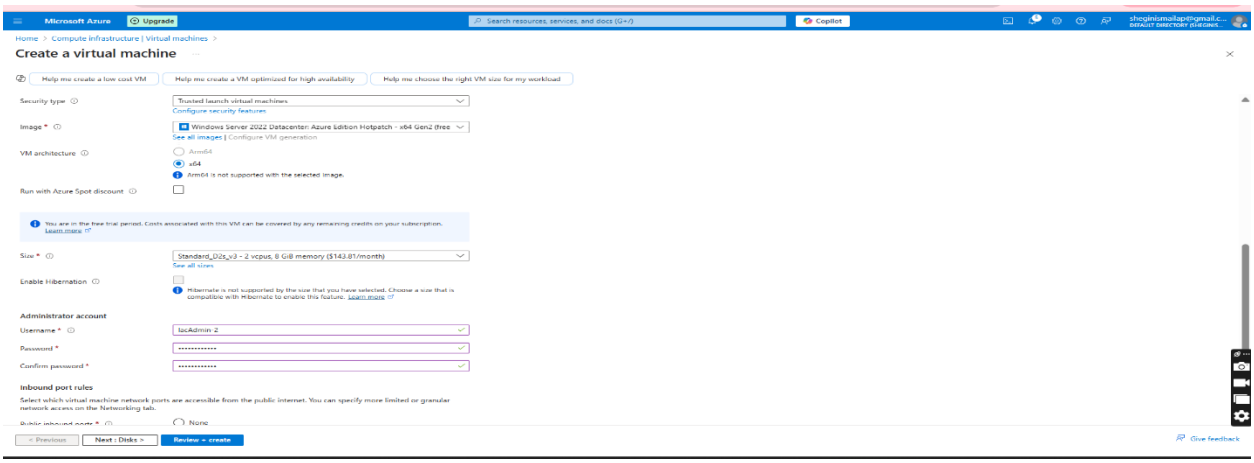
Create a virtual machine
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more
Network interface
When creating a virtual machine, a network interface will be created for you.
Virtual network *
BaseInfra-VNET
Create new
Subnet *
BaseInfra-Subnet (192.168.1.0/24)
Manage subnet configuration
Public IP
(new) WinIaC-VM-1-ip
Create new
NIC network security group
None
Basic
Advanced
Public inbound ports *
None
Allow selected ports
Select inbound ports *
HTTP (80), RDP (3389)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
Delete public IP and NIC when VM is deleted
Enable accelerated networking
NSG
< Previous
Next : Management >
Review + create
Give feedback





Second Windows VM with same configuration

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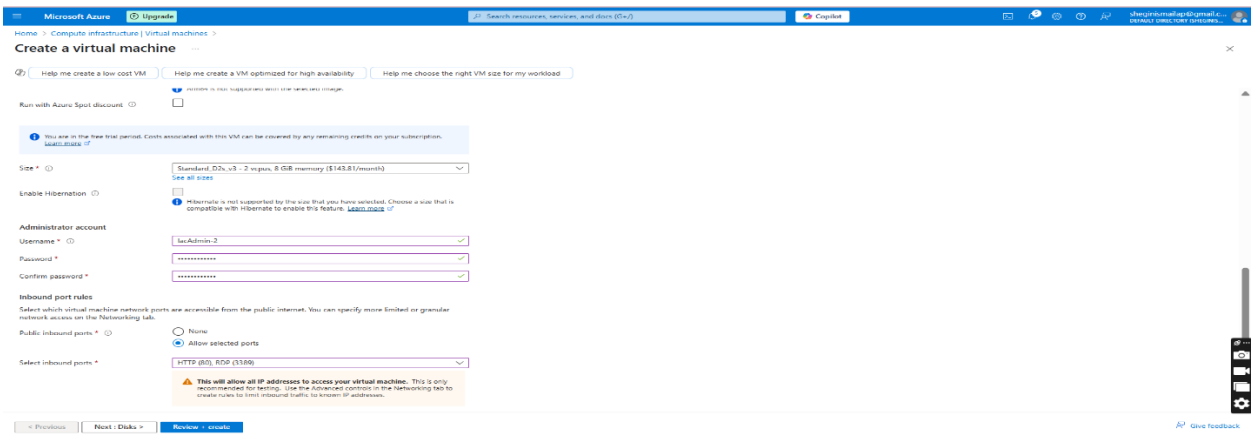
Security type
Trusted launch virtual machines
Configure security features
Image *
Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2 (free
See all images | Configure VM generation
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Run with Azure Spot discount
You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription.
Learn more
Size *
Standard_D2s_v3 - 2 vcpus, 8 GiB memory ($143.81/month)
See all sizes
Enable Hibernation
Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernate to enable this feature. Learn more
Administrator account
Username *
lacAdmin-2
Password *
Confirm password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
Public inbound ports *
None
< Previous
Next : Disks >
Review + create
Give feedback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Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernate to enable this feature. Learn more
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Username *
lacAdmin-2
Password *
Confirm password *
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Public inbound ports *
None
Allow selected ports
Select inbound ports *
HTTP (80), RDP (3389)
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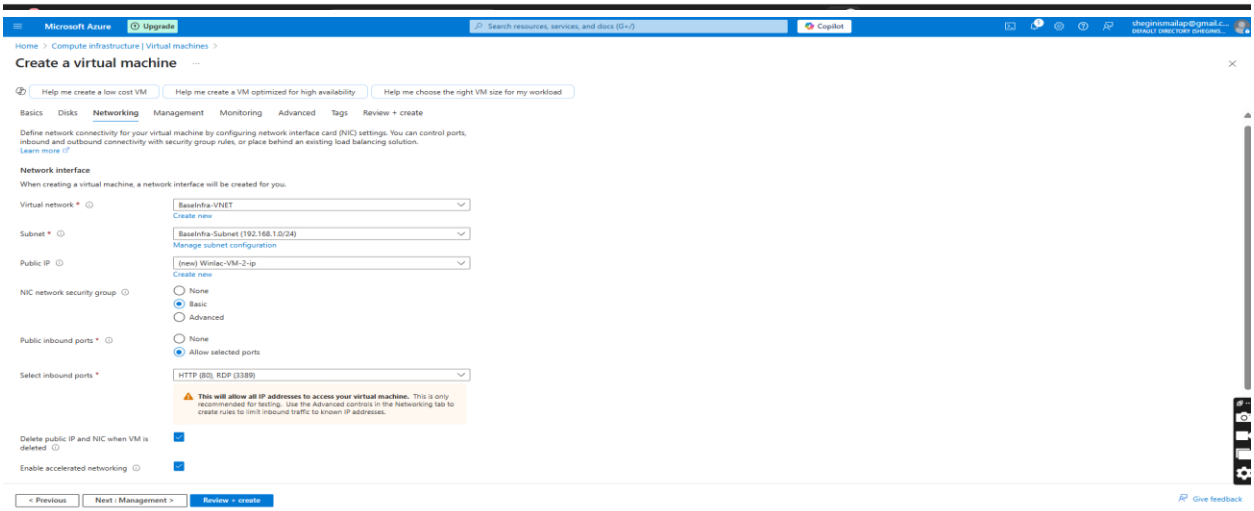
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Help me choose the right VM size for my workload
Basics
Disks
Networking
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Monitoring
Advanced
Tags
Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
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Virtual network *
BaseInfra-VNET
Create new
Subnet *
BaseInfra-Subnet (192.168.1.0/24)
Manage subnet configuration
Public IP
(new) Winlac-VM-2-ip
Create new
NIC network security group
None
Basic
Advanced
Public inbound ports *
None
Allow selected ports
Select inbound ports *
HTTP (80), RDP (3389)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
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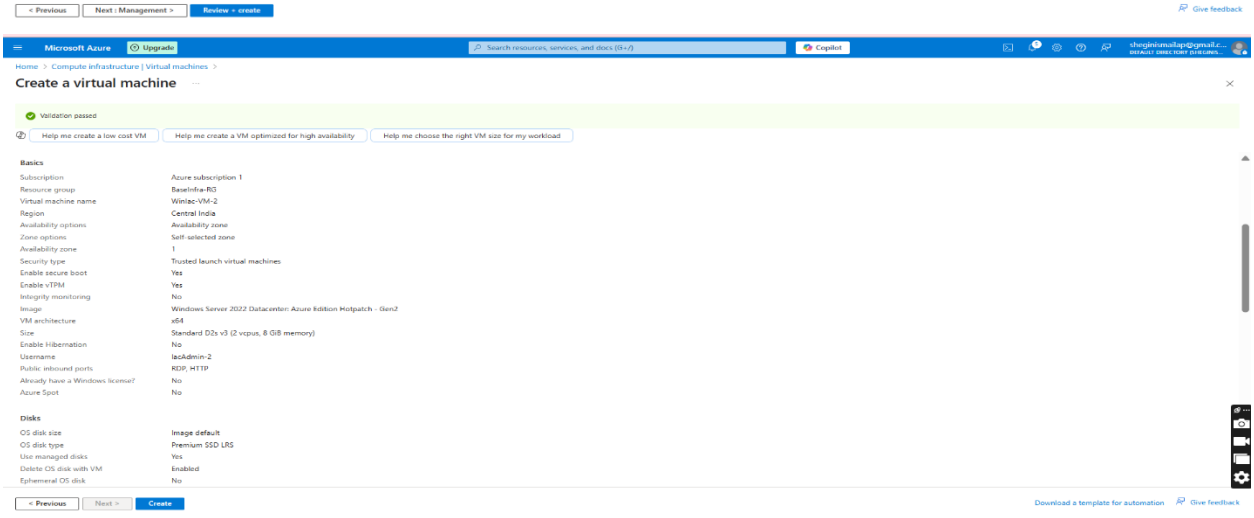
Enable accelerated networking
< Previous
Next : Management >
Review + create
Give feedback


Microsoft Azure
Upgrade
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Copilot
Home > Compute infrastructure | Virtual machines >
Create a virtual machine
Validation passed
Help me create a low cost VM
Help me create a VM optimized for high availability
Help me choose the right VM size for my workload
Basics
Subscription
Azure subscription 1
Resource group
BaseInfra-RG
Virtual machine name
Winlac-VM-2
Region
Central India
Availability options
Availability zone
Zone options
Self-selected zone
Availability zone
1
Security type
Trusted launch virtual machines
Enable secure boot
Yes
Enable vTPM
Yes
Integrity monitoring
No
Image
Windows Server 2022 Datacenter: Azure Edition Hotpatch - Gen2
VM architecture
x64
Size
Standard D2s v3 (2 vcpus, 8 GiB memory)
Enable Hibernation
No
Username
lacAdmin-2
Public inbound ports
RDP, HTTP
Already have a Windows license?
No
Azure Spot
No
Disks
OS disk size
Image default
OS disk type
Premium SSD LRS
Use managed disks
Yes
Delete OS disk with VM
Enabled
Ephemeral OS disk
No
< Previous
Next >
Create
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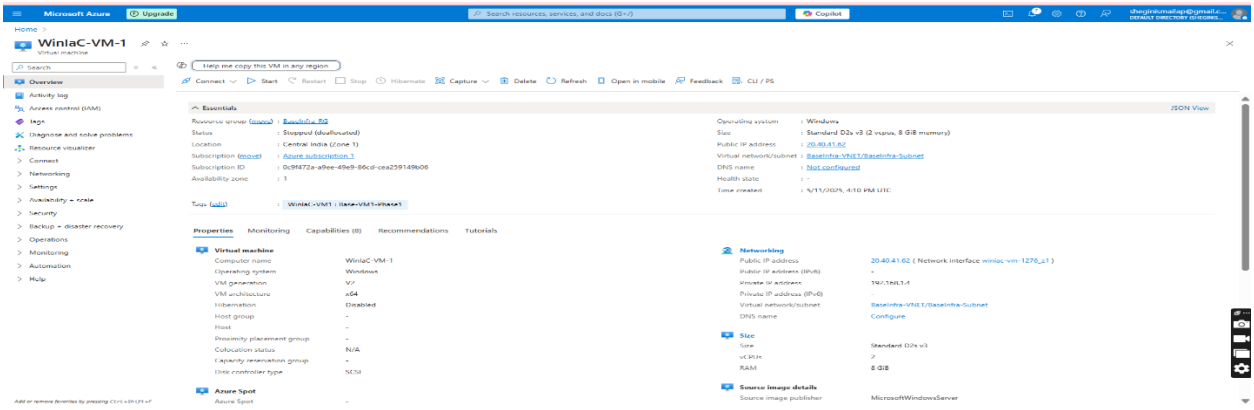
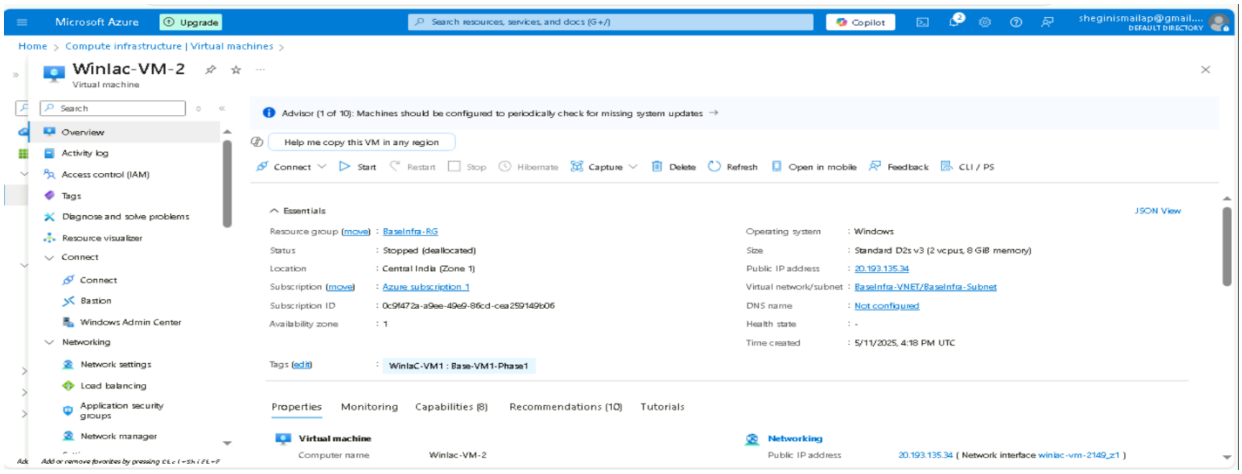
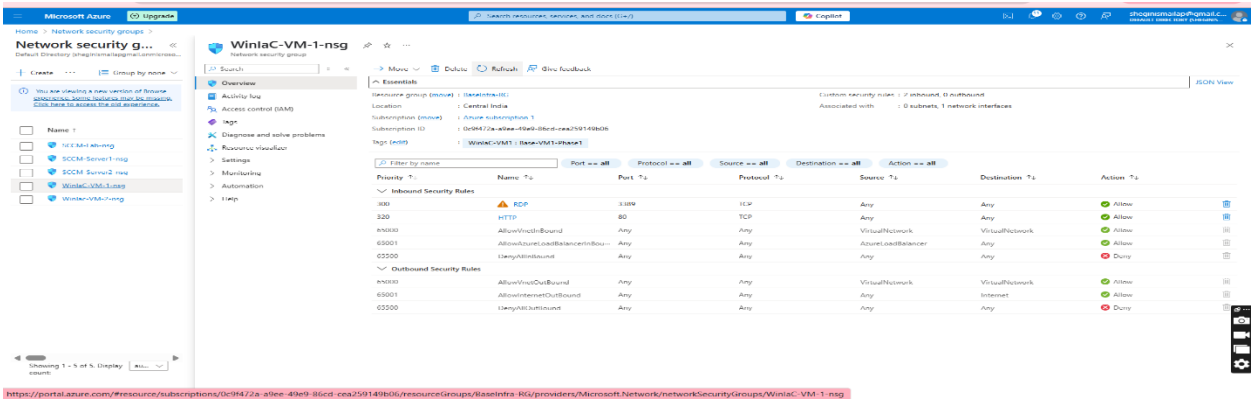
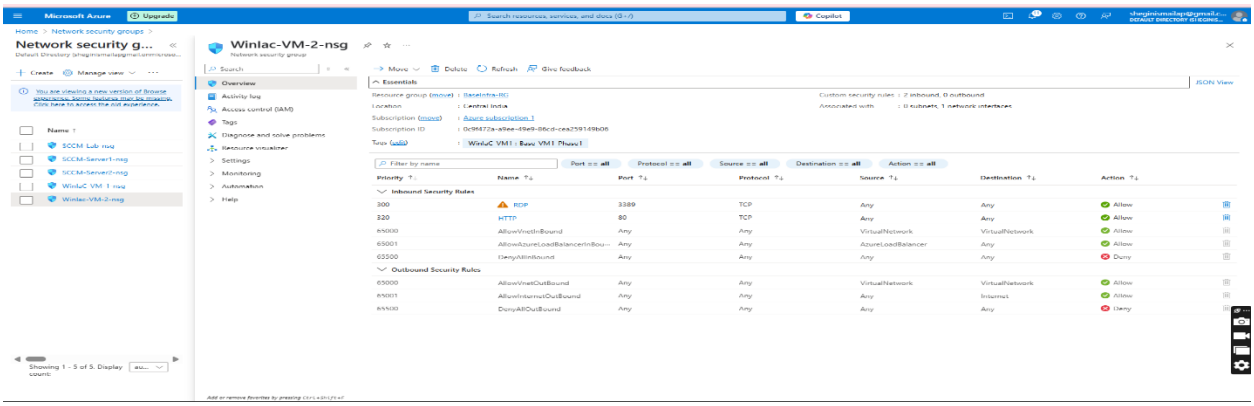
RESULT


Microsoft Azure
WinlaC-VM-1
Virtual machine
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Connect
Networking
Settings
Availability + scale
Security
Backup + disaster recovery
Operations
Monitoring
Automation
Help
Help me copy this VM in any region
Connect
Start
Restart
Stop
Hibernate
Capture
Delete
Refresh
Open in mobile
Feedback
CLI / PS
Essentials
JSON View
Resource group (move) : BaseInfra-RG
Status : Stopped (deallocated)
Location : Central India (Zone 1)
Subscription (move) : Azure subscription 1
Subscription ID : 0c9f472a-a9ee-49e9-86cd-cea259149b06
Availability zone : 1
Tags (edit) : WinlaC-VM1 : Base-VM1-Phase1
Operating system : Windows
Size : Standard D2s v3 (2 vcpus, 8 GiB memory)
Public IP address : 20.40.41.62
Virtual network/subnet : BaseInfra-VNET/BaseInfra-Subnet
DNS name : Not configured
Health state : -
Time created : 5/11/2025, 4:10 PM UTC
Properties
Monitoring
Capabilities (8)
Recommendations
Tutorials
Virtual machine
Computer name WinlaC-VM-1
Operating system Windows
VM generation V2
VM architecture x64
Hibernation Disabled
Host group -
Host -
Proximity placement group -
Colocation status N/A
Capacity reservation group -
Disk controller type SCSI
Azure Spot
Azure Spot -
Networking
Public IP address 20.40.41.62 (Network interface winlac-vm-1276_z1)
Public IP address (IPv6) -
Private IP address 192.168.1.4
Private IP address (IPv6) -
Virtual network/subnet BaseInfra-VNET/BaseInfra-Subnet
DNS name Configure
Size
Size Standard D2s v3
vCPUs 2
RAM 8 GiB
Source image details
Source image publisher MicrosoftWindowsServer


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > Compute infrastructure | Virtual machines >
Winlac-VM-2
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Connect
Connect
Bastion
Windows Admin Center
Networking
Network settings
Load balancing
Application security groups
Network manager
Advisor (1 of 10): Machines should be configured to periodically check for missing system updates
Help me copy this VM in any region
Connect
Start
Restart
Stop
Hibernate
Capture
Delete
Refresh
Open in mobile
Feedback
CLI / PS
Essentials
JSON View
Resource group (move) : BaseInfra-RG
Status : Stopped (deallocated)
Location : Central India (Zone 1)
Subscription (move) : Azure subscription 1
Subscription ID : 0c9f472a-a9ee-49e9-86cd-cea259149b06
Availability zone : 1
Tags (edit) : WinlaC-VM1 : Base-VM1-Phase1
Operating system : Windows
Size : Standard D2s v3 (2 vcpus, 8 GiB memory)
Public IP address : 20.193.135.34
Virtual network/subnet : BaseInfra-VNET/BaseInfra-Subnet
DNS name : Not configured
Health state : -
Time created : 5/11/2025, 4:18 PM UTC
Properties
Monitoring
Capabilities (8)
Recommendations (10)
Tutorials
Virtual machine
Computer name Winlac-VM-2
Networking
Public IP address 20.193.135.34 (Network interface winlac-vm-2149_z1)


Microsoft Azure
Upgrade
Copilot
Network security g...
WinlaC-VM-1-nsg
Network security group
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
Move
Delete
Refresh
Give feedback
Essentials
JSON View
Location : Central India
Associated with : 0 subnets, 1 network interfaces
Tags (edit) : WinlaC-VM1 : Base-VM1-Phase1
Inbound Security Rules
Outbound Security Rules
https://portal.azure.com/#resource/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/BaseInfra-RG/providers/Microsoft.Network/networkSecurityGroups/WinlaC-VM-1-nsg


Microsoft Azure
Upgrade
Copilot
Network security g...
Winlac-VM-2-nsg
Network security group
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
Move
Delete
Refresh
Give feedback
Essentials
JSON View
Location : Central India
Associated with : 0 subnets, 1 network interfaces
Inbound Security Rules
Outbound Security Rules

Steps To implement a **Layer 4 Load Balancer** for virtual machines:

Step 1: Create a Load Balancer

1. Open **Azure Portal** (https://portal.azure.com).
2. Navigate to **Load Balancers**.
3. Click **Create**.
4. Select a **Resource Group**.
5. Choose **Public or Internal Load Balancer** depending on requirements.
6. Configure **Region** and **SKU** (Standard or Basic).
7. Click **Review + Create** → **Create**.

Step 2: Configure Frontend IP

1. Open the **Load Balancer**.
2. Navigate to **Frontend IP Configuration**.
3. Click **Add**.
4. Choose **Public or Private IP**.
5. Click **Save**.

Step 3: Set Up Backend Pool

1. Navigate to **Backend Pools**.
2. Click **Add**.
3. Select **Virtual Machines** to add to the pool.
4. Click **Save**.

Step 4: Create a Health Probe

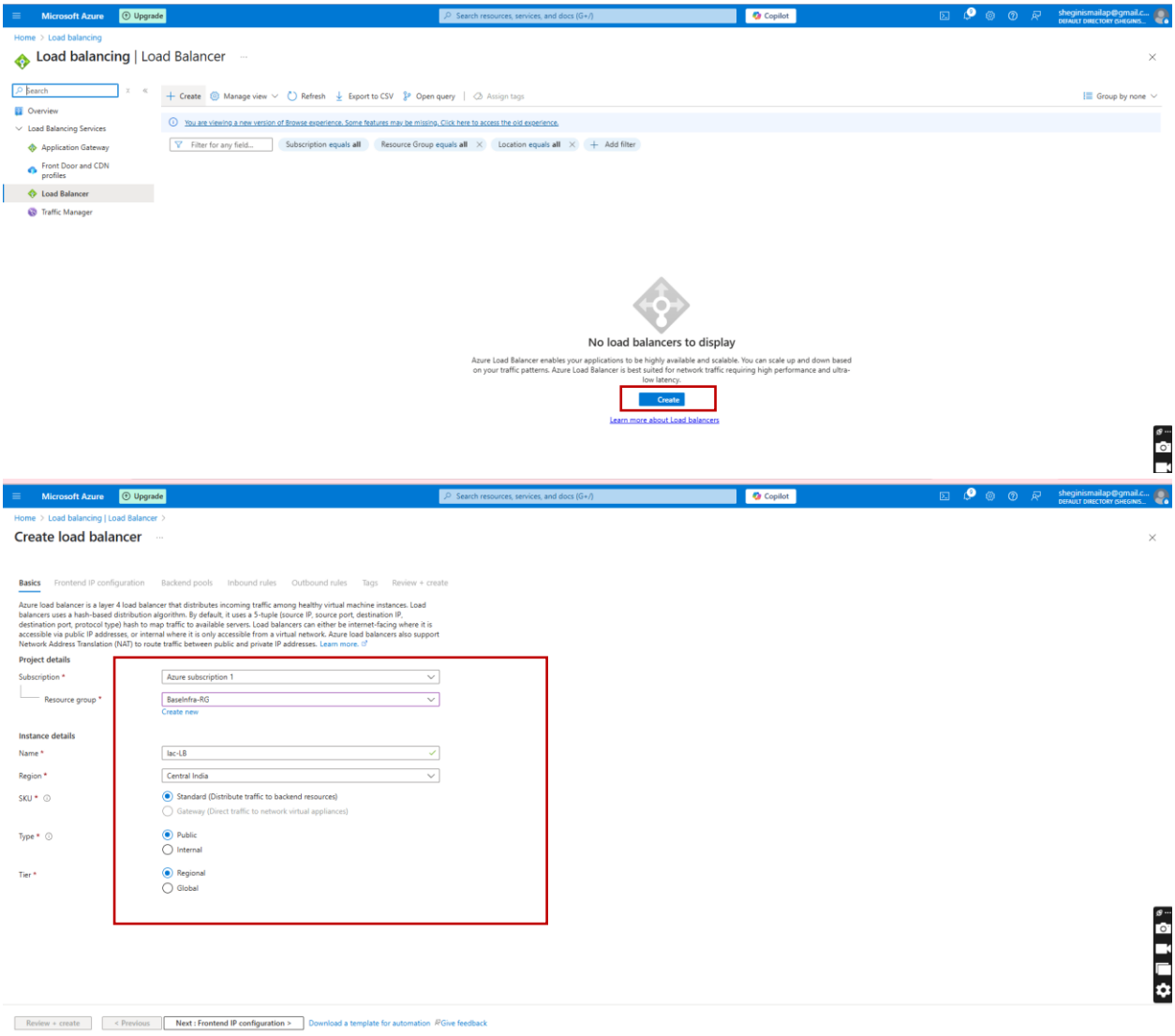
1. Navigate to **Health Probes**.
2. Click **Add**.
3. Choose **Protocol** (TCP, HTTP, or HTTPS).
4. Set **Port** (e.g., 80 for web traffic).
5. Define probe settings (interval, unhealthy threshold).
6. Click **Save**.

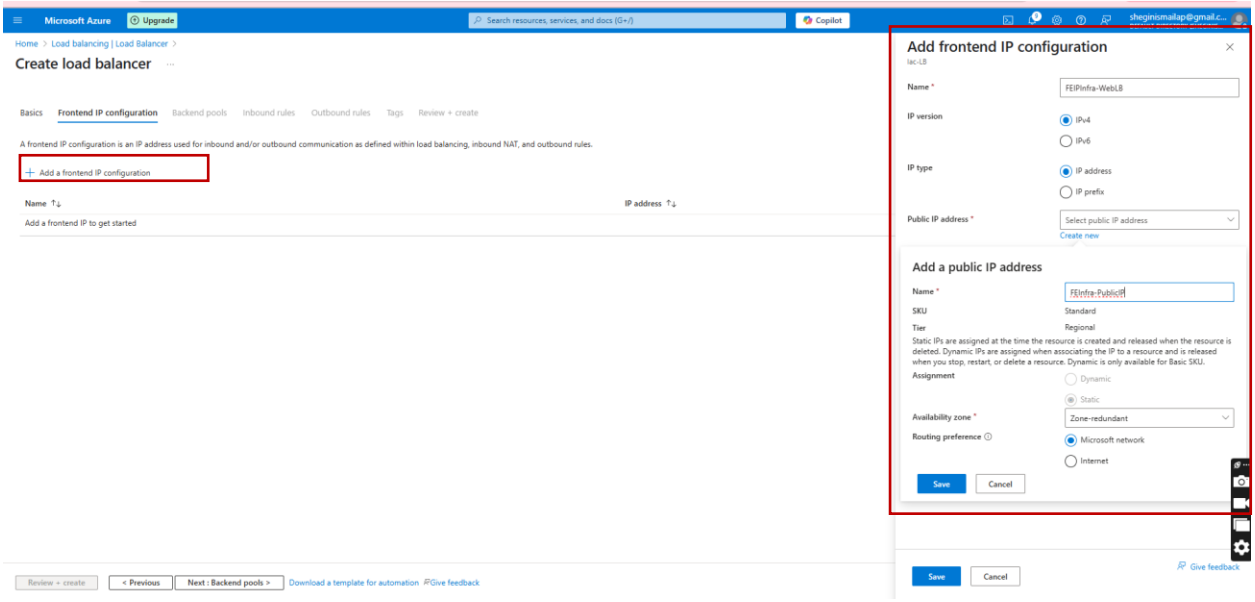
Step 5: Configure Load Balancing Rules

1. Navigate to **Load Balancing Rules**.
2. Click **Add**.
3. Set **Frontend IP, Backend Pool, and Health Probe**.
4. Choose **Protocol (TCP/UDP)** and **Port (e.g., 80, 443, or custom)**.
5. Configure **Session Persistence** if needed.
6. Click **Save**.

Step 6: Test and Validate

- Install Web Server(IIS) on both VM's through Add Roles and Features
- Deploy a **basic HTML page** with different backgrounds on each IIS as default document.
- Access the **Load Balancer Frontend IP** in a browser.
- Refresh to confirm traffic distribution




Microsoft Azure
Upgrade
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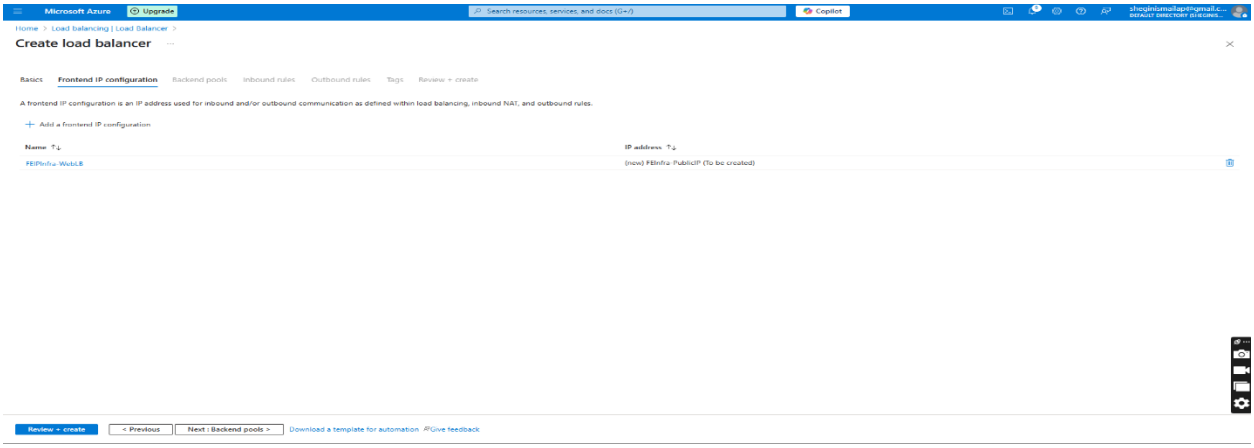
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.
Add a frontend IP configuration
Name
IP address
Add a frontend IP to get started
Review + create
< Previous
Next : Backend pools >
Download a template for automation
Give feedback
Add frontend IP configuration
lac-LB
Name *
FEIPInfra-WebLB
IP version
IPv4
IPv6
IP type
IP address
IP prefix
Public IP address *
Select public IP address
Create new
Add a public IP address
Name *
FEInfra-PublicIP
SKU
Standard
Tier
Regional
Static IPs are assigned at the time the resource is created and released when the resource is deleted. Dynamic IPs are assigned when associating the IP to a resource and is released when you stop, restart, or delete a resource. Dynamic is only available for Basic SKU.
Assignment
Dynamic
Static
Availability zone *
Zone-redundant
Routing preference
Microsoft network
Internet
Save
Cancel
Give feedback


Microsoft Azure
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Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.
Add a frontend IP configuration
Name
IP address
FEIPInfra-WebLB
(new) FEInfra-PublicIP (To be created)
Review + create
< Previous
Next : Backend pools >
Download a template for automation
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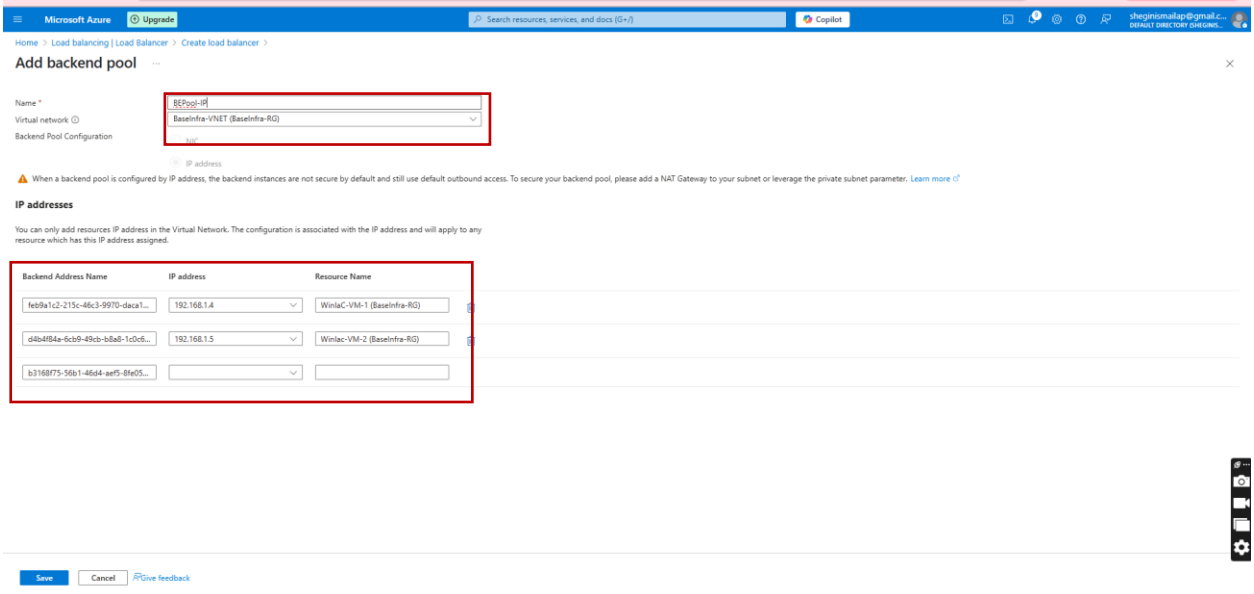
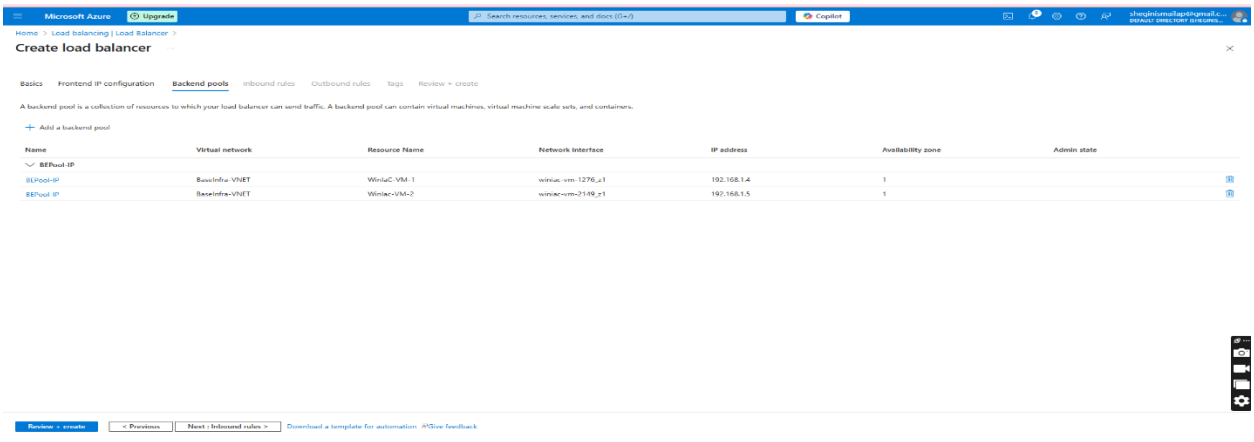

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer > Create load balancer >
Add backend pool
Name *
BEPool-IP
Virtual network
BaseInfra-VNET (BaseInfra-RG)
Backend Pool Configuration
NIC
IP address
When a backend pool is configured by IP address, the backend instances are not secure by default and still use default outbound access. To secure your backend pool, please add a NAT Gateway to your subnet or leverage the private subnet parameter. Learn more
IP addresses
You can only add resources IP address in the Virtual Network. The configuration is associated with the IP address and will apply to any resource which has this IP address assigned.
Backend Address Name
IP address
Resource Name
feb9a1c2-215c-46c3-9970-daca1...
192.168.1.4
WinlaC-VM-1 (BaseInfra-RG)
d4b4f84a-6cb9-49cb-b8a8-1c0c6...
192.168.1.5
Winlac-VM-2 (BaseInfra-RG)
b3168f75-56b1-46d4-aef5-8fe05...
Save
Cancel
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A backend pool is a collection of resources to which your load balancer can send traffic. A backend pool can contain virtual machines, virtual machine scale sets, and containers.
Add a backend pool
Name
Virtual network
Resource Name
Network interface
IP address
Availability zone
Admin state
BEPool-IP
BEPool-IP
BaseInfra-VNET
WinIaC-VM-1
winiac-vm-1276_z1
192.168.1.4
1
BEPool-IP
BaseInfra-VNET
WinIac-VM-2
winiac-vm-2149_z1
192.168.1.5
1
Review + create
< Previous
Next : Inbound rules >
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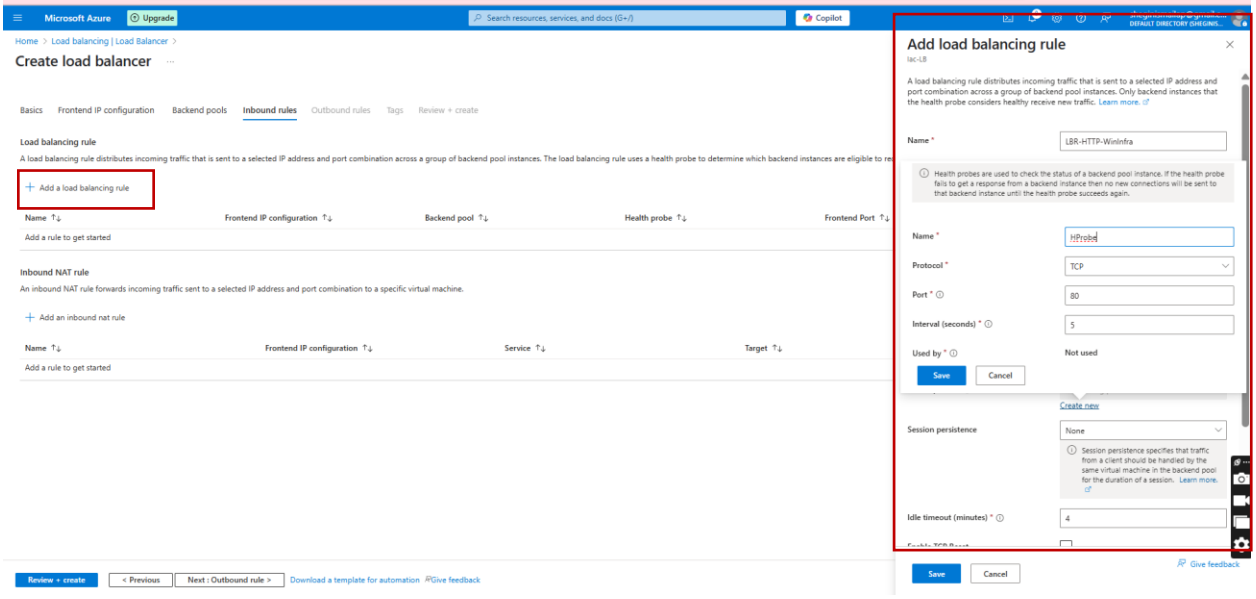
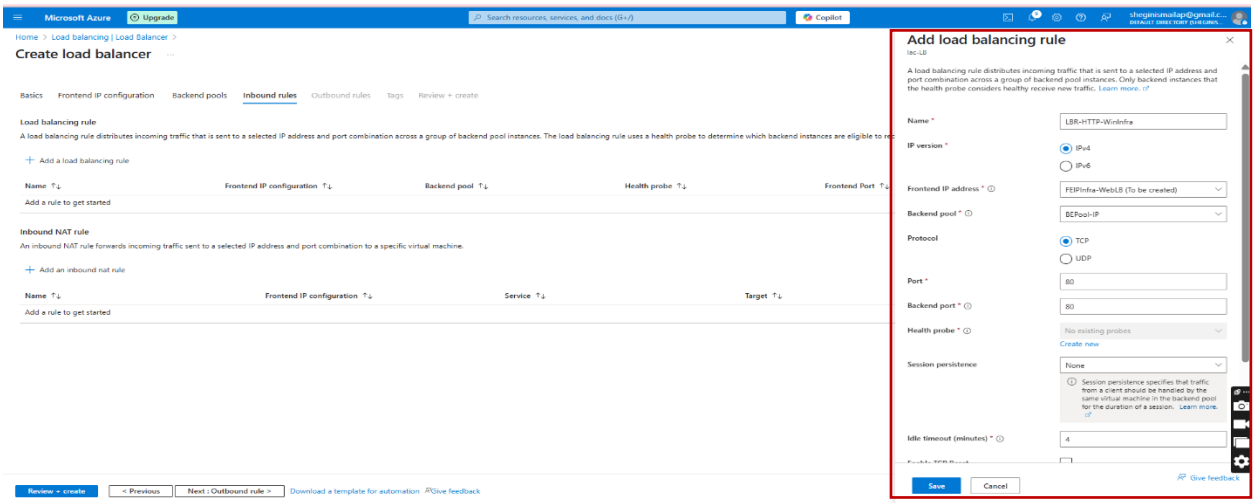

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to re
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Frontend Port
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add load balancing rule
lac-LB
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. Only backend instances that the health probe considers healthy receive new traffic. Learn more.
Name *
LBR-HTTP-WinInfra
Health probes are used to check the status of a backend pool instance. If the health probe fails to get a response from a backend instance then no new connections will be sent to that backend instance until the health probe succeeds again.
Name *
HProbe
Protocol *
TCP
Port *
80
Interval (seconds) *
5
Used by *
Not used
Save
Cancel
Create new
Session persistence
None
Session persistence specifies that traffic from a client should be handled by the same virtual machine in the backend pool for the duration of a session. Learn more.
Idle timeout (minutes) *
4
Save
Cancel
Give feedback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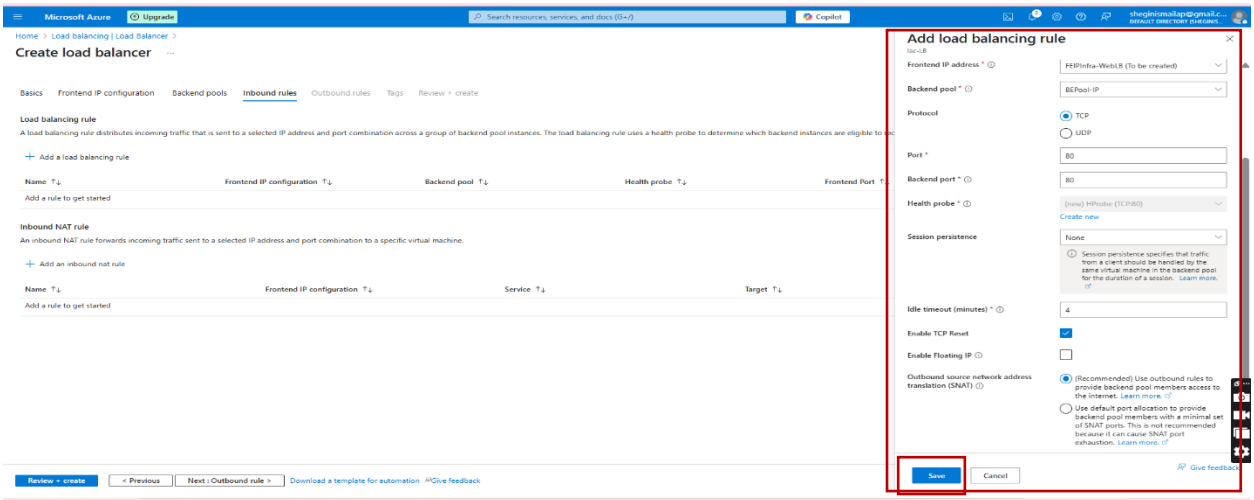
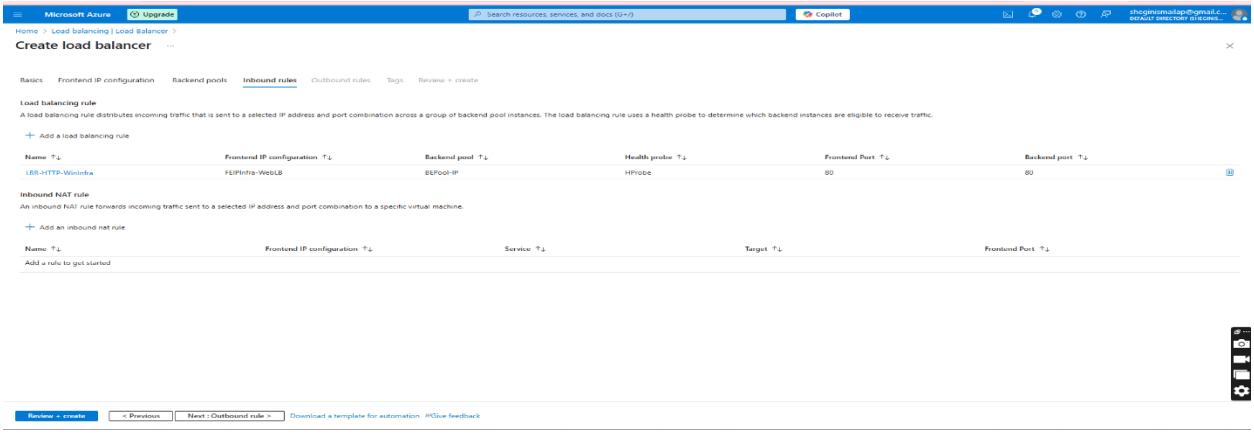
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to re
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Frontend Port
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add load balancing rule
lac-LB
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. Only backend instances that the health probe considers healthy receive new traffic. Learn more.
Name *
LBR-HTTP-WinInfra
IP version *
IPv4
IPv6
Frontend IP address *
FEIPInfra-WebLB (To be created)
Backend pool *
BEPool-IP
Protocol
TCP
UDP
Port *
80
Backend port *
80
Health probe *
No existing probes
Create new
Session persistence
None
Session persistence specifies that traffic from a client should be handled by the same virtual machine in the backend pool for the duration of a session. Learn more.
Idle timeout (minutes) *
4
Save
Cancel
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Frontend Port
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add load balancing rule
lac-LB
Frontend IP address *
FEIPInfra-WebLB (To be created)
Backend pool *
BEPool-IP
Protocol
TCP
UDP
Port *
80
Backend port *
80
Health probe *
(new) HProbe (TCP:80)
Create new
Session persistence
None
Session persistence specifies that traffic from a client should be handled by the same virtual machine in the backend pool for the duration of a session. Learn more.
Idle timeout (minutes) *
4
Enable TCP Reset
Enable Floating IP
Outbound source network address translation (SNAT)
(Recommended) Use outbound rules to provide backend pool members access to the internet. Learn more.
Use default port allocation to provide backend pool members with a minimal set of SNAT ports. This is not recommended because it can cause SNAT port exhaustion. Learn more.
Save
Cancel
Give feedback


Microsoft Azure
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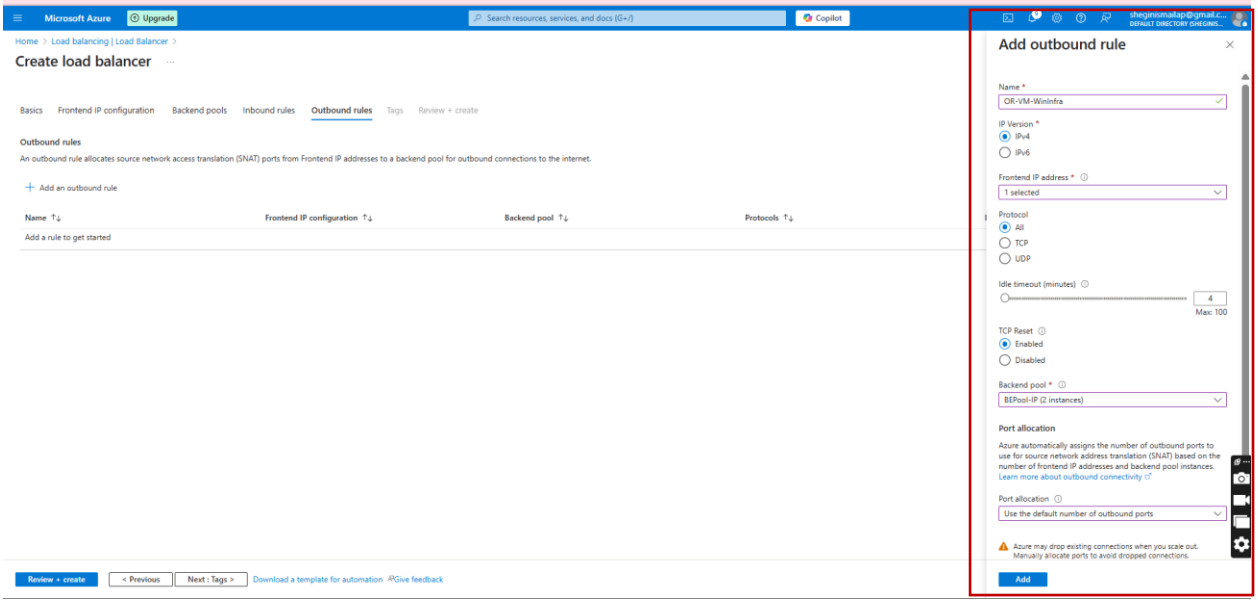
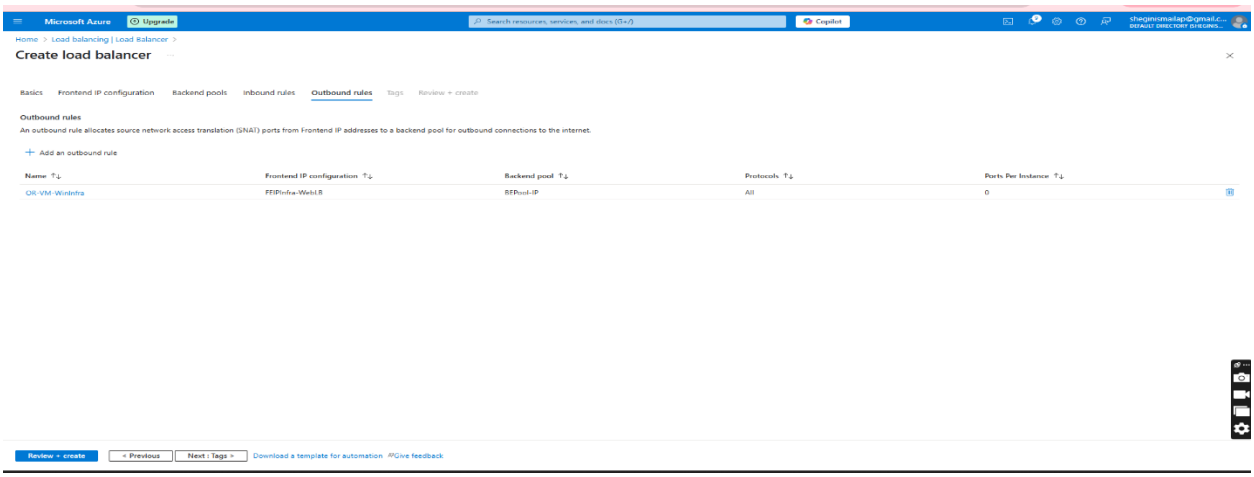
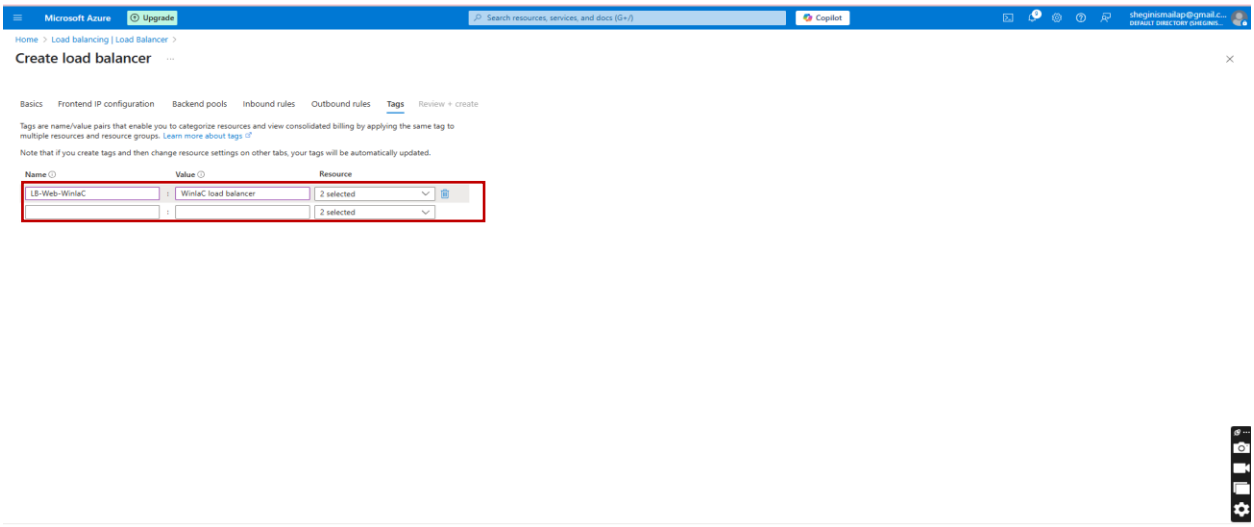
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to receive traffic.
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Frontend Port
Backend port
LBR-HTTP-WinInfra
FEIPInfra-WebLB
BEPool-IP
HProbe
80
80
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Frontend Port
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Outbound rules
An outbound rule allocates source network access translation (SNAT) ports from Frontend IP addresses to a backend pool for outbound connections to the internet.
Add an outbound rule
Name
Frontend IP configuration
Backend pool
Protocols
Add a rule to get started
Review + create
< Previous
Next : Tags >
Download a template for automation
Give feedback
Add outbound rule
Name *
OR-VM-WinInfra
IP Version *
IPv4
IPv6
Frontend IP address *
1 selected
Protocol
All
TCP
UDP
Idle timeout (minutes)
4
Max: 100
TCP Reset
Enabled
Disabled
Backend pool *
BEPool-IP (2 instances)
Port allocation
Azure automatically assigns the number of outbound ports to use for source network address translation (SNAT) based on the number of frontend IP addresses and backend pool instances.
Learn more about outbound connectivity
Port allocation
Use the default number of outbound ports
Azure may drop existing connections when you scale out. Manually allocate ports to avoid dropped connections.
Add


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Outbound rules
An outbound rule allocates source network access translation (SNAT) ports from Frontend IP addresses to a backend pool for outbound connections to the internet.
Add an outbound rule
Name
Frontend IP configuration
Backend pool
Protocols
Ports Per Instance
OR-VM-WinInfra
FEIPInfra-WebLB
BEPool-IP
All
0
Review + create
< Previous
Next : Tags >
Download a template for automation
Give feedback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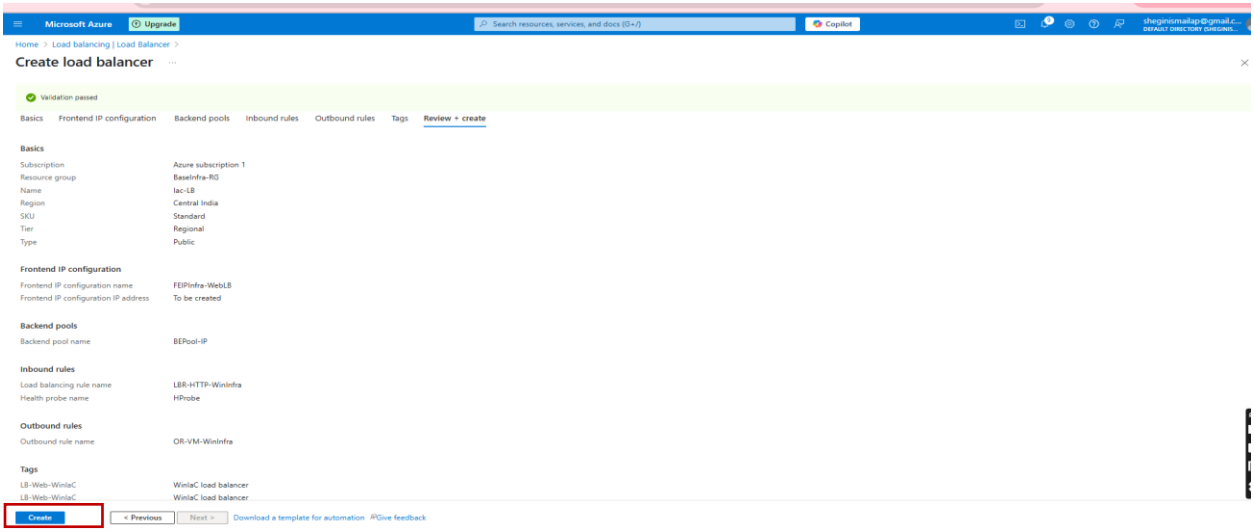
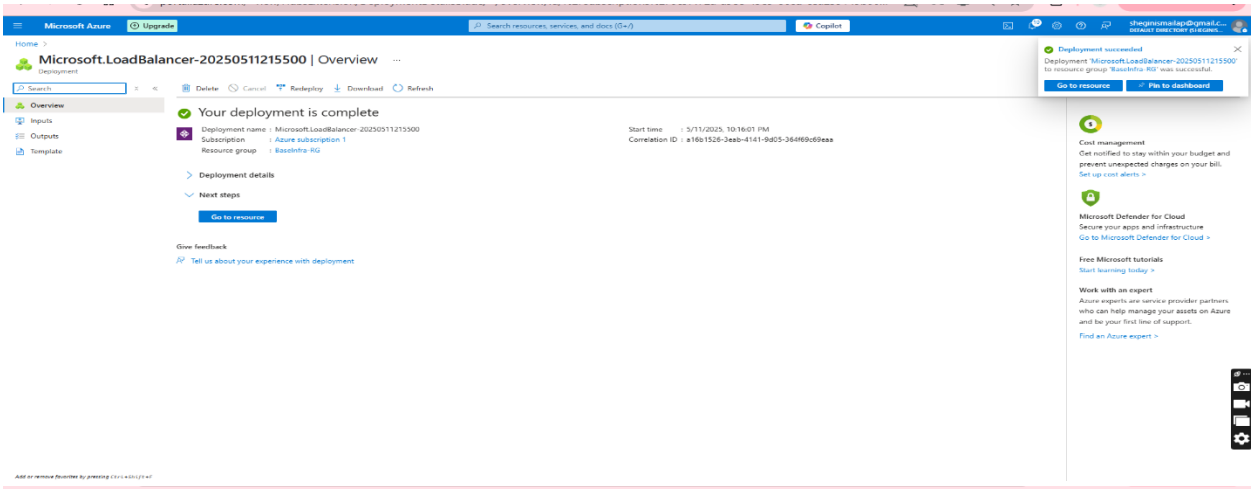
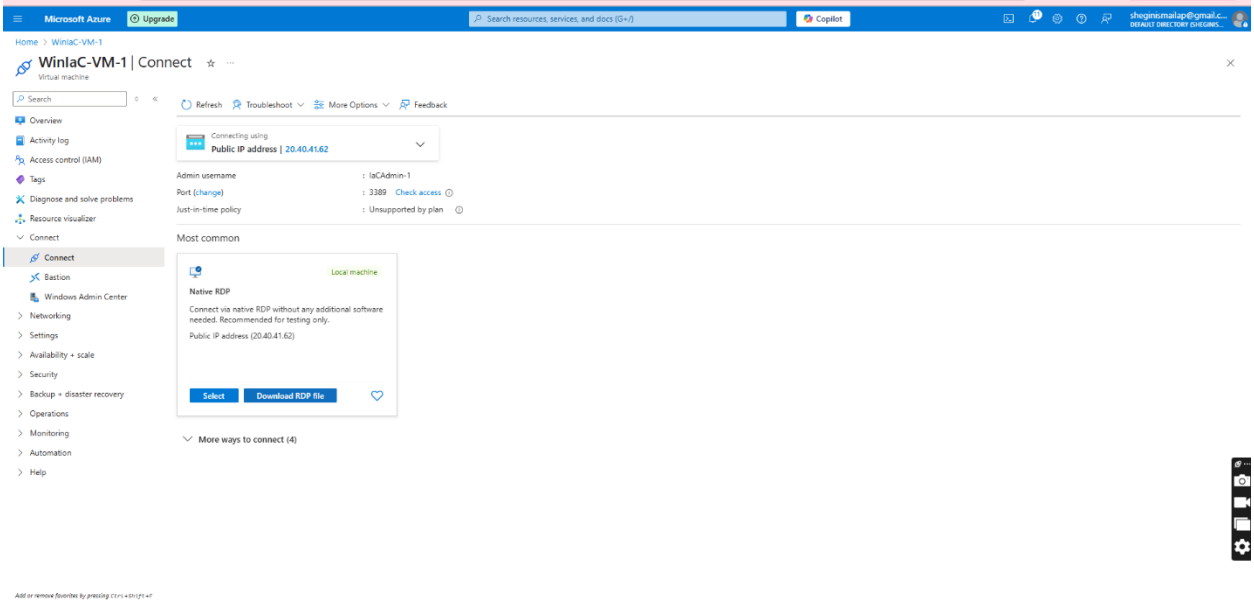
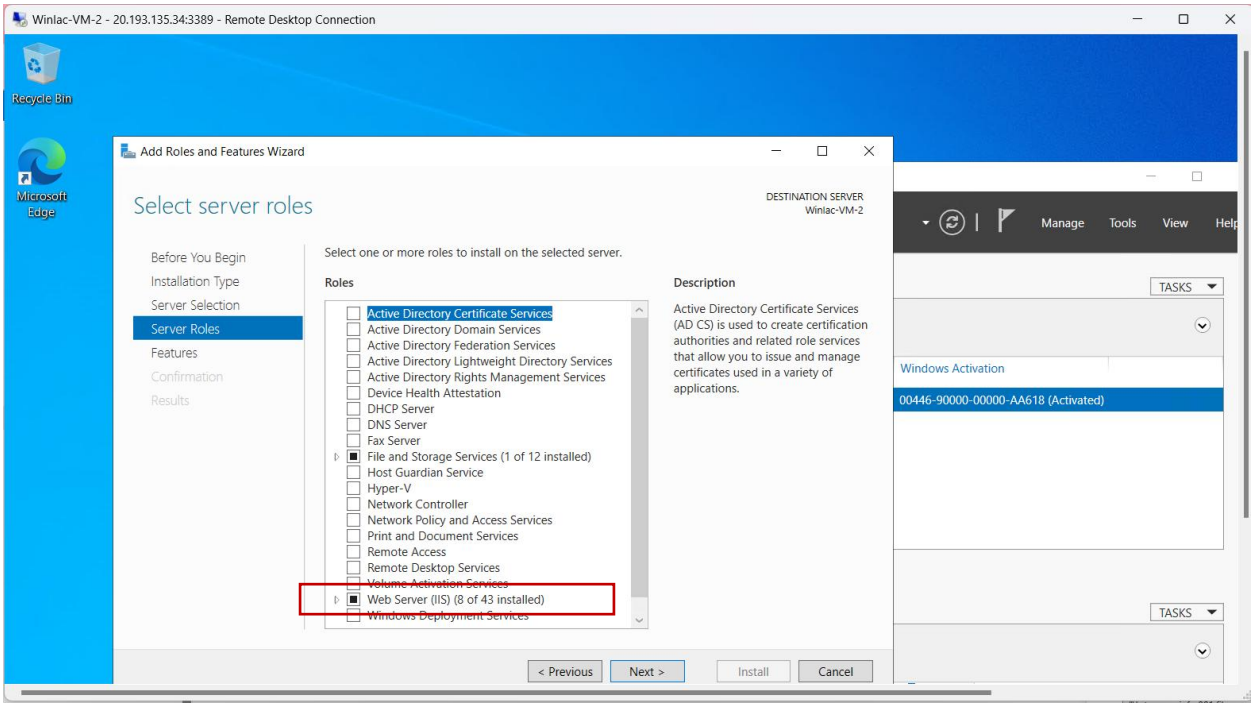
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Learn more about tags
Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.
Name
Value
Resource
LB-Web-WinIaC
WinIaC load balancer
2 selected
2 selected
Review + create
< Previous
Next : Review + create >
Download a template for automation
Give feedback

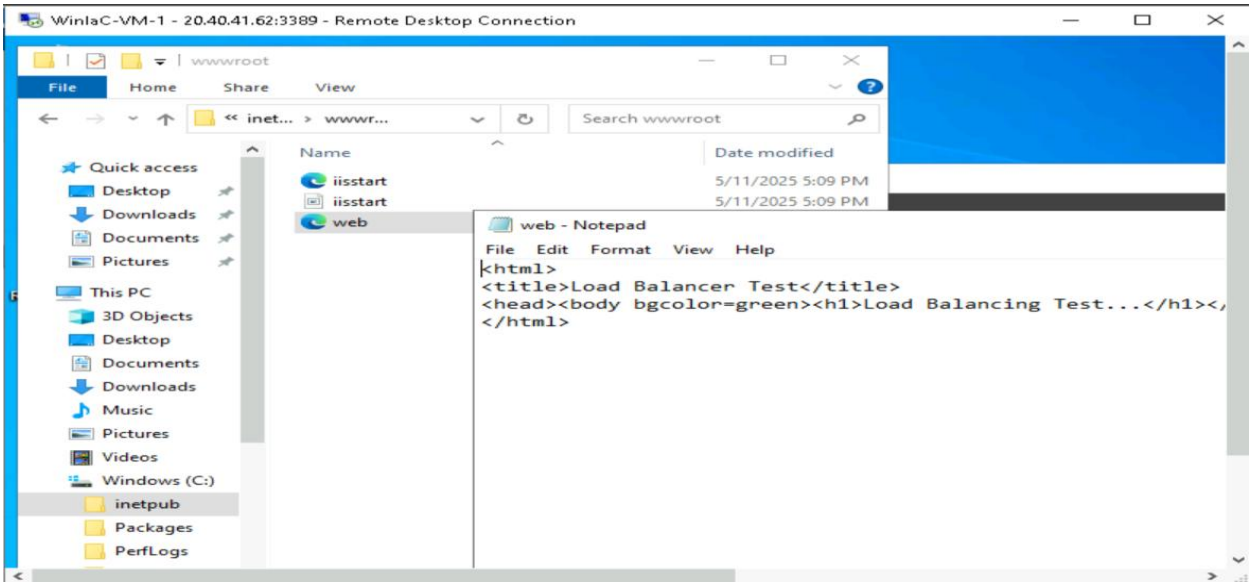

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer >
Create load balancer
Validation passed
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Basics
Subscription
Azure subscription 1
Resource group
BaseInfra-RG
Name
Iac-LB
Region
Central India
SKU
Standard
Tier
Regional
Type
Public
Frontend IP configuration
Frontend IP configuration name
FEIPInfra-WebLB
Frontend IP configuration IP address
To be created
Backend pools
Backend pool name
BEPool-IP
Inbound rules
Load balancing rule name
LBR-HTTP-WinInfra
Health probe name
HProbe
Outbound rules
Outbound rule name
OR-VM-WinInfra
Tags
LB-Web-WinIaC
WinIaC load balancer
LB-Web-WinIaC
WinIaC load balancer
Create
< Previous
Next >
Download a template for automation
Give feedback

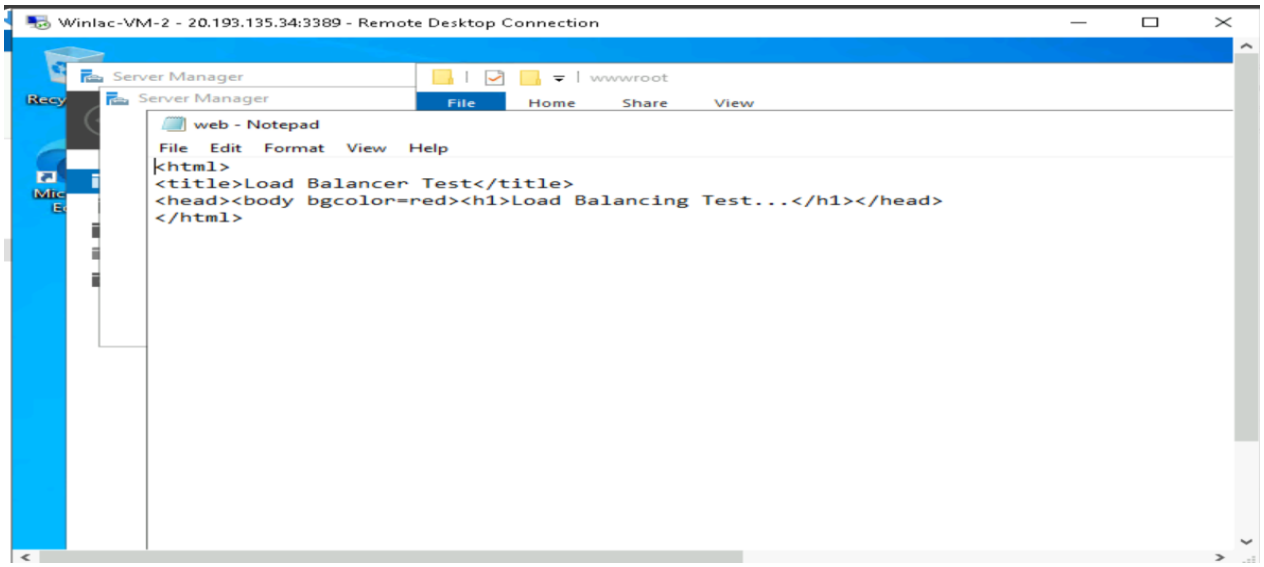


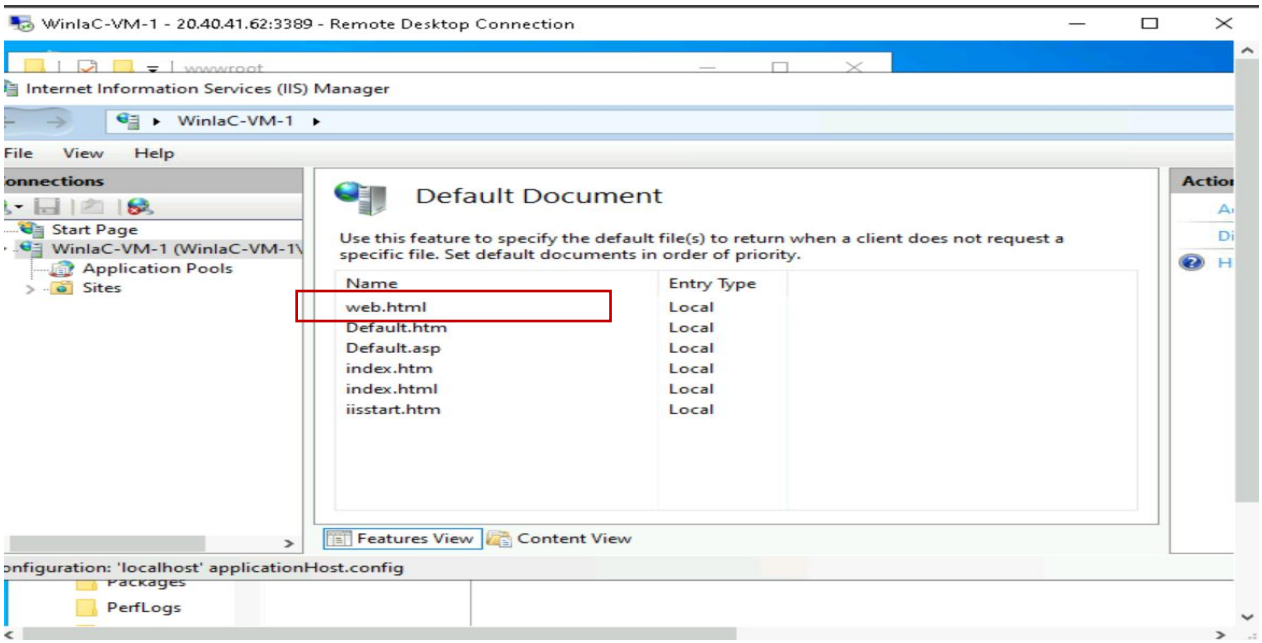


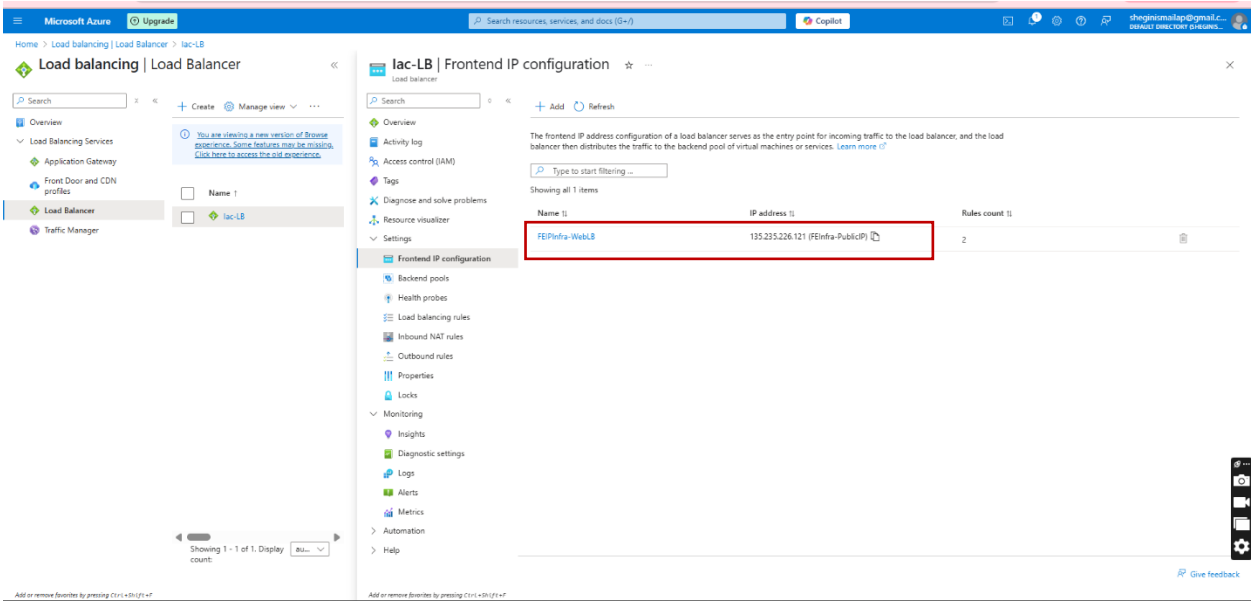
TEST AND VALIDATION




WinlaC-VM-1 - 20.40.41.62:3389 - Remote Desktop Connection
wwwroot
File
Home
Share
View
Search wwwroot
Quick access
Desktop
Downloads
Documents
Pictures
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
inetpub
Packages
PerfLogs
Name
Date modified
iisstart
5/11/2025 5:09 PM
iisstart
5/11/2025 5:09 PM
web
web - Notepad
File Edit Format View Help
<html>
<title>Load Balancer Test</title>
<head><body bgcolor=green><h1>Load Balancing Test...</h1></
</html>


Winlac-VM-2 - 20.193.135.34:3389 - Remote Desktop Connection
Server Manager
Server Manager
wwwroot
File
Home
Share
View
web - Notepad
File Edit Format View Help
<html>
<title>Load Balancer Test</title>
<head><body bgcolor=red><h1>Load Balancing Test...</h1></head>
</html>


WinlaC-VM-1 - 20.40.41.62:3389 - Remote Desktop Connection
Internet Information Services (IIS) Manager
WinlaC-VM-1
File
View
Help
Start Page
WinlaC-VM-1 (WinlaC-VM-1\
Application Pools
Sites
Default Document
Use this feature to specify the default file(s) to return when a client does not request a specific file. Set default documents in order of priority.
Name
Entry Type
web.html
Local
Default.htm
Local
Default.asp
Local
index.htm
Local
index.html
Local
iisstart.htm
Local
Features View
Content View
onfiguration: 'localhost' applicationHost.config
PerfLogs


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Load balancing | Load Balancer > Iac-LB
Load balancing | Load Balancer
Search
Create
Manage view
Overview
Load Balancing Services
Application Gateway
Front Door and CDN profiles
Load Balancer
Traffic Manager
You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.
Name
Iac-LB
Showing 1 - 1 of 1. Display count:
Iac-LB | Frontend IP configuration
Load balancer
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Frontend IP configuration
Backend pools
Health probes
Load balancing rules
Inbound NAT rules
Outbound rules
Properties
Locks
Monitoring
Insights
Diagnostic settings
Logs
Alerts
Metrics
Automation
Help
Add
Refresh
The frontend IP address configuration of a load balancer serves as the entry point for incoming traffic to the load balancer, and the load balancer then distributes the traffic to the backend pool of virtual machines or services. Learn more
Type to start filtering ...
Showing all 1 items
Name
IP address
Rules count
FEIPInfra-WebLB
135.235.226.121 (FEInfra-PublicIP)
2
Give feedback

Not secure
135.235.226.121
New Chrome available
Load Balancing Test...

Not secure
135.235.226.121
New Chrome available
Load Balancing Test...

steps to create Azure Storage with **Blob, File Share, and SAS key**:

1. Create a Storage Account

- Open **Azure Portal** and go to **Storage Accounts**.
- Click **Create**, select **Resource Group**, and set a unique **Storage Account Name**.
- Choose **Region, Performance (Standard/Premium), and Redundancy (LRS/ZRS/GRS)**.
- Click **Review + Create** → **Create**.

2. Configure Blob Storage

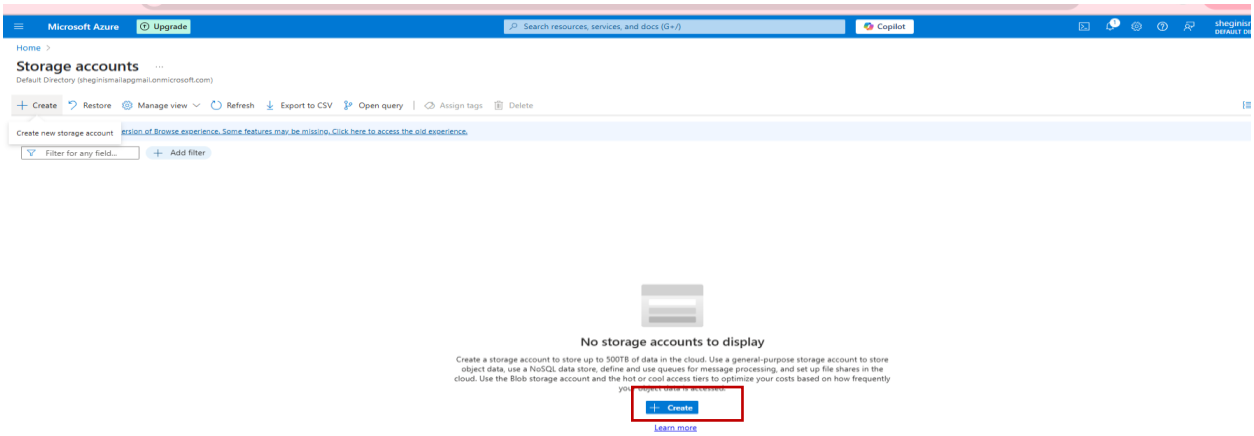
- Open the **Storage Account** and navigate to **Blob Service**.
- Click **Containers** → **Create a new container** (set access level as needed).
- Upload files to the container.

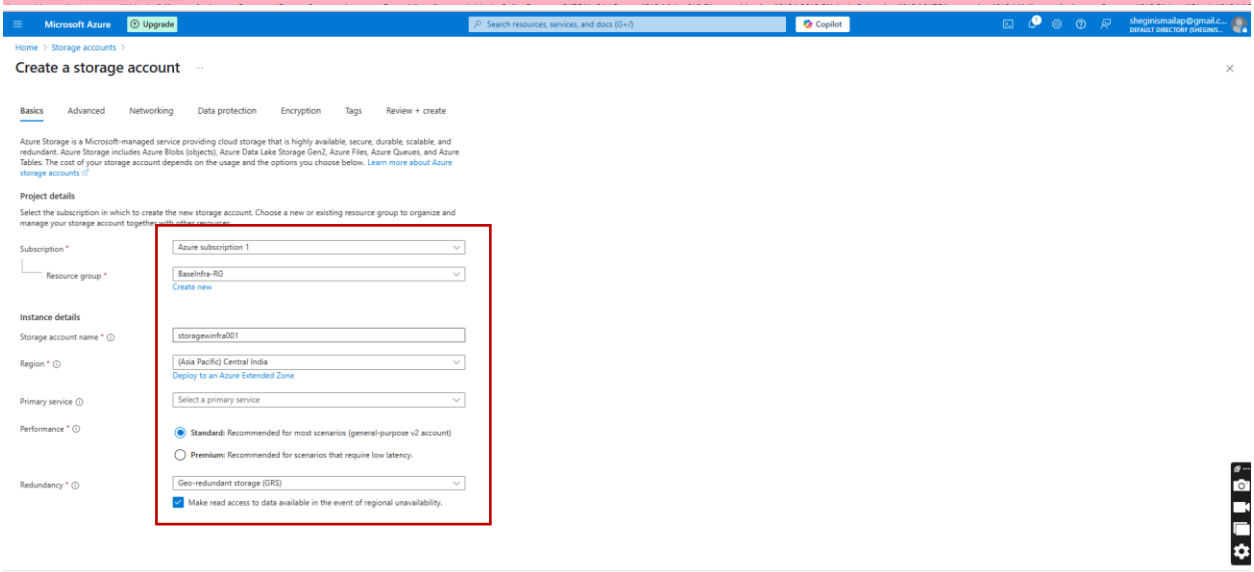
3. Set Up File Share

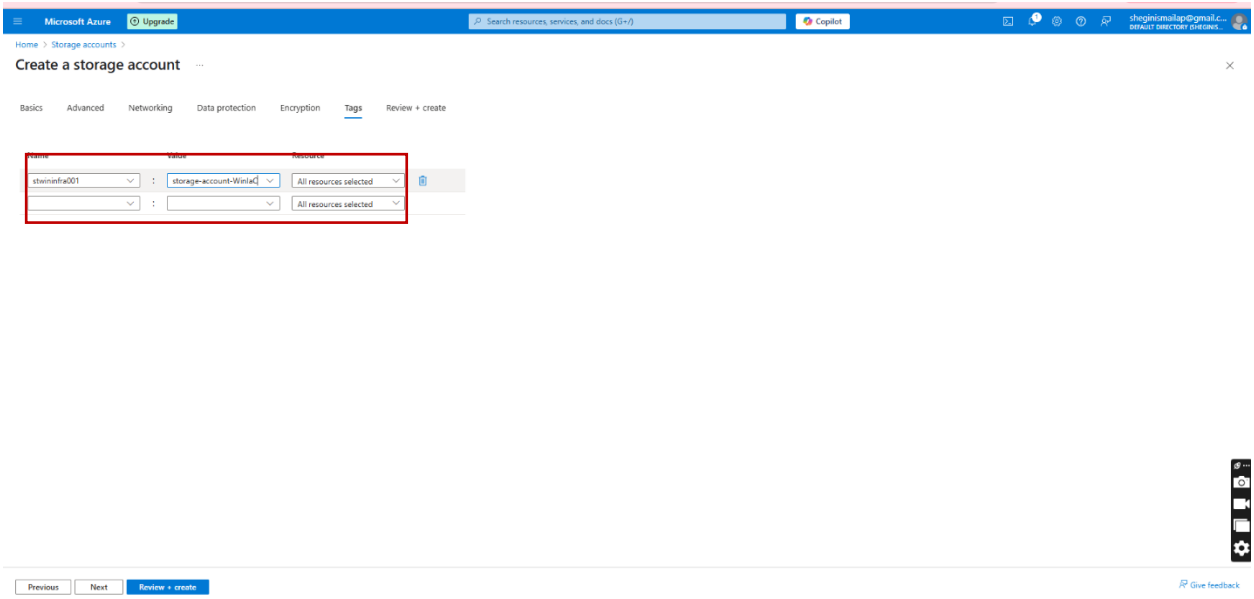
- Go to **File Shares** under the **Storage Account**.
- Click **Create a File Share** and set **Quota (Size Limit)**.
- Upload and manage files.

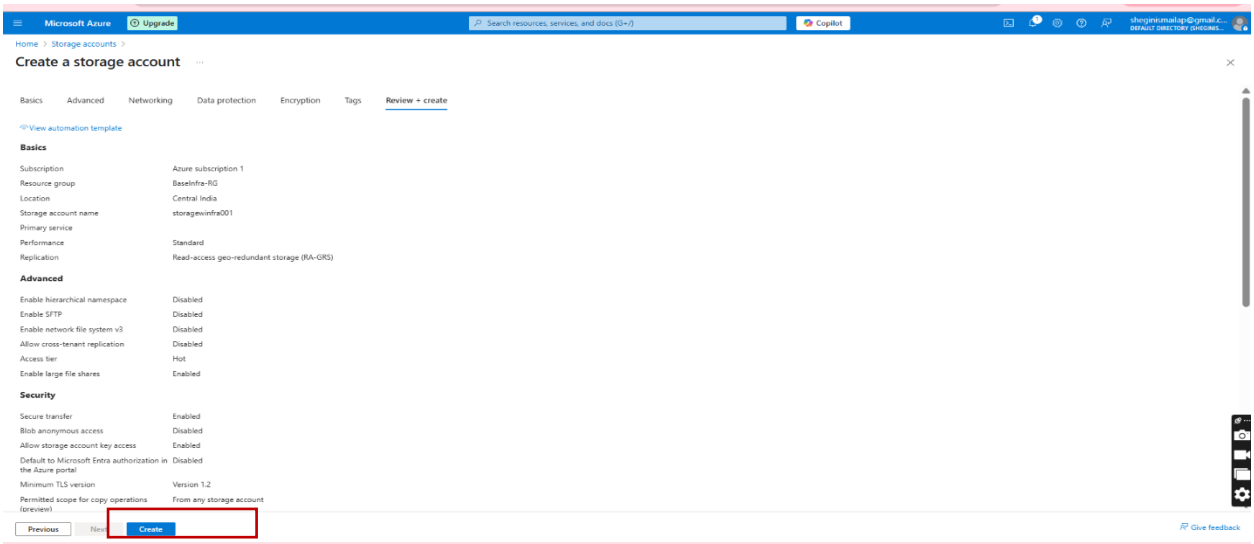
4. Generate SAS Key for Secure Access

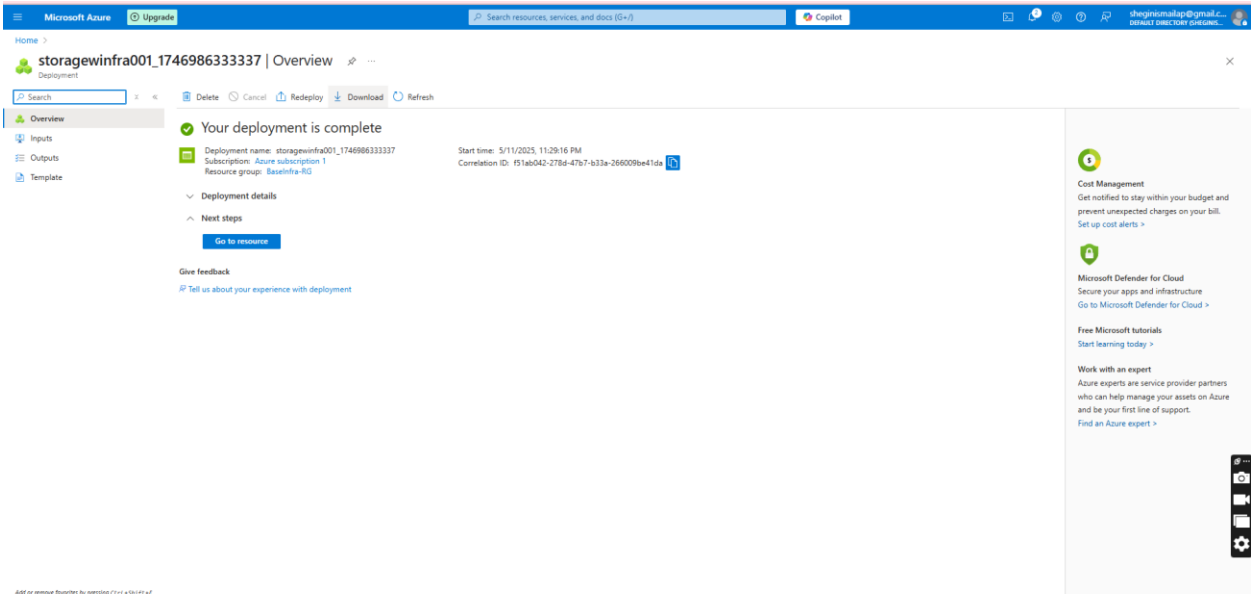
- Navigate to **Storage Account** → **Shared Access Signature (SAS)**.
- Define **permissions** (Read, Write, List, etc.).
- Set **Start/Expiry Time** for access control.
- Click **Generate SAS & Connection String**.
- Use the **SAS URL** for secure access.

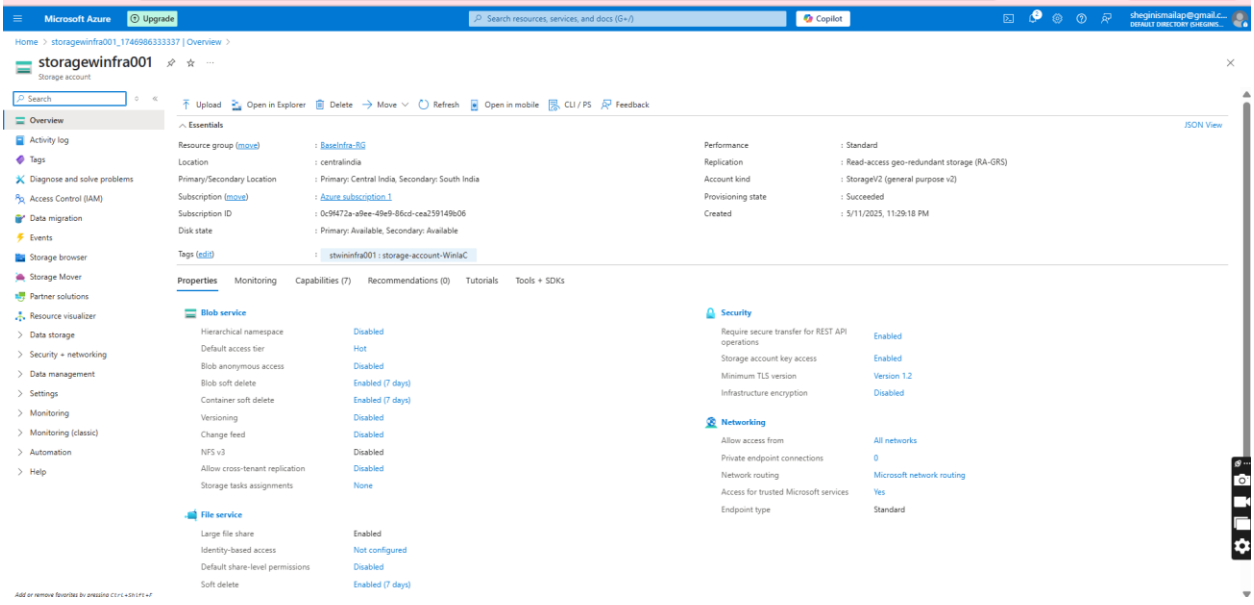

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home
Storage accounts
Default Directory (sheginismailapgmail.onmicrosoft.com)
Create
Restore
Manage view
Refresh
Export to CSV
Open query
Assign tags
Delete
Create new storage account
Filter for any field...
Add filter
No storage accounts to display
Create a storage account to store up to 500TB of data in the cloud. Use a general-purpose storage account to store object data, use a NoSQL data store, define and use queues for message processing, and set up file shares in the cloud. Use the Blob storage account and the hot or cool access tiers to optimize your costs based on how frequently
Create
Learn more

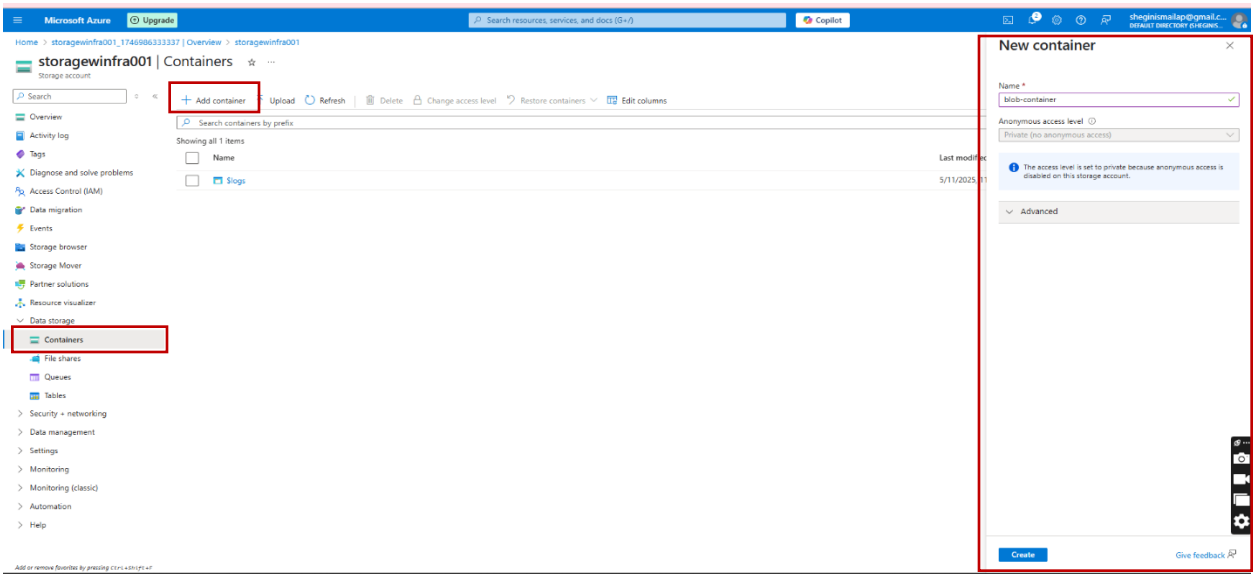

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home
Storage accounts
Create a storage account
Basics
Advanced
Networking
Data protection
Encryption
Tags
Review + create
Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. Learn more about Azure storage accounts
Project details
Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.
Subscription *
Azure subscription 1
Resource group *
BaseInfra-RG
Create new
Instance details
Storage account name *
storagewinfra001
Region *
(Asia Pacific) Central India
Deploy to an Azure Extended Zone
Primary service
Select a primary service
Performance *
Standard: Recommended for most scenarios (general-purpose v2 account)
Premium: Recommended for scenarios that require low latency.
Redundancy *
Geo-redundant storage (GRS)
Make read access to data available in the event of regional unavailability.
Previous
Next
Review + create
Give feedback

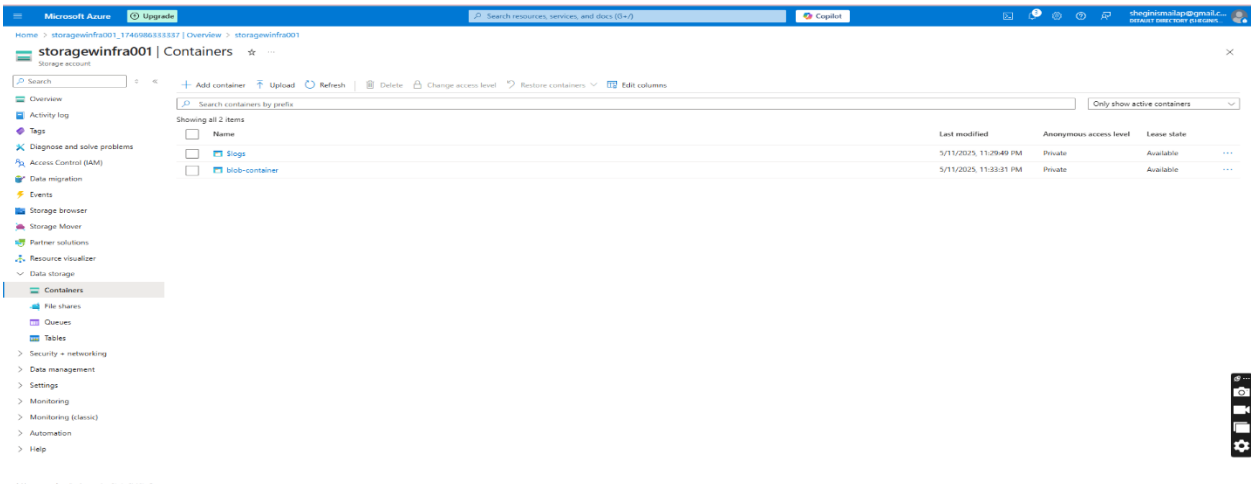

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home
Storage accounts
Create a storage account
Basics
Advanced
Networking
Data protection
Encryption
Tags
Review + create
Name
Value
Resource
stwininfra001
storage-account-WinIac
All resources selected
All resources selected
Previous
Next
Review + create
Give feedback

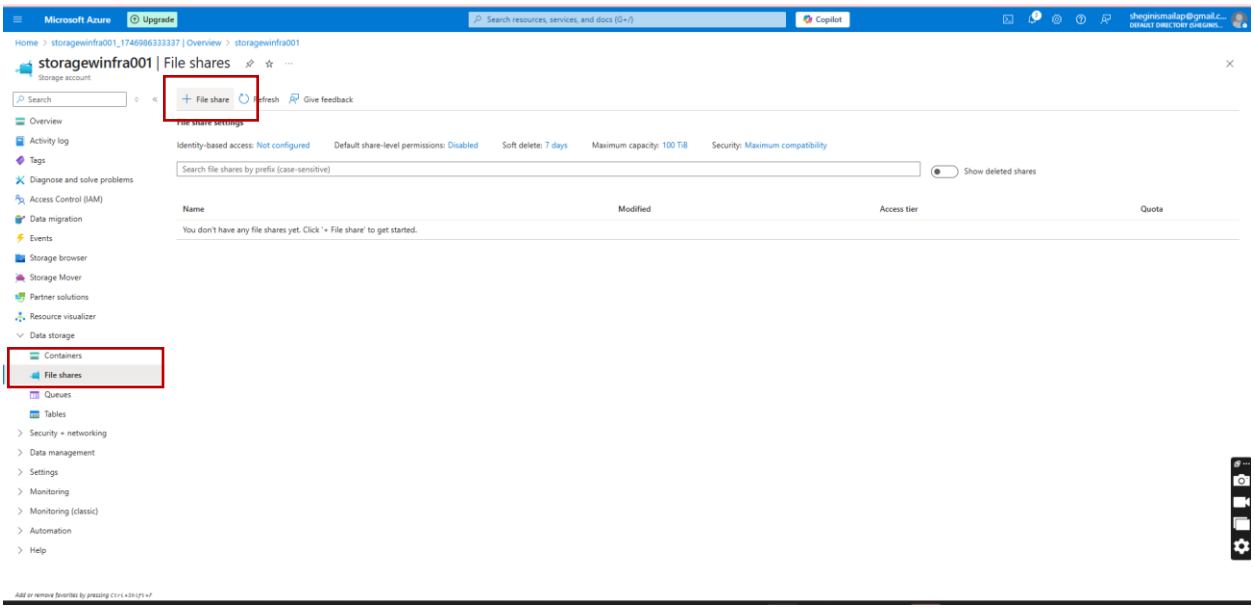

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Storage accounts >
Create a storage account
Basics
Advanced
Networking
Data protection
Encryption
Tags
Review + create
View automation template
Basics
Subscription
Azure subscription 1
Resource group
BaseInfra-RG
Location
Central India
Storage account name
storagewinfra001
Primary service
Performance
Standard
Replication
Read-access geo-redundant storage (RA-GRS)
Advanced
Enable hierarchical namespace
Disabled
Enable SFTP
Disabled
Enable network file system v3
Disabled
Allow cross-tenant replication
Disabled
Access tier
Hot
Enable large file shares
Enabled
Security
Secure transfer
Enabled
Blob anonymous access
Disabled
Allow storage account key access
Enabled
Default to Microsoft Entra authorization in the Azure portal
Disabled
Minimum TLS version
Version 1.2
Permitted scope for copy operations (preview)
From any storage account
Previous
Next
Create
Give feedback

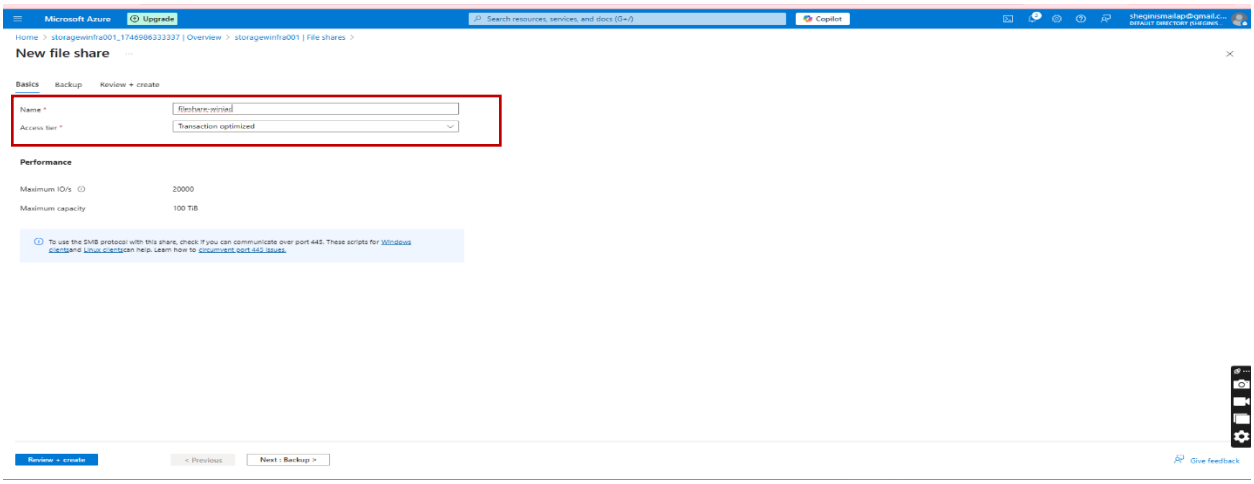

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home >
storagewinfra001_1746986333337 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name: storagewinfra001_1746986333337
Subscription: Azure subscription 1
Resource group: BaseInfra-RG
Start time: 5/11/2025, 11:29:16 PM
Correlation ID: f51ab042-278d-47b7-b33a-266009be41da
Deployment details
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment
Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

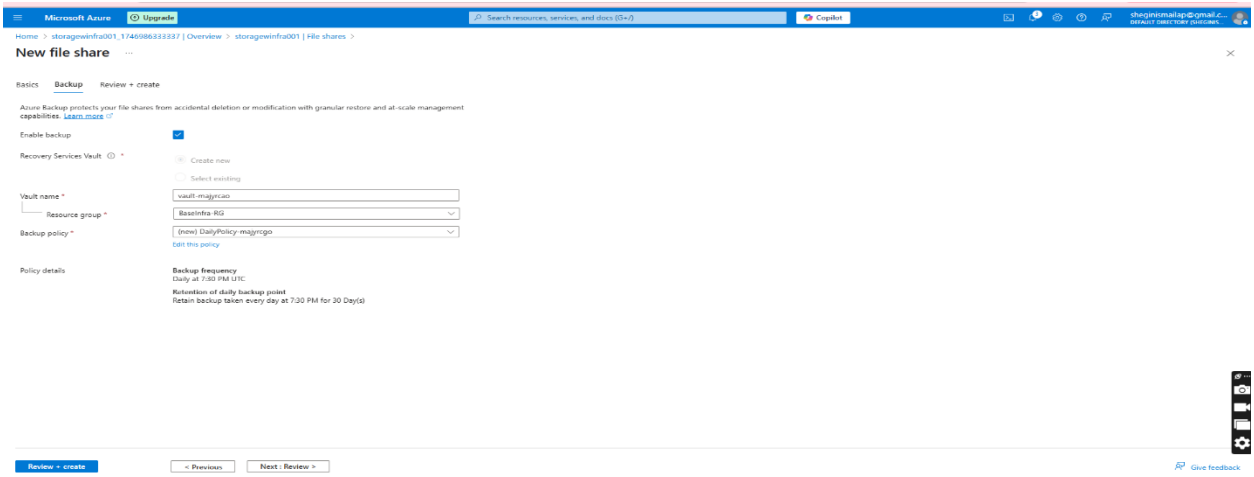

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > storagewinfra001_1746986333337 | Overview >
storagewinfra001
Storage account
Search
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Security + networking
Data management
Settings
Monitoring
Monitoring (classic)
Automation
Help
Upload
Open in Explorer
Delete
Move
Refresh
Open in mobile
CLI / PS
Feedback
Essentials
JSON View
Resource group (move) : BaseInfra-RG
Location : centralindia
Primary/Secondary Location : Primary: Central India, Secondary: South India
Subscription (move) : Azure subscription 1
Subscription ID : 0c9f472a-a9ee-49e9-86cd-cea259149b06
Disk state : Primary: Available, Secondary: Available
Tags (edit) : stwininfra001 : storage-account-WinIaC
Performance : Standard
Replication : Read-access geo-redundant storage (RA-GRS)
Account kind : StorageV2 (general purpose v2)
Provisioning state : Succeeded
Created : 5/11/2025, 11:29:18 PM
Properties
Monitoring
Capabilities (7)
Recommendations (0)
Tutorials
Tools + SDKs
Blob service
Hierarchical namespace Disabled
Default access tier Hot
Blob anonymous access Disabled
Blob soft delete Enabled (7 days)
Container soft delete Enabled (7 days)
Versioning Disabled
Change feed Disabled
NFS v3 Disabled
Allow cross-tenant replication Disabled
Storage tasks assignments None
File service
Large file share Enabled
Identity-based access Not configured
Default share-level permissions Disabled
Soft delete Enabled (7 days)
Security
Require secure transfer for REST API operations Enabled
Storage account key access Enabled
Minimum TLS version Version 1.2
Infrastructure encryption Disabled
Networking
Allow access from All networks
Private endpoint connections 0
Network routing Microsoft network routing
Access for trusted Microsoft services Yes
Endpoint type Standard

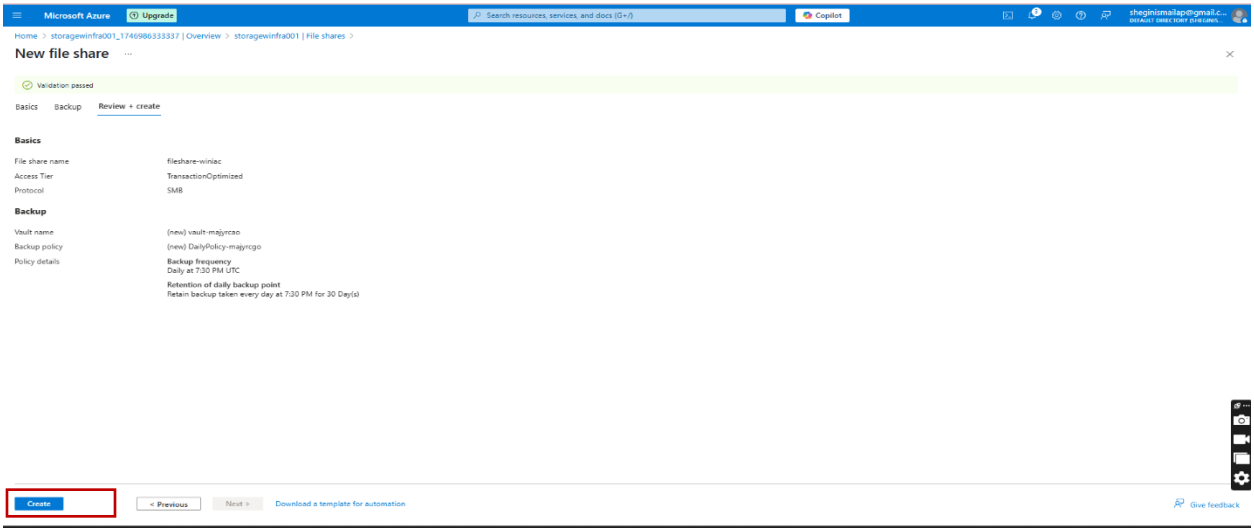

Microsoft Azure
Upgrade
Copilot
Home
storagewinfra001 | Containers
Storage account
Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Showing all 1 items
Name
$logs
New container
Name *
blob-container
Anonymous access level
Private (no anonymous access)
The access level is set to private because anonymous access is disabled on this storage account.
Advanced
Create
Give feedback
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
File shares
Queues
Tables
Security + networking
Data management
Settings
Monitoring
Monitoring (classic)
Automation
Help

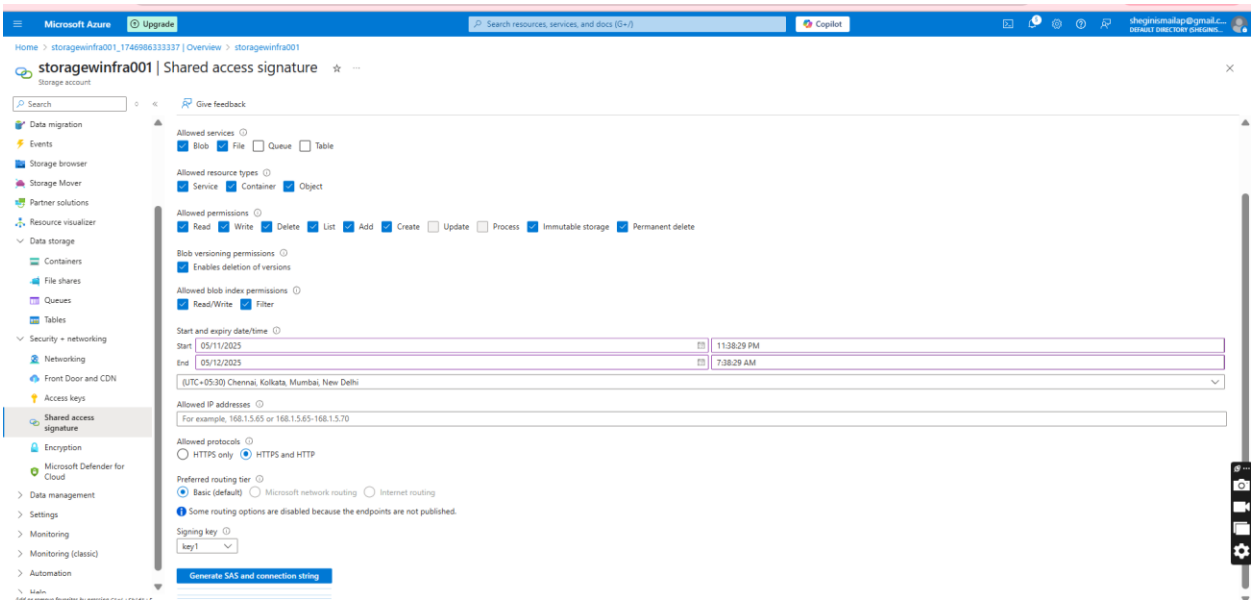

Microsoft Azure
Upgrade
Copilot
Home
storagewinfra001 | Containers
Storage account
Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Only show active containers
Showing all 2 items
Name
Last modified
Anonymous access level
Lease state
$logs
5/11/2025, 11:29:49 PM
Private
Available
blob-container
5/11/2025, 11:33:31 PM
Private
Available
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
File shares
Queues
Tables
Security + networking
Data management
Settings
Monitoring
Monitoring (classic)
Automation
Help


Microsoft Azure
Upgrade
Copilot
Home
storagewinfra001 | File shares
Storage account
File share
Refresh
Give feedback
File share settings
Identity-based access: Not configured
Default share-level permissions: Disabled
Soft delete: 7 days
Maximum capacity: 100 TiB
Security: Maximum compatibility
Search file shares by prefix (case-sensitive)
Show deleted shares
Name
Modified
Access tier
Quota
You don't have any file shares yet. Click '+ File share' to get started.
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
File shares
Queues
Tables
Security + networking
Data management
Settings
Monitoring
Monitoring (classic)
Automation
Help


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > storagewinfra001_1746986333337 | Overview > storagewinfra001 | File shares >
New file share
Basics
Backup
Review + create
Name *
fileshare-winiac
Access tier *
Transaction optimized
Performance
Maximum IO/s
20000
Maximum capacity
100 TiB
To use the SMB protocol with this share, check if you can communicate over port 445. These scripts for Windows clients and Linux clients can help. Learn how to circumvent port 445 issues.
Review + create
< Previous
Next : Backup >
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > storagewinfra001_1746986333337 | Overview > storagewinfra001 | File shares >
New file share
Basics
Backup
Review + create
Azure Backup protects your file shares from accidental deletion or modification with granular restore and at-scale management capabilities. Learn more
Enable backup
Recovery Services Vault *
Create new
Select existing
Vault name *
vault-majyrcao
Resource group *
BaseInfra-RG
Backup policy *
(new) DailyPolicy-majyrcgo
Edit this policy
Policy details
Backup frequency
Daily at 7:30 PM UTC
Retention of daily backup point
Retain backup taken every day at 7:30 PM for 30 Day(s)
Review + create
< Previous
Next : Review >
Give feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > storagewinfra001_1746986333337 | Overview > storagewinfra001 | File shares >
New file share
Validation passed
Basics
Backup
Review + create
Basics
File share name
fileshare-winiac
Access Tier
TransactionOptimized
Protocol
SMB
Backup
Vault name
(new) vault-majyrcao
Backup policy
(new) DailyPolicy-majyrcgo
Policy details
Backup frequency
Daily at 7:30 PM UTC
Retention of daily backup point
Retain backup taken every day at 7:30 PM for 30 Day(s)
Create
< Previous
Next >
Download a template for automation
Give feedback

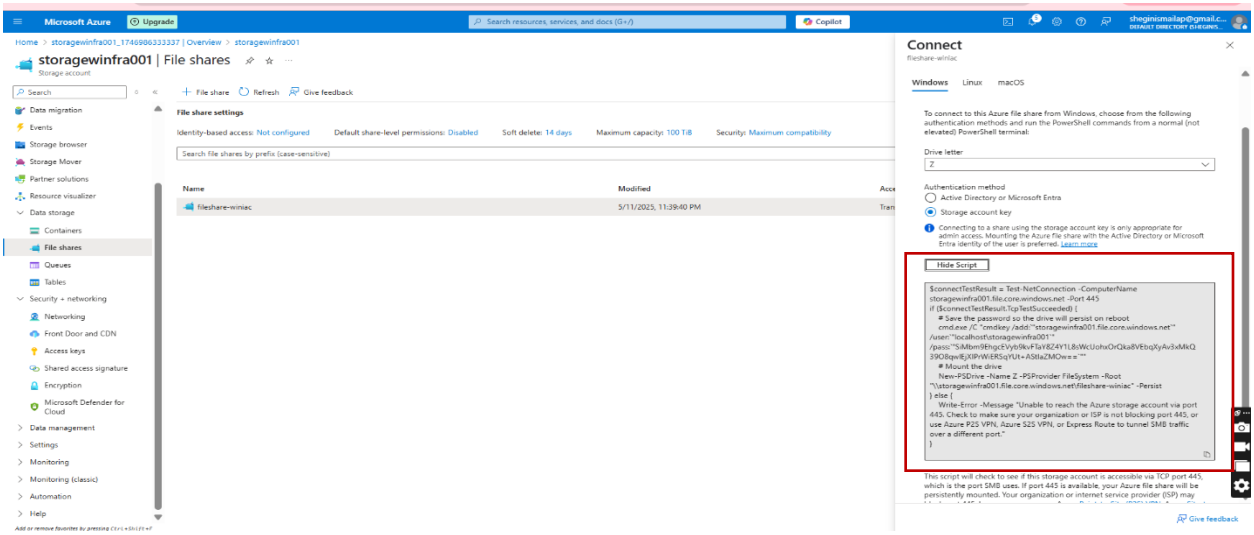


Method 1: Map via Custom Script (Z Drive)

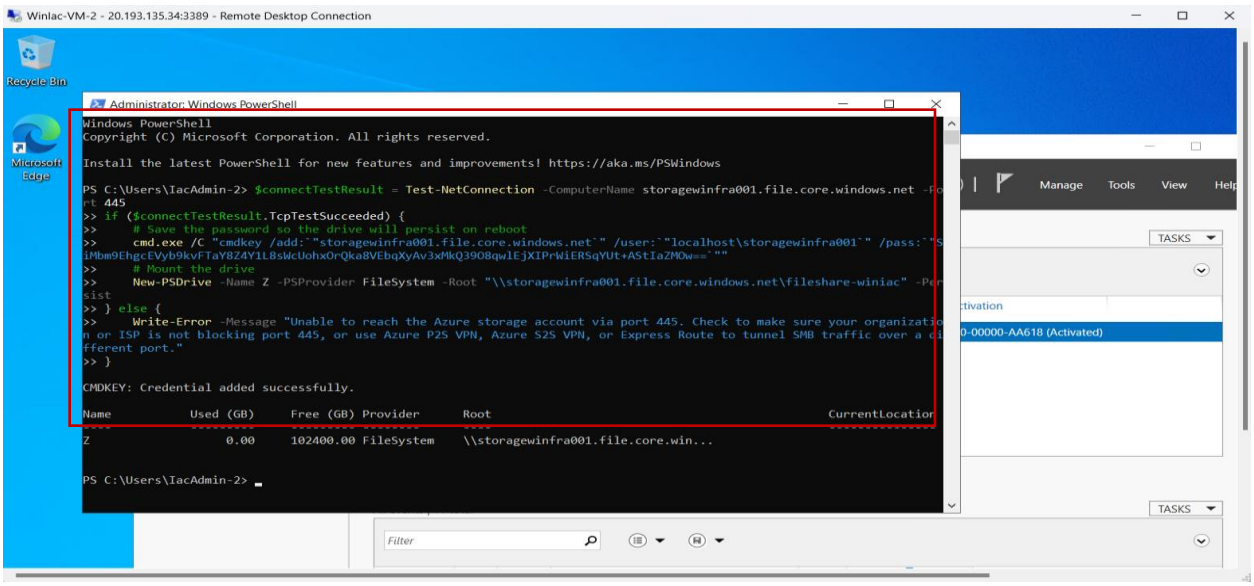
1. Open **Azure Portal**, navigate to **Storage Account** > **File Shares**.
2. Click **Connect** and select **Windows**.
3. Copy the PowerShell script and run it on the VM.
4. This will map the **Z drive** automatically.

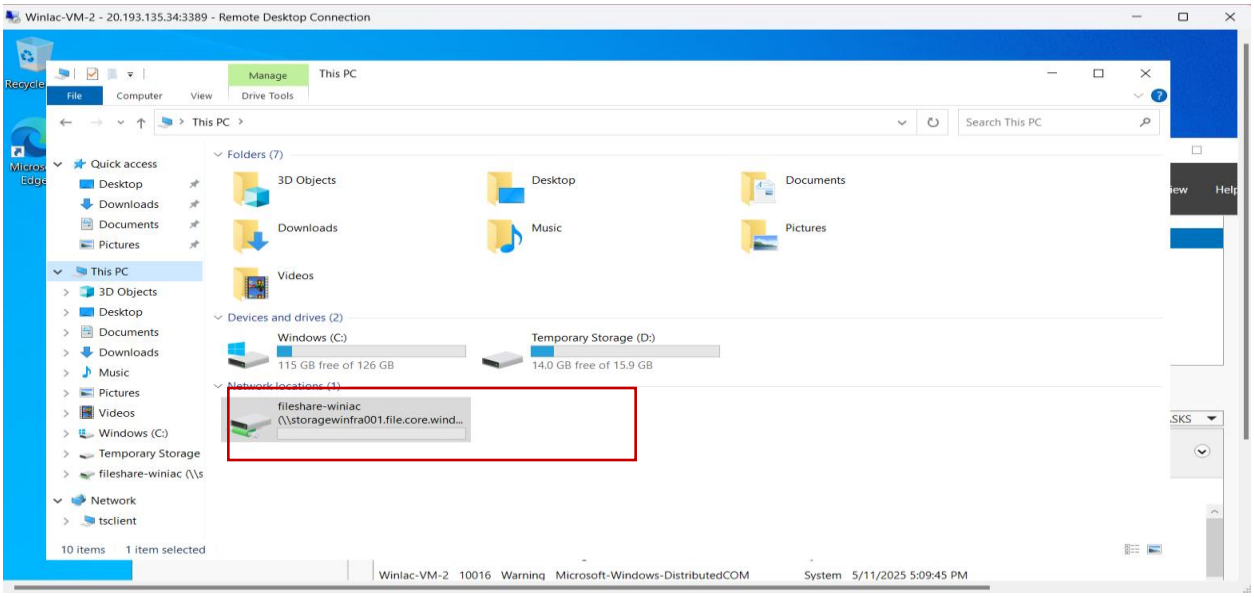
Method 2: Connect with Map using Network Drive

1. Open **File Explorer**, right-click **This PC** > **Map Network Drive**.
2. Enter the **Azure File Share path**.
3. Check **Connect using different credentials**.
4. Enter:
    - **Username** = **Storage Account Name**.
    - **Password** = **Access Key** (from Azure Storage Account).
5. Click **OK** to establish the connection.

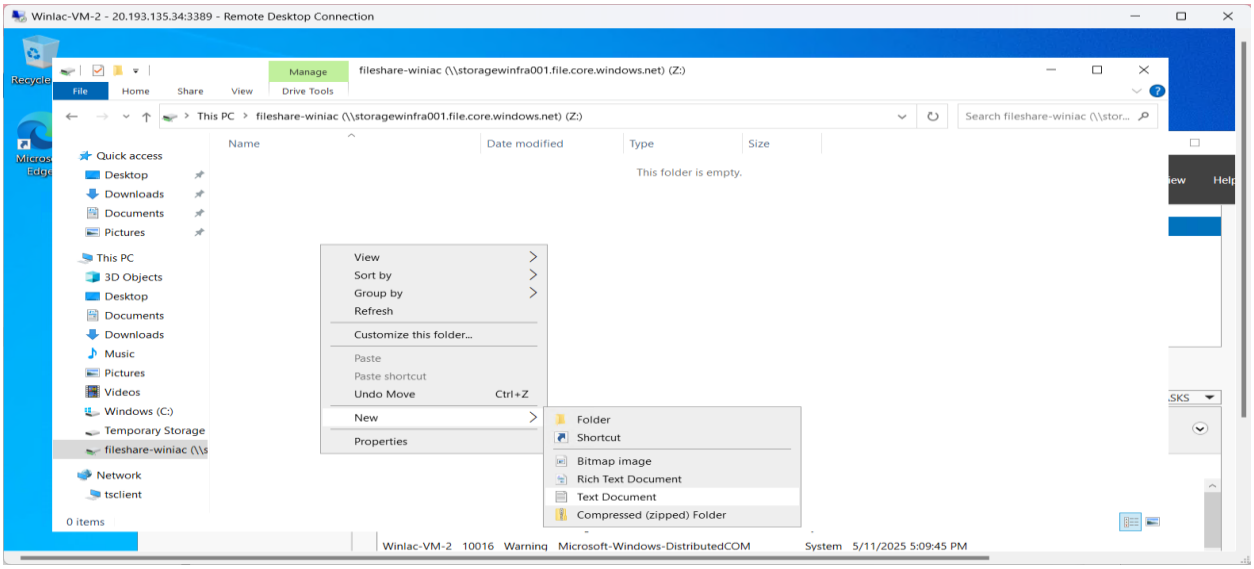

Microsoft Azure
storagewinfra001 | File shares
Connect
Windows
Linux
macOS
Hide Script

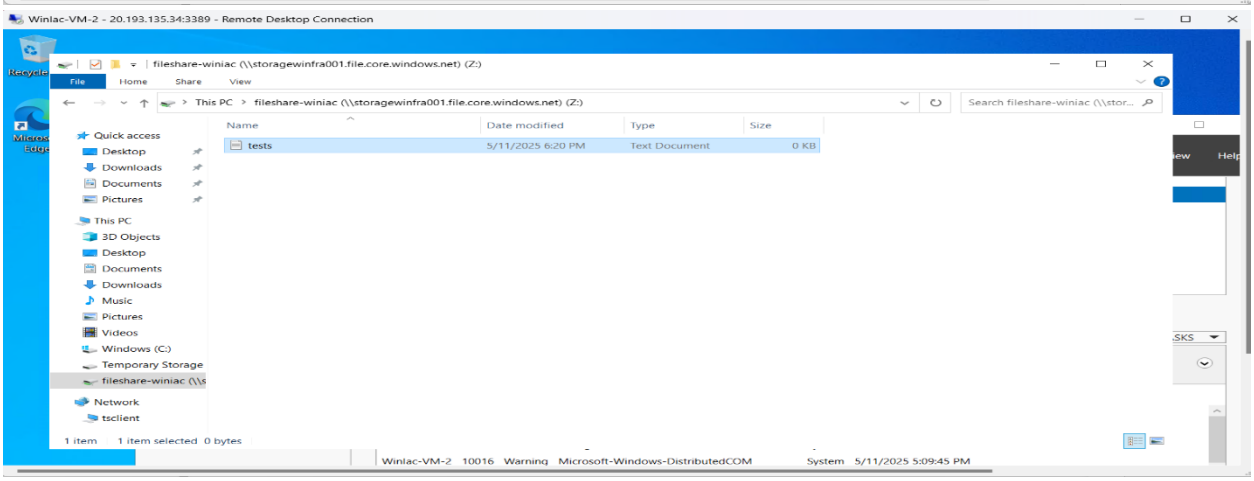
Method 1:


Winlac-VM-2 - 20.193.135.34:3389 - Remote Desktop Connection
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\IacAdmin-2> $connectTestResult = Test-NetConnection -ComputerName storagewinfra001.file.core.windows.net -Port 445
CMDKEY: Credential added successfully.
Name Used (GB) Free (GB) Provider Root CurrentLocation
Z 0.00 102400.00 FileSystem \\storagewinfra001.file.core.win...
PS C:\Users\IacAdmin-2>
Recycle Bin
Microsoft Edge

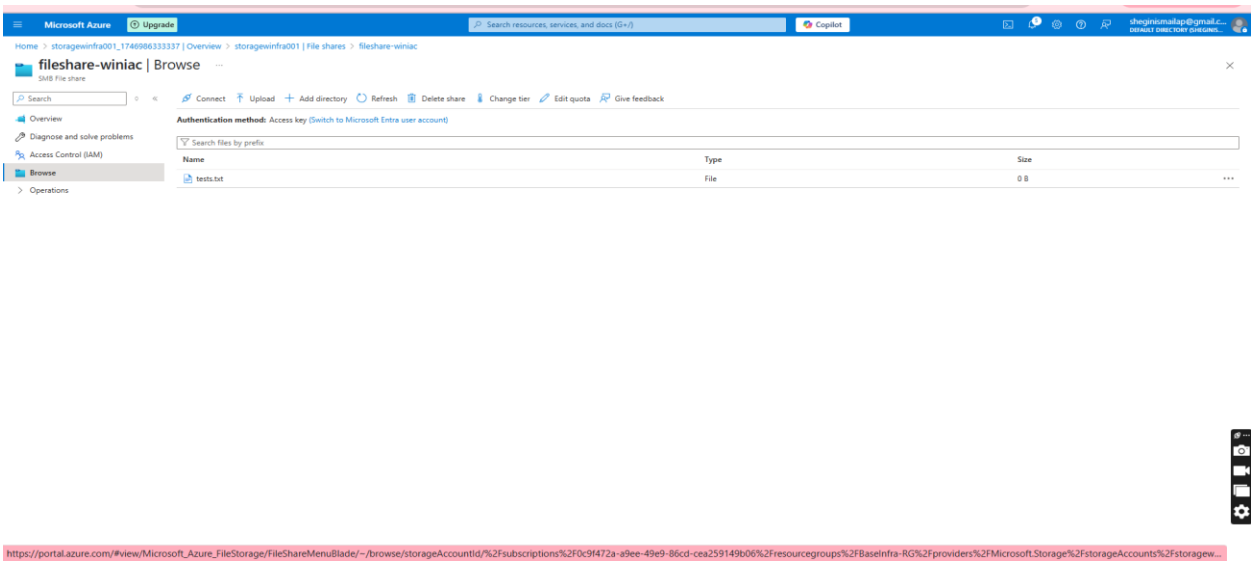

Winlac-VM-2 - 20.193.135.34:3389 - Remote Desktop Connection
This PC
Folders (7)
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Devices and drives (2)
Windows (C:)
115 GB free of 126 GB
Temporary Storage (D:)
14.0 GB free of 15.9 GB
Network locations (1)
fileshare-winiac
(\\storagewinfra001.file.core.wind...
10 items
1 item selected
Winlac-VM-2 10016 Warning Microsoft-Windows-DistributedCOM System 5/11/2025 5:09:45 PM

Verifying file created is reflected on azure portal

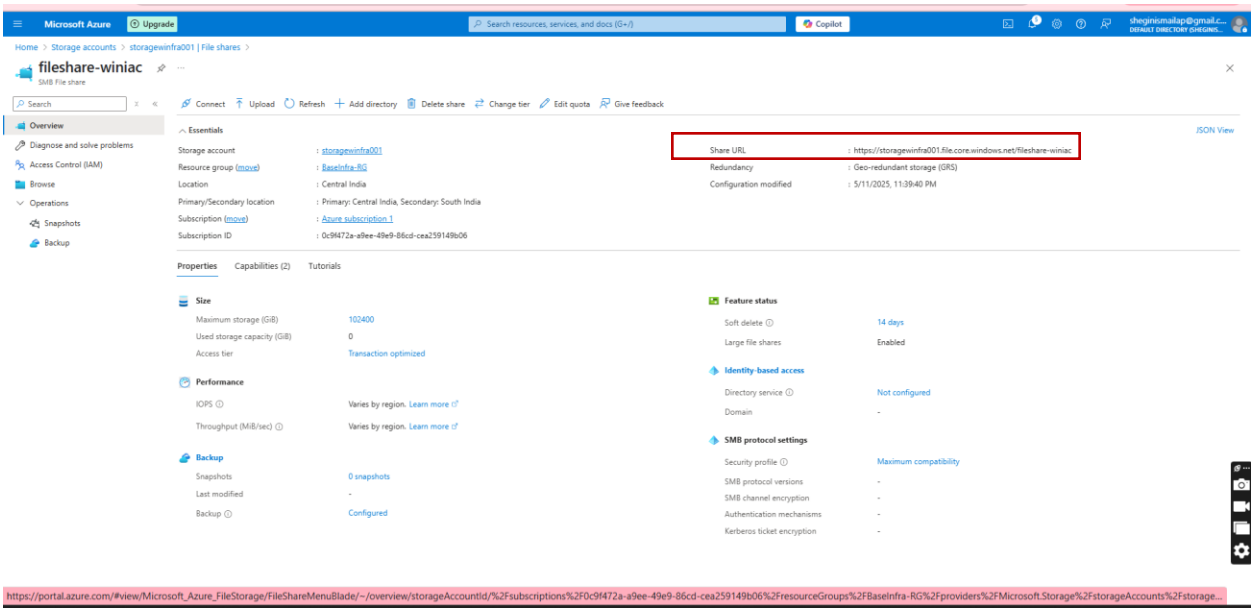


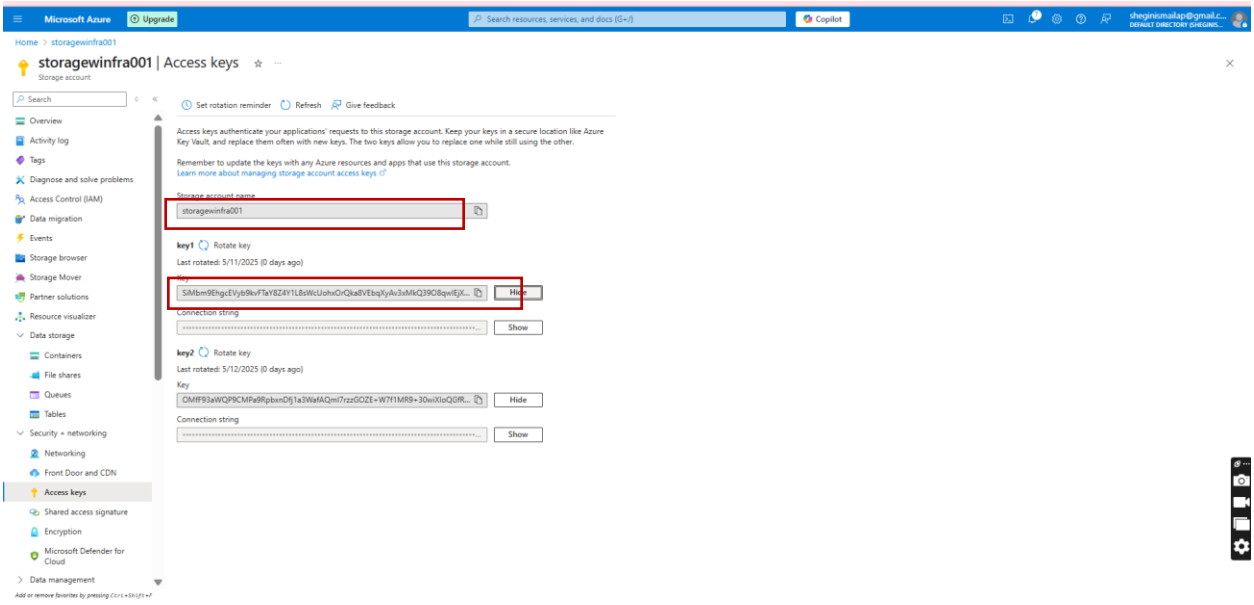


RESULT

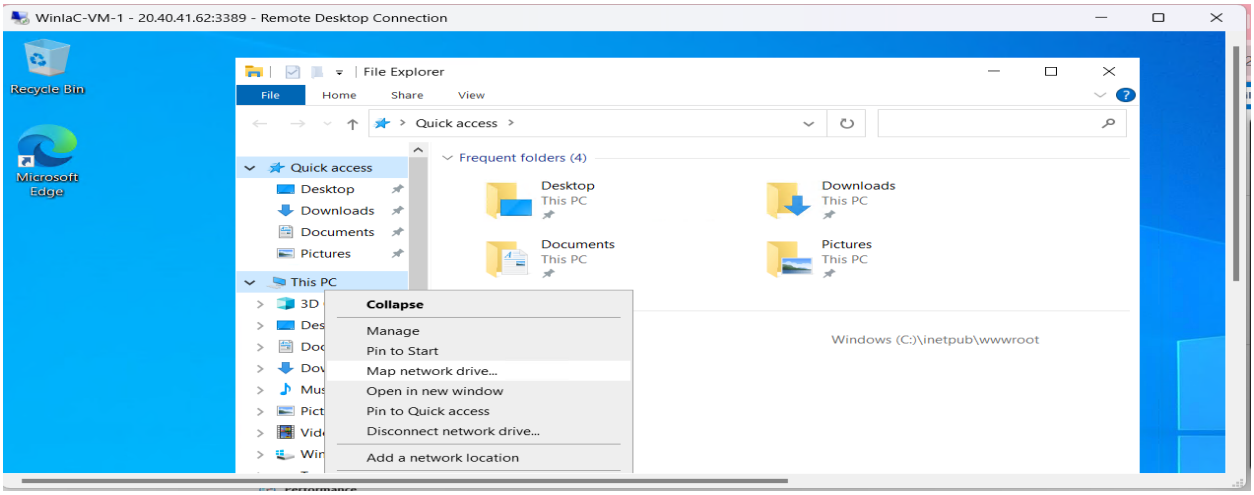


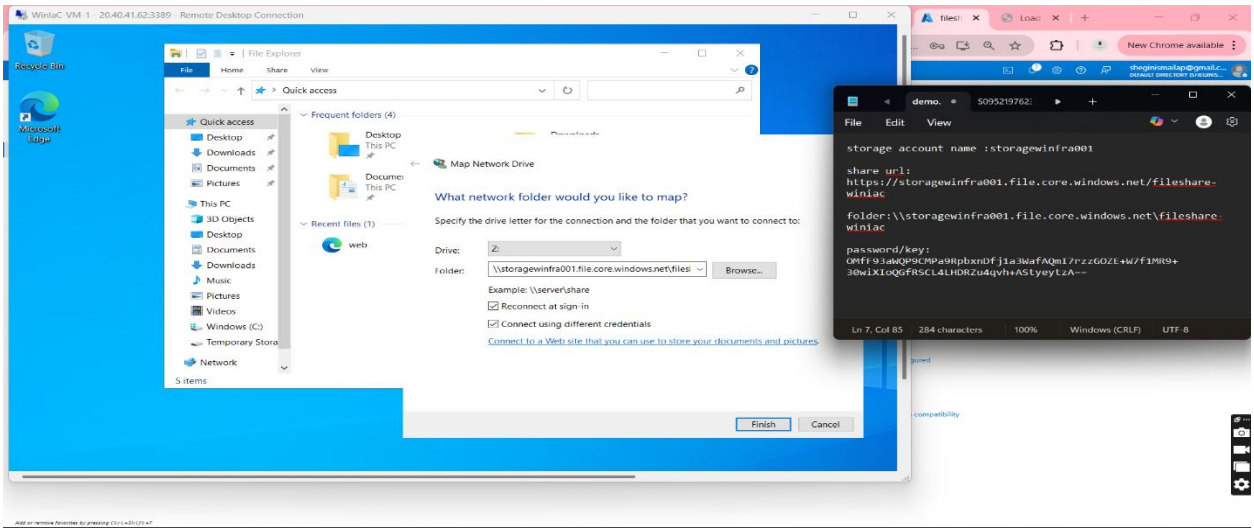
Method 2:

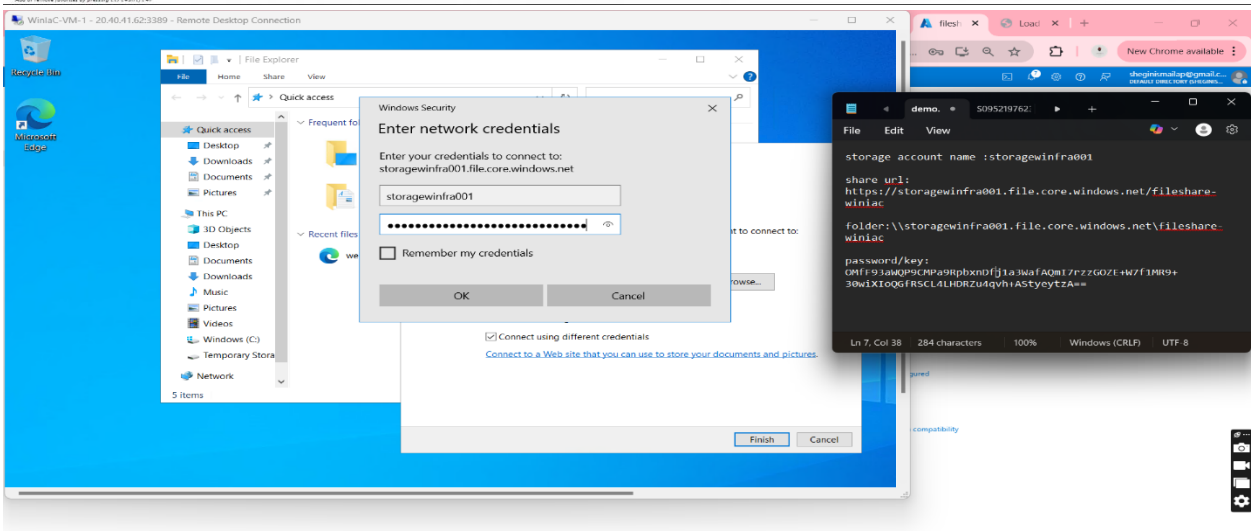




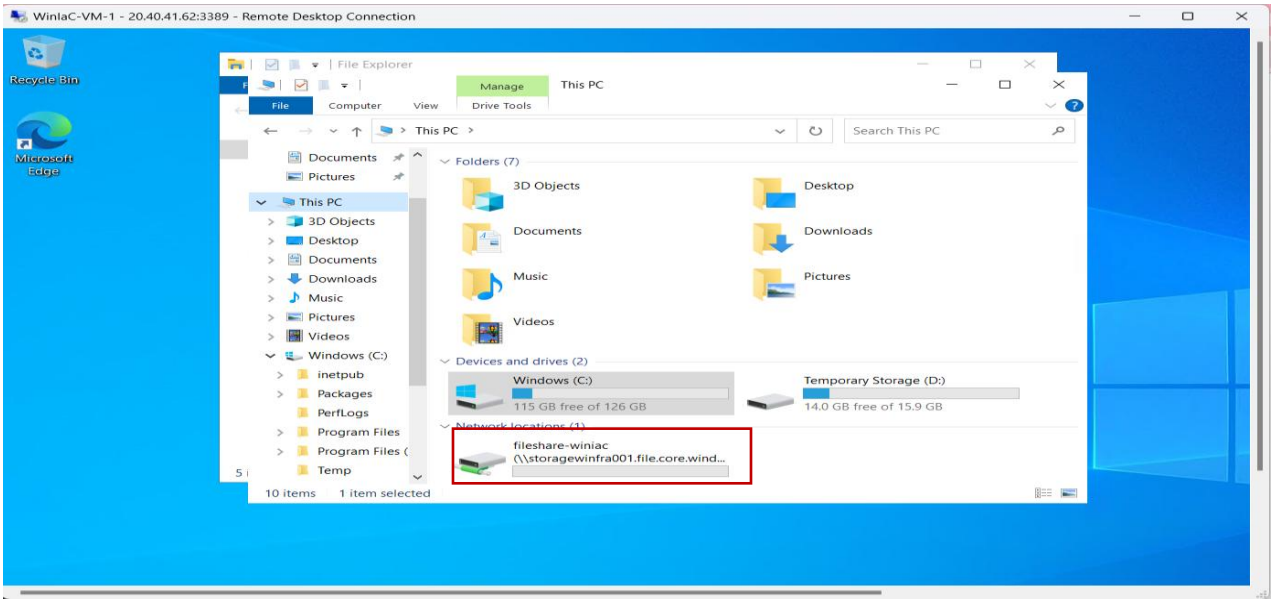
Note the storage account name(acts as username),access key(password) and share-url(the share-url's part on which http part removed and the backslash reversed will act as path)

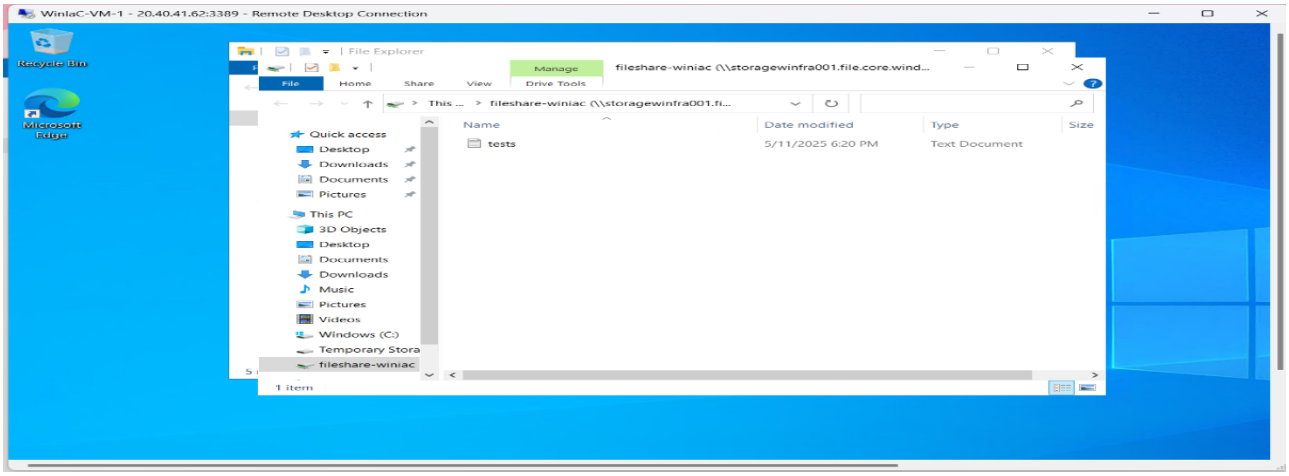



Map Network Drive
What network folder would you like to map?
Specify the drive letter for the connection and the folder that you want to connect to:
Drive:
Folder:
Browse...
Example: \\server\share
Reconnect at sign-in
Connect using different credentials
Connect to a Web site that you can use to store your documents and pictures.
Finish
Cancel
storage account name :storagewinfra001
share url:
https://storagewinfra001.file.core.windows.net/fileshare-winiac
folder:\\storagewinfra001.file.core.windows.net\fileshare-winiac
password/key:
OMfF93aWQP9CMPa9RpbxnDfj1a3WafAQmI7rzzGOZE+W7f1MR9+30wiXIoQGfRSCL4LHDRZu4qvh+AStyeytzA==
Ln 7, Col 85
284 characters
100%
Windows (CRLF)
UTF-8


Windows Security
Enter network credentials
Enter your credentials to connect to:
storagewinfra001.file.core.windows.net
storagewinfra001
Remember my credentials
OK
Cancel
Connect using different credentials
Connect to a Web site that you can use to store your documents and pictures.
Finish
Cancel
storage account name :storagewinfra001
share url:
https://storagewinfra001.file.core.windows.net/fileshare-winiac
folder:\\storagewinfra001.file.core.windows.net\fileshare-winiac
password/key:
OMfF93aWQP9CMPa9RpbxnDfj1a3WafAQmI7rzzGOZE+W7f1MR9+30wiXIoQGfRSCL4LHDRZu4qvh+AStyeytzA==
Ln 7, Col 38
284 characters
100%
Windows (CRLF)
UTF-8

RESULT


WinIaC-VM-1 - 20.40.41.62:3389 - Remote Desktop Connection
Recycle Bin
Microsoft Edge
File Explorer
Manage
This PC
File
Computer
View
Drive Tools
Search This PC
Documents
Pictures
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
inetpub
Packages
PerfLogs
Program Files
Program Files (
Temp
Folders (7)
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Devices and drives (2)
Windows (C:)
115 GB free of 126 GB
Temporary Storage (D:)
14.0 GB free of 15.9 GB
Network locations (1)
fileshare-winiac
(\\storagewinfra001.file.core.wind...
10 items
1 item selected



steps to provision an **Azure SQL Database (Basic Tier)**:

1. Create an Azure SQL Server

- Open **Azure Portal** and navigate to **Azure SQL**.
- Click **Create SQL Server**.
- Select **Resource Group** and define **Server Name**.
- Set **Region** and configure **Authentication (SQL or Azure AD)**.
- Click **Review + Create** → **Create**.

2. Create a SQL Database

- Navigate to **Azure SQL Databases**.
- Click **Create Database**.
- Select the **SQL Server** created earlier.
- Choose **Basic Tier** under **Compute + Storage**.
- Define **Collation** and **Backup options** as needed.
- Click **Review + Create** → **Create**.

3. Configure Networking & Firewall

- Open **SQL Server** and navigate to **Networking**.
- Set **Public access** or **Private endpoint**.
- Configure **Firewall Rules** to allow VM/IP access.

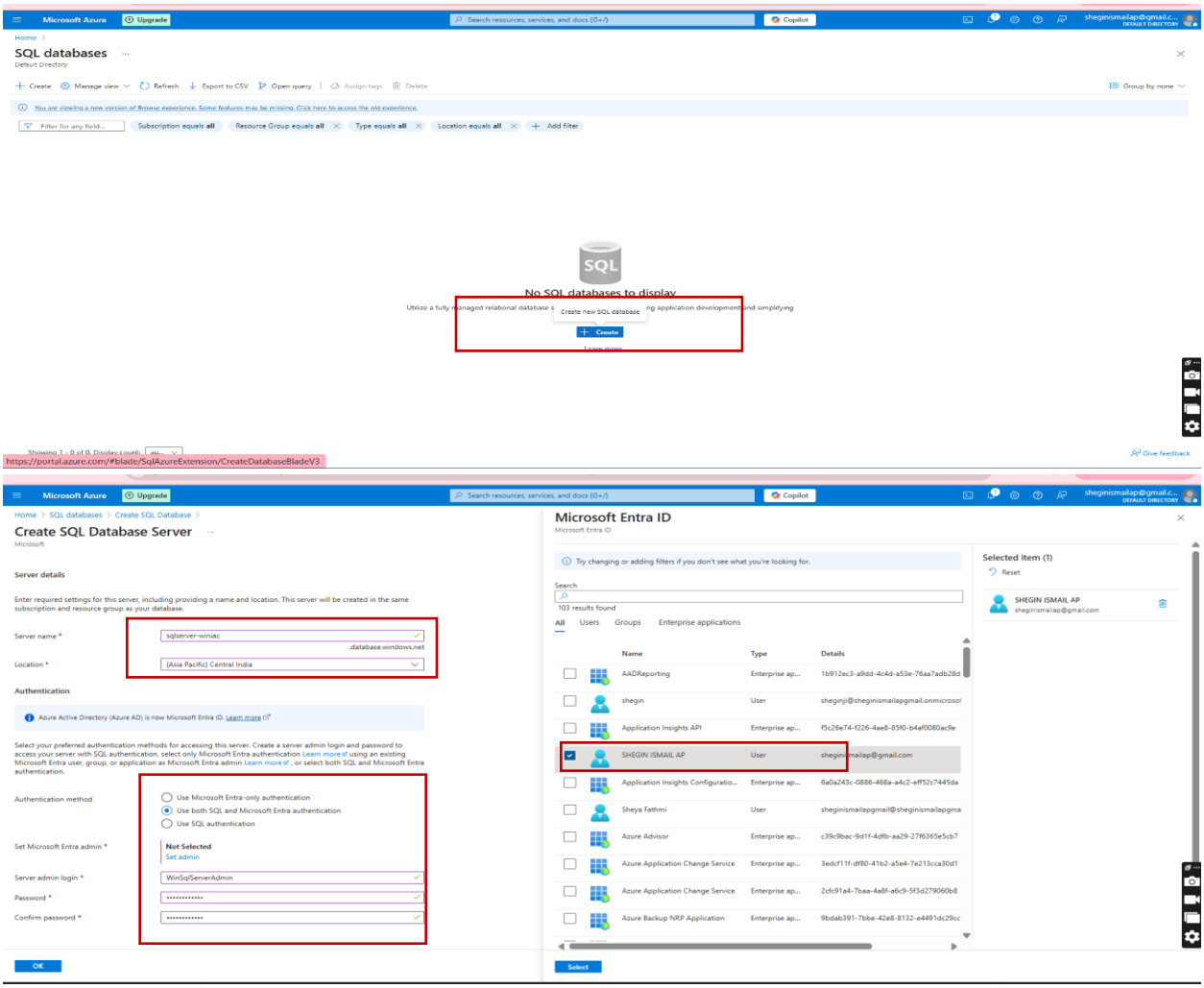
4. Connect to the SQL Database

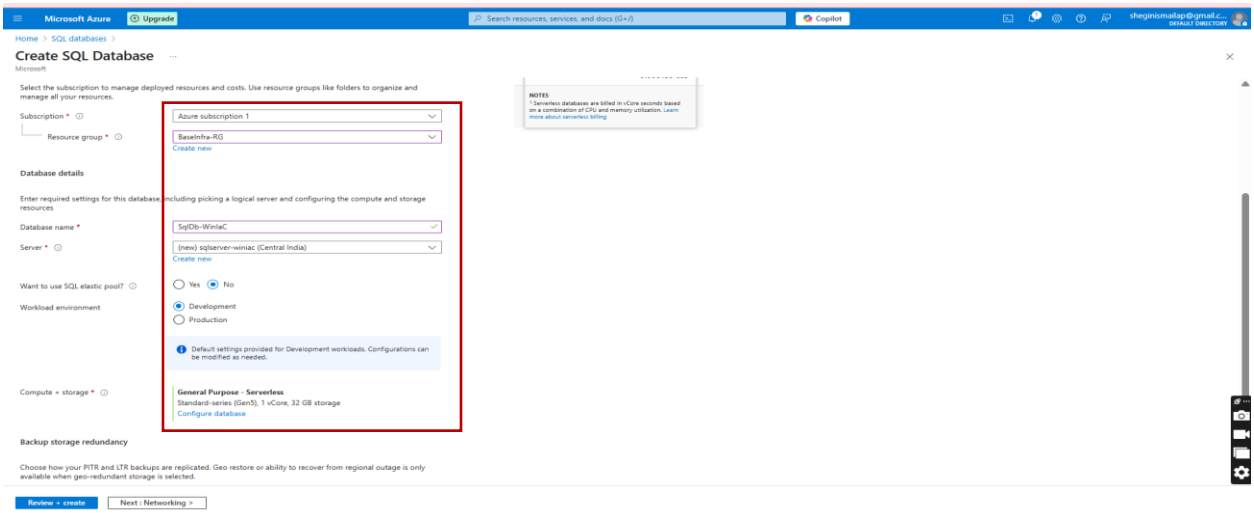
- Use **Azure Data Studio** or **SQL Server Management Studio (SSMS)**.
- Enter **Server Name**, **Database Name**, and **Authentication method**.

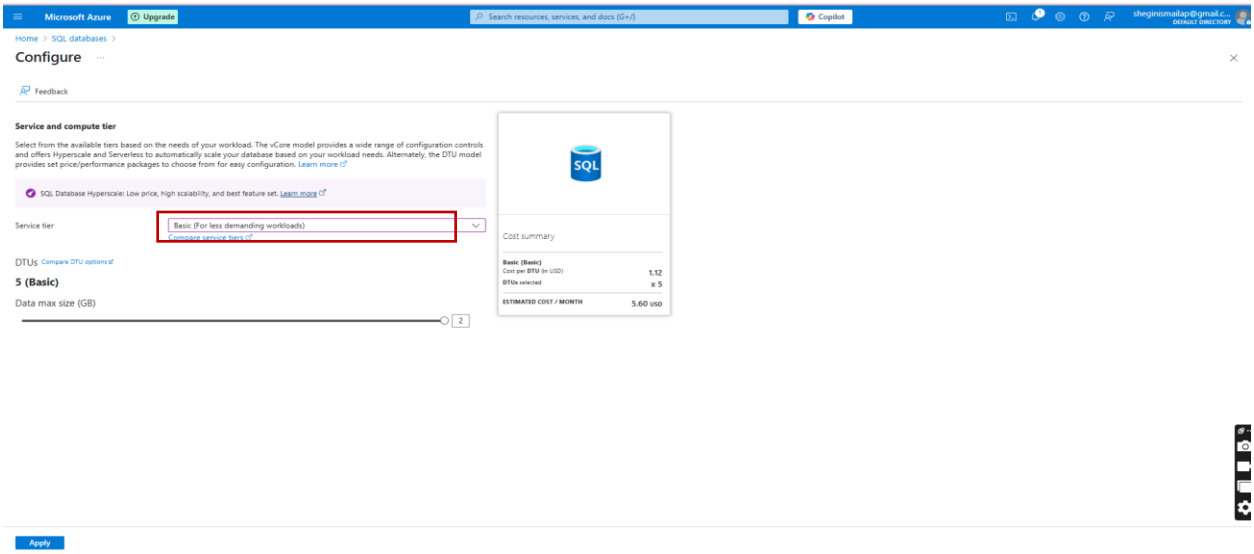
- Validate connection and start running queries.

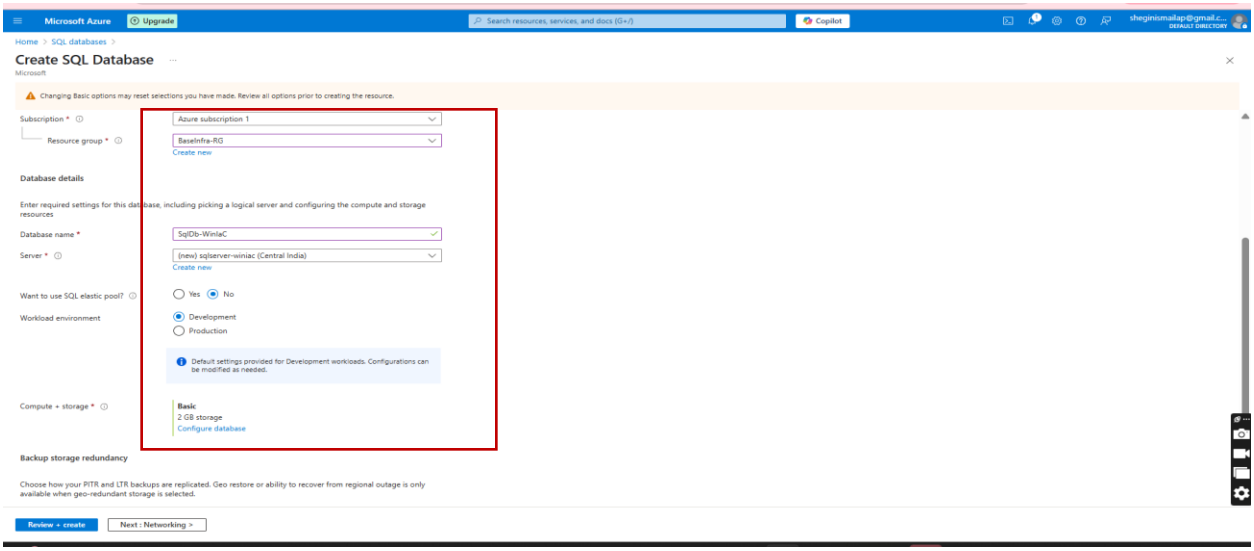
5. Test & Secure the Database

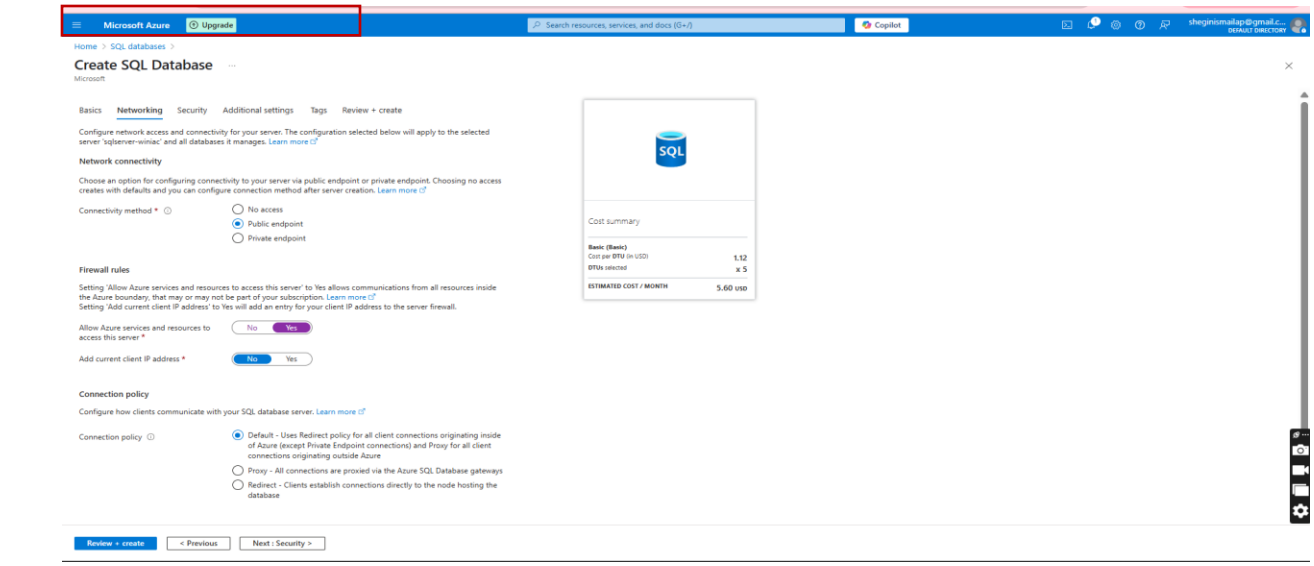
- Run basic queries to confirm database availability.
- Configure **Azure Defender** for security monitoring.
- Set up **Automated Backups** for recovery.

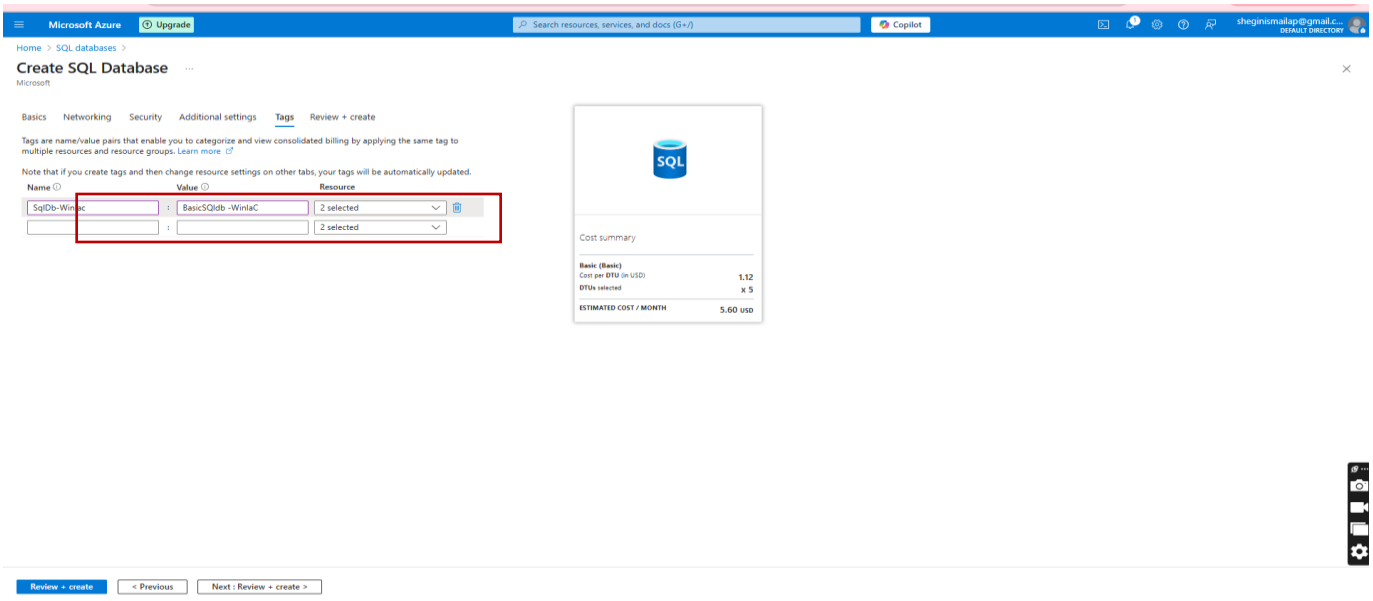


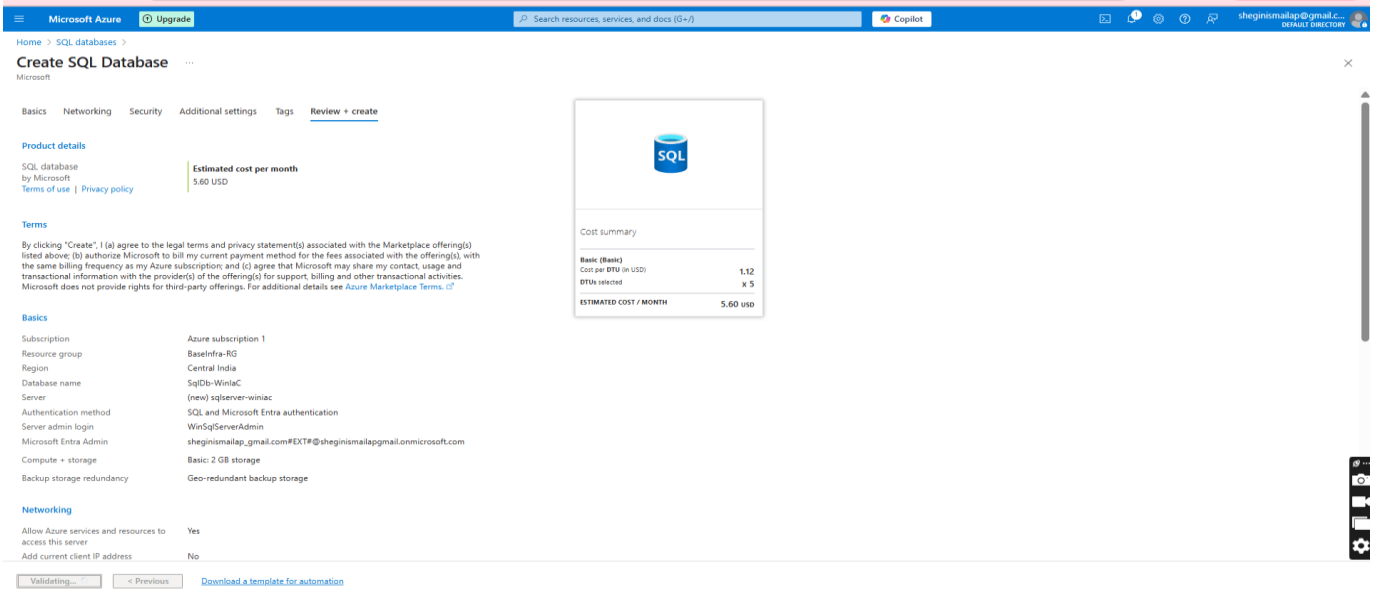

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > SQL databases >
Create SQL Database
Microsoft
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
BaseInfra-RG
Create new
Database details
Enter required settings for this database, including picking a logical server and configuring the compute and storage resources
Database name *
SqlDb-WinIaC
Server *
(new) sqlserver-winiac (Central India)
Create new
Want to use SQL elastic pool?
Yes
No
Workload environment
Development
Production
Default settings provided for Development workloads. Configurations can be modified as needed.
Compute + storage *
General Purpose - Serverless
Standard-series (Gen5), 1 vCore, 32 GB storage
Configure database
Backup storage redundancy
Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.
Review + create
Next : Networking >

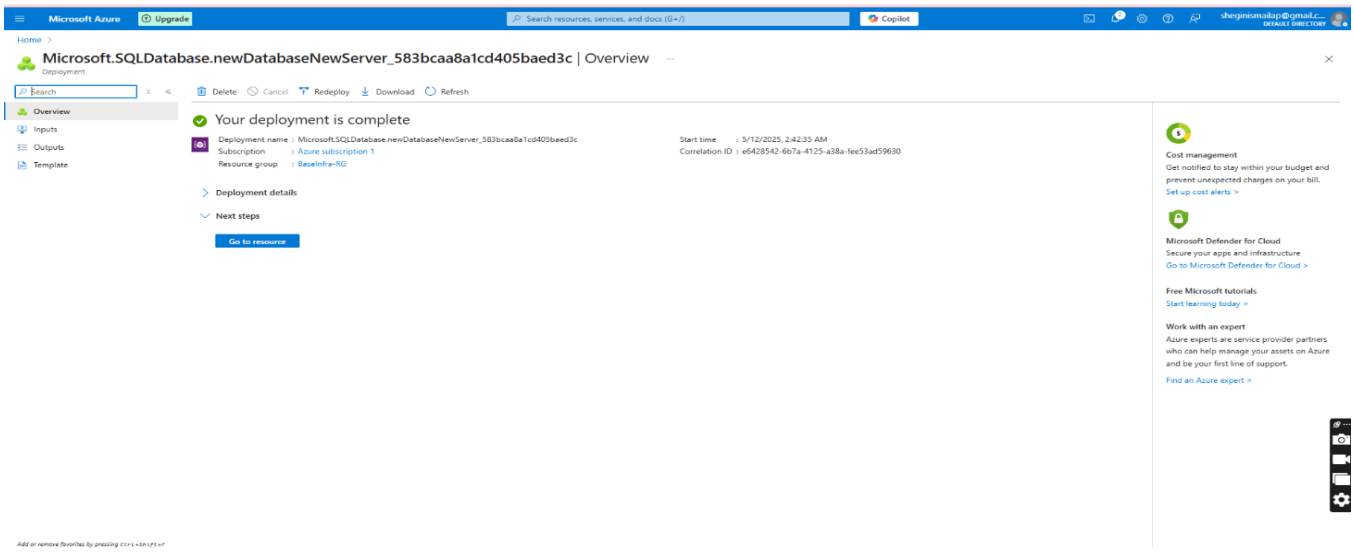

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > SQL databases >
Configure
Feedback
Service and compute tier
Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. Learn more
SQL Database Hyperscale: Low price, high scalability, and best feature set. Learn more
Service tier
Basic (For less demanding workloads)
Compare service tiers
DTUs Compare DTU options
5 (Basic)
Data max size (GB)
2
SQL
Cost summary
Basic (Basic)
Cost per DTU (in USD)
1.12
DTUs selected
x 5
ESTIMATED COST / MONTH
5.60 USD
Apply


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > SQL databases >
Create SQL Database
Microsoft
Changing Basic options may reset selections you have made. Review all options prior to creating the resource.
Subscription *
Azure subscription 1
Resource group *
BaseInfra-RG
Create new
Database details
Enter required settings for this database, including picking a logical server and configuring the compute and storage resources
Database name *
SqlDb-WinIaC
Server *
(new) sqlserver-winiac (Central India)
Create new
Want to use SQL elastic pool?
Yes
No
Workload environment
Development
Production
Default settings provided for Development workloads. Configurations can be modified as needed.
Compute + storage *
Basic
2 GB storage
Configure database
Backup storage redundancy
Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.
Review + create
Next : Networking >

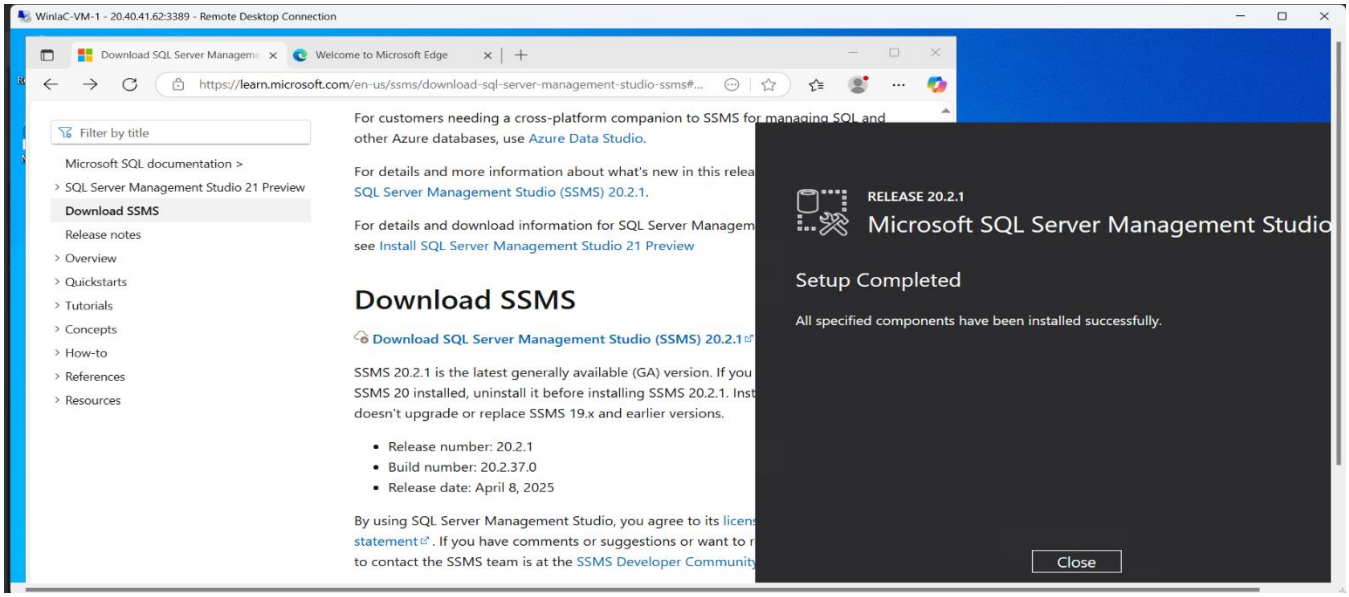

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > SQL databases >
Create SQL Database
Microsoft
Basics Networking Security Additional settings Tags Review + create
Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'sqlserver-winiac' and all databases it manages. Learn more
Network connectivity
Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. Learn more
Connectivity method *
No access
Public endpoint
Private endpoint
Firewall rules
Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. Learn more
Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.
Allow Azure services and resources to access this server *
No
Yes
Add current client IP address *
No
Yes
Connection policy
Configure how clients communicate with your SQL database server. Learn more
Connection policy
Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure
Proxy - All connections are proxied via the Azure SQL Database gateways
Redirect - Clients establish connections directly to the node hosting the database
Cost summary
Basic (Basic)
Cost per DTU (in USD)
1.12
DTUs selected
x 5
ESTIMATED COST / MONTH
5.60 USD
Review + create
< Previous
Next : Security >

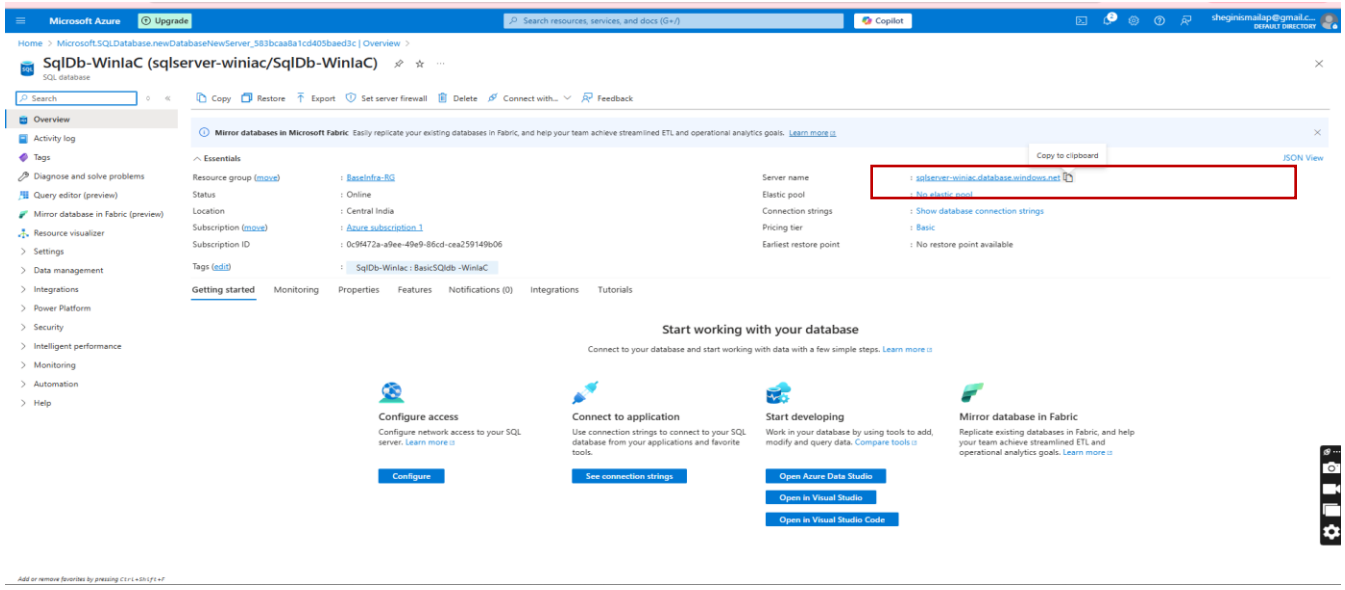

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > SQL databases >
Create SQL Database
Microsoft
Basics Networking Security Additional settings Tags Review + create
Tags are name/value pairs that enable you to categorize and view consolidated billing by applying the same tag to multiple resources and resource groups. Learn more
Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.
Name
Value
Resource
SqlDb-Winiac
BasicSQldb -WinlaC
2 selected
2 selected
Cost summary
Basic (Basic)
Cost per DTU (in USD)
1.12
DTUs selected
x 5
ESTIMATED COST / MONTH
5.60 USD
Review + create
< Previous
Next : Review + create >

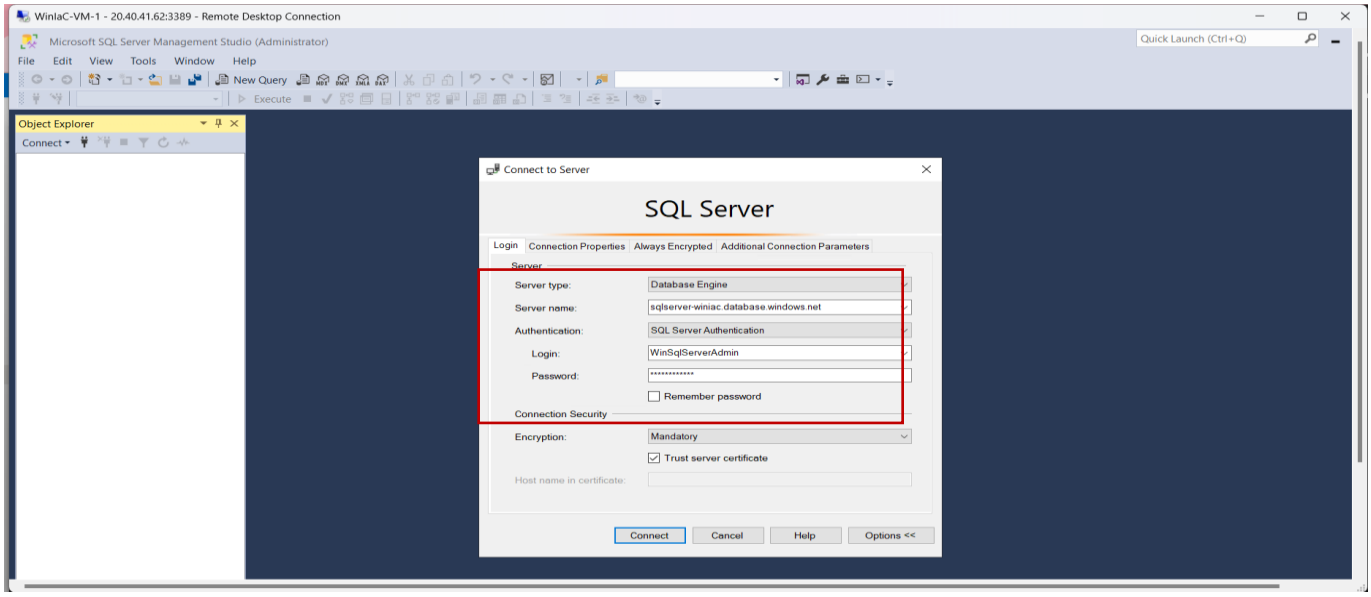

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > SQL databases >
Create SQL Database
Microsoft
Basics Networking Security Additional settings Tags Review + create
Product details
SQL database
by Microsoft
Terms of use | Privacy policy
Estimated cost per month
5.60 USD
Terms
By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see Azure Marketplace Terms.
Basics
Subscription
Azure subscription 1
Resource group
BaseInfra-RG
Region
Central India
Database name
SqlDb-WinlaC
Server
(new) sqlserver-winiac
Authentication method
SQL and Microsoft Entra authentication
Server admin login
WinSqlServerAdmin
Microsoft Entra Admin
sheginismailap_gmail.com#EXT#@sheginismailapgmail.onmicrosoft.com
Compute + storage
Basic: 2 GB storage
Backup storage redundancy
Geo-redundant backup storage
Networking
Allow Azure services and resources to access this server
Yes
Add current client IP address
No
Cost summary
Basic (Basic)
Cost per DTU (in USD)
1.12
DTUs selected
x 5
ESTIMATED COST / MONTH
5.60 USD
Validating...
< Previous
Download a template for automation



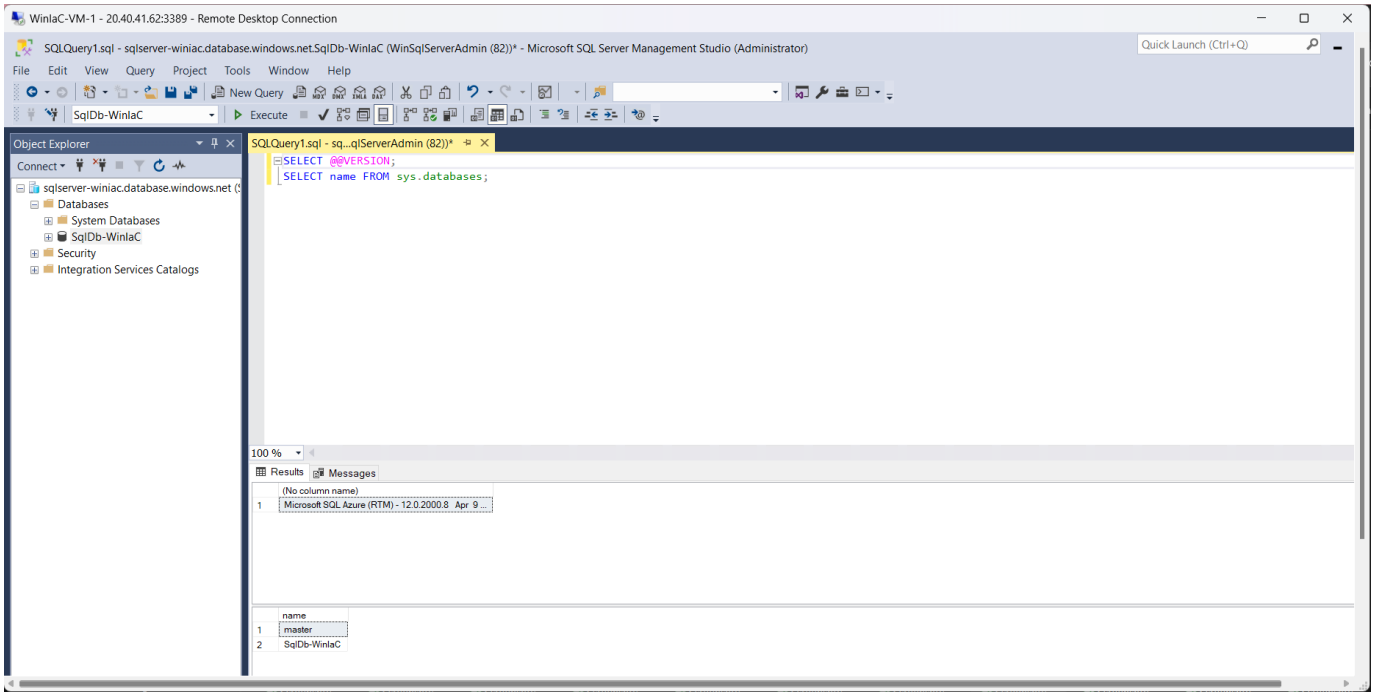
TESTING AND VALIDATION



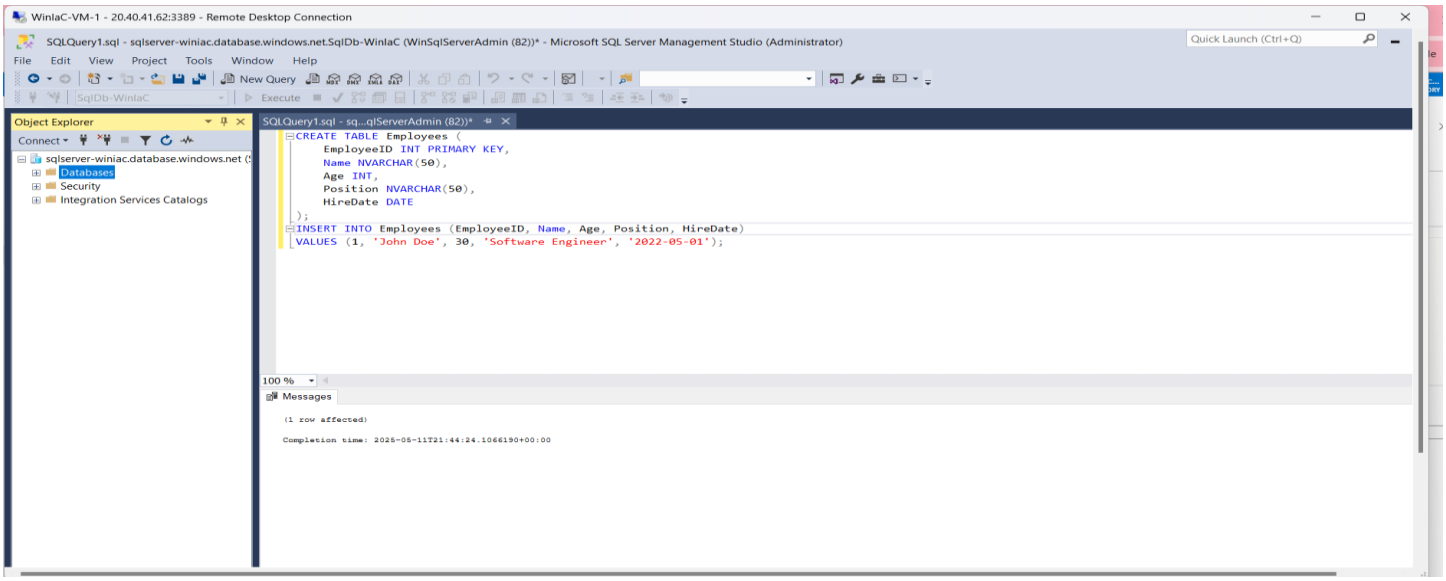




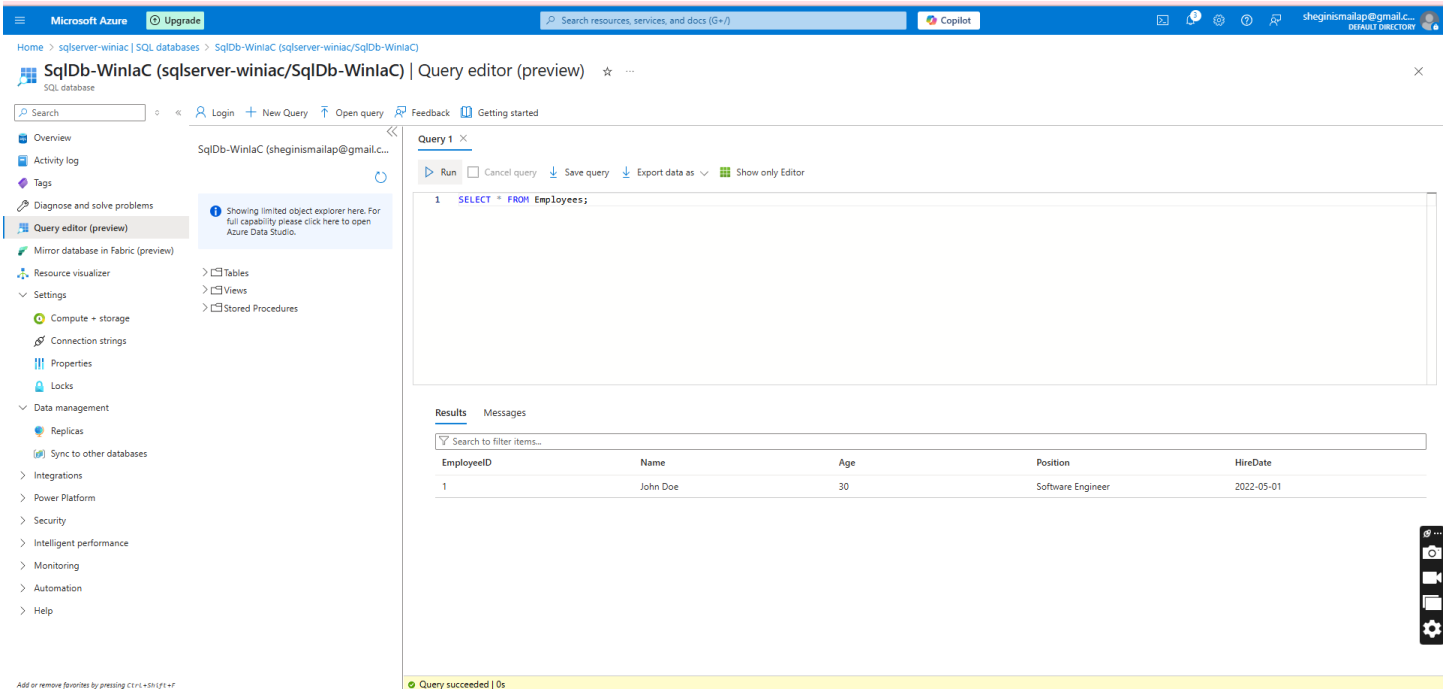
RESULT



Creating table and adding values



RESULT -REFLECTED ON AZURE PORTAL



steps to set up a **Recovery Services Vault** for VM backups in Azure:

1. Create a Recovery Services Vault

- Open **Azure Portal**, go to **Recovery Services Vaults**.
- Click **Create**, select **Resource Group**, and enter a vault name.
- Choose **Region** and click **Review + Create** → **Create**.

2. Configure Backup for Virtual Machines

- Open the **Recovery Services Vault**.
- Navigate to **Backup** > Click **Configure Backup**.
- Select **Azure Virtual Machine** as the workload.

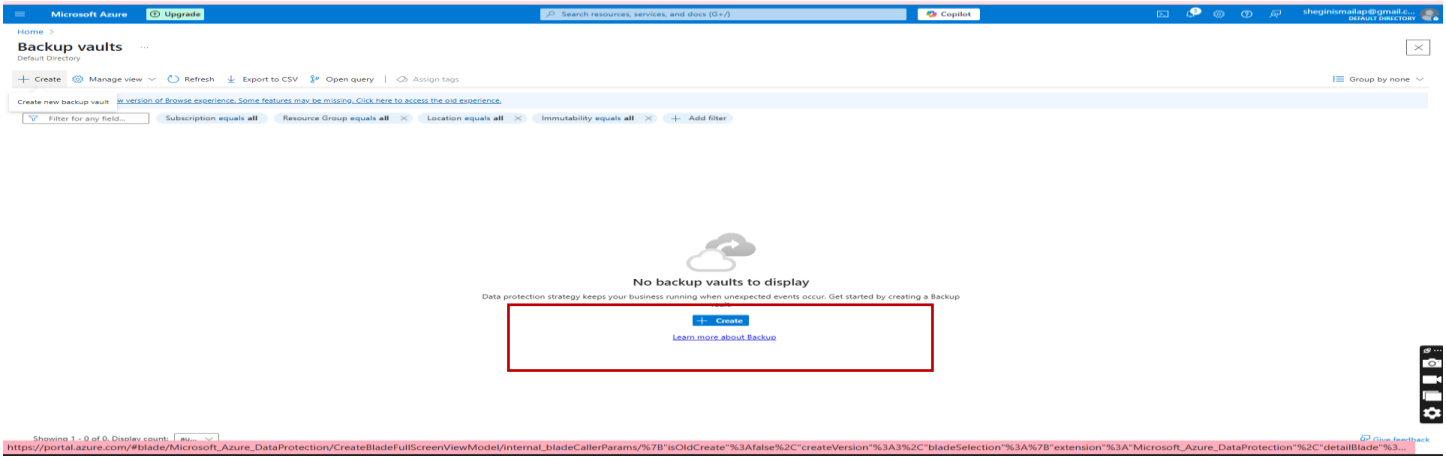
- Choose the VM(s) to be backed up.
- Click **Enable Backup**.

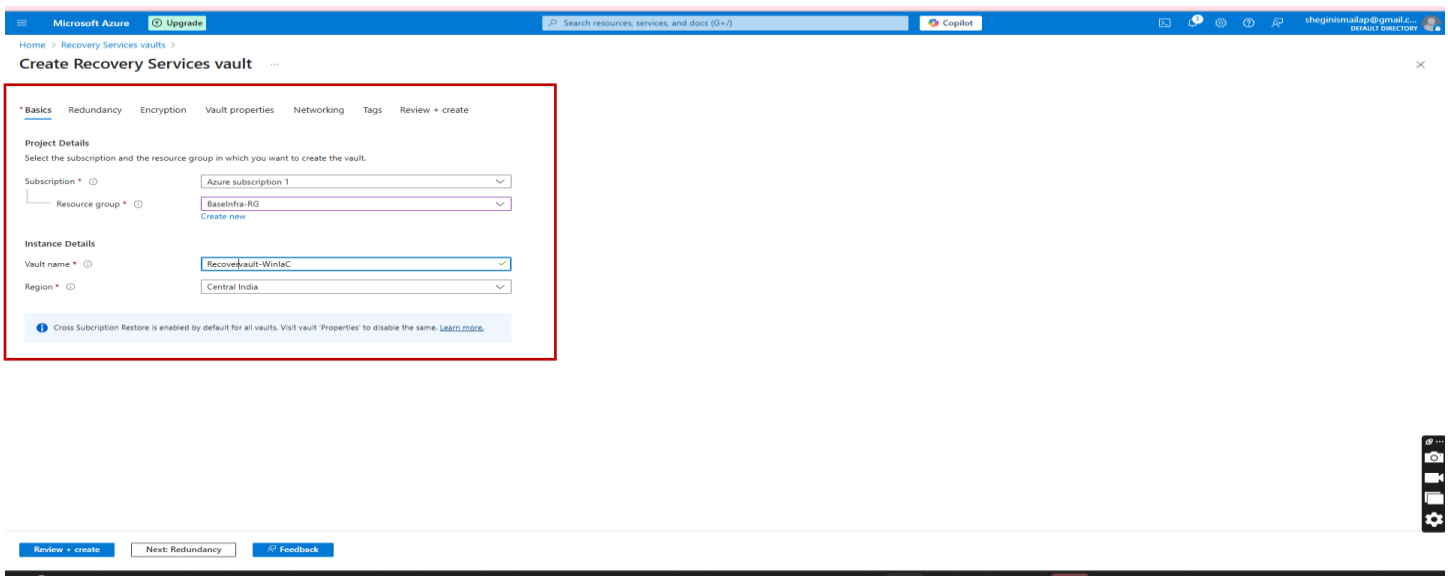
3. Define Backup Policy

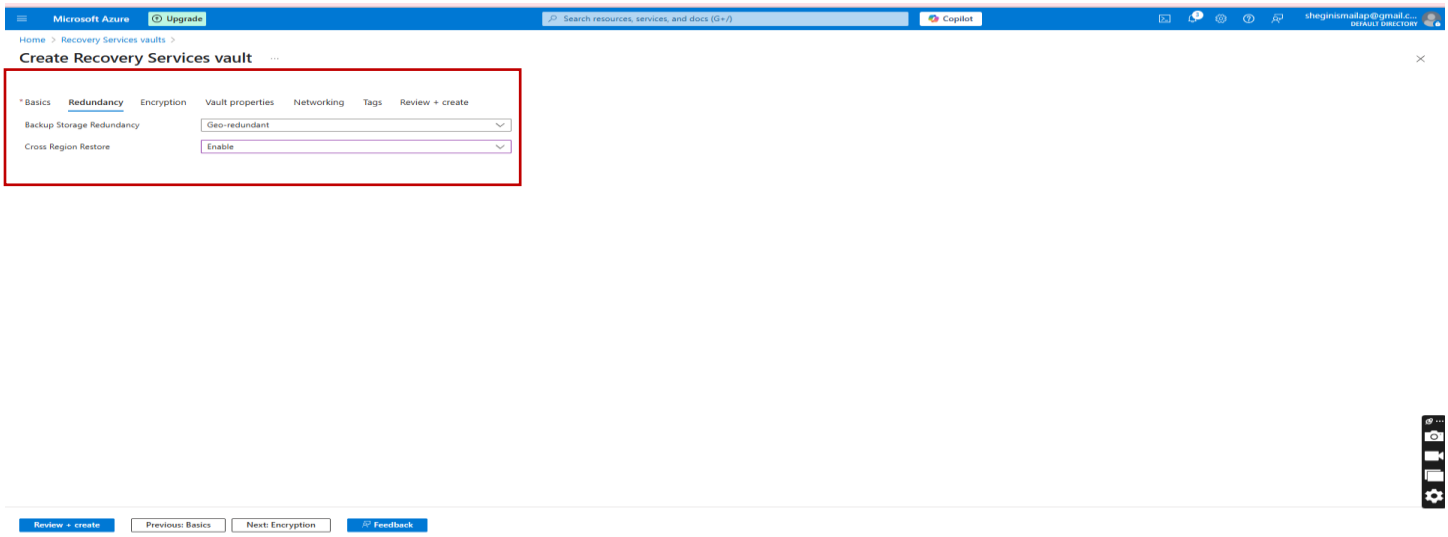
- In the **Vault**, go to **Backup Policies**.
- Click **Create New Policy** or use an existing one.
- Set **Retention Period, Backup Frequency**, and **Time Zone**.
- Apply the policy to selected VMs.

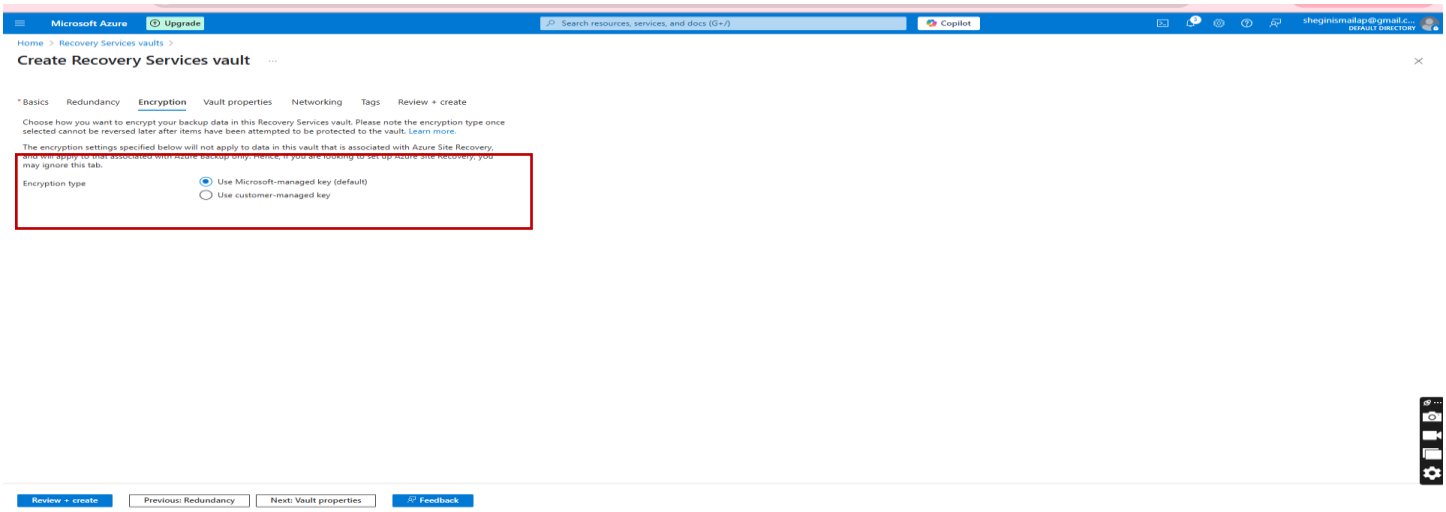
4. Trigger & Monitor Backups

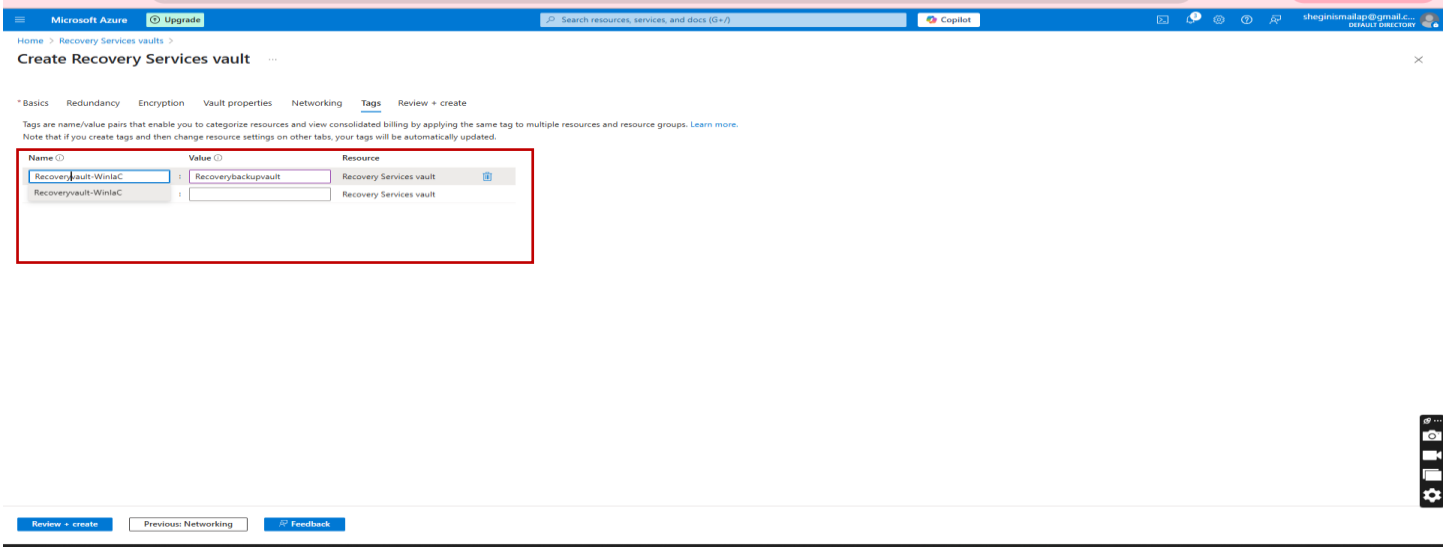
- Manually start an **on-demand backup** to verify setup.
- Monitor backup jobs under the **Vault > Backup Jobs** section.
- Review alerts and failure logs if needed.

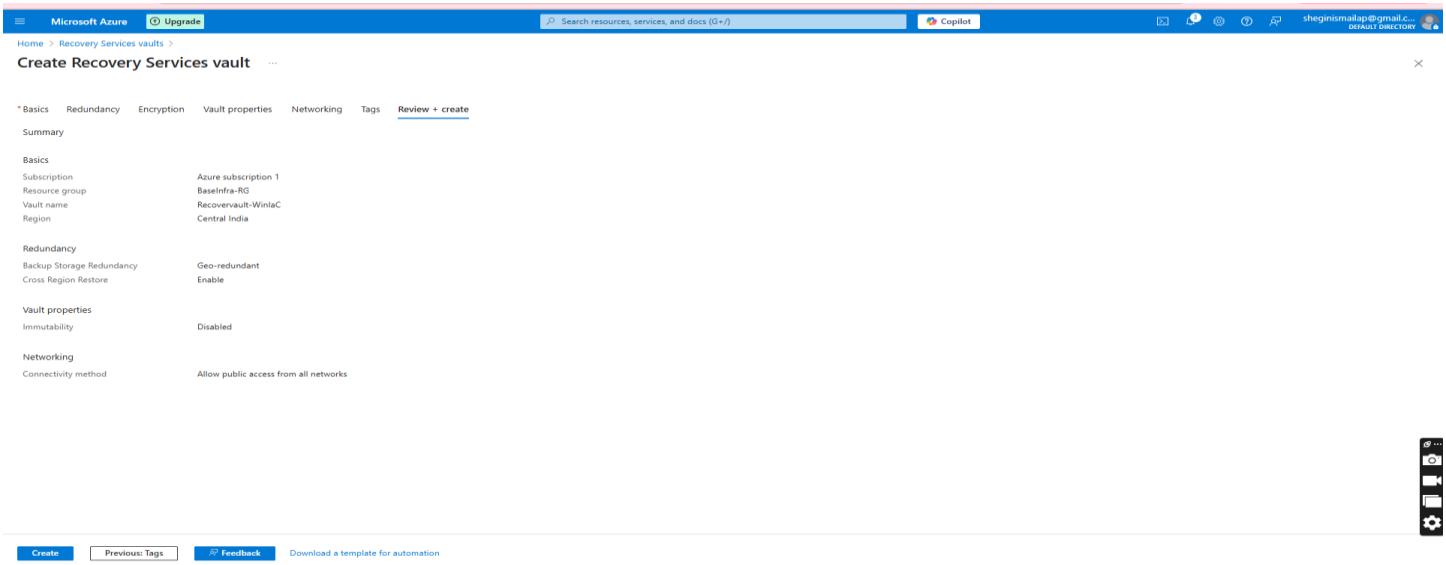


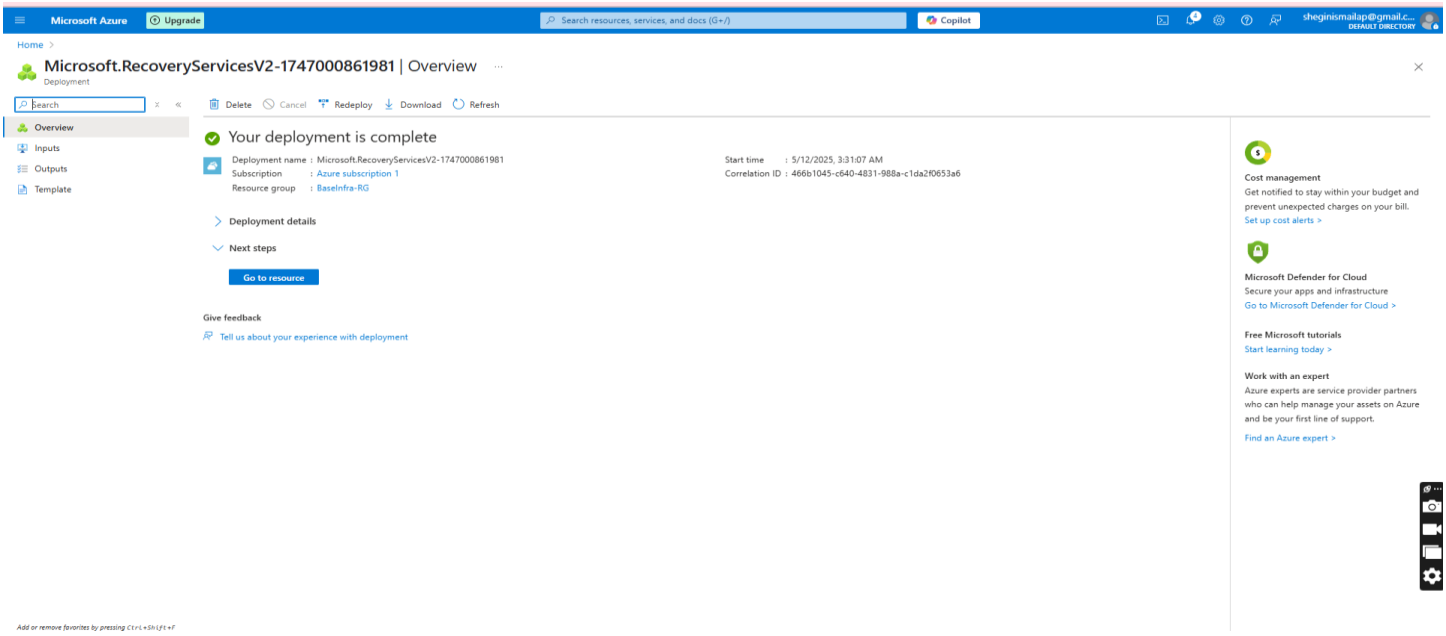



Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Recovery Services vaults >
Create Recovery Services vault
*Basics
Redundancy
Encryption
Vault properties
Networking
Tags
Review + create
Backup Storage Redundancy
Geo-redundant
Cross Region Restore
Enable
Review + create
Previous: Basics
Next: Encryption
Feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Recovery Services vaults >
Create Recovery Services vault
*Basics
Redundancy
Encryption
Vault properties
Networking
Tags
Review + create
Choose how you want to encrypt your backup data in this Recovery Services vault. Please note the encryption type once selected cannot be reversed later after items have been attempted to be protected to the vault. Learn more.
The encryption settings specified below will not apply to data in this vault that is associated with Azure Site Recovery,
may ignore this tab.
Encryption type
Use Microsoft-managed key (default)
Use customer-managed key
Review + create
Previous: Redundancy
Next: Vault properties
Feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Recovery Services vaults >
Create Recovery Services vault
*Basics
Redundancy
Encryption
Vault properties
Networking
Tags
Review + create
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Learn more.
Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.
Name
Value
Resource
Recoveryvault-WinlaC
Recoverybackupvault
Recovery Services vault
Recoveryvault-WinlaC
Recovery Services vault
Review + create
Previous: Networking
Feedback


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Recovery Services vaults >
Create Recovery Services vault
Basics Redundancy Encryption Vault properties Networking Tags Review + create
Summary
Basics
Subscription Azure subscription 1
Resource group BaseInfra-RG
Vault name Recovervault-WinlaC
Region Central India
Redundancy
Backup Storage Redundancy Geo-redundant
Cross Region Restore Enable
Vault properties
Immutability Disabled
Networking
Connectivity method Allow public access from all networks
Create
Previous: Tags
Feedback
Download a template for automation


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home >
Microsoft.RecoveryServicesV2-1747000861981 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete Cancel Redeploy Download Refresh
Your deployment is complete
Deployment name : Microsoft.RecoveryServicesV2-1747000861981
Subscription : Azure subscription 1
Resource group : BaseInfra-RG
Start time : 5/12/2025, 3:31:07 AM
Correlation ID : 466b1045-c640-4831-988a-c1da2f0653a6
Deployment details
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
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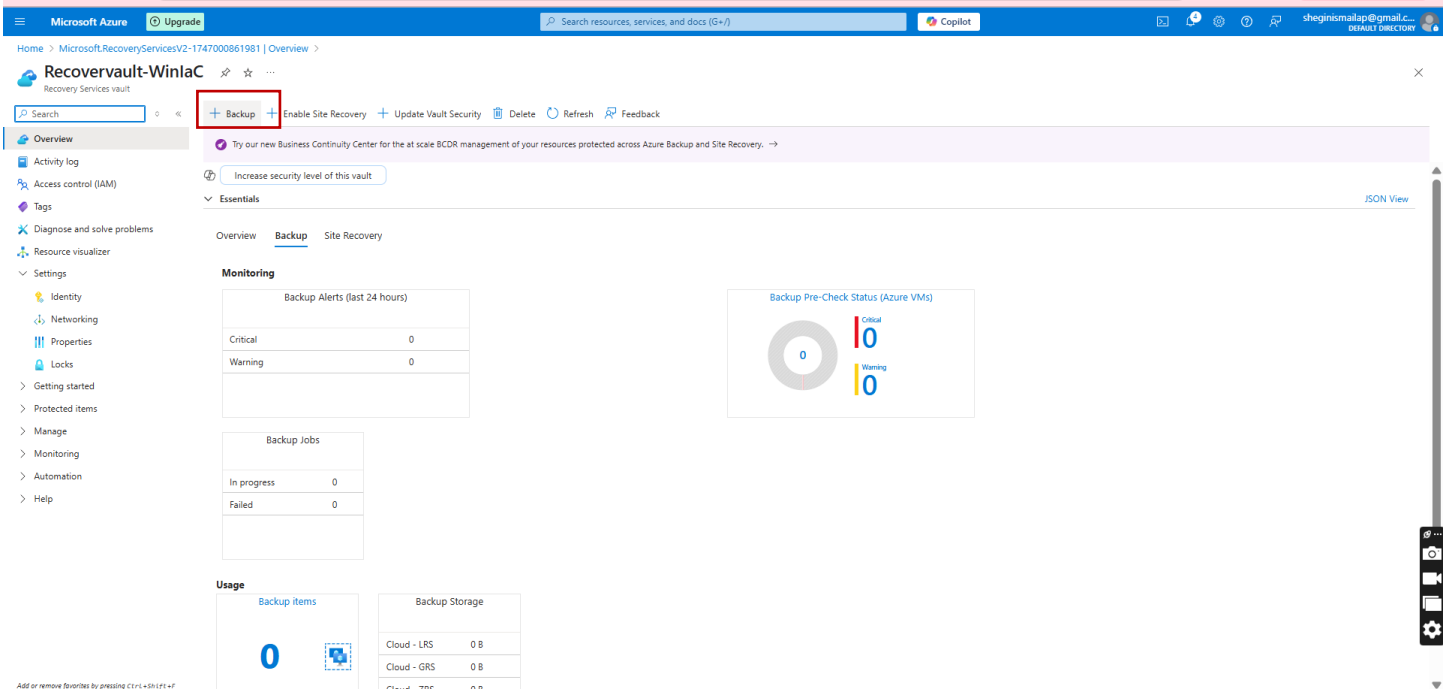
Microsoft Azure
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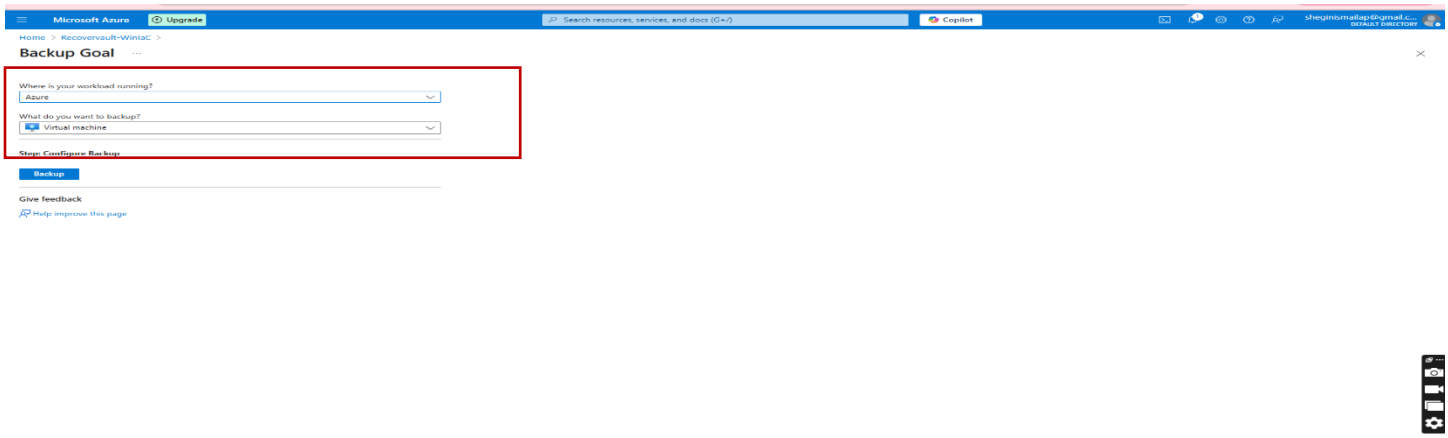
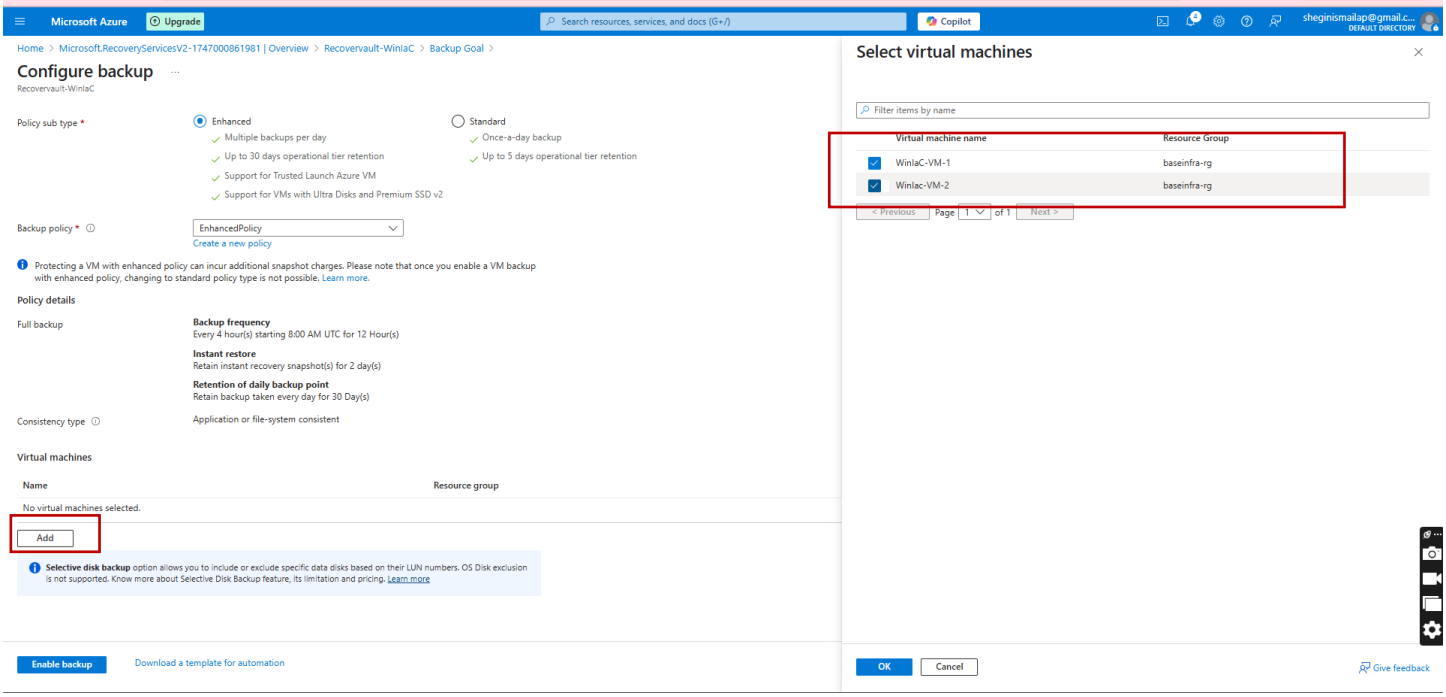
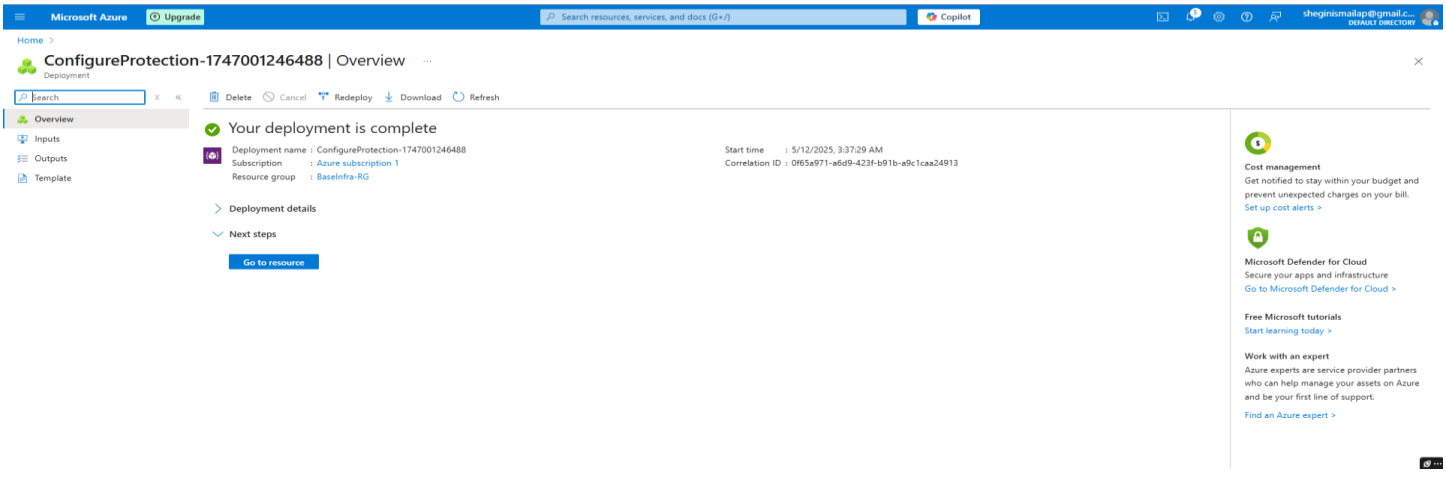
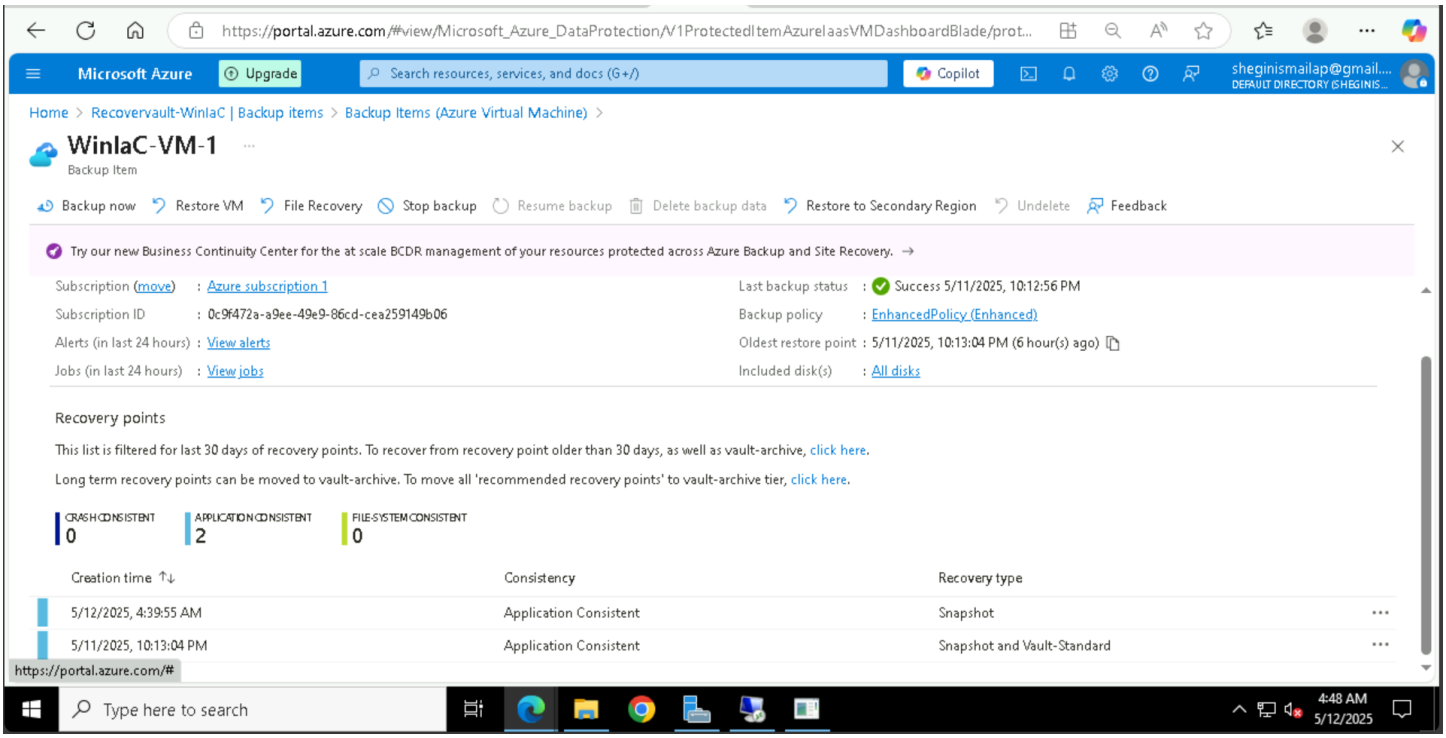
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Microsoft.RecoveryServicesV2-1747000861981 | Overview >
Recovervault-WinlaC
Recovery Services vault
Search
Backup
Enable Site Recovery
Update Vault Security
Delete
Refresh
Feedback
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Identity
Networking
Properties
Locks
Getting started
Protected items
Manage
Monitoring
Automation
Help
Try our new Business Continuity Center for the at scale BCDR management of your resources protected across Azure Backup and Site Recovery.
Increase security level of this vault
Essentials
JSON View
Overview Backup Site Recovery
Monitoring
Backup Alerts (last 24 hours)
Critical 0
Warning 0
Backup Pre-Check Status (Azure VMs)
Critical
0
Warning
0
Backup Jobs
In progress 0
Failed 0
Usage
Backup items
0
Backup Storage
Cloud - LRS 0 B
Cloud - GRS 0 B
Add or remove favorites by pressing Ctrl+Shift+F



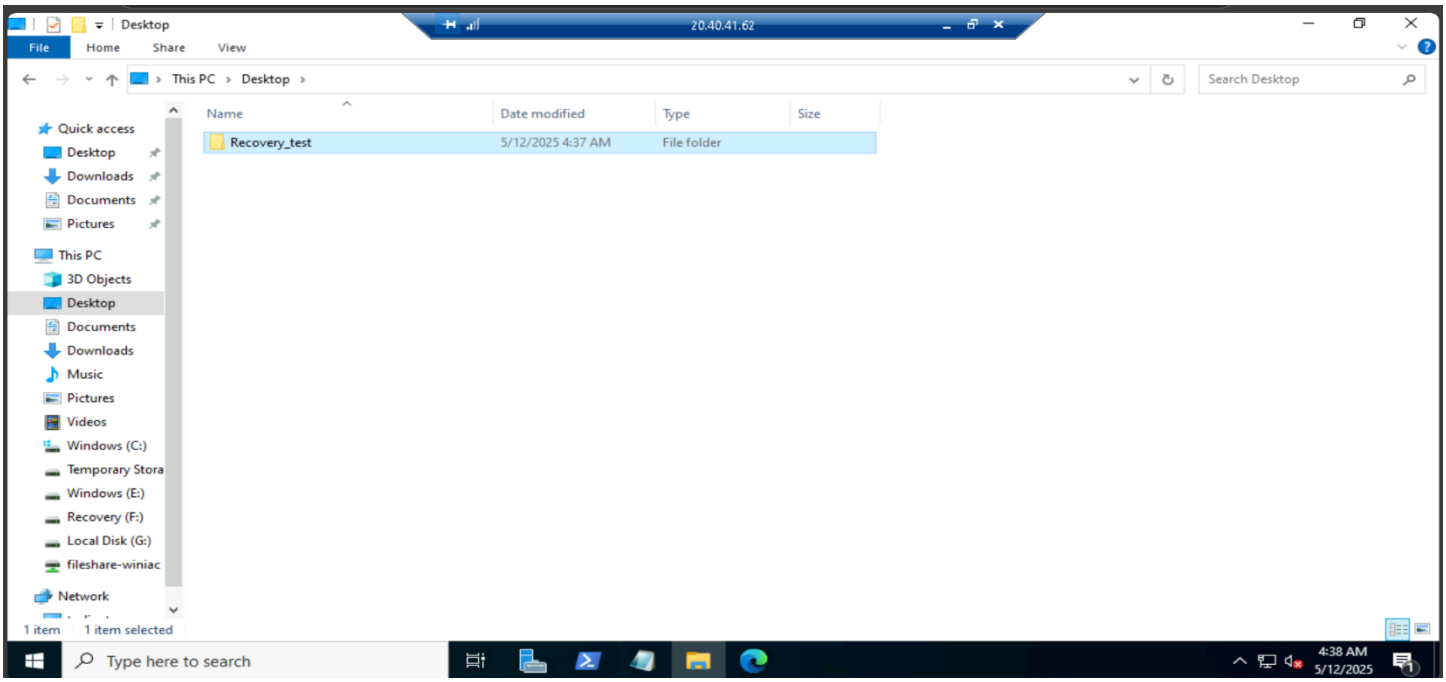


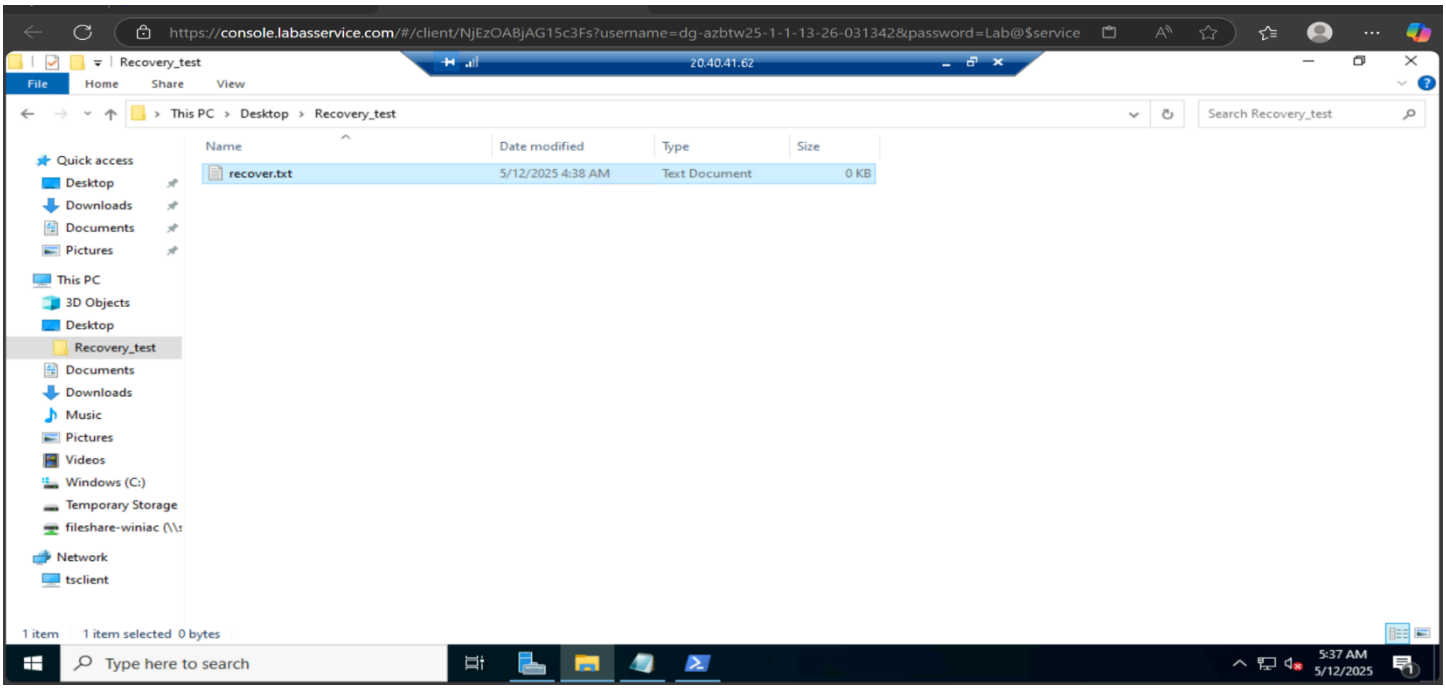


TESTING AND VALIDATION

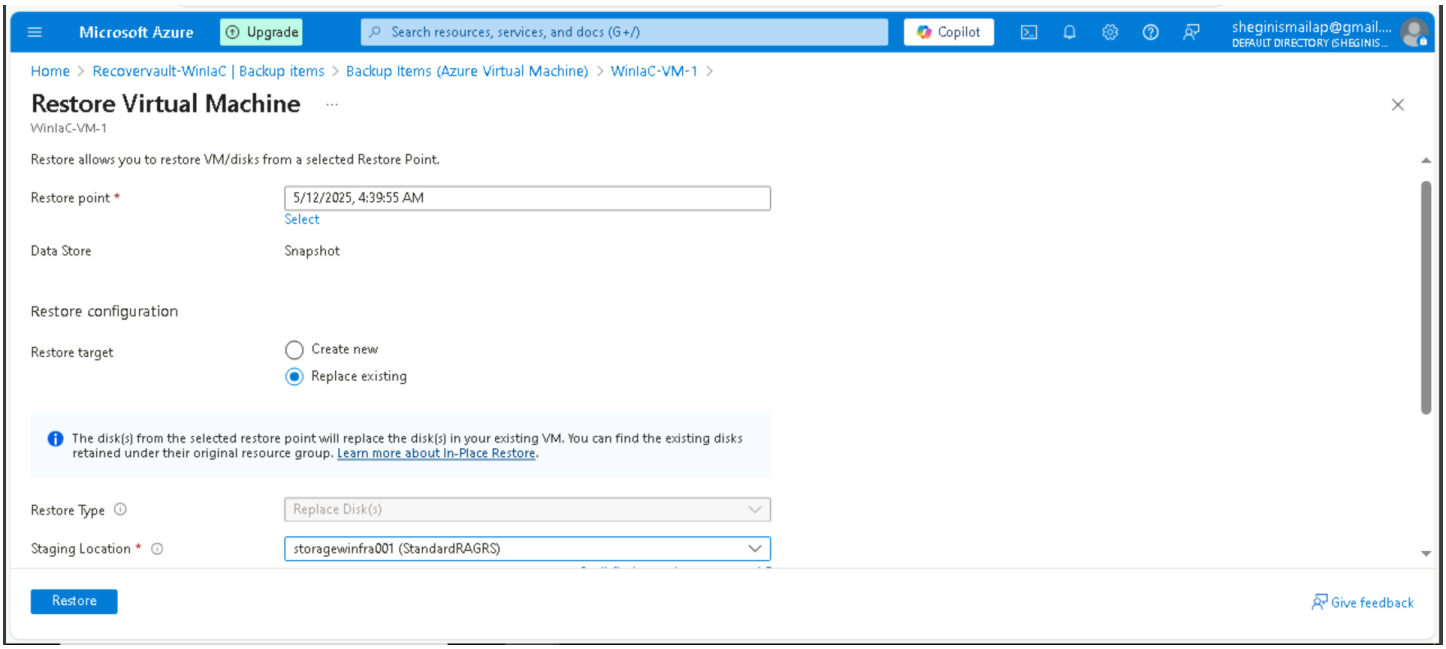


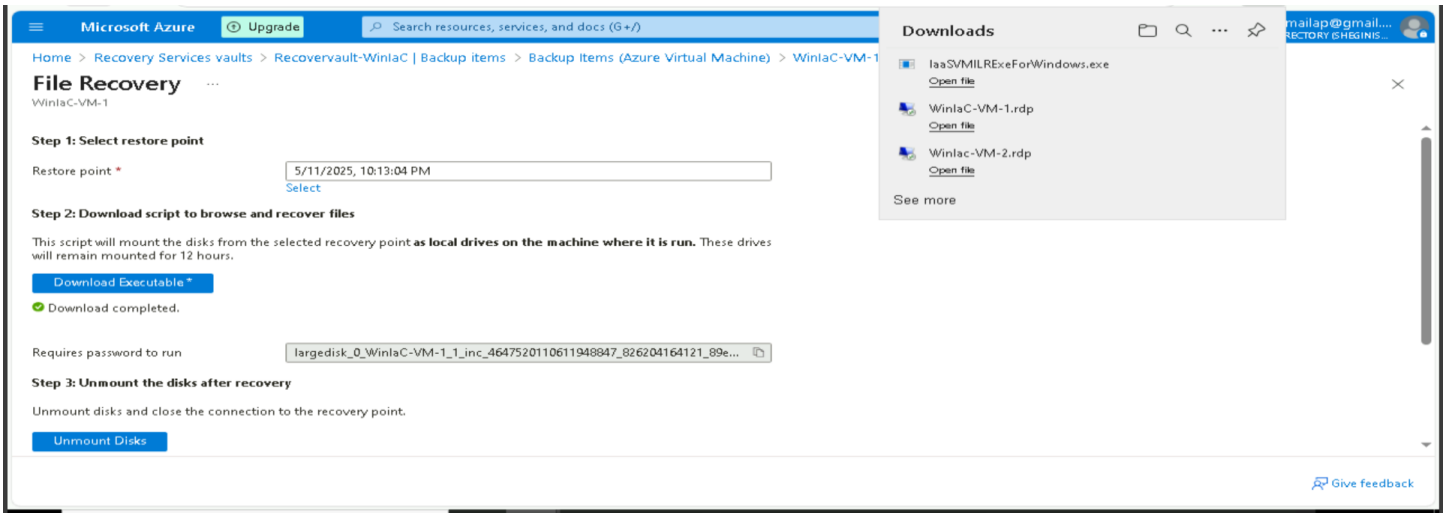
Create a folder recovery-test with file recover.txt which shall be deleted after backup and tried for restored

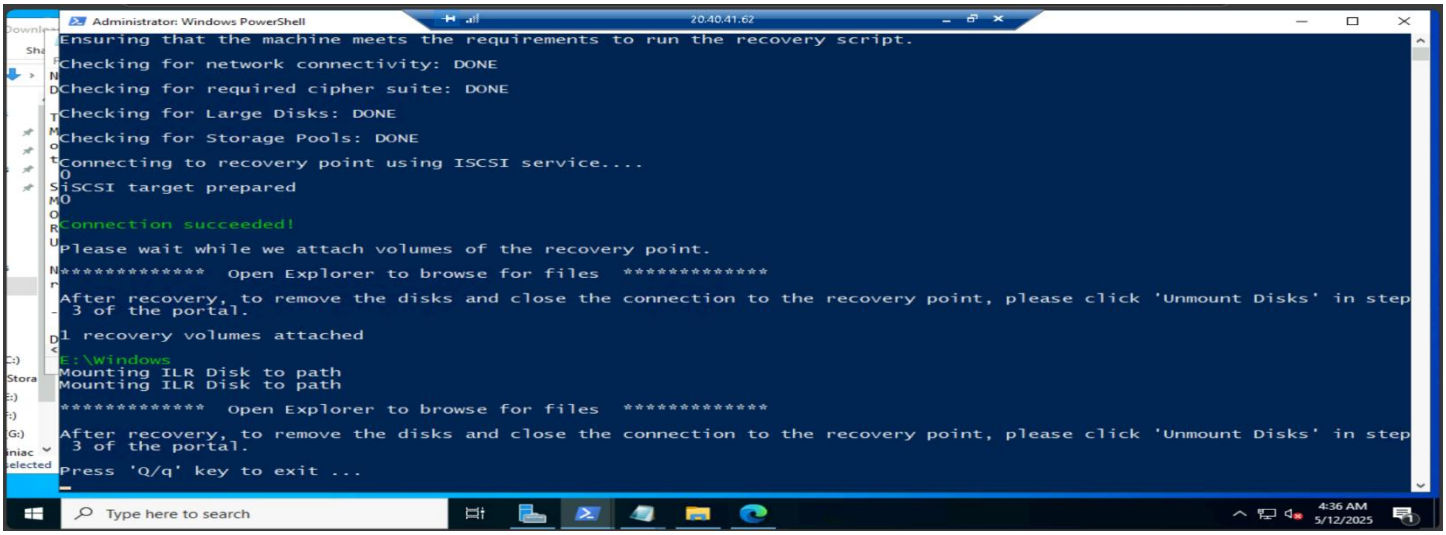




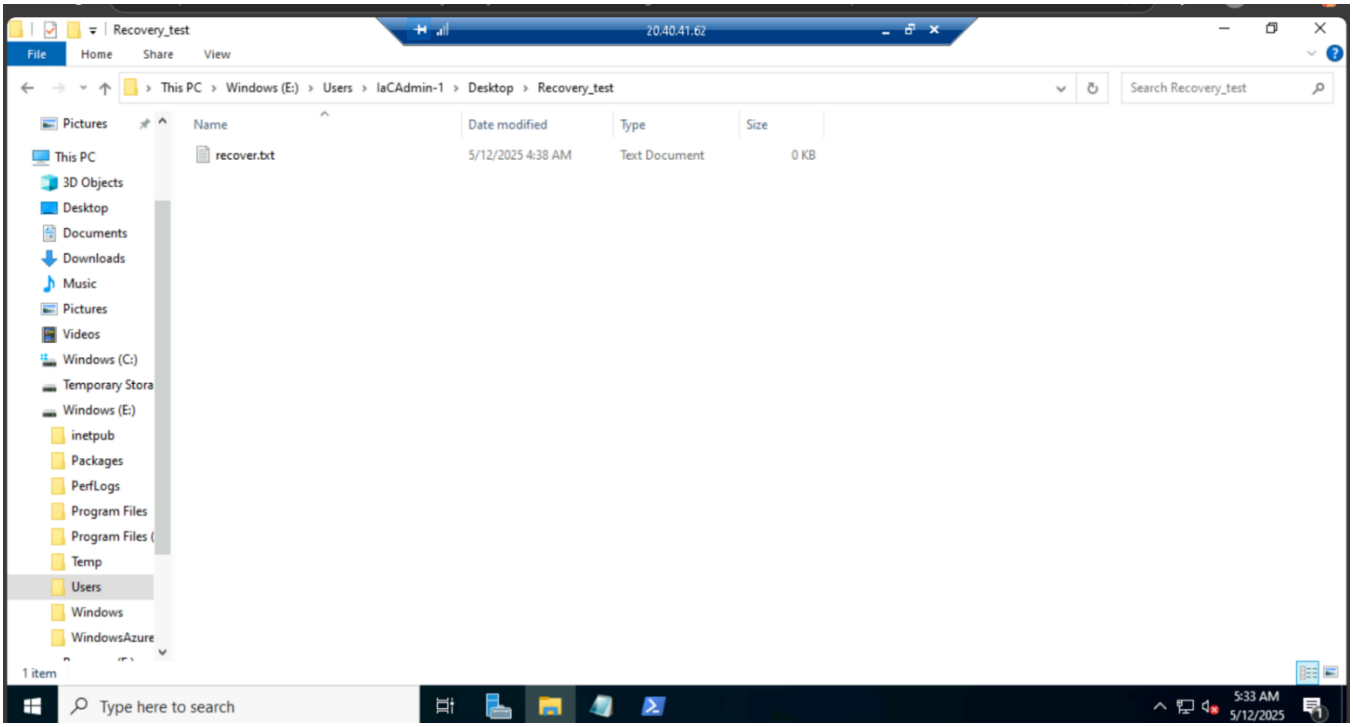
Performing file restoration




Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Home > Recovery Services vaults > Recovervault-WinlaC | Backup items > Backup Items (Azure Virtual Machine) > WinlaC-VM-1
File Recovery
WinlaC-VM-1
Step 1: Select restore point
Restore point *
5/11/2025, 10:13:04 PM
Select
Step 2: Download script to browse and recover files
This script will mount the disks from the selected recovery point as local drives on the machine where it is run. These drives will remain mounted for 12 hours.
Download Executable *
Download completed.
Requires password to run
largedisk_0_WinlaC-VM-1_1_inc_4647520110611948847_826204164121_89e...
Step 3: Unmount the disks after recovery
Unmount disks and close the connection to the recovery point.
Unmount Disks
Give feedback
Downloads
IaaSVMILRExeForWindows.exe
Open file
WinlaC-VM-1.rdp
Open file
Winlac-VM-2.rdp
Open file
See more


Administrator: Windows PowerShell
20.40.41.62
Ensuring that the machine meets the requirements to run the recovery script.
Checking for network connectivity: DONE
Checking for required cipher suite: DONE
Checking for Large Disks: DONE
Checking for Storage Pools: DONE
Connecting to recovery point using ISCSI service....
iSCSI target prepared
Connection succeeded!
Please wait while we attach volumes of the recovery point.
********** Open Explorer to browse for files **********
After recovery, to remove the disks and close the connection to the recovery point, please click 'Unmount Disks' in step 3 of the portal.
1 recovery volumes attached
E:\Windows
Mounting ILR Disk to path
Mounting ILR Disk to path
********** Open Explorer to browse for files **********
After recovery, to remove the disks and close the connection to the recovery point, please click 'Unmount Disks' in step 3 of the portal.
Press 'Q/q' key to exit ...
Type here to search
4:36 AM
5/12/2025

File found


Recovery_test
20.40.41.62
File
Home
Share
View
This PC > Windows (E:) > Users > laCAdmin-1 > Desktop > Recovery_test
Search Recovery_test
Pictures
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
Temporary Stora
Windows (E:)
inetpub
Packages
PerfLogs
Program Files
Program Files (
Temp
Users
Windows
WindowsAzure
Name
Date modified
Type
Size
recover.txt
5/12/2025 4:38 AM
Text Document
0 KB
1 item
Type here to search
5:33 AM
5/12/2025

steps to configure a **CPU alert (80%) via Azure Monitor**:

1. Open Azure Monitor

- Navigate to **Azure Portal** → **Monitor**.
- Click **Alerts** → **Create Alert Rule**.

2. Define Scope

- Click **Scope** → Select the **VM** to monitor.
- Click **Next: Condition**.

3. Configure Alert Condition

- Select **CPU Percentage** as the **Signal Name**.
- Set **Threshold** to **80%**.
- Configure **Aggregation Period** (e.g., 5 minutes).
- Click **Next: Actions**.

4. Create Action Group Within Alert Rule Setup

- Click **Create New Action Group** (since it wasn't set up before).
- Define **Action Group Name** and specify **Notification Type** (Email, SMS, Azure App).
- Configure **Receivers** for alerts.
- Click **Save**.

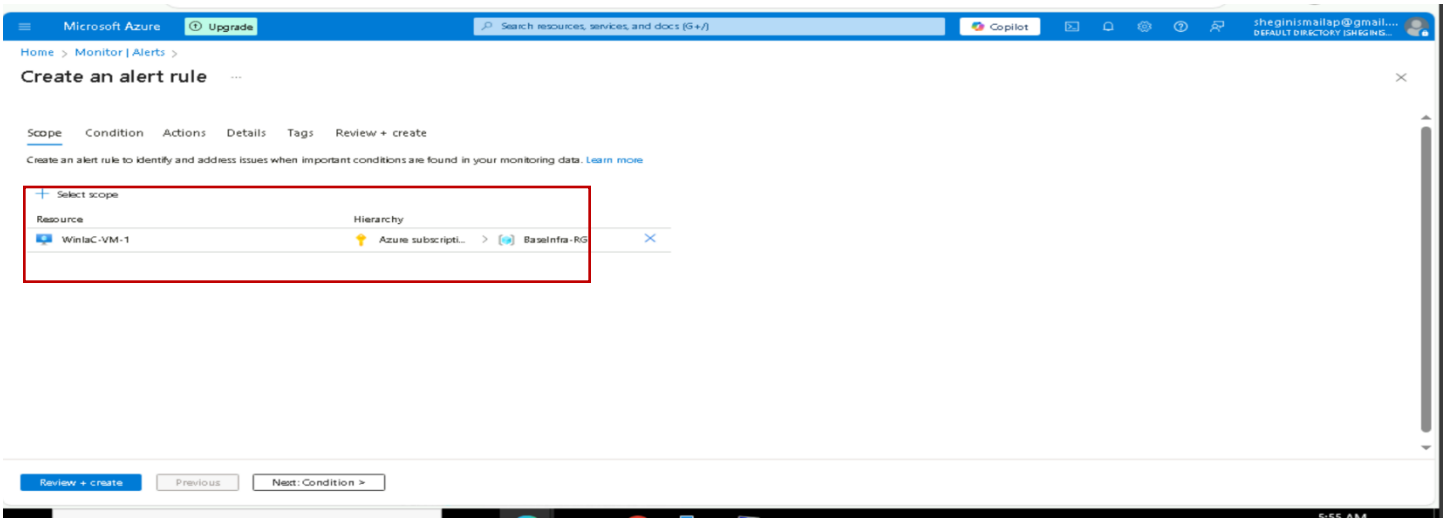
5. Finalize Alert Rule

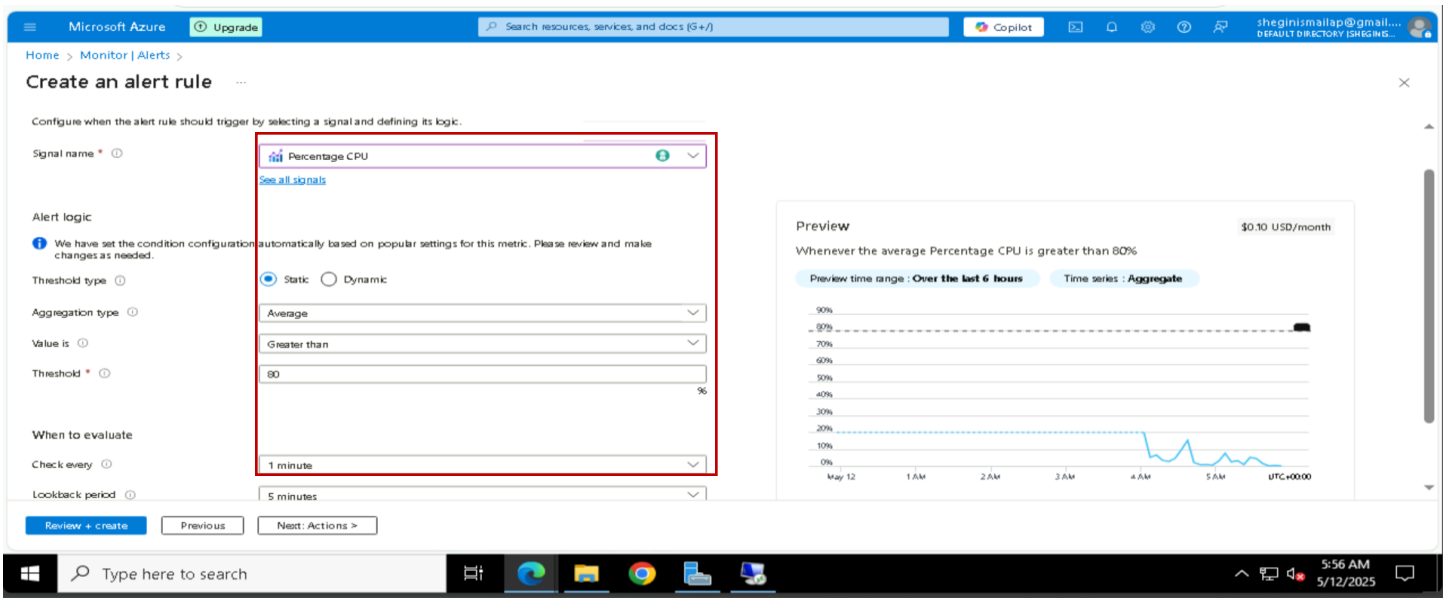
- Set **Severity Level** (Warning or Critical).
- Enter **Rule Name** and **Description**.
- Click **Next: Review + Create** → **Create**.

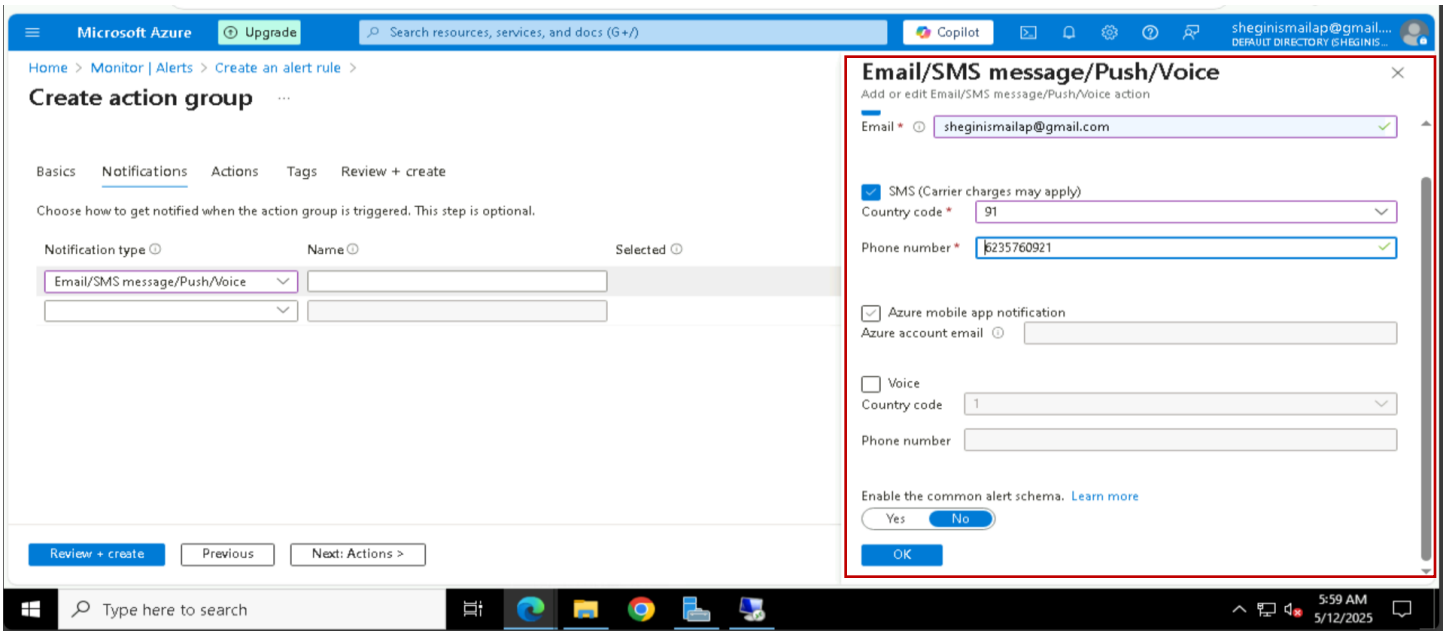
6. Validate and Test

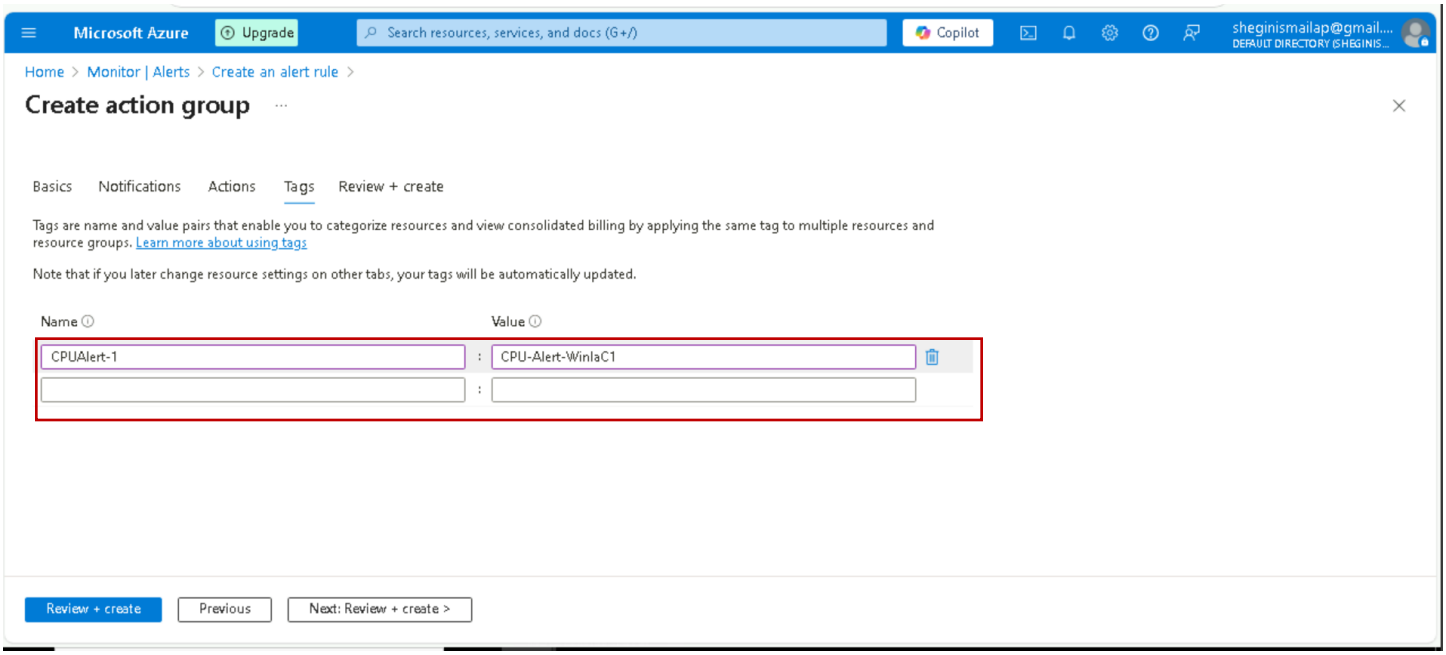
- Generate high CPU usage or monitor naturally occurring spikes.
- Ensure alerts are sent to the newly created action group.

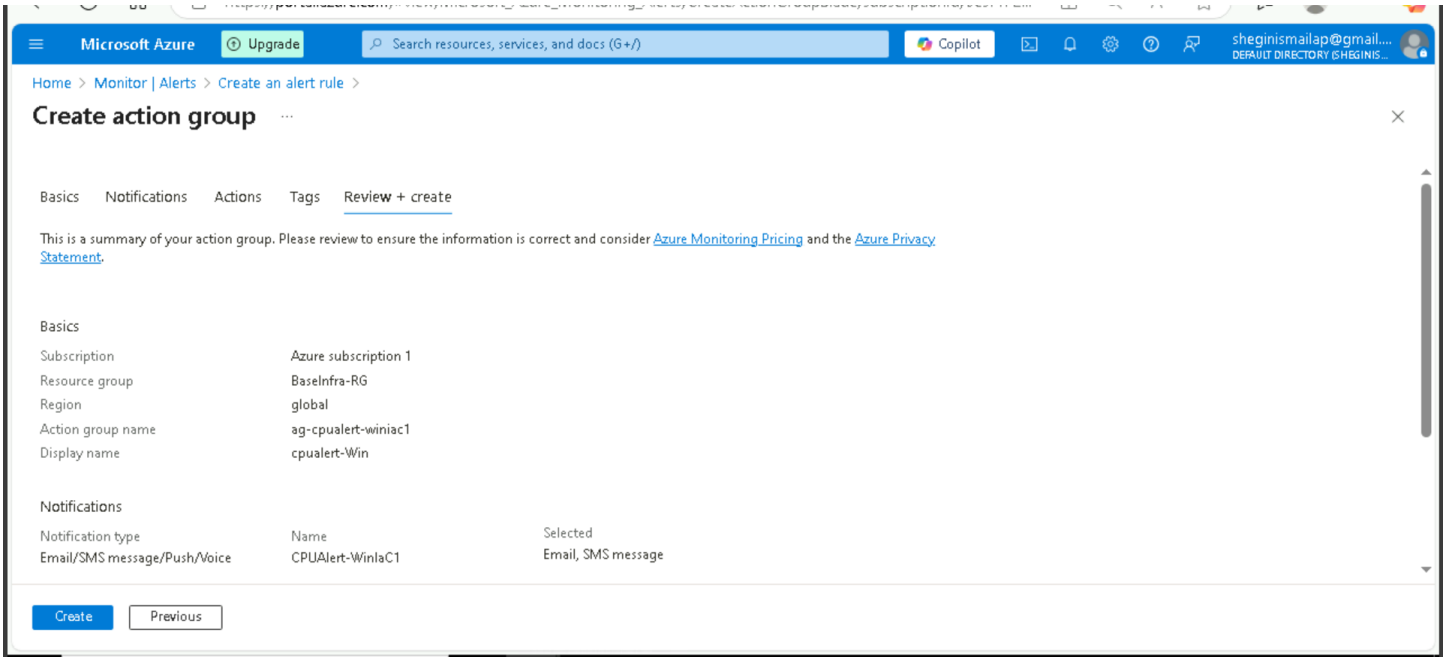
Setting Alert Rule for First VMI

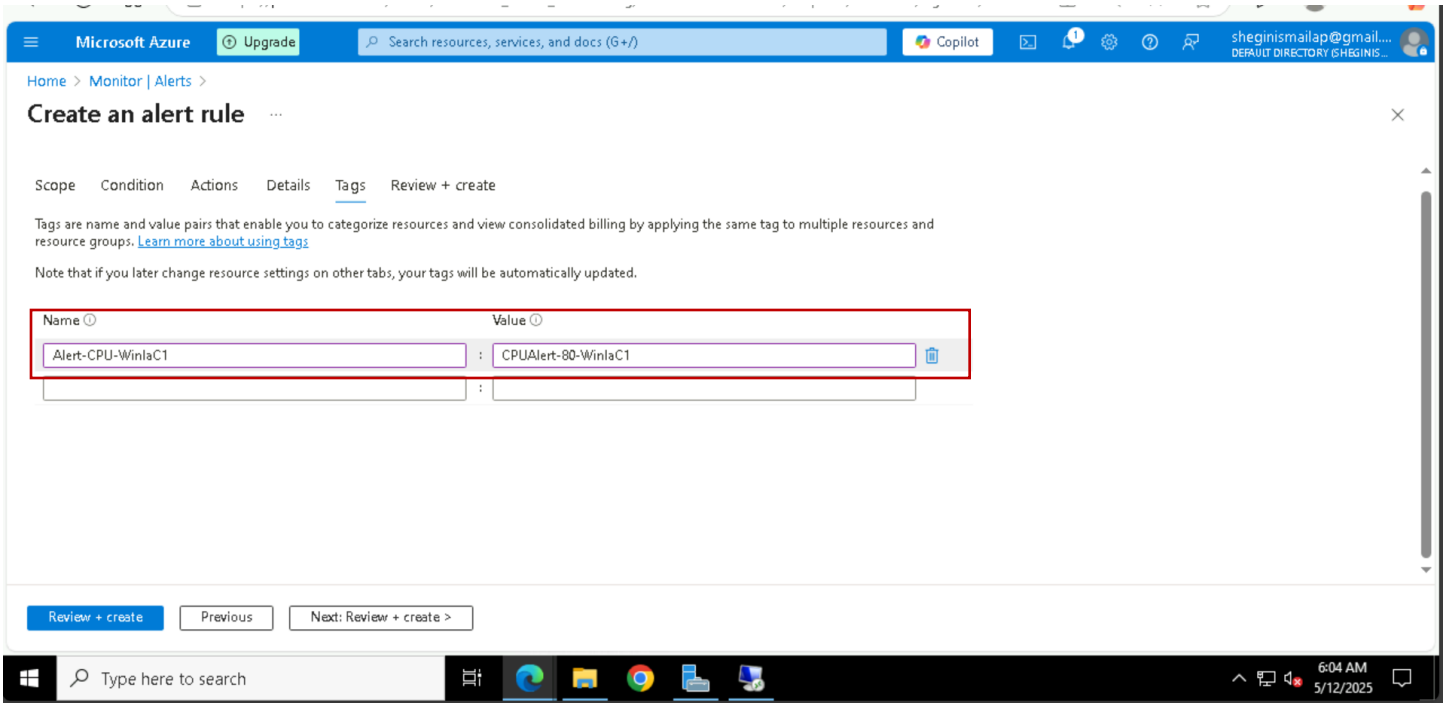


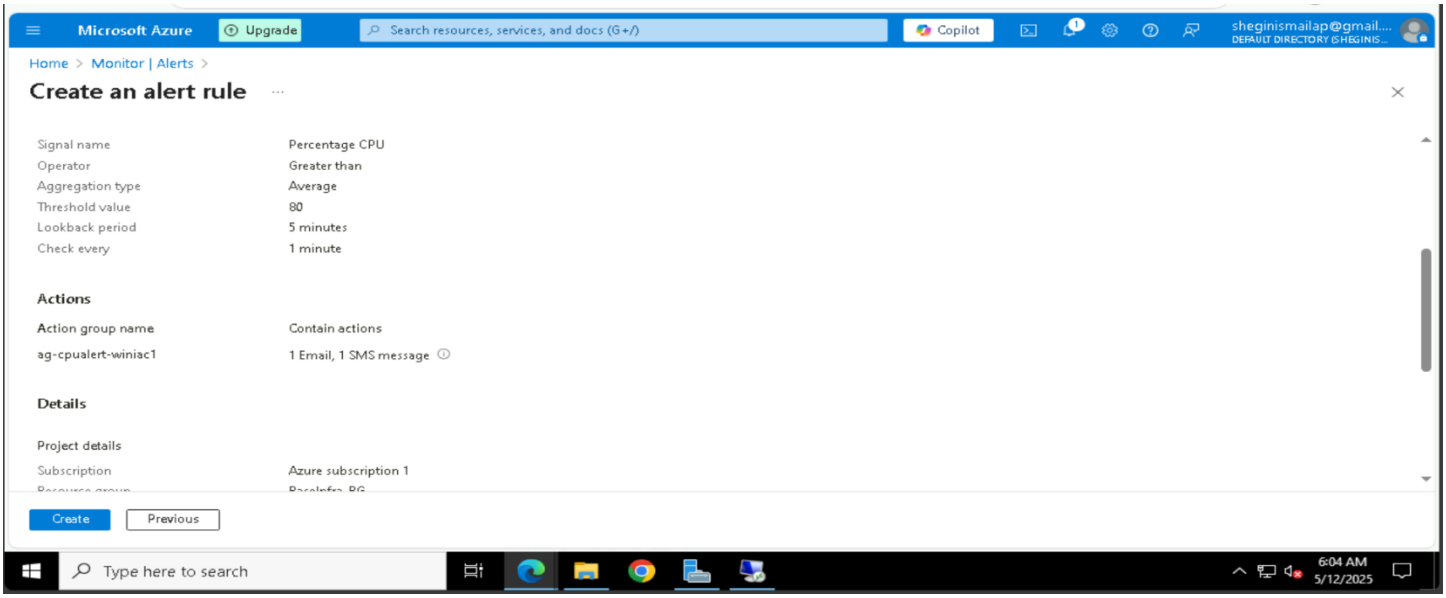




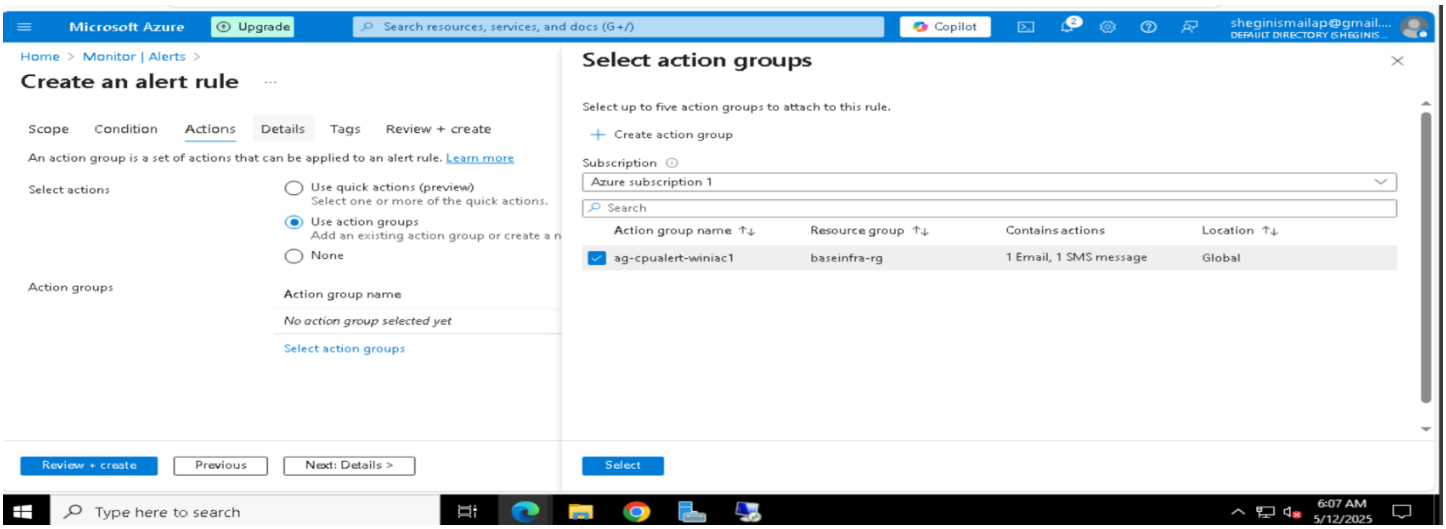

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY (SHEGINIS...
Home > Monitor | Alerts > Create an alert rule >
Create action group
Basics
Notifications
Actions
Tags
Review + create
Tags are name and value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Learn more about using tags
Note that if you later change resource settings on other tabs, your tags will be automatically updated.
Name
Value
CPUAlert-1
CPU-Alert-WinIaC1
Review + create
Previous
Next: Review + create >

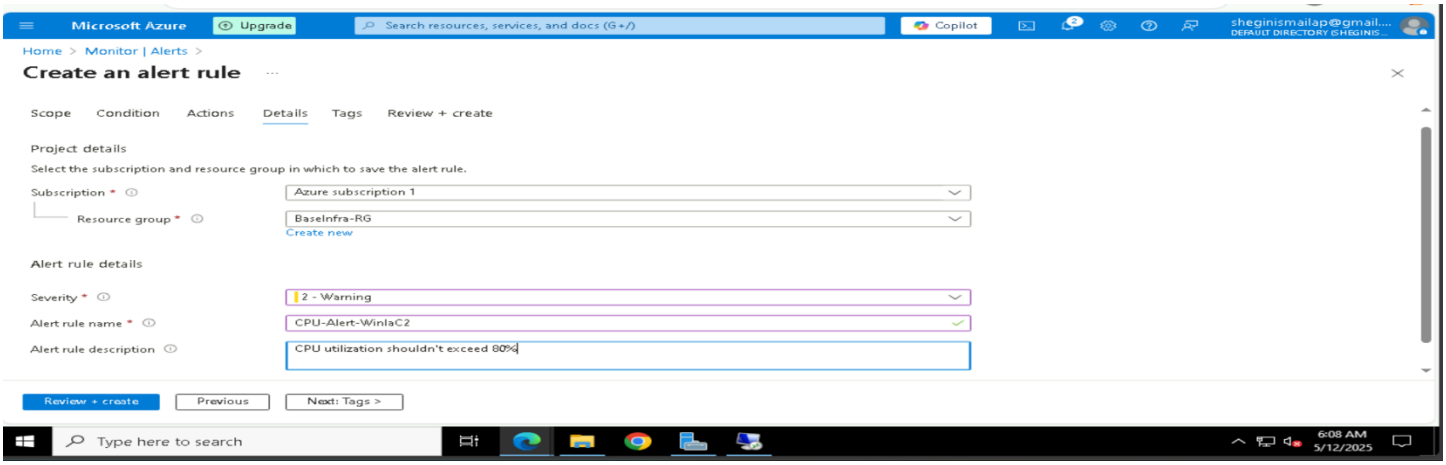

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY (SHEGINIS...
Home > Monitor | Alerts > Create an alert rule >
Create action group
Basics
Notifications
Actions
Tags
Review + create
This is a summary of your action group. Please review to ensure the information is correct and consider Azure Monitoring Pricing and the Azure Privacy Statement.
Basics
Subscription
Azure subscription 1
Resource group
BaseInfra-RG
Region
global
Action group name
ag-cpualert-winiac1
Display name
cpualert-Win
Notifications
Notification type
Email/SMS message/Push/Voice
Name
CPUAlert-WinIaC1
Selected
Email, SMS message
Create
Previous





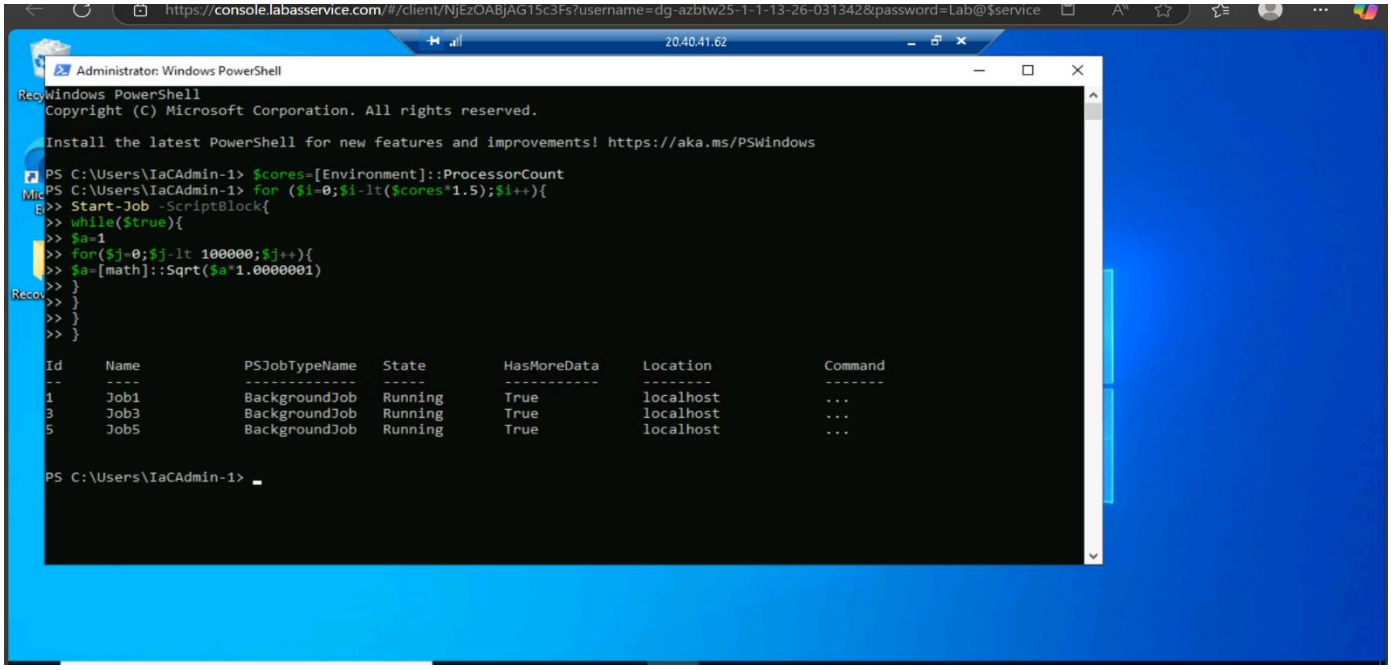
Use existing action group for second VM



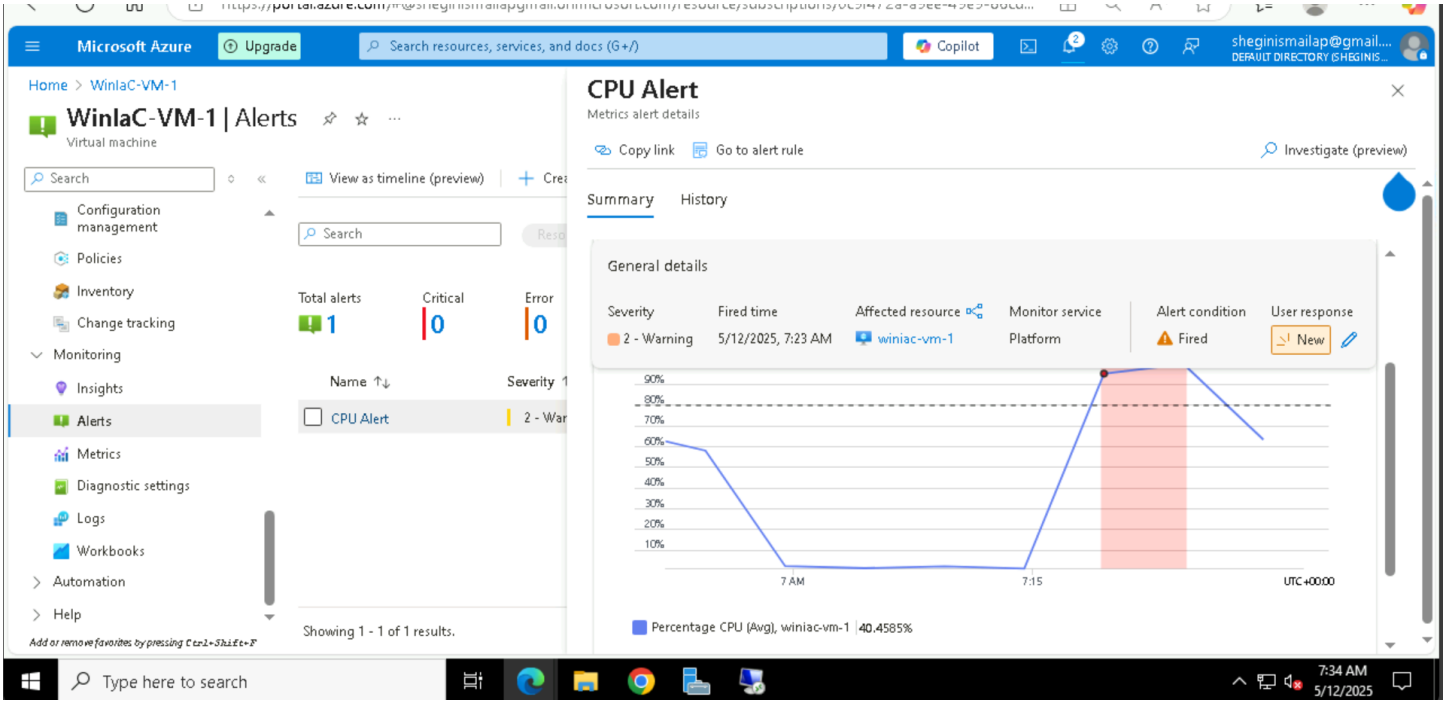


TESTING AND VALIDATION

VM1- CPU overwhelming by running process intensive jobs



RESULT



⚠ Your Azure Monitor alert was triggered

Azure monitor alert rule CPU Alert was triggered for WinIaC-VM-1 at May 12, 2025 7:23 UTC.

Alert rule description	Average CPU Usage should't exceed 80%
Rule ID	/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/BaseInfra-RG/providers/microsoft.insights/metricAlerts/CPU Alert View Rule >
Resource ID	/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/BaseInfra-RG/providers/Microsoft.Compute/virtualMachines/WinIaC-VM-1 View Resource >

Alert Activated Because:

Metric name	Percentage CPU
Metric namespace	virtualMachines/WinIaC-VM-1

← 56767457

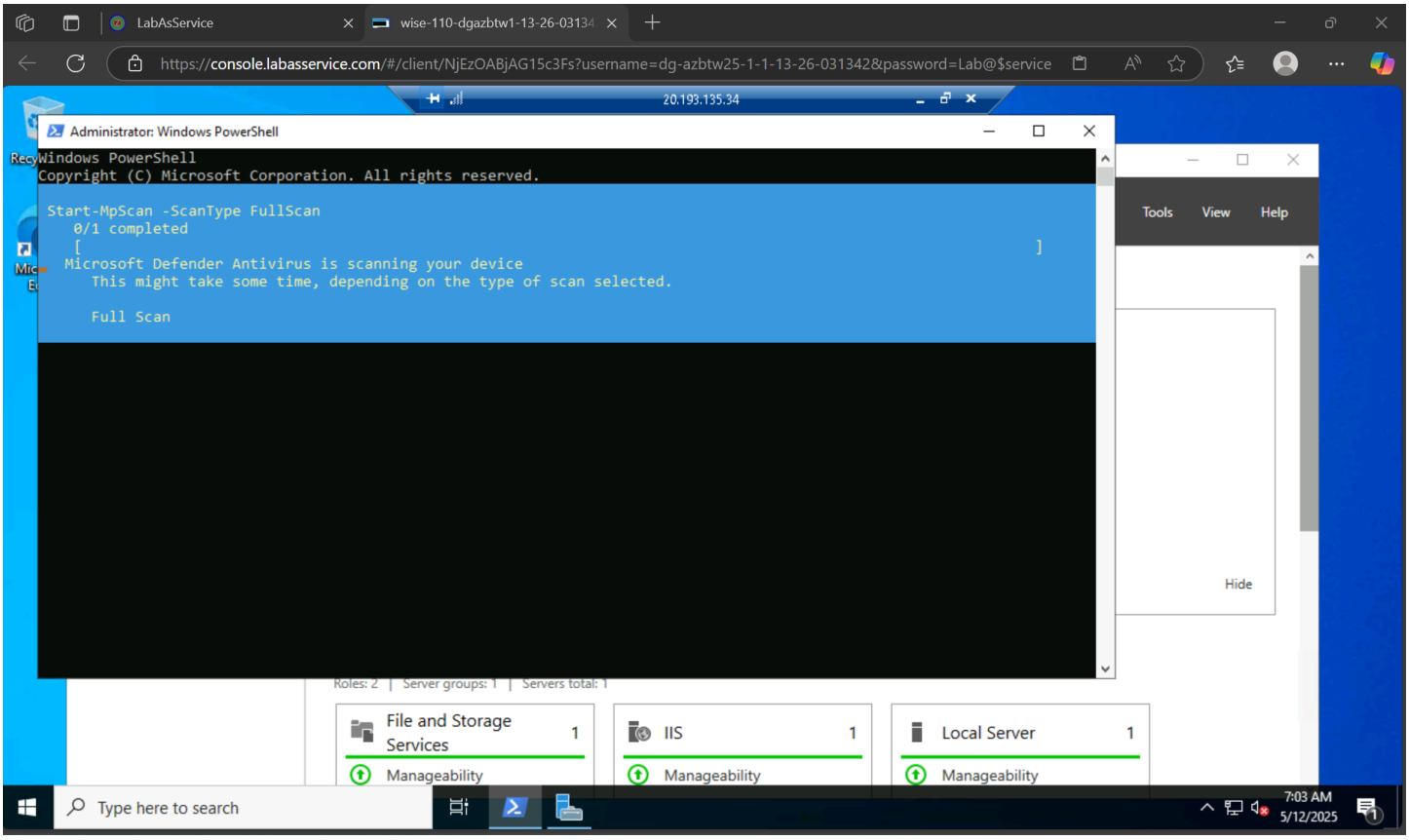
5–12 11:32 AM

All done. You're in cpualert–Win group. Reply 'STOP' to stop all, 'Disable cpualert–Win' to stop grp, 'HELP' for info. Msg&data rates apply.

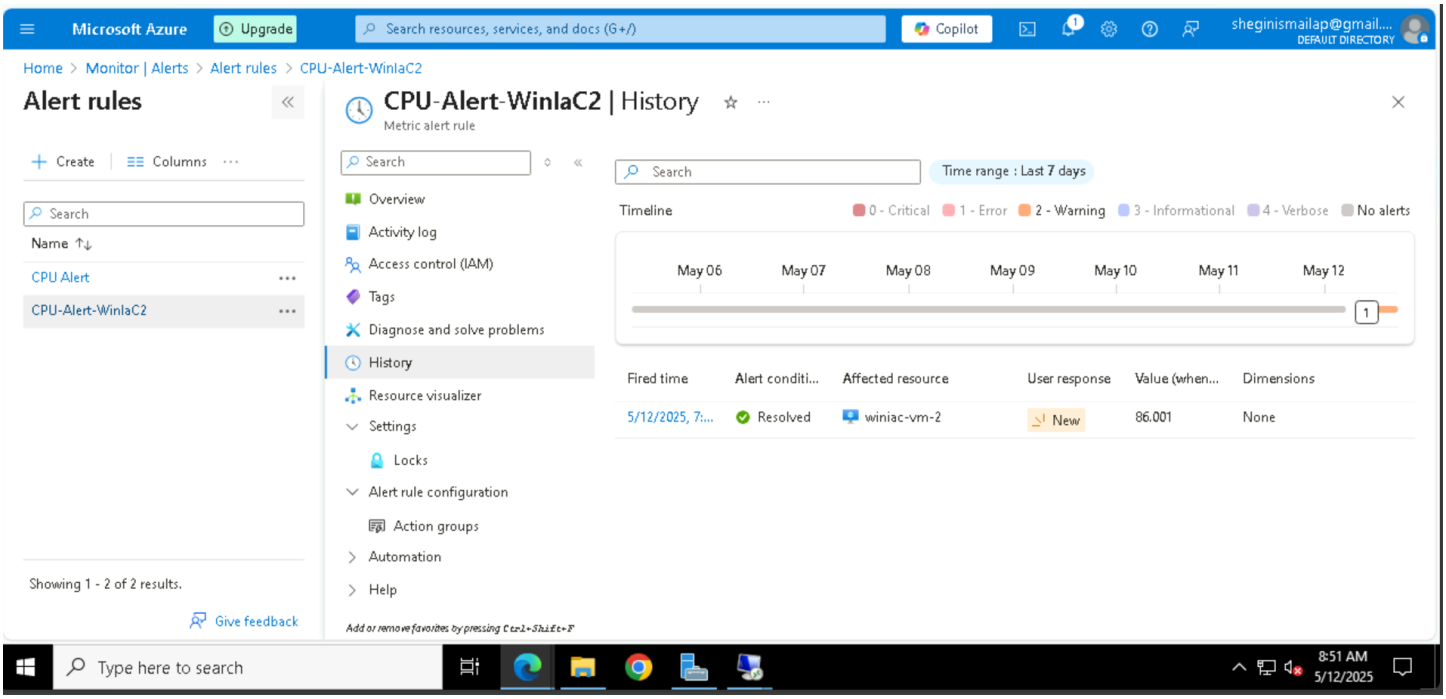
5–12 12:25 PM

cpualert–Win:Fired:Sev2 Azure Monitor Alert CPU Alert on winiac–vm–1

VM2-CPU overwhelming by running full scan



RESULT



Microsoft Azure

⚠ Your Azure Monitor alert was triggered

Azure monitor alert rule CPU-Alert-WinIaC2 was triggered for WinIac-VM-2 at May 12, 2025 7:10 UTC.

Alert rule description	CPU utilization shouldn't exceed 80%
Rule ID	/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/BaseInfra-RG/providers/microsoft.insights/metricAlerts/CPU-Alert-WinIaC2 View Rule >
Resource ID	/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/BaseInfra-RG/providers/Microsoft.Compute/virtualMachines/WinIac-VM-2 View Resource >

Alert Activated Because:

Metric name	Percentage CPU
Metric namespace	virtualMachines/WinIac-VM-2

← 56767457

5–12 12:40 PM

cpualert–Win:Fired:Sev2 Azure Monitor Alert CPU–Alert–WinIaC2 on winiac–vm–2

steps to create an **Azure AD Group** and assign **RBAC (Role-Based Access Control)**:

1. Create Azure AD Group

- Open **Azure Portal** → Navigate to **Azure Active Directory**.
- Click **Groups** → **New Group**.
- Choose **Group Type** (Security or Microsoft 365).
- Enter **Group Name** (e.g., ITAdmins).
- Set **Membership Type** (Assigned/Dynamic).
- Click **Create**.

2. Assign Members to Group

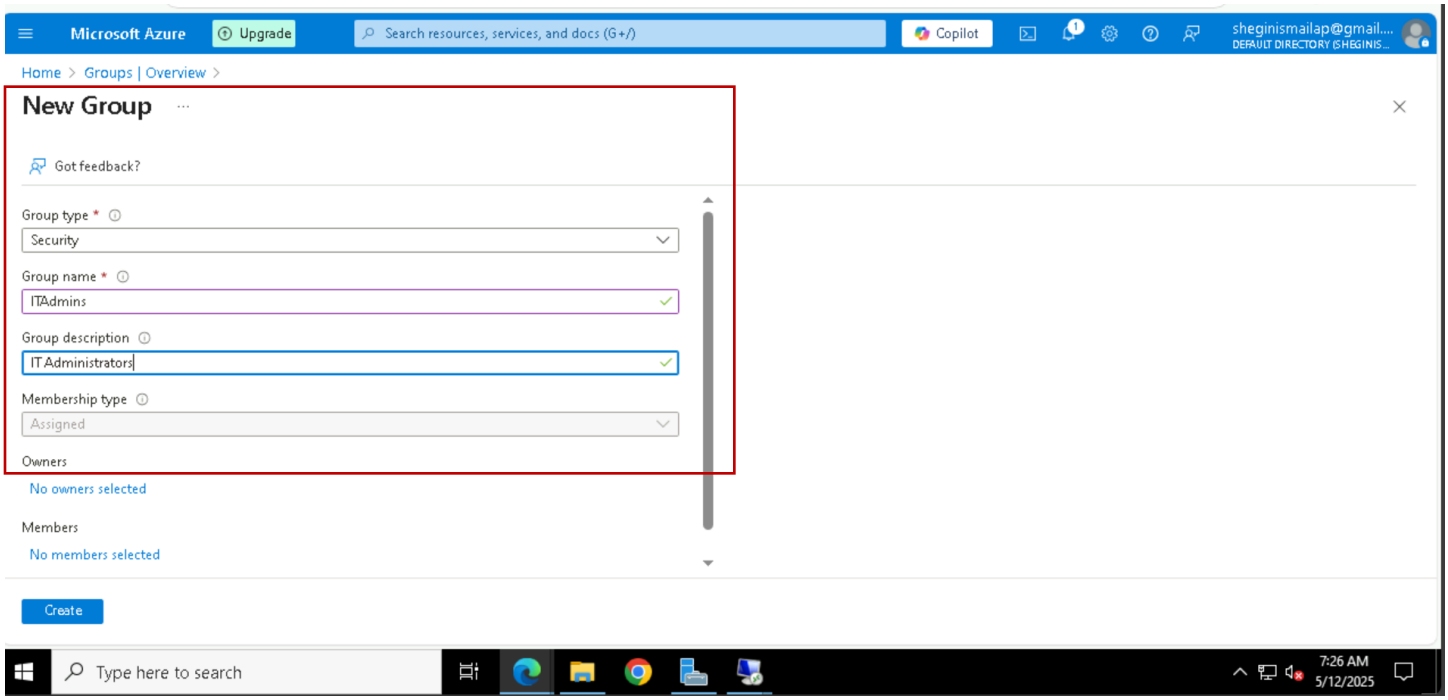
- Open the newly created **Group**.
- Navigate to **Members** → **Add Members**.
- Select users to assign and click **Save**.

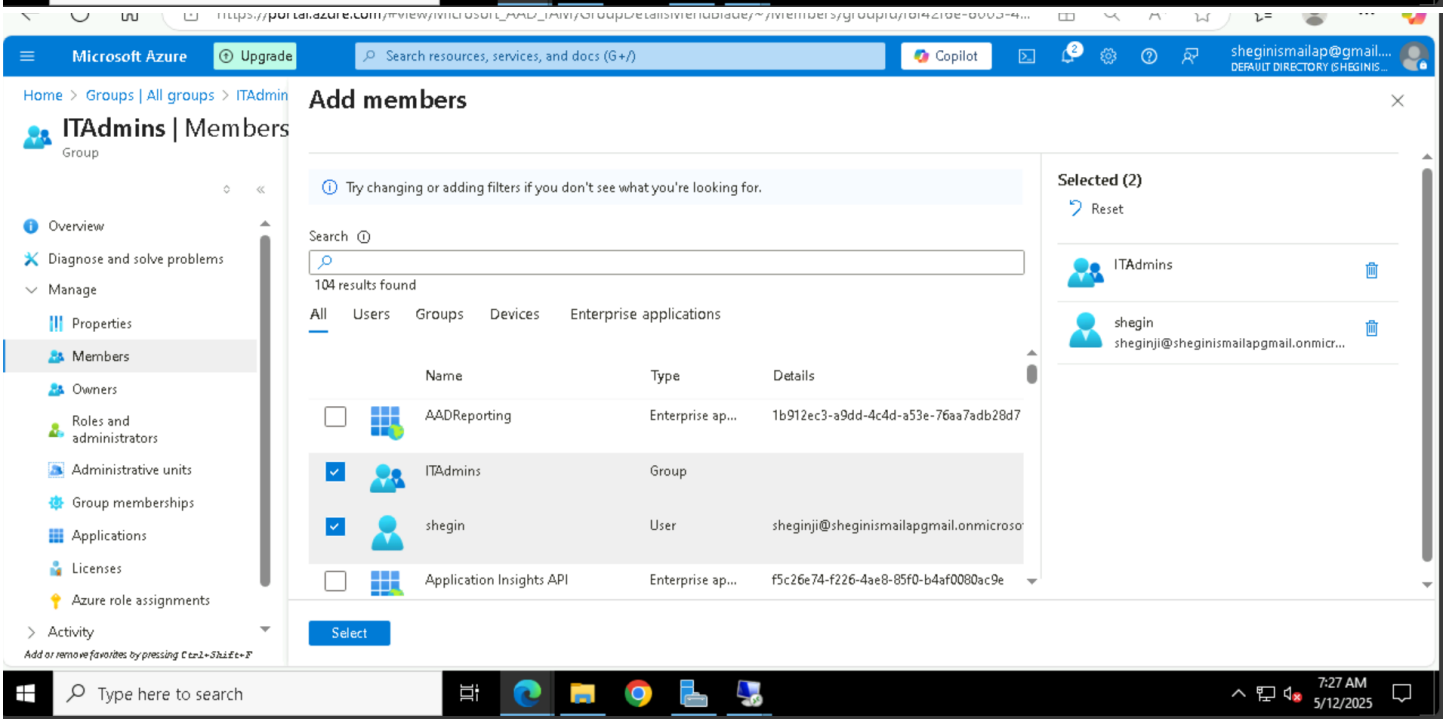
3. Assign RBAC Role to Group

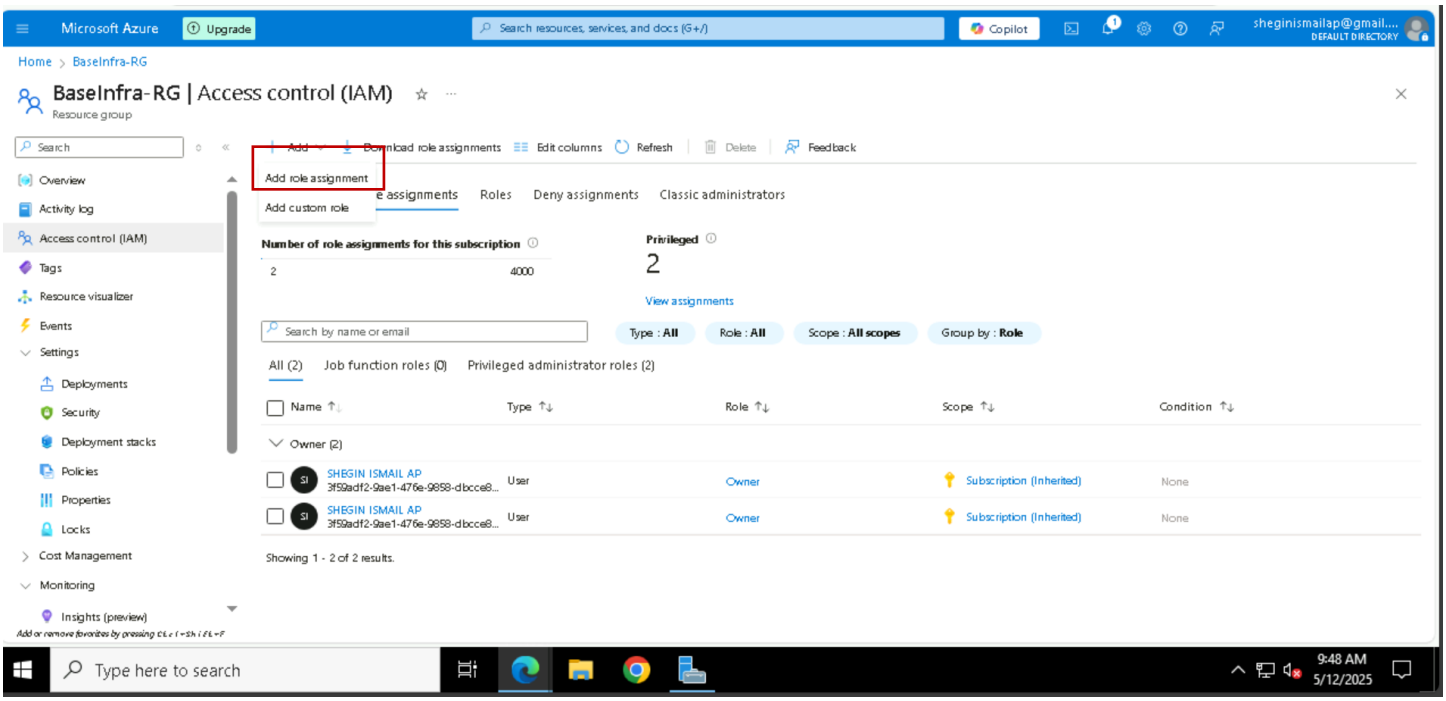
- Navigate to **Subscriptions / Resource Group / Specific Resource**.
- Click **Access Control (IAM)** → **Add Role Assignment**.
- Select the **Role** (e.g., Contributor, Reader, Owner).
- Choose **Azure AD Group** (ITAdmins) as the **Assign To** option.
- Click **Review + Assign**.

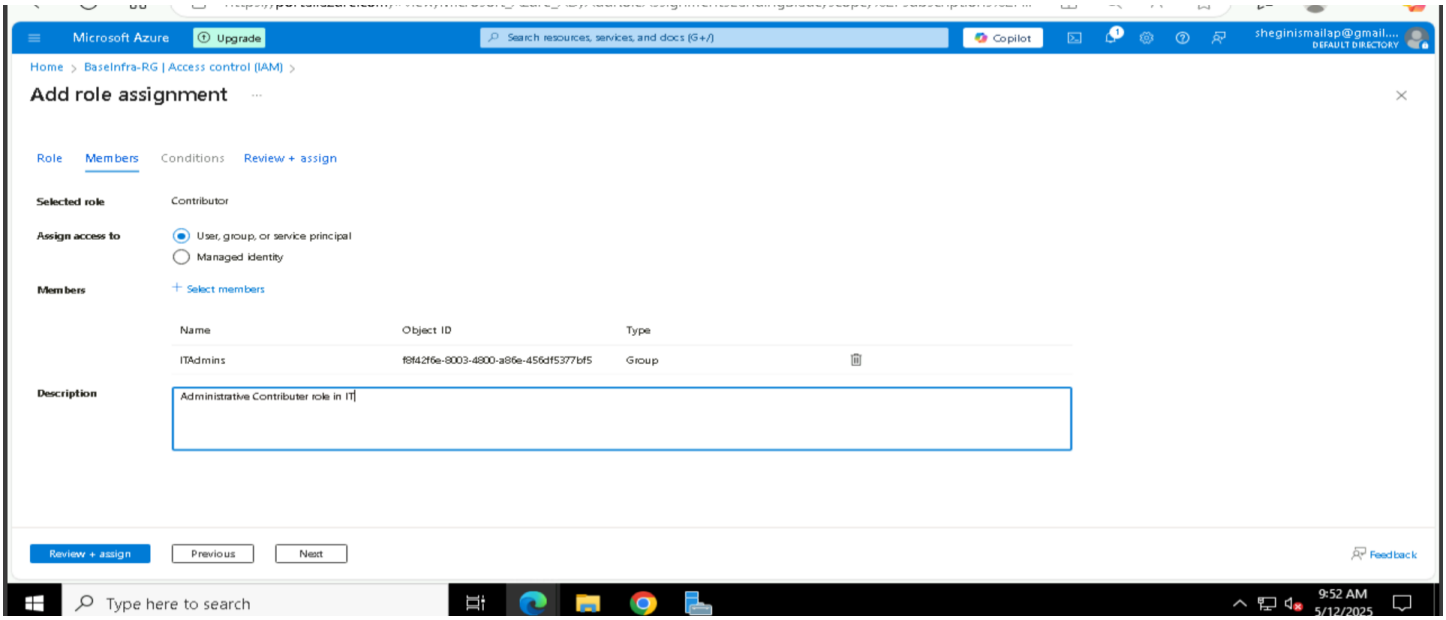
4. Validate Access

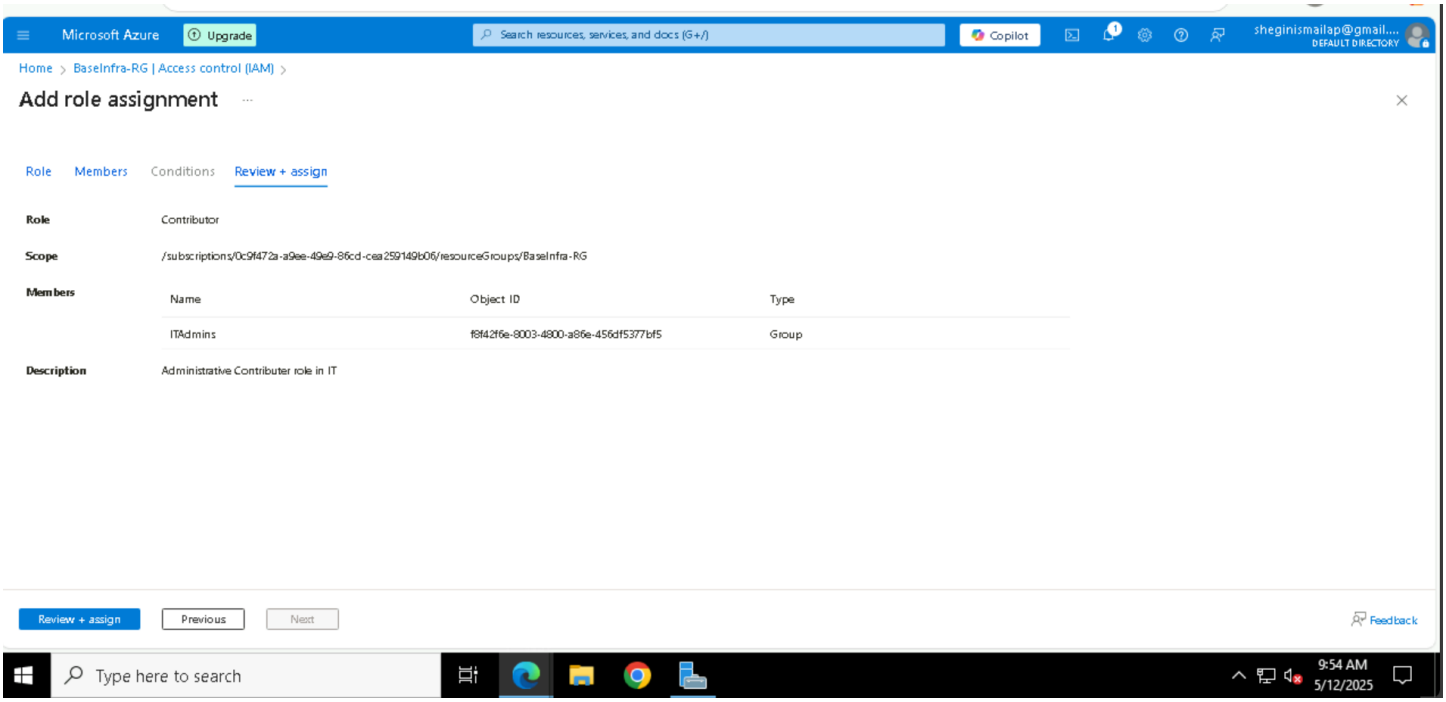
- Verify group members can access resources with assigned roles.
- Review permissions under **Access Control (IAM)**.

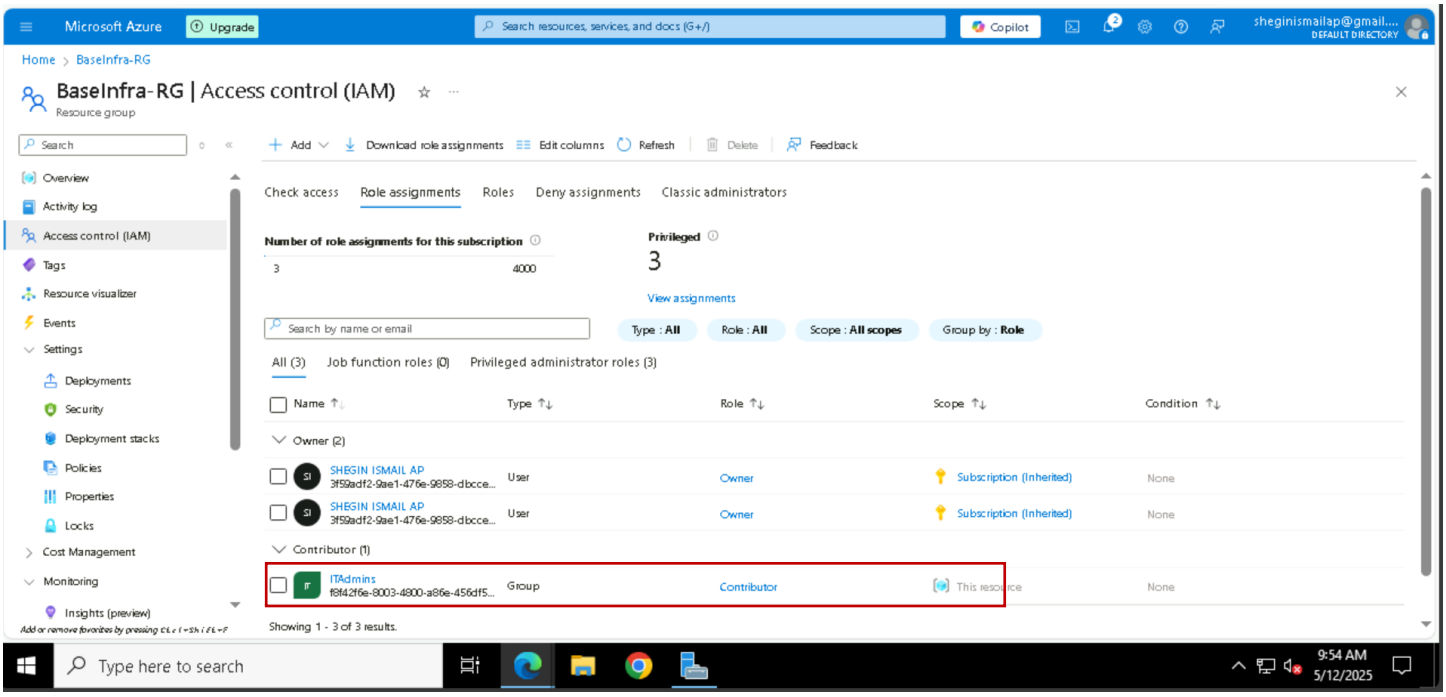

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY (SHEGINIS...
Home > Groups | Overview >
New Group
Got feedback?
Group type *
Security
Group name *
ITAdmins
Group description
IT Administrators
Membership type
Assigned
Owners
No owners selected
Members
No members selected
Create
Type here to search
7:26 AM
5/12/2025


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY (SHEGINIS...
Home > Groups | All groups > ITAdmin
ITAdmins | Members
Group
Overview
Diagnose and solve problems
Manage
Properties
Members
Owners
Roles and administrators
Administrative units
Group memberships
Applications
Licenses
Azure role assignments
Activity
Add or remove favorites by pressing Ctrl+Shift+F
Add members
Try changing or adding filters if you don't see what you're looking for.
Search
104 results found
All
Users
Groups
Devices
Enterprise applications
Name
Type
Details
AADReporting
Enterprise ap...
1b912ec3-a9dd-4c4d-a53e-76aa7adb28d7
ITAdmins
Group
shegin
User
sheginji@sheginismailapgmail.onmicroso
Application Insights API
Enterprise ap...
f5c26e74-f226-4ae8-85f0-b4af0080ac9e
Selected (2)
Reset
ITAdmins
shegin
sheginji@sheginismailapgmail.onmicr...
Select
Type here to search
7:27 AM
5/12/2025


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > BaseInfra-RG
BaseInfra-RG | Access control (IAM)
Resource group
Search
Add
Download role assignments
Edit columns
Refresh
Delete
Feedback
Add role assignment
Add custom role
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Deployment stacks
Policies
Properties
Locks
Cost Management
Monitoring
Insights (preview)
Roles
Deny assignments
Classic administrators
Number of role assignments for this subscription
2
4000
Privileged
2
View assignments
Search by name or email
Type : All
Role : All
Scope : All scopes
Group by : Role
All (2)
Job function roles (0)
Privileged administrator roles (2)
Name
Type
Role
Scope
Condition
Owner (2)
SHEGIN ISMAIL AP
3f59adf2-9ae1-476e-9858-dbcce8...
User
Owner
Subscription (Inherited)
None
Showing 1 - 2 of 2 results.
Type here to search
9:48 AM
5/12/2025


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > BaseInfra-RG | Access control (IAM) >
Add role assignment
Role
Members
Conditions
Review + assign
Selected role
Contributor
Assign access to
User, group, or service principal
Managed identity
Members
Select members
Name
Object ID
Type
ITAdmins
f8f42f6e-8003-4800-a86e-456df5377bf5
Group
Description
Administrative Contributer role in IT
Review + assign
Previous
Next
Feedback
Type here to search
9:52 AM
5/12/2025


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > BaseInfra-RG | Access control (IAM) >
Add role assignment
Role
Members
Conditions
Review + assign
Role
Contributor
Scope
/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/BaseInfra-RG
Members
Name
Object ID
Type
ITAdmins
f8f42f6e-8003-4800-a86e-456df5377bf5
Group
Description
Administrative Contributer role in IT
Review + assign
Previous
Next
Feedback
Type here to search
9:54 AM
5/12/2025


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > BaseInfra-RG
BaseInfra-RG | Access control (IAM)
Resource group
Search
Add
Download role assignments
Edit columns
Refresh
Delete
Feedback
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Deployment stacks
Policies
Properties
Locks
Cost Management
Monitoring
Insights (preview)
Check access
Role assignments
Roles
Deny assignments
Classic administrators
Number of role assignments for this subscription
3
4000
Privileged
3
View assignments
Search by name or email
Type : All
Role : All
Scope : All scopes
Group by : Role
All (3)
Job function roles (0)
Privileged administrator roles (3)
Name
Type
Role
Scope
Condition
Owner (2)
SHEGIN ISMAIL AP
3f59adf2-9ae1-476e-9858-dbcce...
User
Owner
Subscription (Inherited)
None
SHEGIN ISMAIL AP
3f59adf2-9ae1-476e-9858-dbcce...
User
Owner
Subscription (Inherited)
None
Contributor (1)
ITAdmins
f8f42f6e-8003-4800-a86e-456df5...
Group
Contributor
This resource
None
Showing 1 - 3 of 3 results.
Type here to search
9:54 AM
5/12/2025

Phase 2: Automated Infrastructure Deployment

Business Requirements

The DevOps team wants to standardize deployments and minimize manual efforts by using Infrastructure as

Code.

Implementation Flow

1. Write ARM/Bicep templates for Phase 1 setup.

2. Add parameterization for environment variables.

3. Deploy via Azure CLI.

4. Validate using Azure Resource Manager logs.

5. Store templates in GitHub for reuse.

Takeaway Learnings

• Building repeatable environments using IaC.

• Simplifying version control and change tracking.

Tools and Technologies Used

• ARM Templates, Bicep, GitHub, Azure CLI

Best Practices

• Modularize templates.

• Follow naming and tagging standards.

Common Pitfalls to Avoid

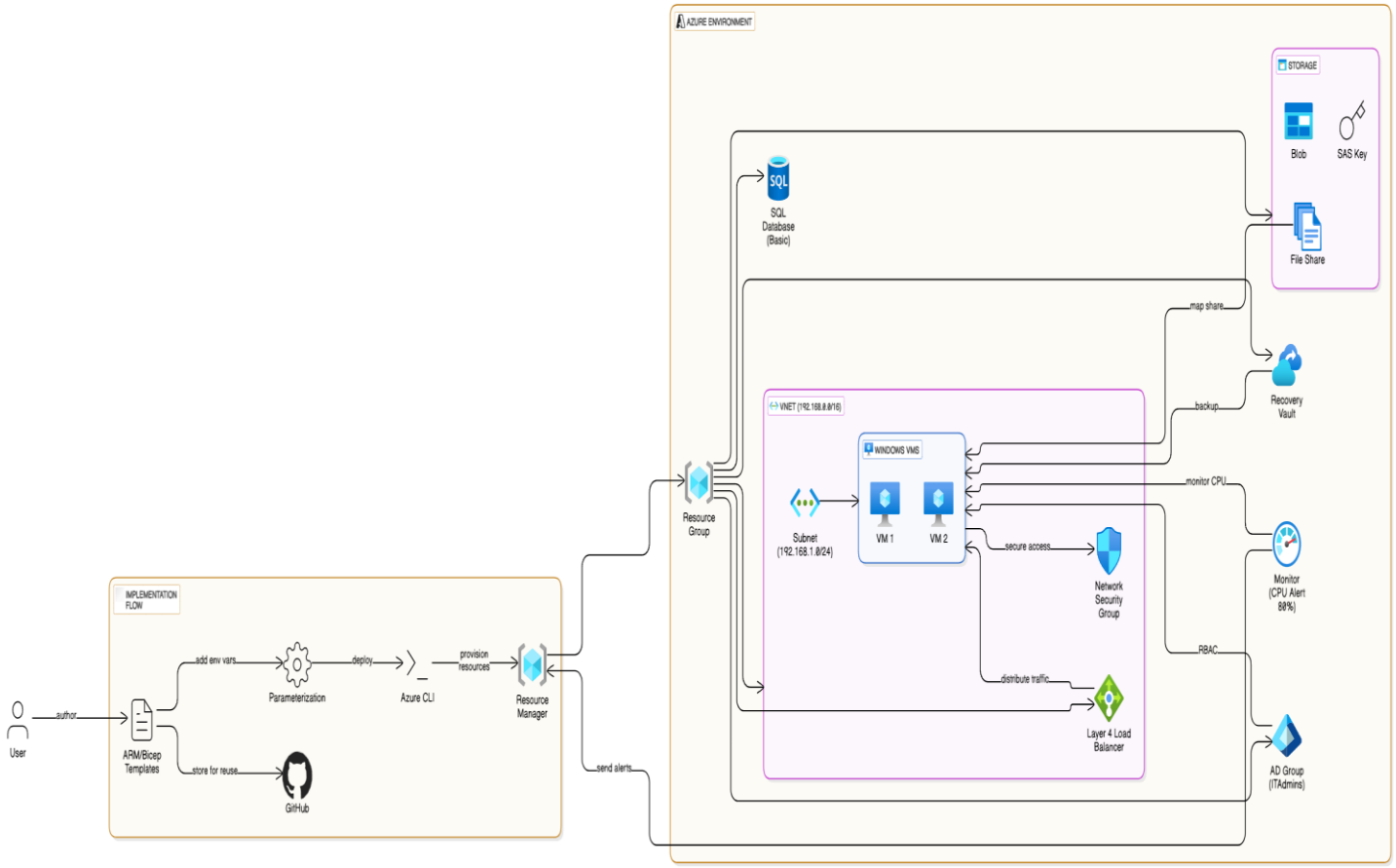
• Hardcoding sensitive values.

• No rollback or error handling logic.

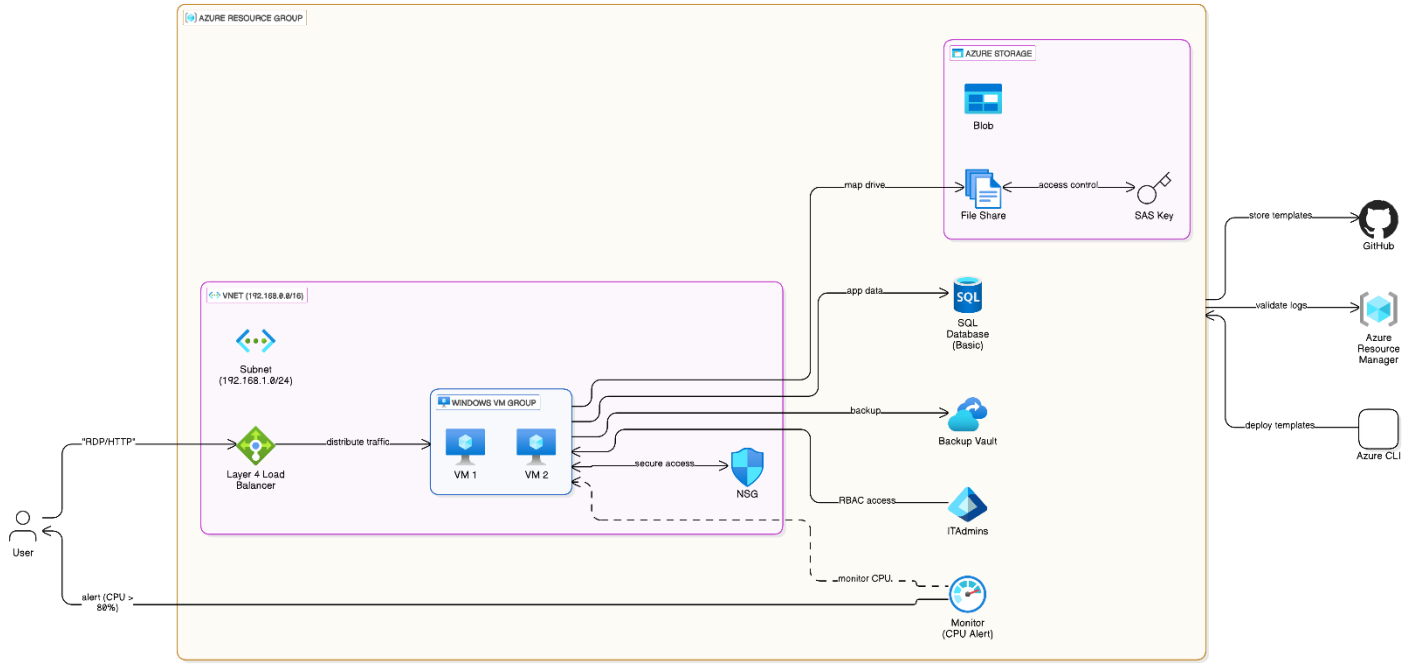
Free Tier Compatibility Notes

• Fully compatible as long as deployed resources respect free tier limits.

Network Diagram

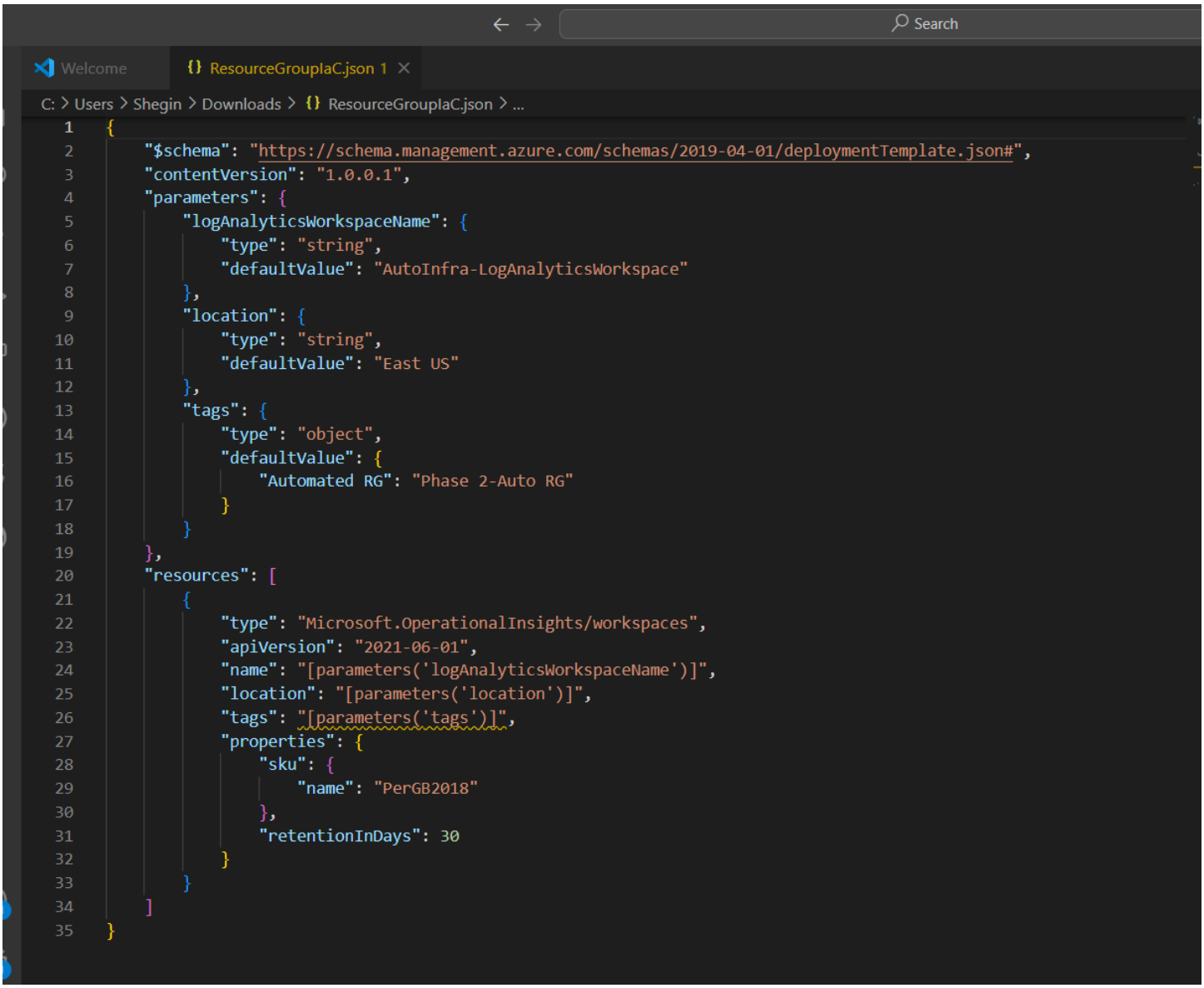


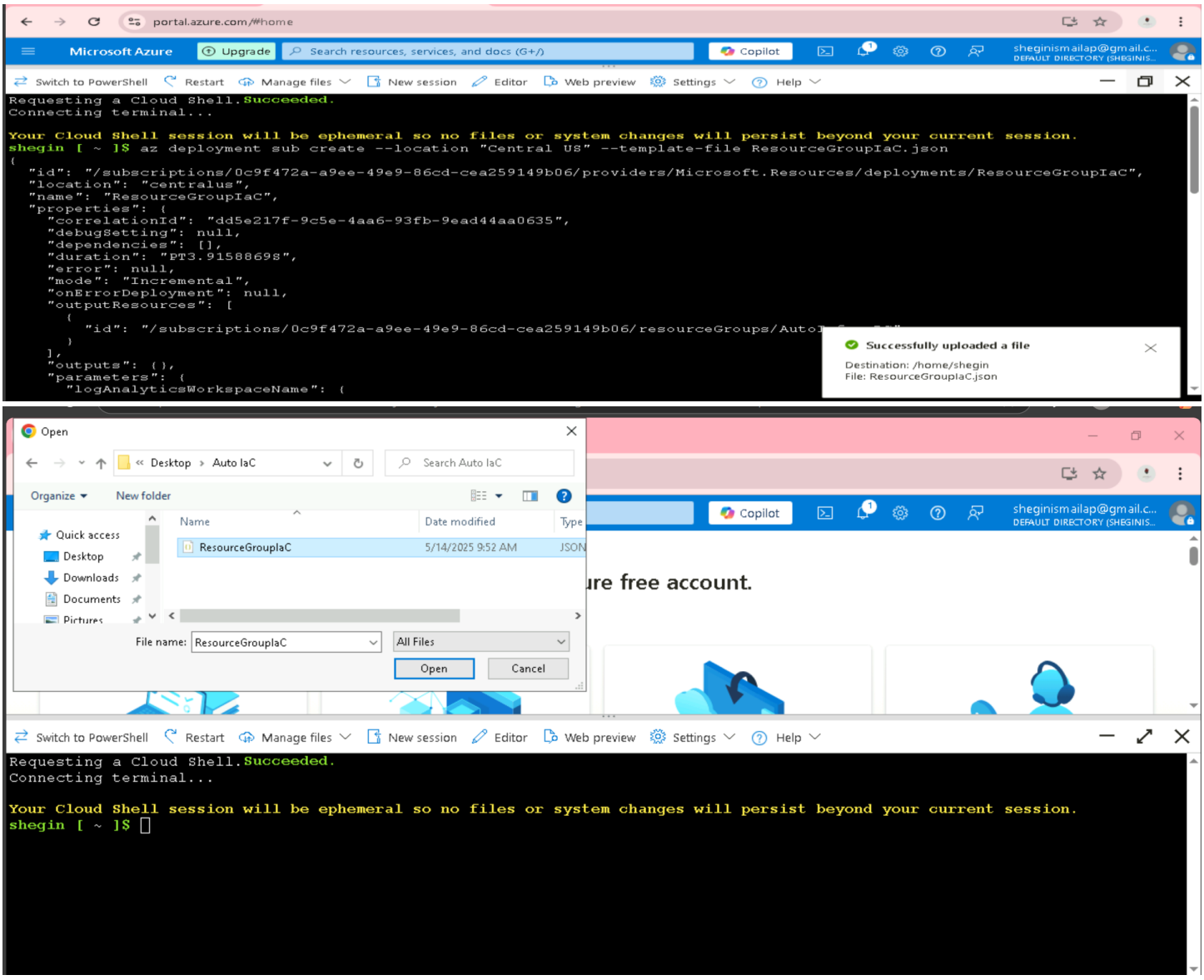
Architecture Diagram



Implementation demonstration:

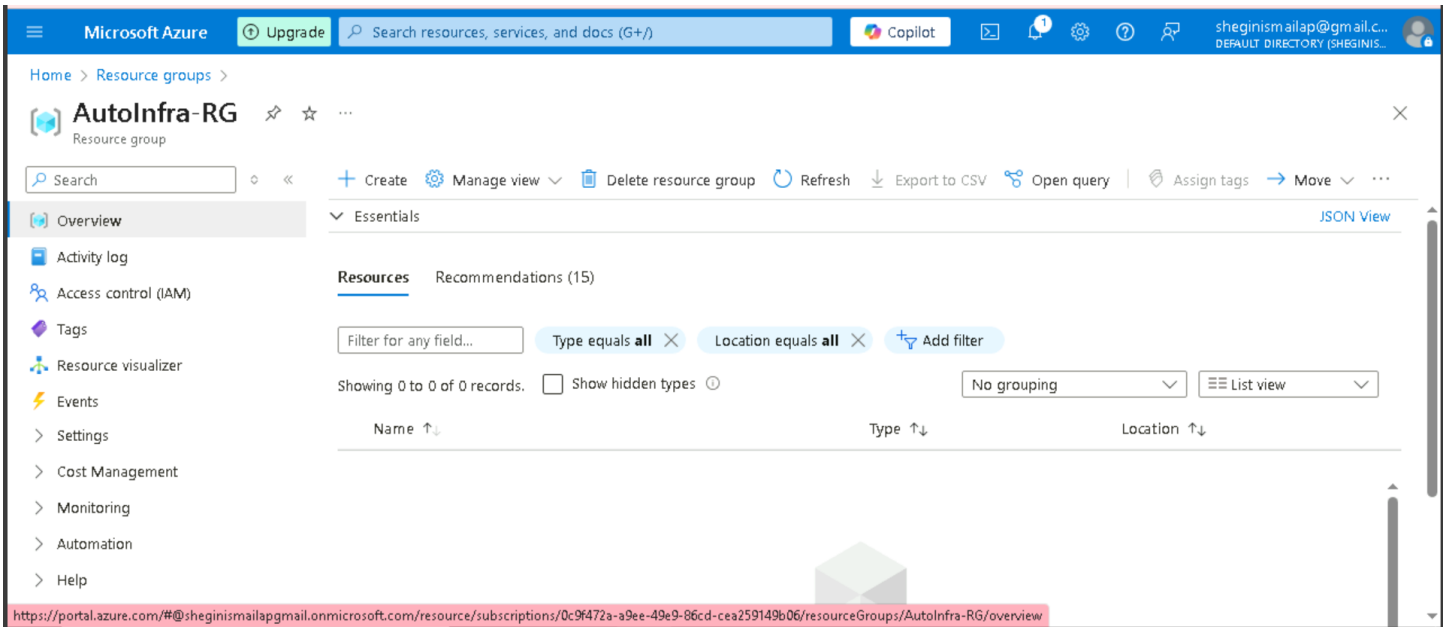
1)The ARM template for creation of Resource Group is given in ResourceGroupIaC.json,the snippets from template,azure cli and outputs are shown


Search
Welcome
ResourceGroupIaC.json 1
C: > Users > Shegin > Downloads > ResourceGroupIaC.json > ...
{
"$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
"contentVersion": "1.0.0.1",
"parameters": {
"logAnalyticsWorkspaceName": {
"type": "string",
"defaultValue": "AutoInfra-LogAnalyticsWorkspace"
},
"location": {
"type": "string",
"defaultValue": "East US"
},
"tags": {
"type": "object",
"defaultValue": {
"Automated RG": "Phase 2-Auto RG"
}
}
},
"resources": [
{
"type": "Microsoft.OperationalInsights/workspaces",
"apiVersion": "2021-06-01",
"name": "[parameters('logAnalyticsWorkspaceName')]",
"location": "[parameters('location')]",
"tags": "[parameters('tags')]",
"properties": {
"sku": {
"name": "PerGB2018"
},
"retentionInDays": 30
}
}
]
}

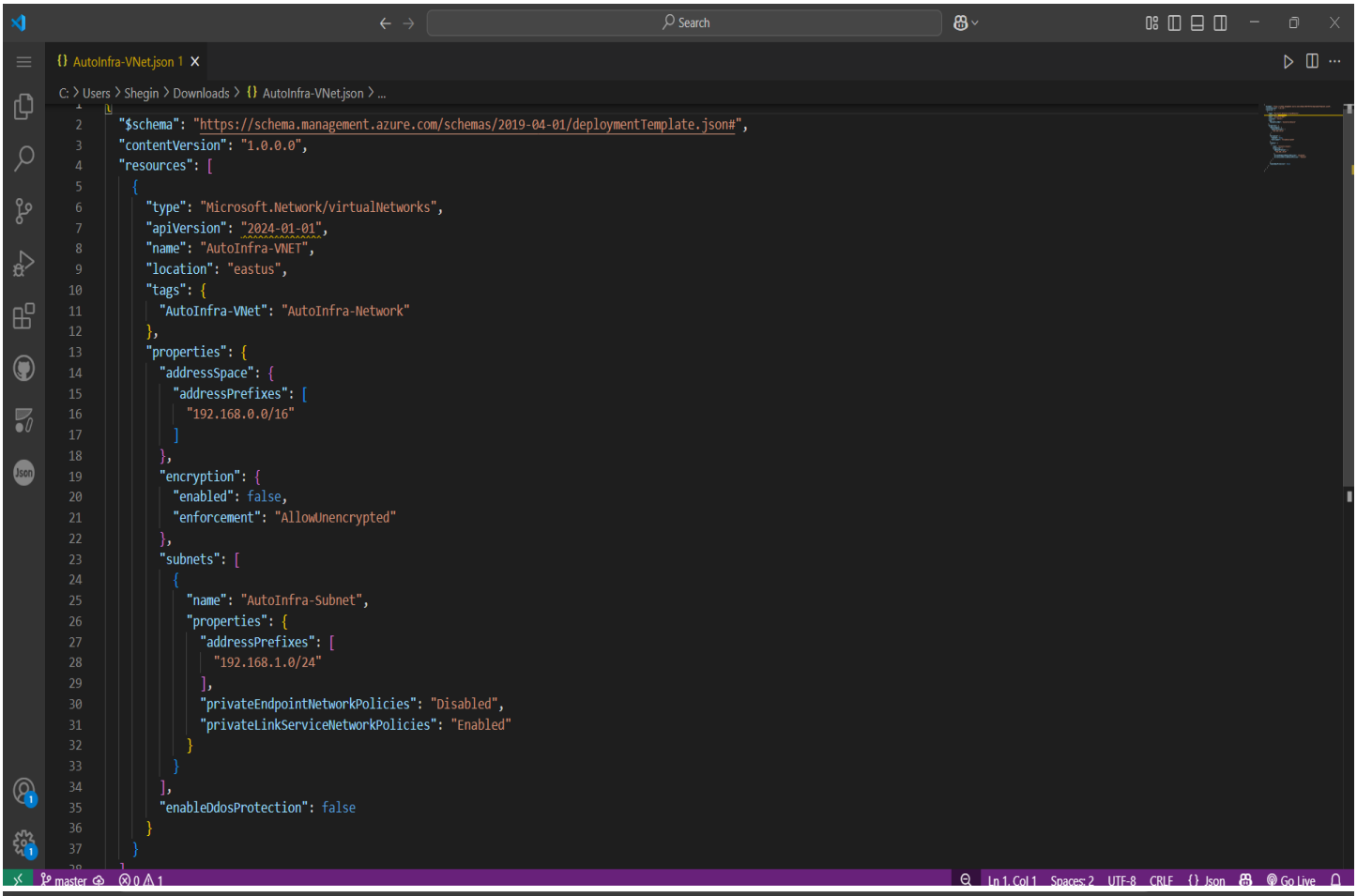

portal.azure.com/#home
Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY (SHEGINIS...
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ az deployment sub create --location "Central US" --template-file ResourceGroupIaC.json
{
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/providers/Microsoft.Resources/deployments/ResourceGroupIaC",
"location": "centralus",
"name": "ResourceGroupIaC",
"properties": {
"correlationId": "dd5e217f-9c5e-4aa6-93fb-9ead44aa0635",
"debugSetting": null,
"dependencies": [],
"duration": "PT3.9158869S",
"error": null,
"mode": "Incremental",
"onErrorDeployment": null,
"outputResources": [
{
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/Auto
}
],
"outputs": {},
"parameters": {
"logAnalyticsWorkspaceName": {
Successfully uploaded a file
Destination: /home/shegin
File: ResourceGroupIaC.json

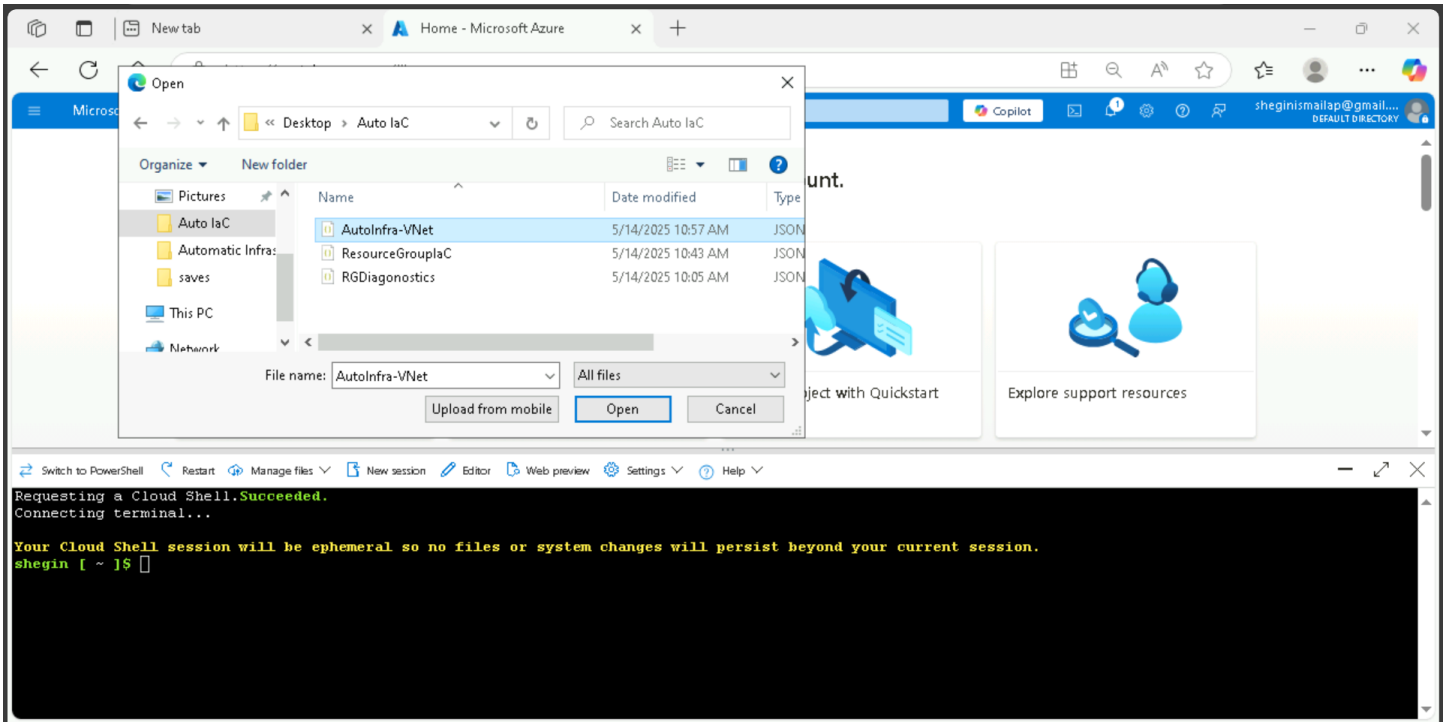
Open
Desktop › Auto IaC
Search Auto IaC
Organize
New folder
Quick access
Desktop
Downloads
Documents
Pictures
Name
Date modified
Type
ResourceGroupIaC
5/14/2025 9:52 AM
JSON
File name: ResourceGroupIaC
All Files
Open
Cancel
ure free account.
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$

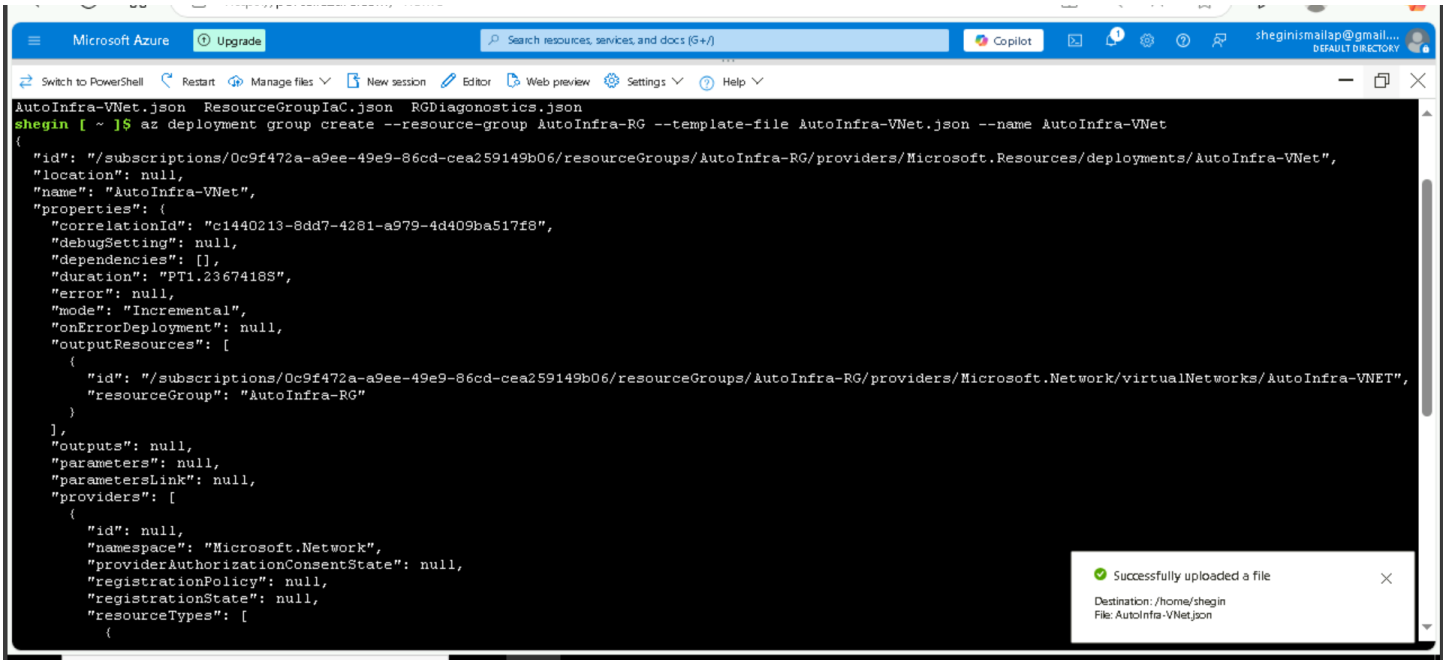
RESULT


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY (SHEGINIS...
Home > Resource groups >
AutoInfra-RG
Resource group
Search
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Cost Management
Monitoring
Automation
Help
Create
Manage view
Delete resource group
Refresh
Export to CSV
Open query
Assign tags
Move
Essentials
JSON View
Resources
Recommendations (15)
Filter for any field...
Type equals all
Location equals all
Add filter
Showing 0 to 0 of 0 records.
Show hidden types
No grouping
List view
Name
Type
Location
https://portal.azure.com/#@sheginismailapgmail.onmicrosoft.com/resource/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/overview

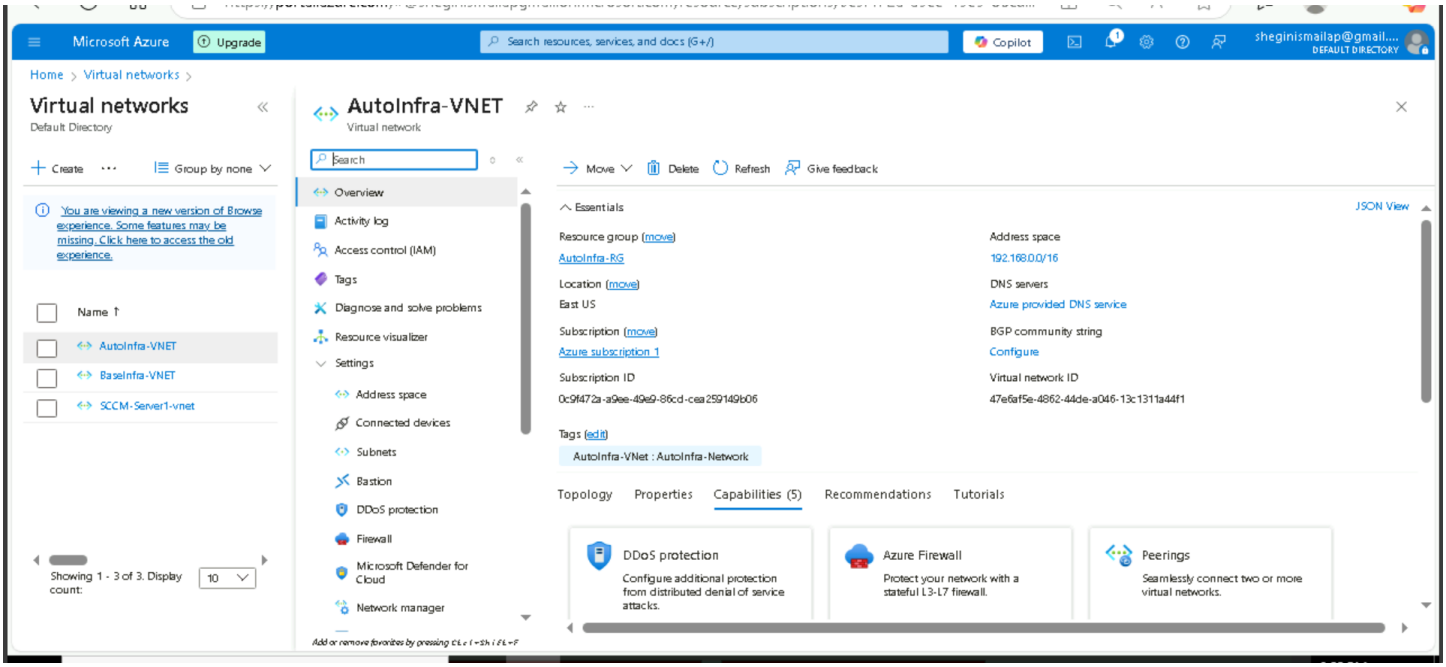
2)The ARM template for creation of VNet (192.168.0.0/16) with Subnet (192.168.1.0/24) is given in AutoInfra-VNet.json,the snippets from template,azure cli and outputs are shown





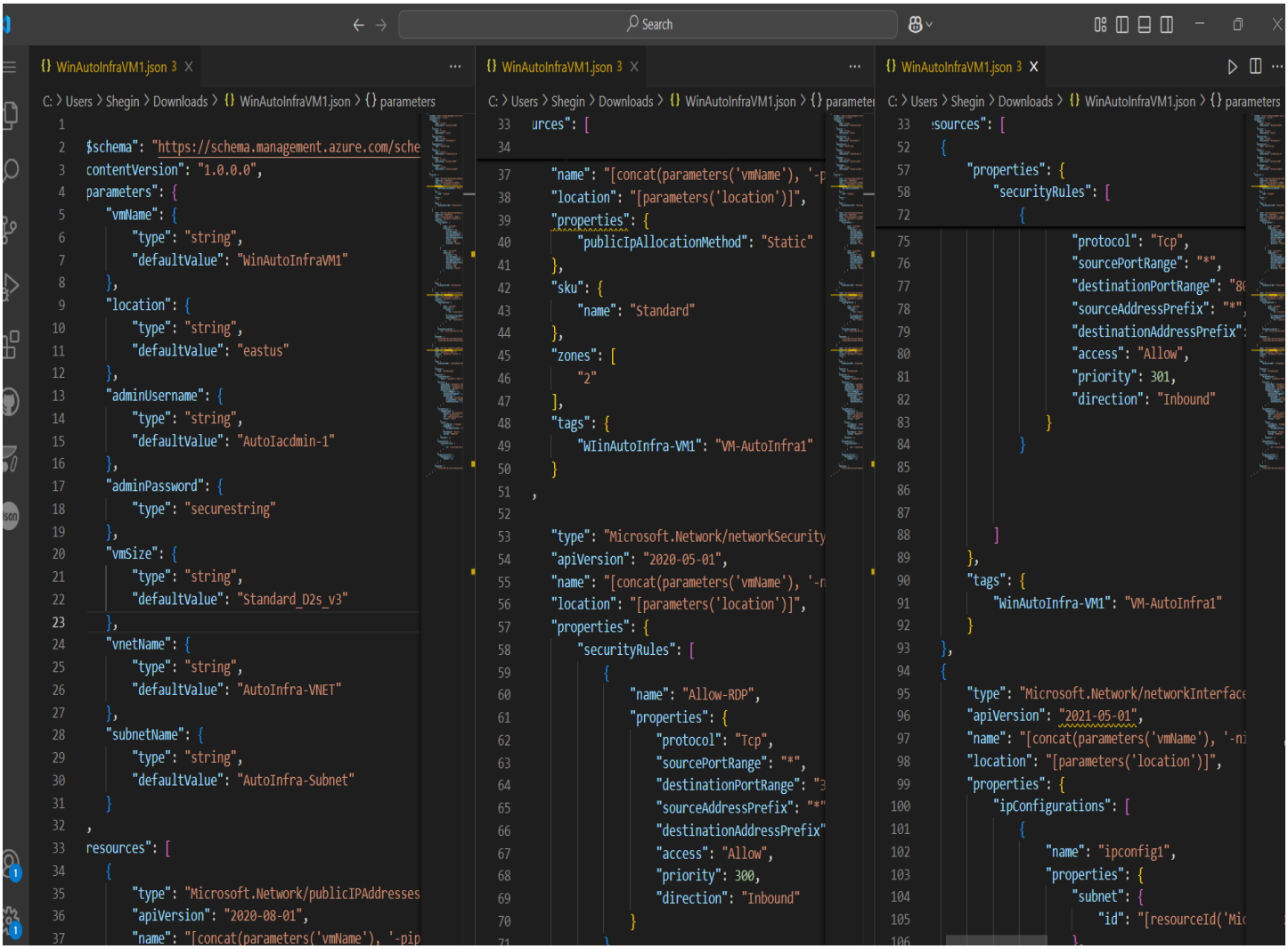


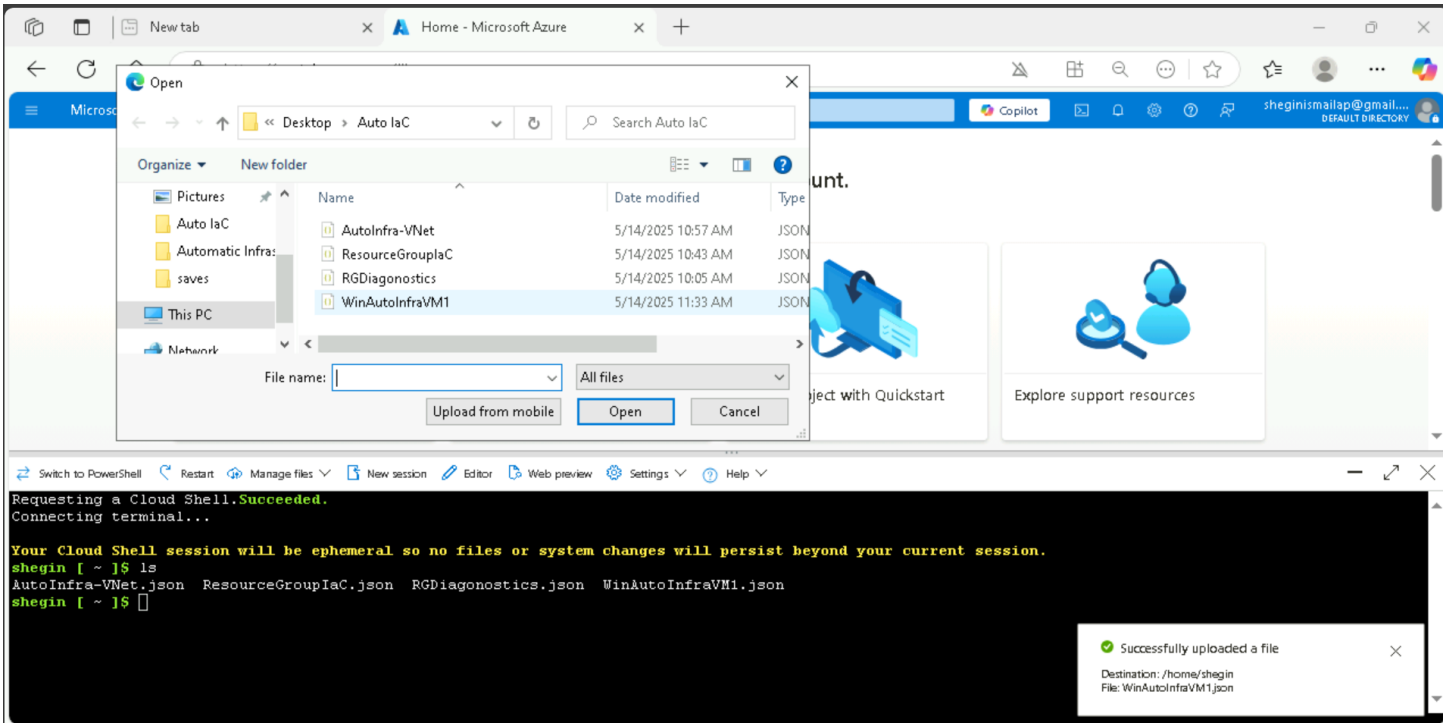
RESULT

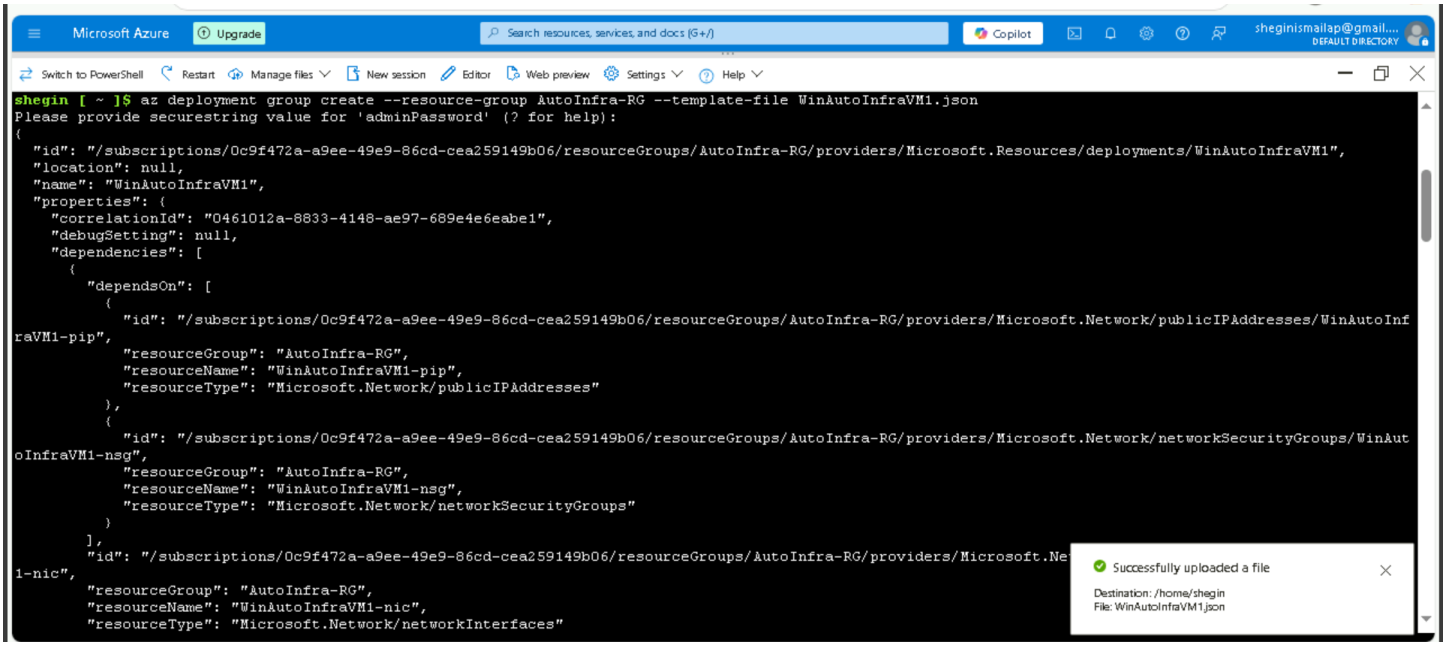


3)The ARM template for Deploy 2 VMs (Windows), secure with NSG is given in WinAutoInfraVM1.json and WinAutoInfraVM2.json,the snippets from template,azure cli and outputs are shown

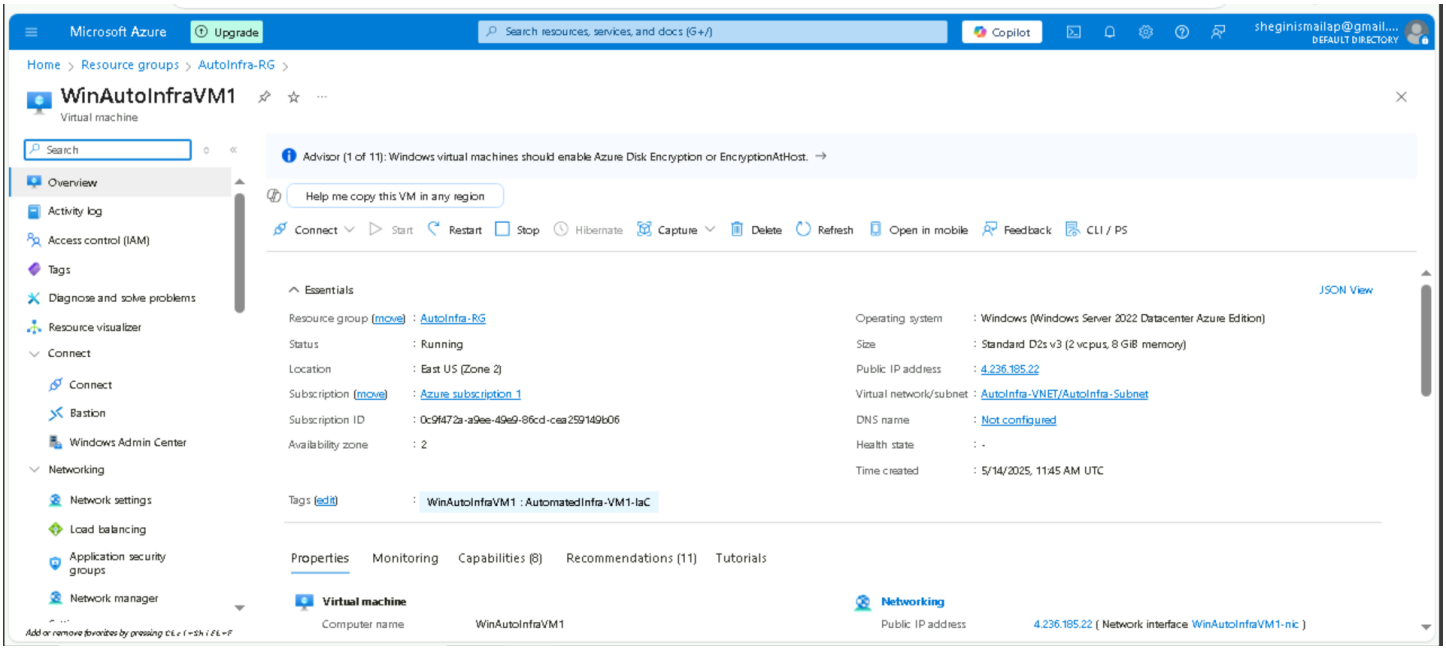
WinAutoInfraVM1


WinAutoInfraVM1.json 3
C: > Users > Shegin > Downloads > {} WinAutoInfraVM1.json > {} parameters
"$schema": "https://schema.management.azure.com/sche
"contentVersion": "1.0.0.0",
"parameters": {
"vmName": {
"type": "string",
"defaultValue": "WinAutoInfraVM1"
"location": {
"type": "string",
"defaultValue": "eastus"
"adminUsername": {
"type": "string",
"defaultValue": "AutoIacdmin-1"
"adminPassword": {
"type": "securestring"
"vmSize": {
"type": "string",
"defaultValue": "Standard_D2s_v3"
"vnetName": {
"type": "string",
"defaultValue": "AutoInfra-VNET"
"subnetName": {
"type": "string",
"defaultValue": "AutoInfra-Subnet"
"resources": [
"type": "Microsoft.Network/publicIPAddresses
"apiVersion": "2020-08-01",
"name": "[concat(parameters('vmName'), '-pip
"name": "[concat(parameters('vmName'), '-p
"location": "[parameters('location')]",
"properties": {
"publicIpAllocationMethod": "Static"
"sku": {
"name": "Standard"
"zones": [
"2"
"tags": {
"WIinAutoInfra-VM1": "VM-AutoInfra1"
"type": "Microsoft.Network/networkSecurity
"apiVersion": "2020-05-01",
"name": "[concat(parameters('vmName'), '-n
"location": "[parameters('location')]",
"properties": {
"securityRules": [
"name": "Allow-RDP",
"properties": {
"protocol": "Tcp",
"sourcePortRange": "*",
"destinationPortRange": "3
"sourceAddressPrefix": "*"
"destinationAddressPrefix"
"access": "Allow",
"priority": 300,
"direction": "Inbound"
"sources": [
"properties": {
"securityRules": [
"protocol": "Tcp",
"sourcePortRange": "*",
"destinationPortRange": "8
"sourceAddressPrefix": "*",
"destinationAddressPrefix":
"access": "Allow",
"priority": 301,
"direction": "Inbound"
"tags": {
"WinAutoInfra-VM1": "VM-AutoInfra1"
"type": "Microsoft.Network/networkInterfac
"apiVersion": "2021-05-01",
"name": "[concat(parameters('vmName'), '-ni
"location": "[parameters('location')]",
"properties": {
"ipConfigurations": [
"name": "ipconfig1",
"properties": {
"subnet": {
"id": "[resourceId('Mic

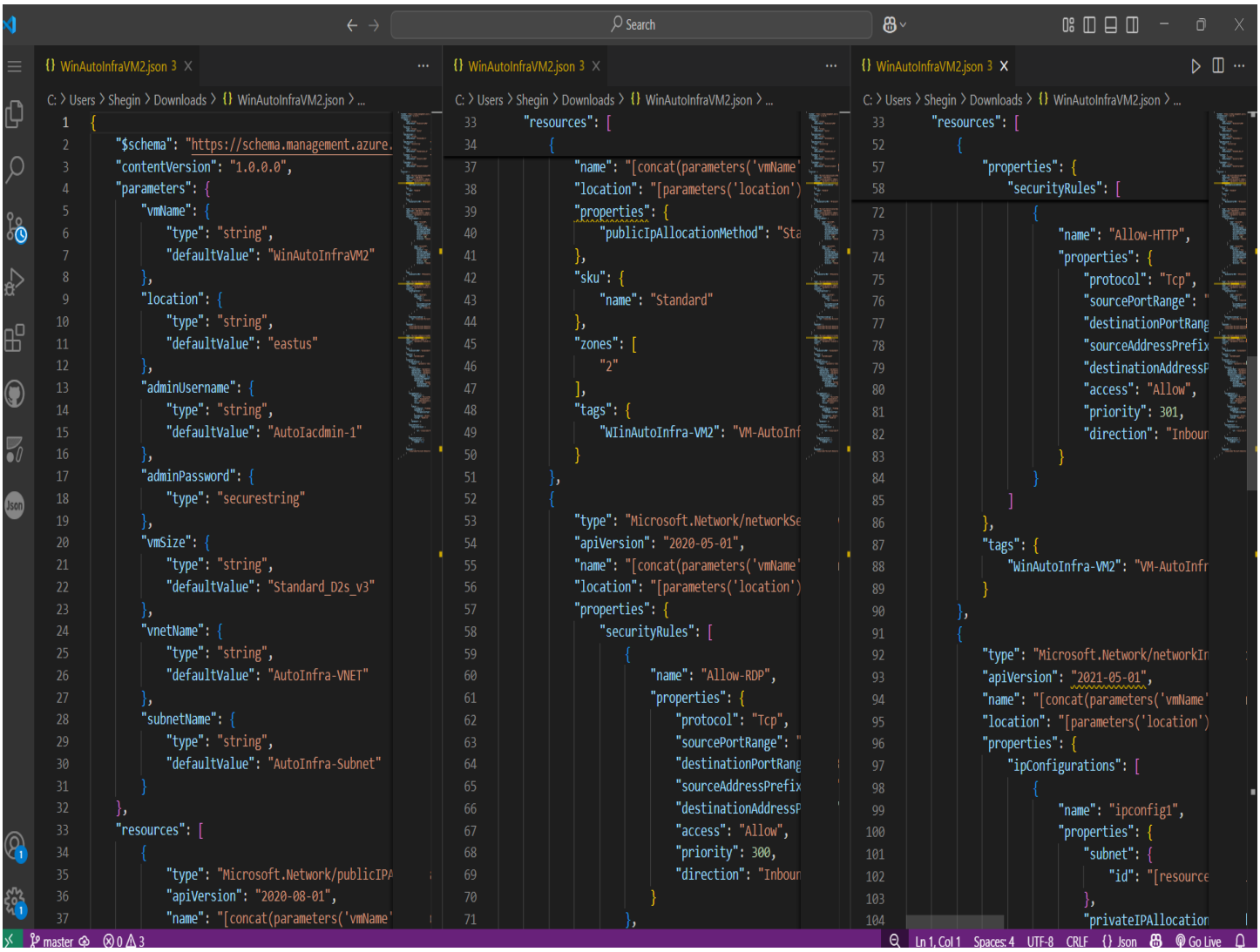

New tab
Home - Microsoft Azure
Open
Desktop > Auto IaC
Search Auto IaC
Organize
New folder
Pictures
Auto IaC
Automatic Infra
saves
This PC
Network
Name
Date modified
Type
AutoInfra-VNet 5/14/2025 10:57 AM JSON
ResourceGroupIaC 5/14/2025 10:43 AM JSON
RGDiagonostics 5/14/2025 10:05 AM JSON
WinAutoInfraVM1 5/14/2025 11:33 AM JSON
File name:
All files
Upload from mobile
Open
Cancel
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
unt.
ject with Quickstart
Explore support resources
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
AutoInfra-VNet.json ResourceGroupIaC.json RGDiagonostics.json WinAutoInfraVM1.json
shegin [~]$
Successfully uploaded a file
Destination: /home/shegin
File: WinAutoInfraVM1.json

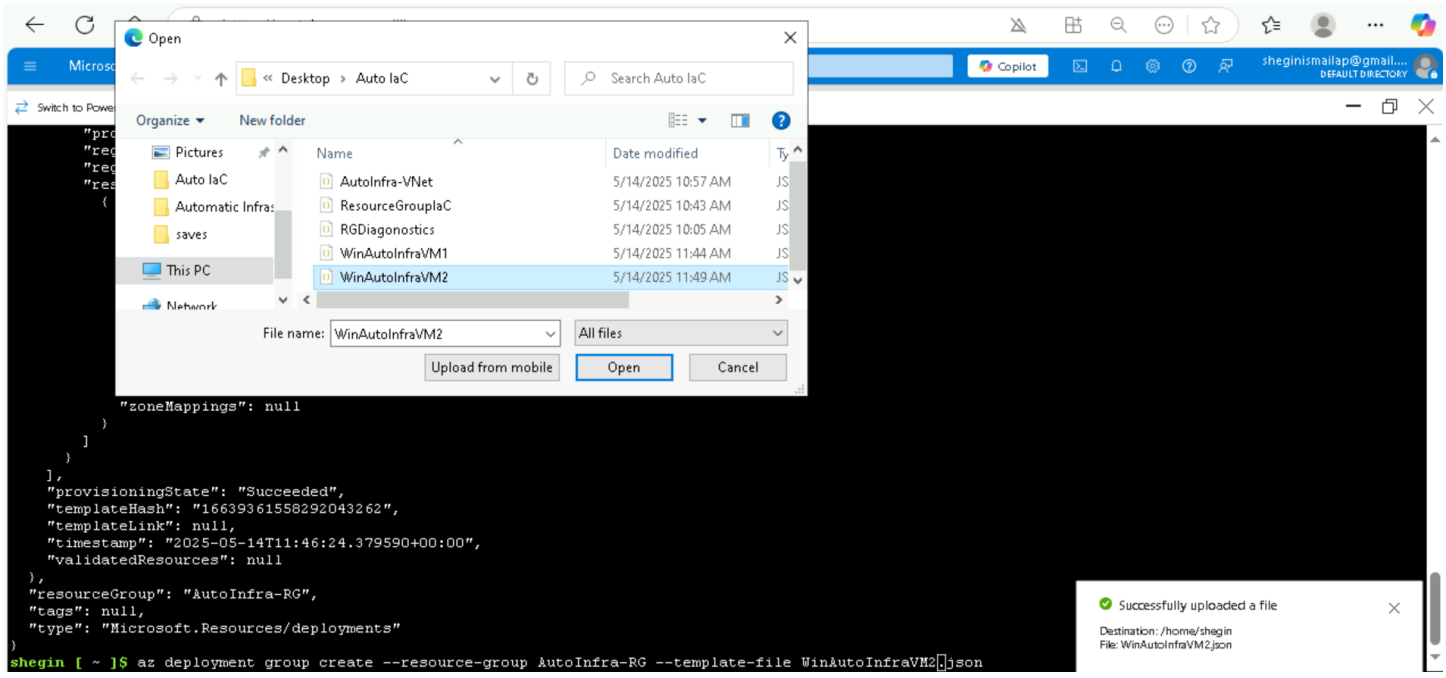


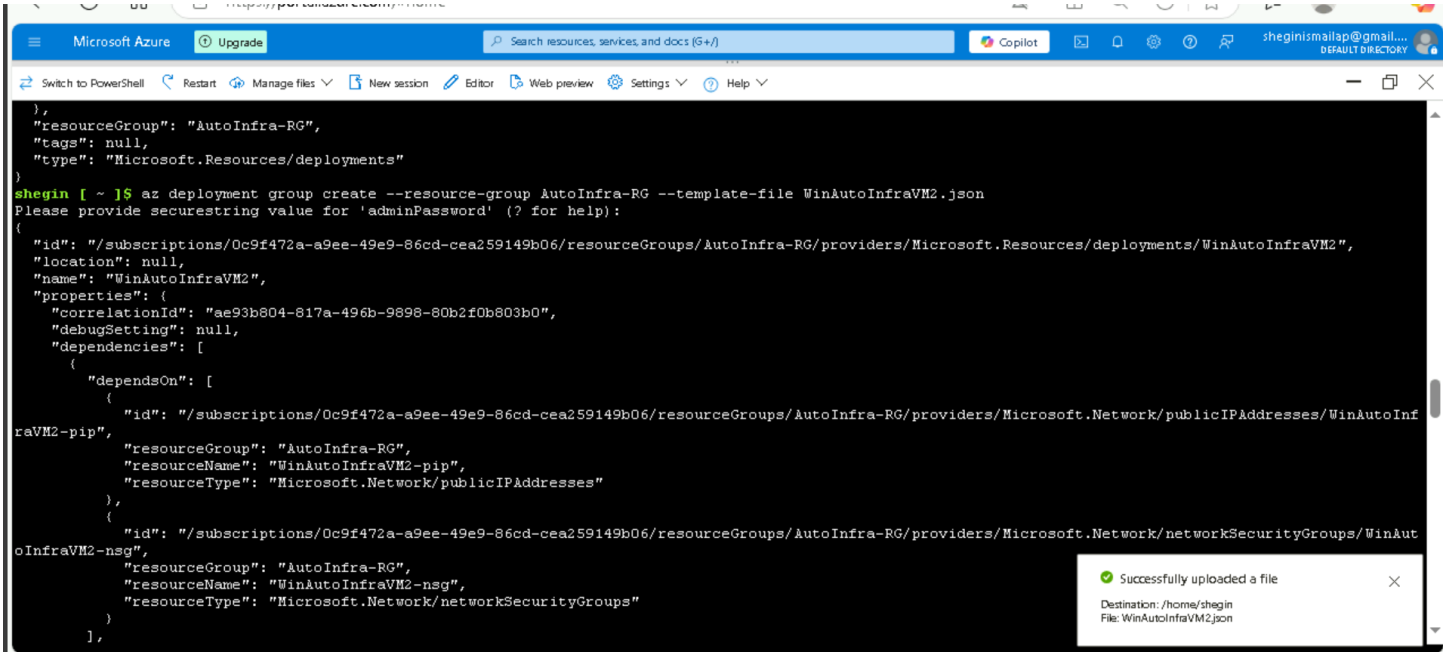
RESULT



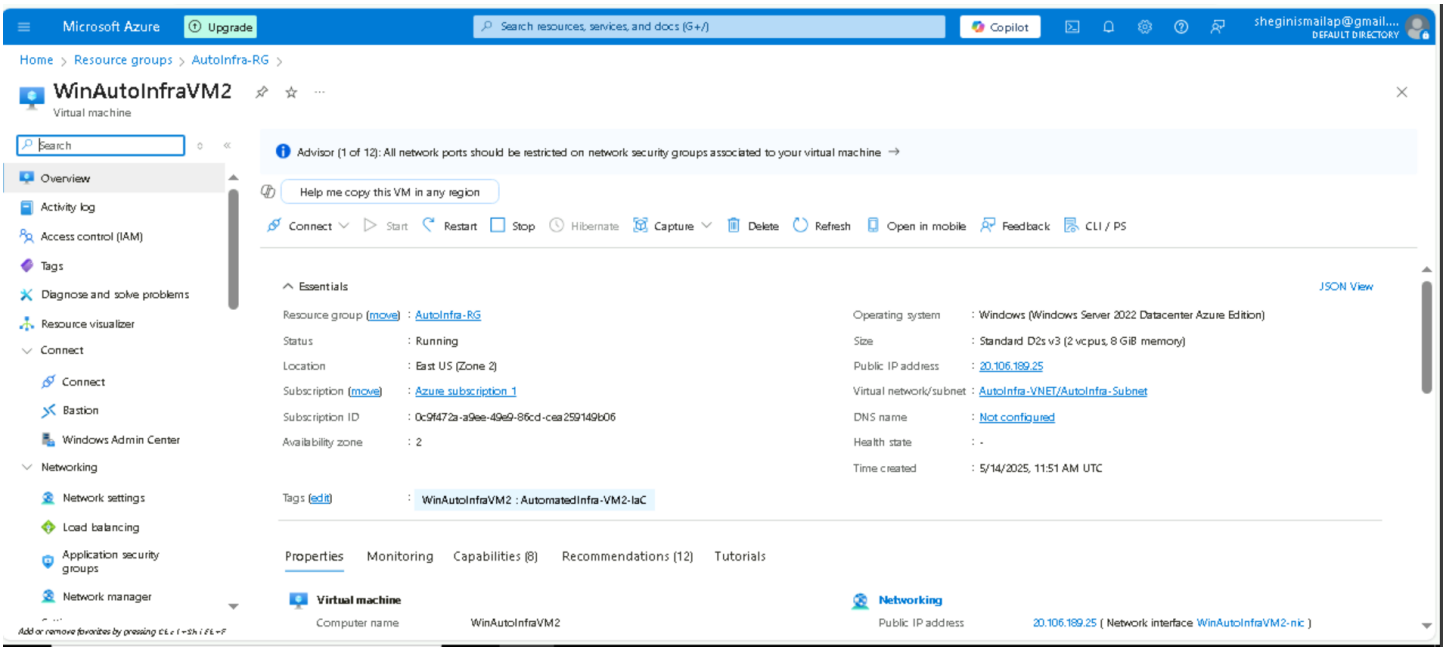
WinAutoInfraVM2


WinAutoInfraVM2.json 3
C: > Users > Shegin > Downloads > {} WinAutoInfraVM2.json > ...
"$schema": "https://schema.management.azure.
"contentVersion": "1.0.0.0",
"parameters": {
"vmName": {
"type": "string",
"defaultValue": "WinAutoInfraVM2"
"location": {
"type": "string",
"defaultValue": "eastus"
"adminUsername": {
"type": "string",
"defaultValue": "AutoIacdmin-1"
"adminPassword": {
"type": "securestring"
"vmSize": {
"type": "string",
"defaultValue": "Standard_D2s_v3"
"vnetName": {
"type": "string",
"defaultValue": "AutoInfra-VNET"
"subnetName": {
"type": "string",
"defaultValue": "AutoInfra-Subnet"
"resources": [
"type": "Microsoft.Network/publicIPA
"apiVersion": "2020-08-01",
"name": "[concat(parameters('vmName'
"resources": [
"name": "[concat(parameters('vmName'
"location": "[parameters('location')
"properties": {
"publicIpAllocationMethod": "Sta
"sku": {
"name": "Standard"
"zones": [
"2"
"tags": {
"WIinAutoInfra-VM2": "VM-AutoInf
"type": "Microsoft.Network/networkSe
"apiVersion": "2020-05-01",
"name": "[concat(parameters('vmName'
"location": "[parameters('location')
"properties": {
"securityRules": [
"name": "Allow-RDP",
"properties": {
"protocol": "Tcp",
"sourcePortRange": "
"destinationPortRang
"sourceAddressPrefix
"destinationAddressP
"access": "Allow",
"priority": 300,
"direction": "Inbour
"resources": [
"properties": {
"securityRules": [
"name": "Allow-HTTP",
"properties": {
"protocol": "Tcp",
"sourcePortRange": "
"destinationPortRang
"sourceAddressPrefix
"destinationAddressP
"access": "Allow",
"priority": 301,
"direction": "Inbour
"tags": {
"WinAutoInfra-VM2": "VM-AutoInfr
"type": "Microsoft.Network/networkIr
"apiVersion": "2021-05-01",
"name": "[concat(parameters('vmName'
"location": "[parameters('location')
"properties": {
"ipConfigurations": [
"name": "ipconfig1",
"properties": {
"subnet": {
"id": "[resource
"privateIPAllocation
master
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
{} Json
Go Live

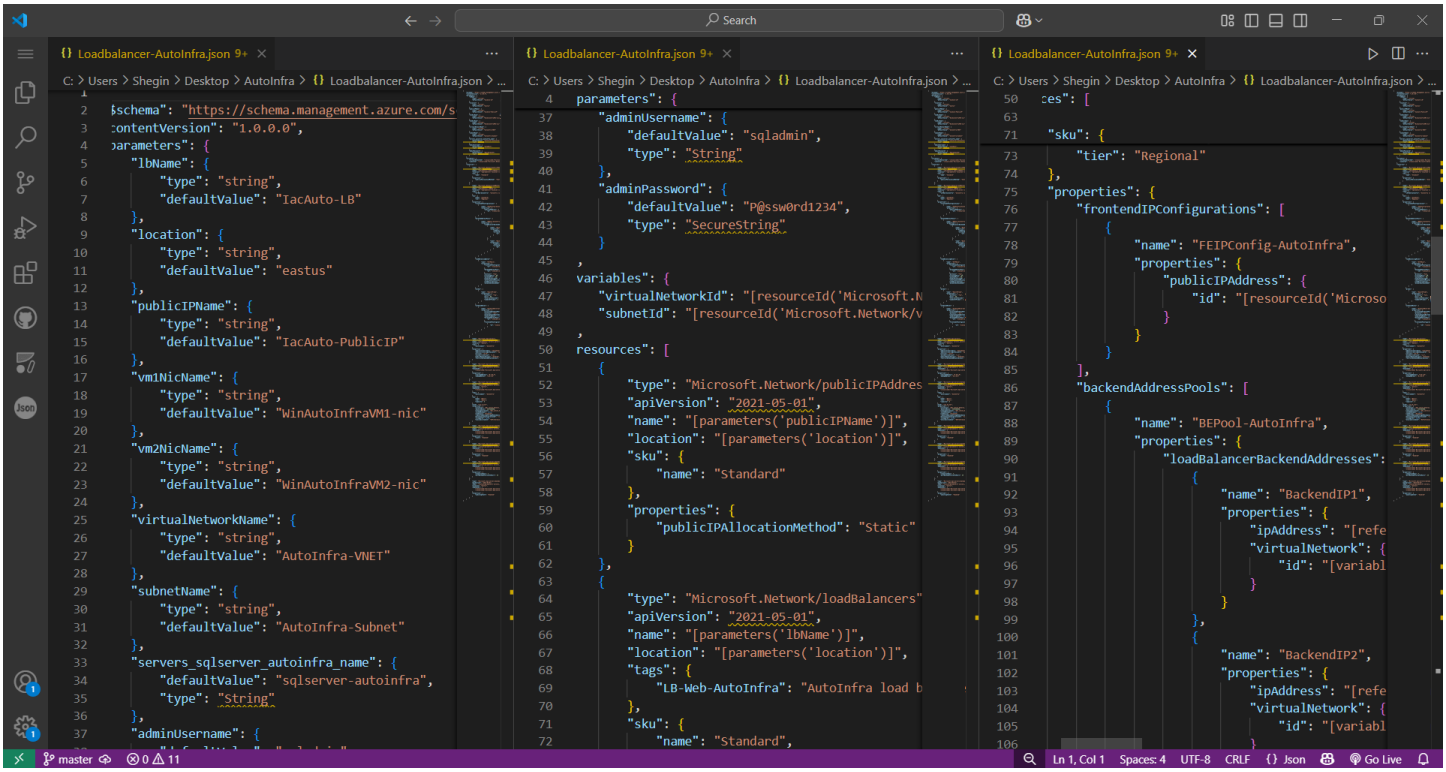

Open
Desktop > Auto IaC
Search Auto IaC
Organize
New folder
Pictures
Auto IaC
Automatic Infra
saves
This PC
Network
Name
Date modified
AutoInfra-VNet 5/14/2025 10:57 AM
ResourceGroupIaC 5/14/2025 10:43 AM
RGDiagonostics 5/14/2025 10:05 AM
WinAutoInfraVM1 5/14/2025 11:44 AM
WinAutoInfraVM2 5/14/2025 11:49 AM
File name: WinAutoInfraVM2
All files
Upload from mobile
Open
Cancel
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
"zoneMappings": null
"provisioningState": "Succeeded",
"templateHash": "16639361558292043262",
"templateLink": null,
"timestamp": "2025-05-14T11:46:24.379590+00:00",
"validatedResources": null
"resourceGroup": "AutoInfra-RG",
"tags": null,
"type": "Microsoft.Resources/deployments"
shegin [~]$ az deployment group create --resource-group AutoInfra-RG --template-file WinAutoInfraVM2.json
Successfully uploaded a file
Destination: /home/shegin
File: WinAutoInfraVM2.json

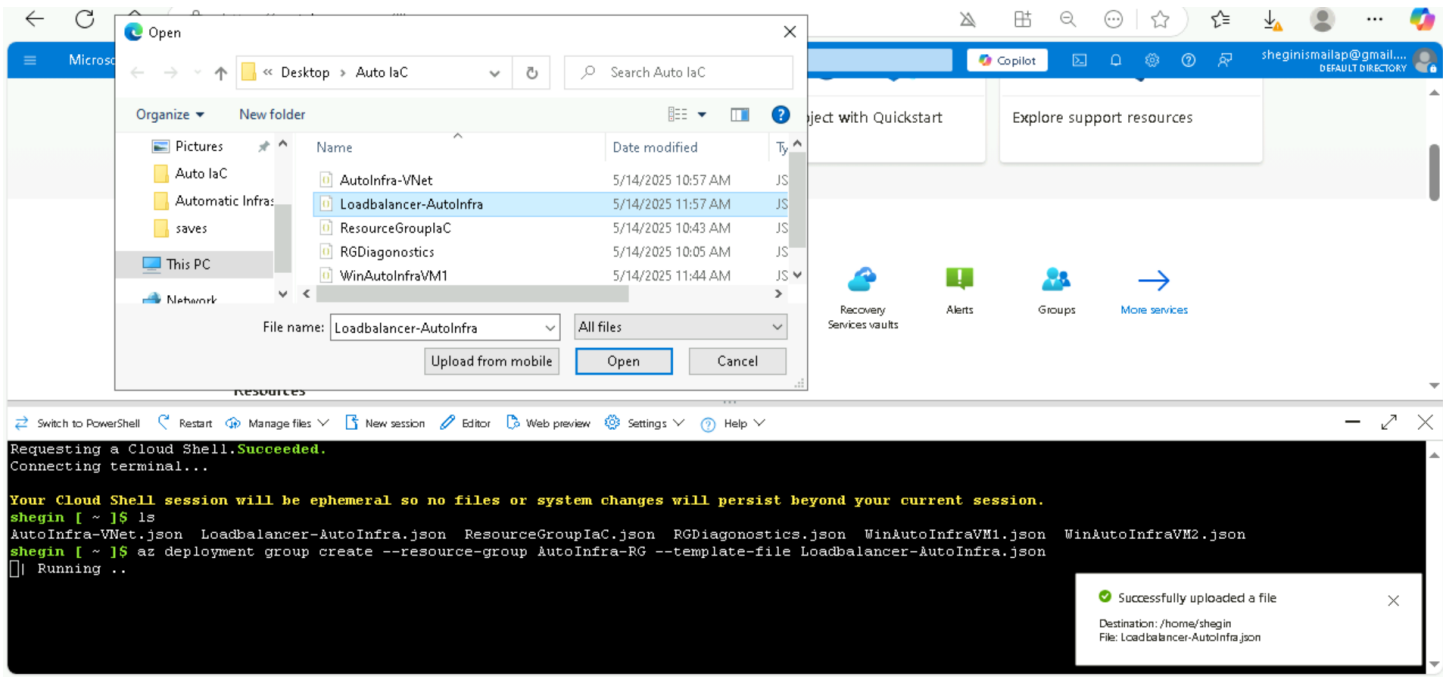


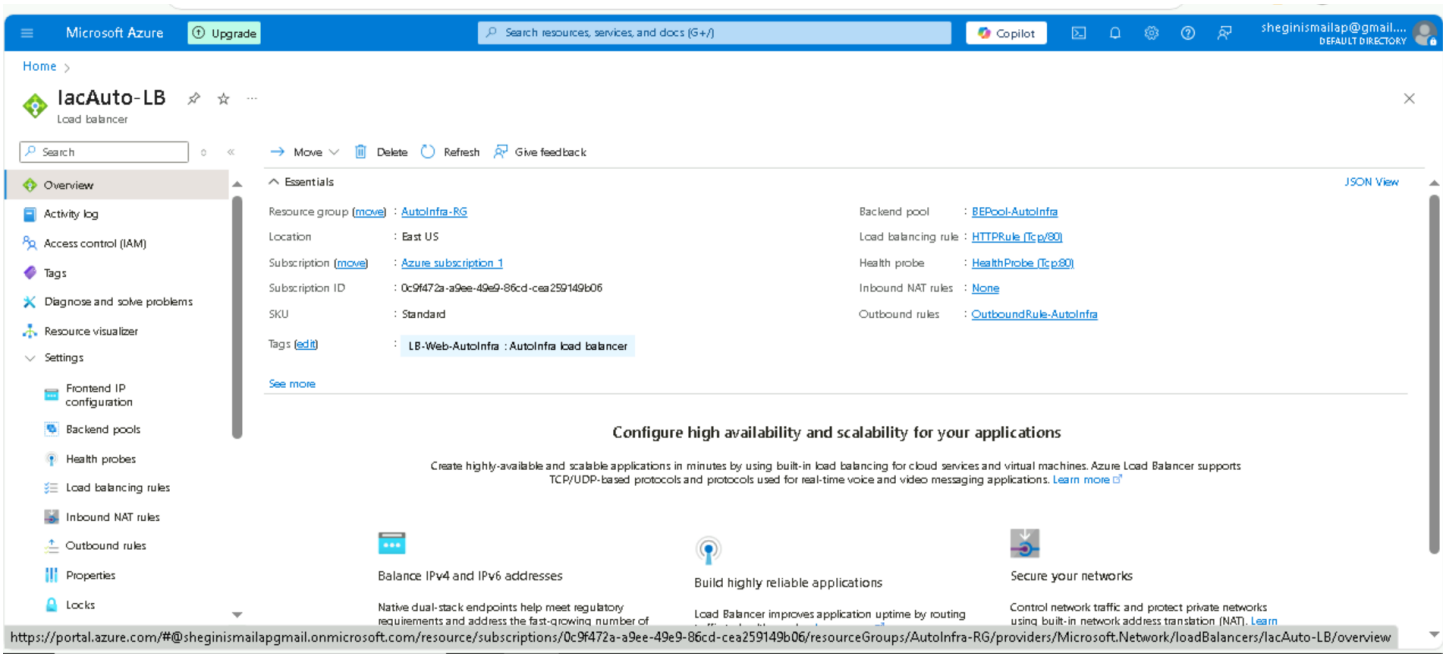
RESULT



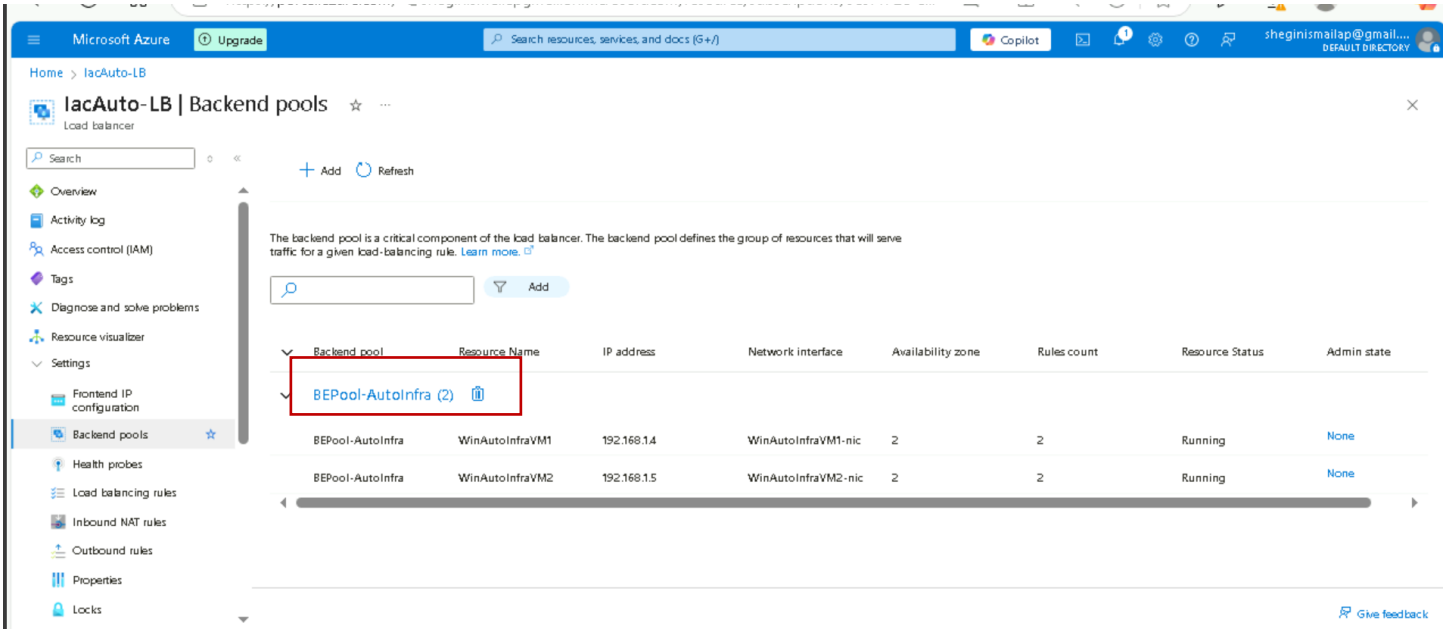
4) The ARM template for Implement Load Balancer (Layer 4) for the VMs at Loadbalancer-AutoInfra.json,the snippets from template,azure cli and outputs are shown

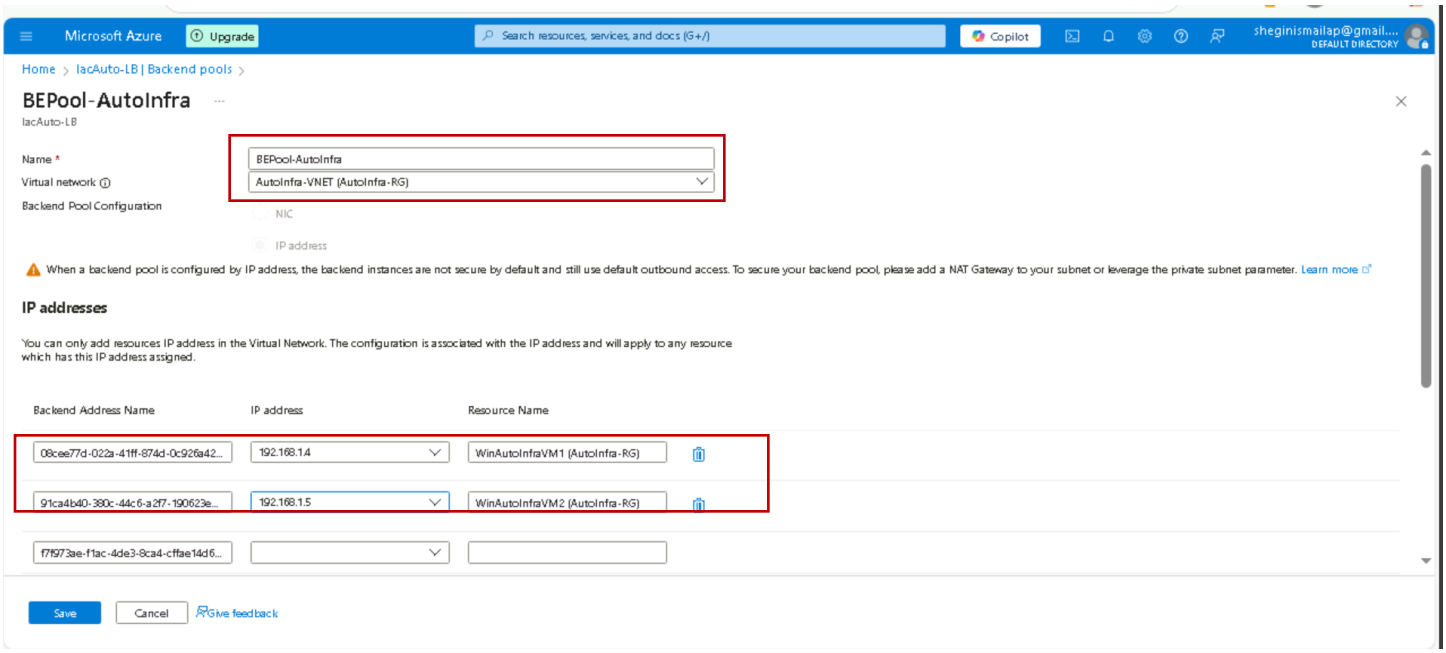

Loadbalancer-AutoInfra.json 9+
C: > Users > Shegin > Desktop > AutoInfra > Loadbalancer-AutoInfra.json
{
"$schema": "https://schema.management.azure.com/s
"contentVersion": "1.0.0.0",
"parameters": {
"lbName": {
"type": "string",
"defaultValue": "IacAuto-LB"
},
"location": {
"type": "string",
"defaultValue": "eastus"
},
"publicIPName": {
"type": "string",
"defaultValue": "IacAuto-PublicIP"
},
"vm1NicName": {
"type": "string",
"defaultValue": "WinAutoInfraVM1-nic"
},
"vm2NicName": {
"type": "string",
"defaultValue": "WinAutoInfraVM2-nic"
},
"virtualNetworkName": {
"type": "string",
"defaultValue": "AutoInfra-VNET"
},
"subnetName": {
"type": "string",
"defaultValue": "AutoInfra-Subnet"
},
"servers_sqlserver_autoinfra_name": {
"defaultValue": "sqlserver-autoinfra",
"type": "String"
},
"adminUsername": {
"parameters": {
"adminUsername": {
"defaultValue": "sqladmin",
"type": "String"
},
"adminPassword": {
"defaultValue": "P@ssw0rd1234",
"type": "SecureString"
}
},
"variables": {
"virtualNetworkId": "[resourceId('Microsoft.N
"subnetId": "[resourceId('Microsoft.Network/v
},
"resources": [
{
"type": "Microsoft.Network/publicIPAddres
"apiVersion": "2021-05-01",
"name": "[parameters('publicIPName')]",
"location": "[parameters('location')]",
"sku": {
"name": "Standard"
},
"properties": {
"publicIPAllocationMethod": "Static"
}
},
{
"type": "Microsoft.Network/loadBalancers"
"apiVersion": "2021-05-01",
"name": "[parameters('lbName')]",
"location": "[parameters('location')]",
"tags": {
"LB-Web-AutoInfra": "AutoInfra load b
},
"sku": {
"name": "Standard",
"ces": [
"sku": {
"tier": "Regional"
},
"properties": {
"frontendIPConfigurations": [
{
"name": "FEIPConfig-AutoInfra",
"properties": {
"publicIPAddress": {
"id": "[resourceId('Microso
}
}
}
],
"backendAddressPools": [
{
"name": "BEPool-AutoInfra",
"properties": {
"loadBalancerBackendAddresses":
{
"name": "BackendIP1",
"properties": {
"ipAddress": "[refe
"virtualNetwork": {
"id": "[variabl
}
}
},
{
"name": "BackendIP2",
"properties": {
"ipAddress": "[refe
"virtualNetwork": {
"id": "[variabl
master
0 11
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
Json
Go Live

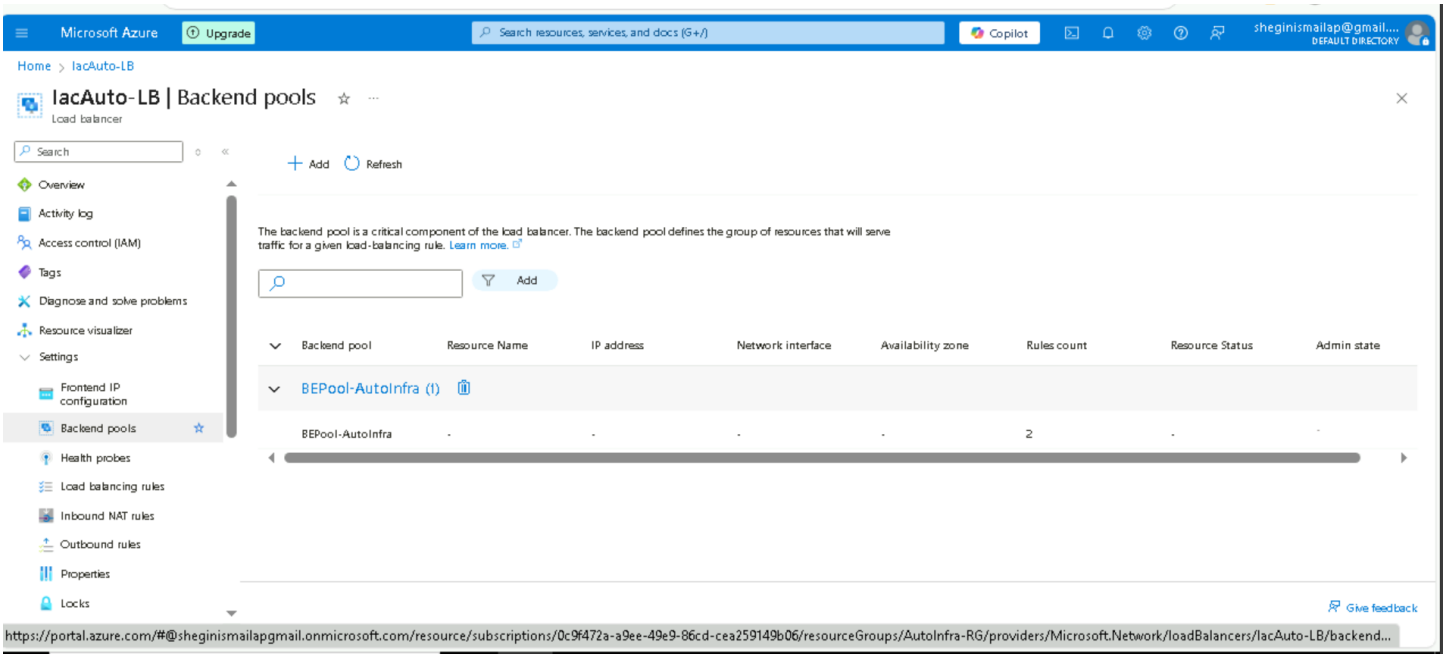

Open
Desktop > Auto IaC
Search Auto IaC
Organize
New folder
Pictures
Auto IaC
Automatic Infra
saves
This PC
Network
Name
Date modified
AutoInfra-VNet 5/14/2025 10:57 AM
Loadbalancer-AutoInfra 5/14/2025 11:57 AM
ResourceGroupIaC 5/14/2025 10:43 AM
RGDiagonostics 5/14/2025 10:05 AM
WinAutoInfraVM1 5/14/2025 11:44 AM
File name: Loadbalancer-AutoInfra
All files
Upload from mobile
Open
Cancel
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Explore support resources
Recovery Services vaults
Alerts
Groups
More services
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
AutoInfra-VNet.json Loadbalancer-AutoInfra.json ResourceGroupIaC.json RGDiagonostics.json WinAutoInfraVM1.json WinAutoInfraVM2.json
shegin [~]$ az deployment group create --resource-group AutoInfra-RG --template-file Loadbalancer-AutoInfra.json
Running ..
Successfully uploaded a file
Destination: /home/shegin
File: Loadbalancer-AutoInfra.json



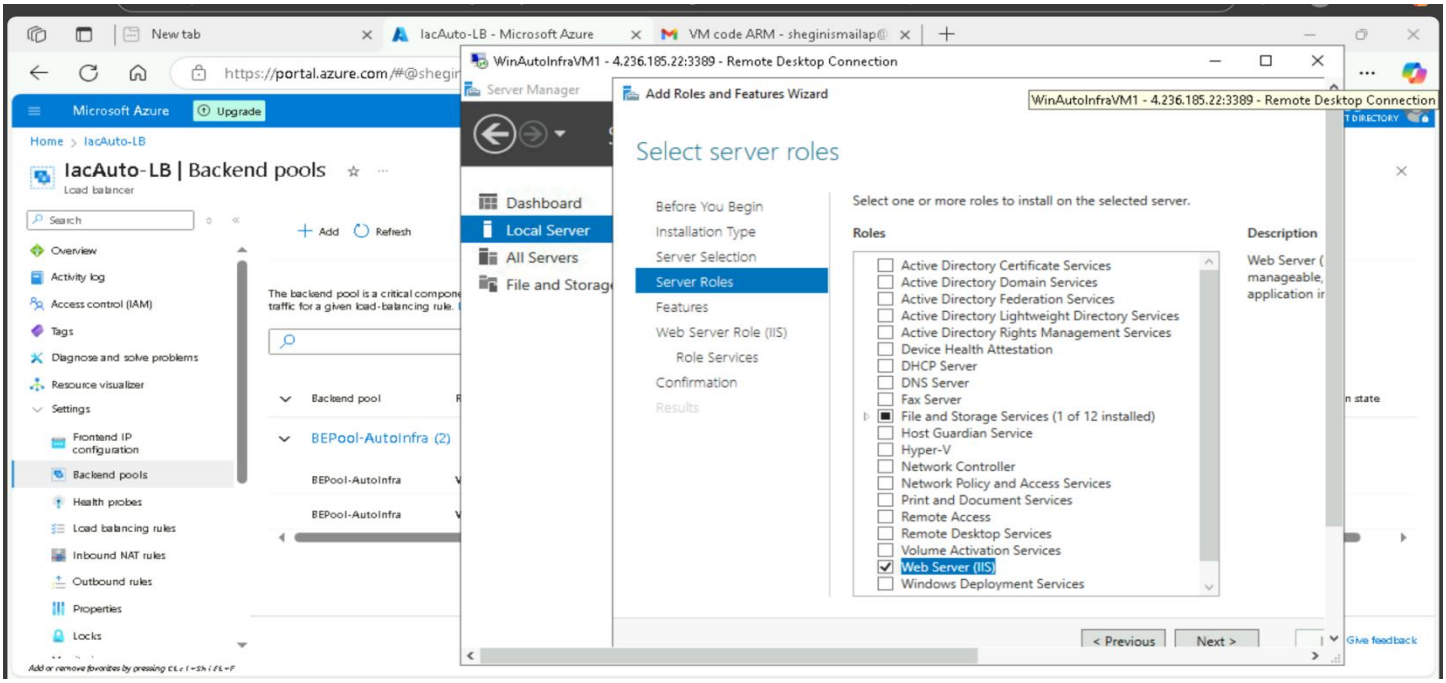
Inorder to prevent hardcoding sensitive values and automation,the direct backend pool selection of VM is not done,which needs to be progressed manually as shown

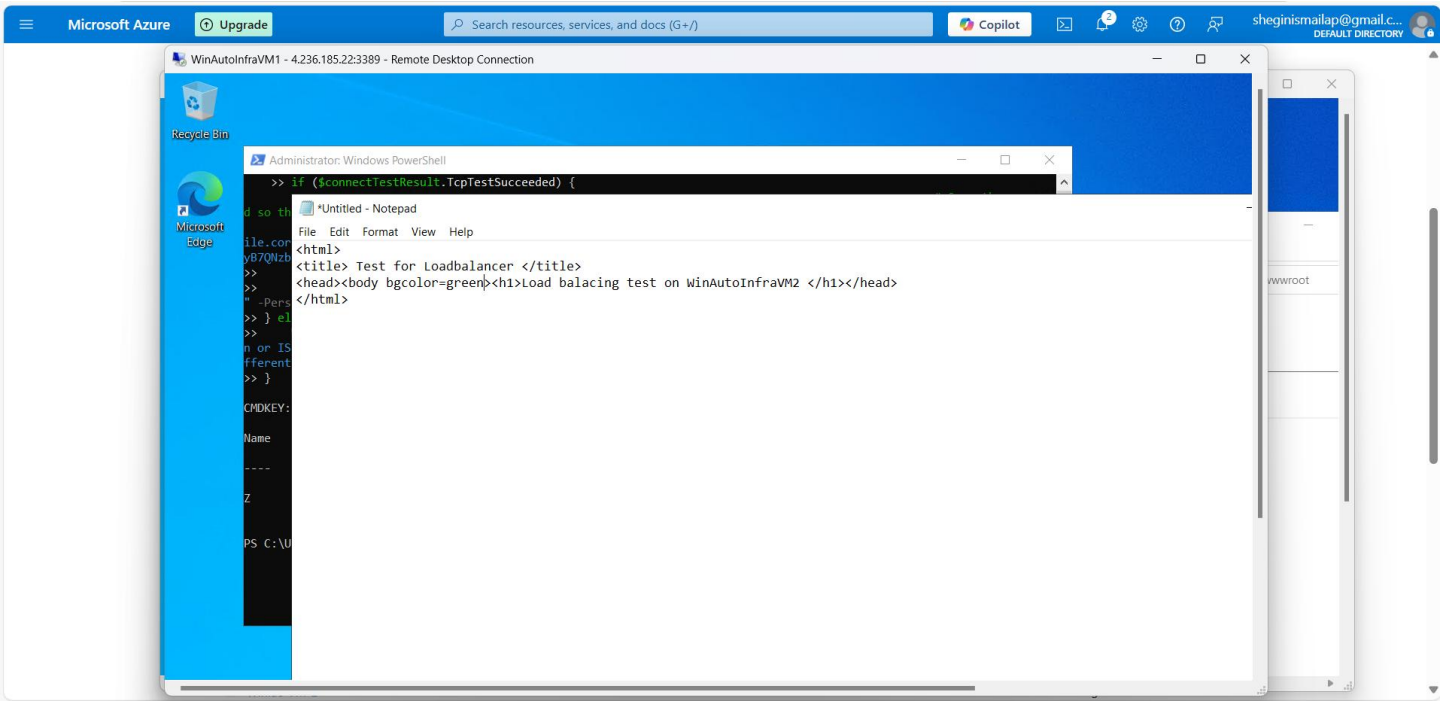


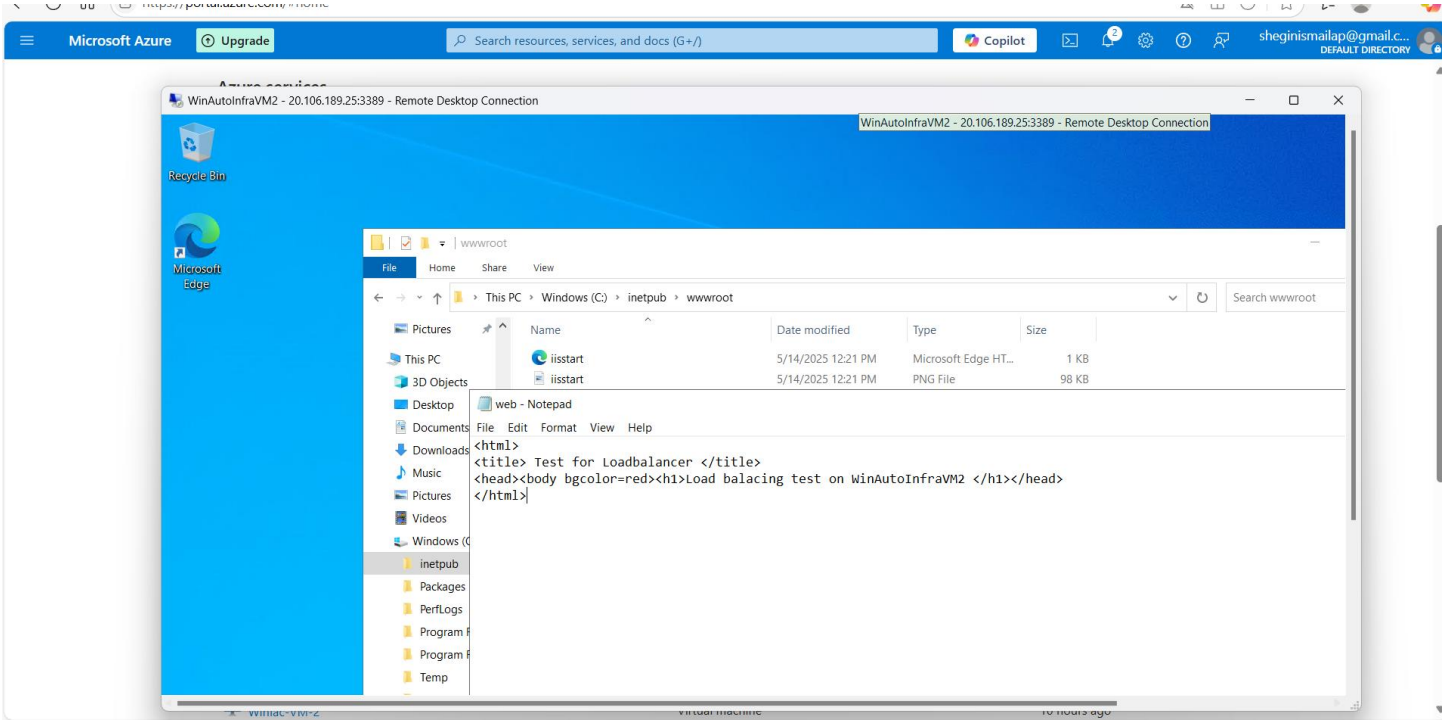

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > lacAuto-LB | Backend pools >
BEPool-AutoInfra
lacAuto-LB
Name *
BEPool-AutoInfra
Virtual network
AutoInfra-VNET (AutoInfra-RG)
Backend Pool Configuration
NIC
IP address
When a backend pool is configured by IP address, the backend instances are not secure by default and still use default outbound access. To secure your backend pool, please add a NAT Gateway to your subnet or leverage the private subnet parameter. Learn more
IP addresses
You can only add resources IP address in the Virtual Network. The configuration is associated with the IP address and will apply to any resource which has this IP address assigned.
Backend Address Name
IP address
Resource Name
08cee77d-022a-41ff-874d-0c926a42...
192.168.1.4
WinAutoInfraVM1 (AutoInfra-RG)
91ca4b40-380c-44c6-a2f7-190623e...
192.168.1.5
WinAutoInfraVM2 (AutoInfra-RG)
f7f973ae-f1ac-4de3-8ca4-cffae14d6...
Save
Cancel
Give feedback

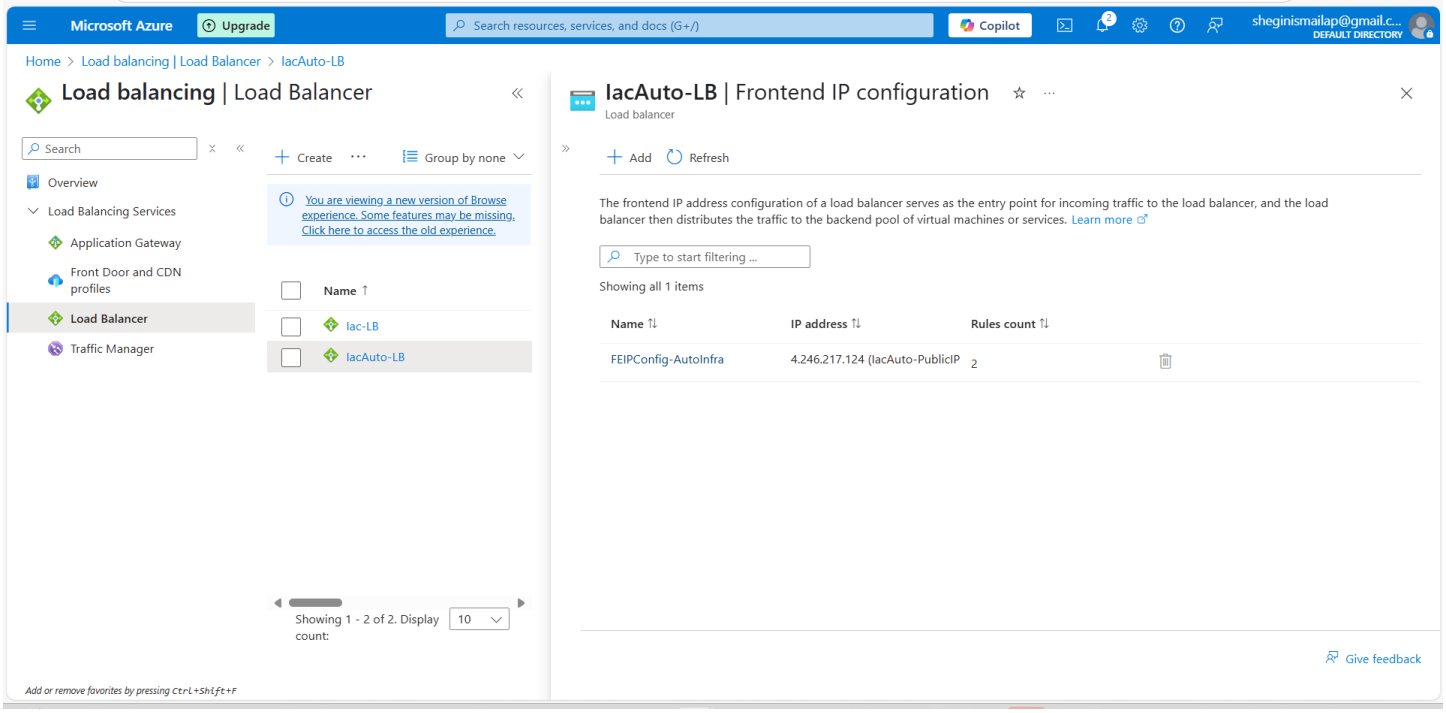

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > lacAuto-LB
lacAuto-LB | Backend pools
Load balancer
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Frontend IP configuration
Backend pools
Health probes
Load balancing rules
Inbound NAT rules
Outbound rules
Properties
Locks
Add
Refresh
The backend pool is a critical component of the load balancer. The backend pool defines the group of resources that will serve traffic for a given load-balancing rule. Learn more.
Add
Backend pool
Resource Name
IP address
Network interface
Availability zone
Rules count
Resource Status
Admin state
BEPool-AutoInfra (1)
BEPool-AutoInfra
2
Give feedback
https://portal.azure.com/#@sheginismailapgmail.onmicrosoft.com/resource/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/Microsoft.Network/loadBalancers/lacAuto-LB/backend...

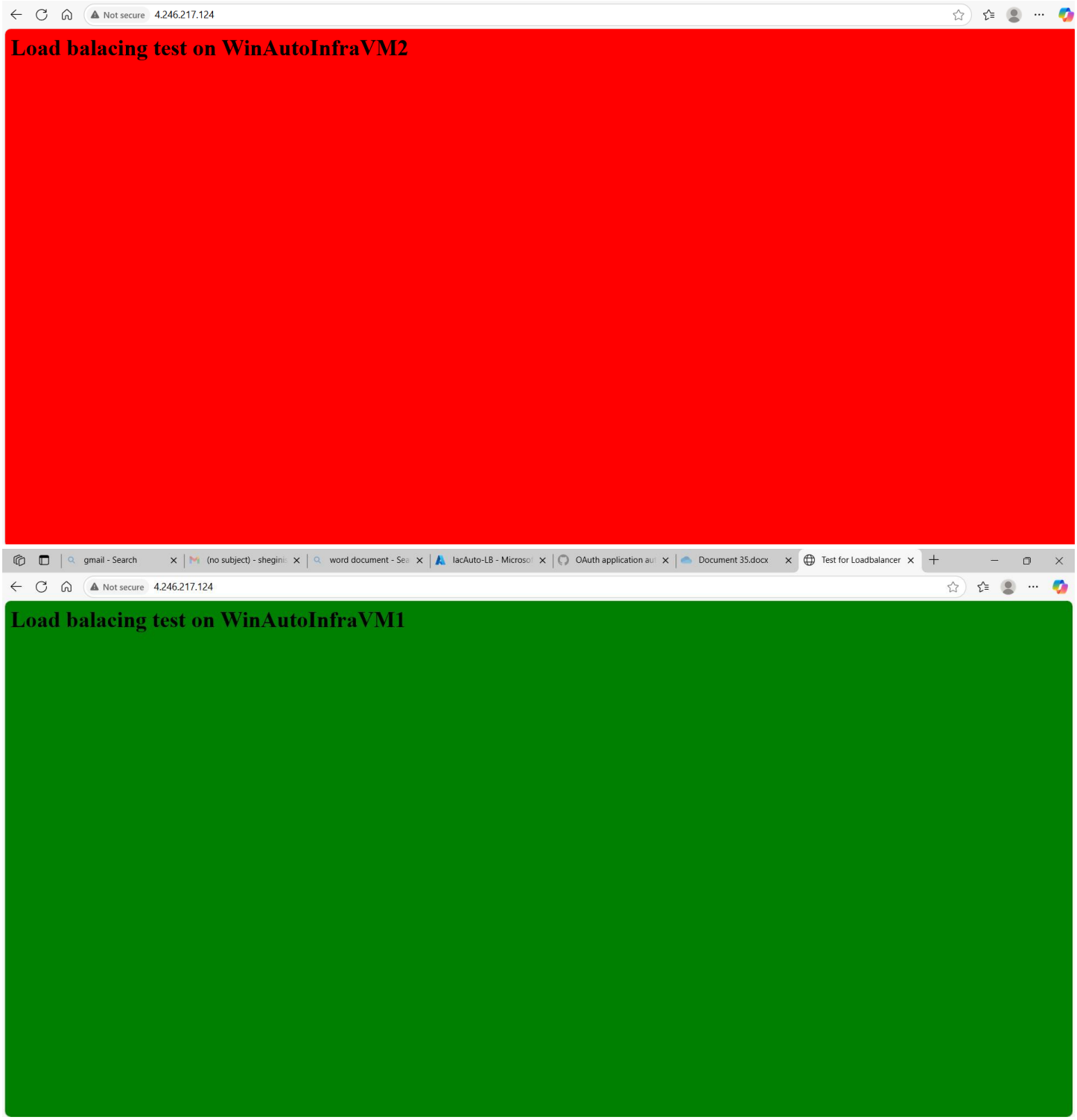
TEST AND RESULT BY IIS



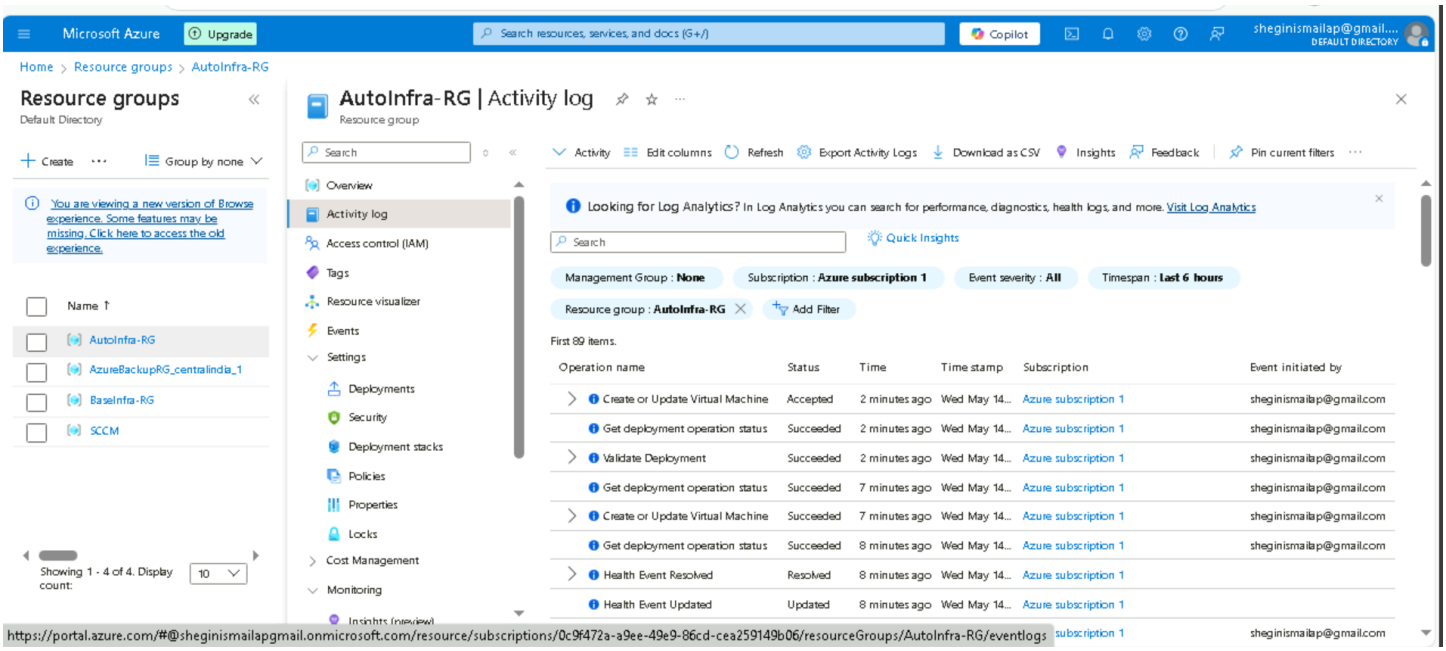


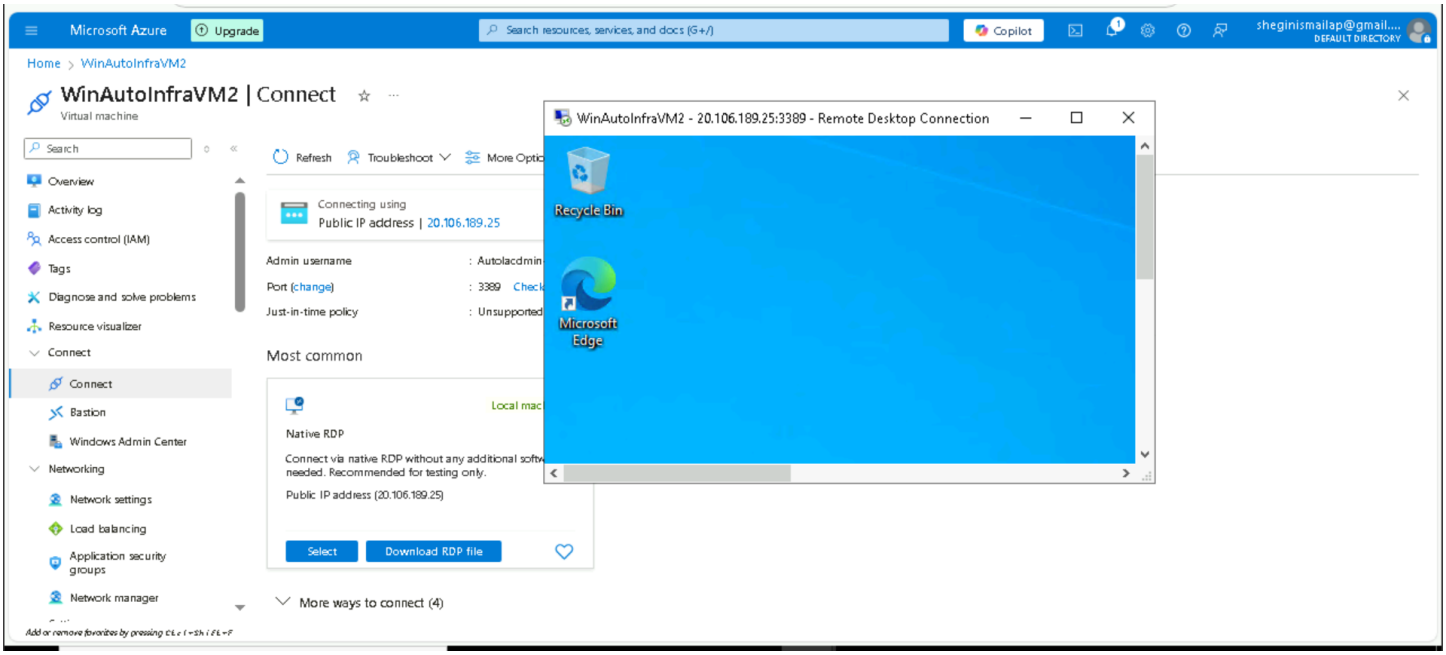

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
WinAutoInfraVM2 - 20.106.189.25:3389 - Remote Desktop Connection
Recycle Bin
Microsoft Edge
wwwroot
File Home Share View
This PC › Windows (C:) › inetpub › wwwroot
Search wwwroot
Name Date modified Type Size
iisstart 5/14/2025 12:21 PM Microsoft Edge HT... 1 KB
iisstart 5/14/2025 12:21 PM PNG File 98 KB
web - Notepad
File Edit Format View Help
<html>
<title> Test for Loadbalancer </title>
<head><body bgcolor=red><h1>Load balacing test on WinAutoInfraVM2 </h1></head>
</html>

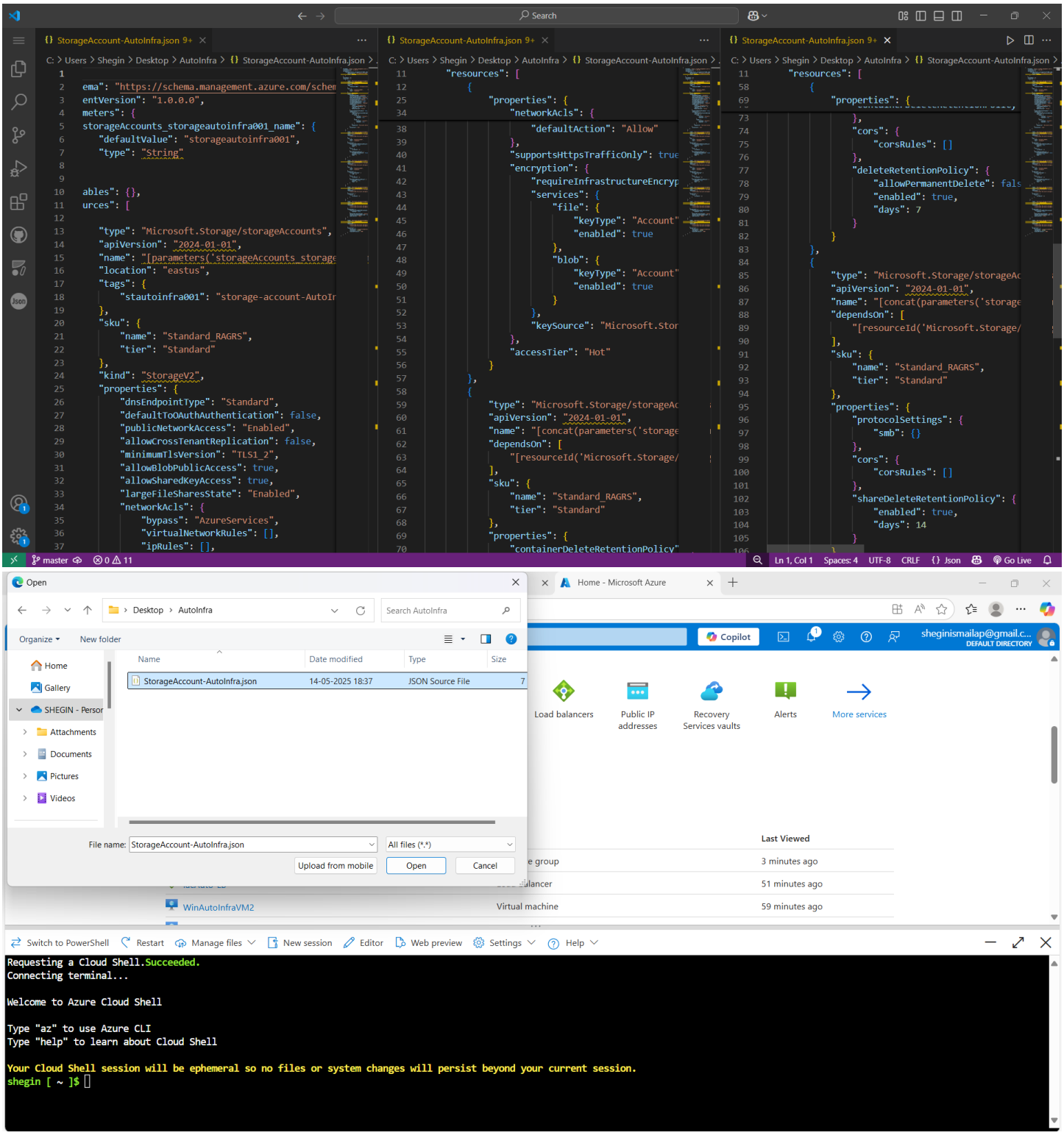

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Load balancing | Load Balancer > lacAuto-LB
Load balancing | Load Balancer
Search
Overview
Load Balancing Services
Application Gateway
Front Door and CDN profiles
Load Balancer
Traffic Manager
Create
Group by none
You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.
Name
lac-LB
lacAuto-LB
Showing 1 - 2 of 2. Display count: 10
Add or remove favorites by pressing Ctrl+Shift+F
lacAuto-LB | Frontend IP configuration
Load balancer
Add
Refresh
The frontend IP address configuration of a load balancer serves as the entry point for incoming traffic to the load balancer, and the load balancer then distributes the traffic to the backend pool of virtual machines or services. Learn more
Type to start filtering ...
Showing all 1 items
Name
IP address
Rules count
FEIPConfig-AutoInfra
4.246.217.124 (lacAuto-PublicIP
2
Give feedback

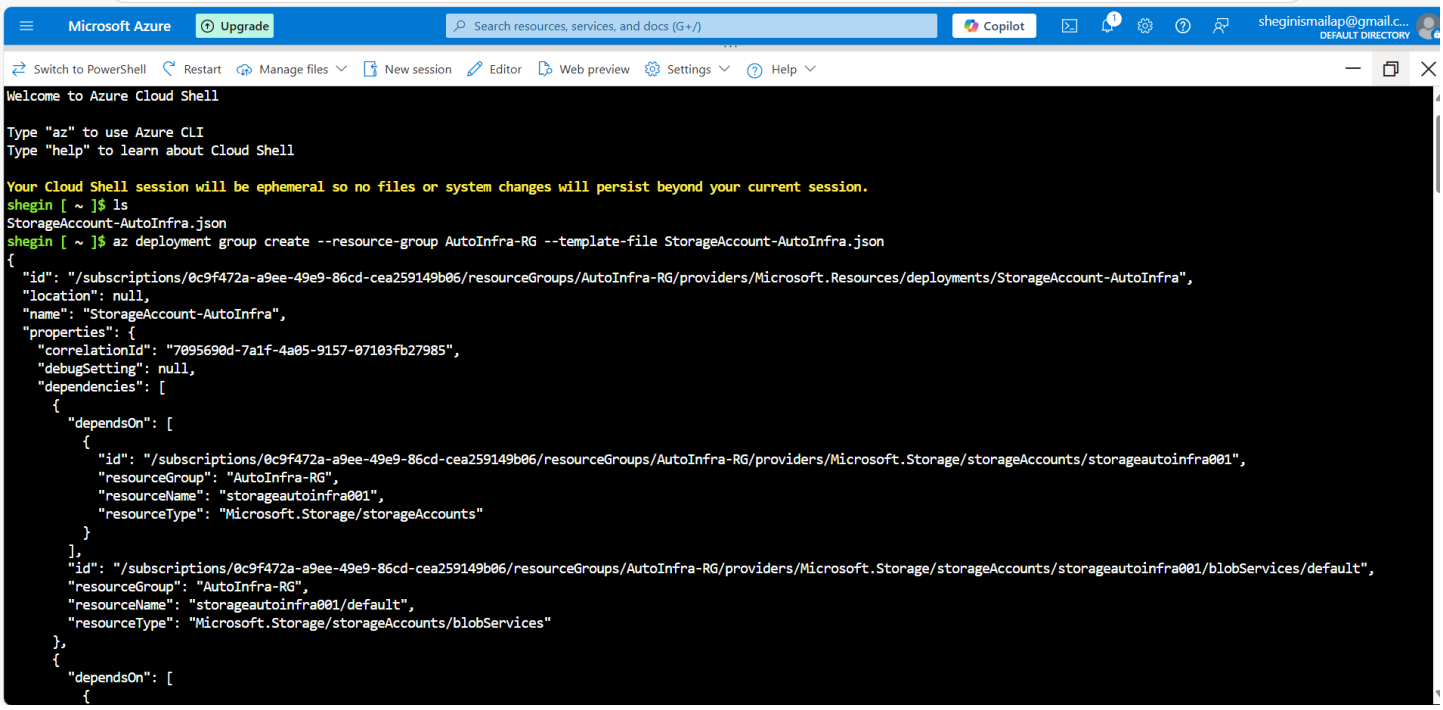


5) The ARM template for Create Azure Storage (Blob, File Share, SAS key) at StorageAccount-AutoInfra.json,the snippets from template,azure cli and outputs are shown

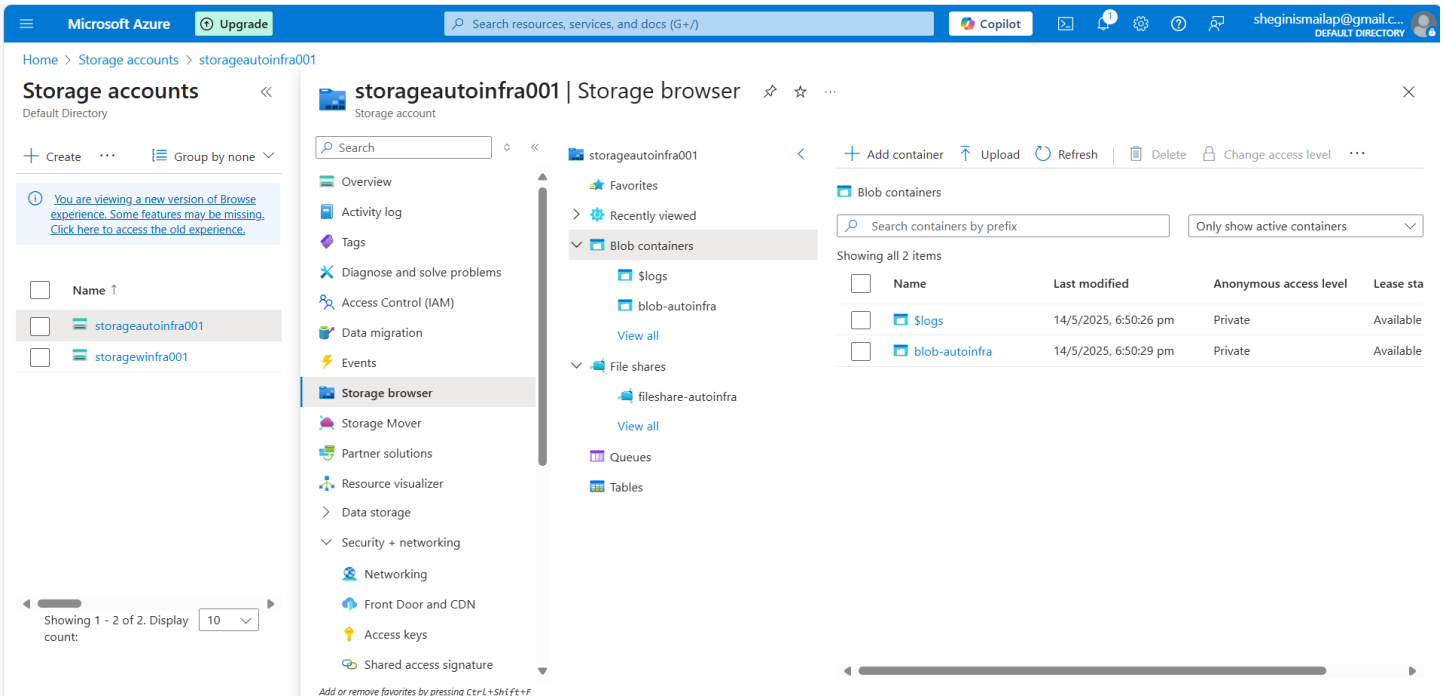

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > Resource groups > AutoInfra-RG
Resource groups
Default Directory
Create
Group by none
You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.
Name
AutoInfra-RG
AzureBackupRG_centralindia_1
BaseInfra-RG
SCCM
Showing 1 - 4 of 4. Display count: 10
AutoInfra-RG | Activity log
Resource group
Search
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Deployment stacks
Policies
Properties
Locks
Cost Management
Monitoring
Activity
Edit columns
Refresh
Export Activity Logs
Download as CSV
Insights
Feedback
Pin current filters
Looking for Log Analytics? In Log Analytics you can search for performance, diagnostics, health logs, and more. Visit Log Analytics
Search
Quick Insights
Management Group : None
Subscription : Azure subscription 1
Event severity : All
Timespan : Last 6 hours
Resource group : AutoInfra-RG
Add Filter
First 89 items.
Operation name
Status
Time
Time stamp
Subscription
Event initiated by
Create or Update Virtual Machine
Accepted
2 minutes ago
Wed May 14...
Azure subscription 1
sheginismailap@gmail.com
Get deployment operation status
Succeeded
2 minutes ago
Wed May 14...
Azure subscription 1
sheginismailap@gmail.com
Validate Deployment
Succeeded
2 minutes ago
Wed May 14...
Azure subscription 1
sheginismailap@gmail.com
Get deployment operation status
Succeeded
7 minutes ago
Wed May 14...
Azure subscription 1
sheginismailap@gmail.com
Create or Update Virtual Machine
Succeeded
7 minutes ago
Wed May 14...
Azure subscription 1
sheginismailap@gmail.com
Get deployment operation status
Succeeded
8 minutes ago
Wed May 14...
Azure subscription 1
sheginismailap@gmail.com
Health Event Resolved
Resolved
8 minutes ago
Wed May 14...
Azure subscription 1
Health Event Updated
Updated
8 minutes ago
Wed May 14...
Azure subscription 1
https://portal.azure.com/#@sheginismailapgmail.onmicrosoft.com/resource/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/eventlogs
subscription 1
sheginismailap@gmail.com

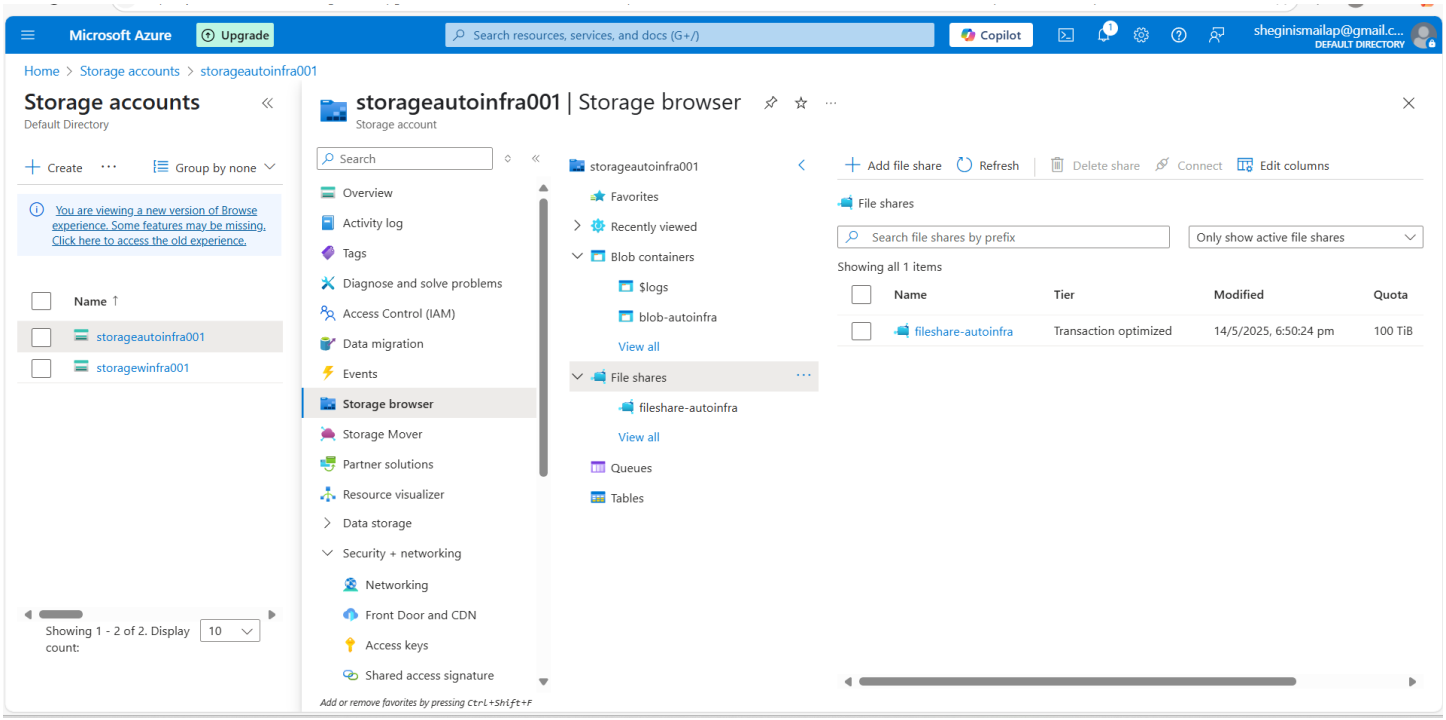

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail....
DEFAULT DIRECTORY
Home > WinAutoInfraVM2
WinAutoInfraVM2 | Connect
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Connect
Connect
Bastion
Windows Admin Center
Networking
Network settings
Load balancing
Application security groups
Network manager
Refresh
Troubleshoot
Connecting using
Public IP address | 20.106.189.25
Admin username : Autolacdmin
Port (change) : 3389 Check
Just-in-time policy : Unsupported
Most common
Native RDP
Connect via native RDP without any additional softw
needed. Recommended for testing only.
Public IP address (20.106.189.25)
Select
Download RDP file
More ways to connect (4)
WinAutoInfraVM2 - 20.106.189.25:3389 - Remote Desktop Connection
Recycle Bin
Microsoft Edge


StorageAccount-AutoInfra.json 9+
C: > Users > Shegin > Desktop > AutoInfra > StorageAccount-AutoInfra.json
"$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
"contentVersion": "1.0.0.0",
"parameters": {
"storageAccounts_storageautoinfra001_name": {
"defaultValue": "storageautoinfra001",
"type": "String"
},
"variables": {},
"resources": [
{
"type": "Microsoft.Storage/storageAccounts",
"apiVersion": "2024-01-01",
"name": "[parameters('storageAccounts_storageautoinfra001_name')]",
"location": "eastus",
"tags": {
"stautoinfra001": "storage-account-AutoInfra"
},
"sku": {
"name": "Standard_RAGRS",
"tier": "Standard"
},
"kind": "StorageV2",
"properties": {
"dnsEndpointType": "Standard",
"defaultToOAuthAuthentication": false,
"publicNetworkAccess": "Enabled",
"allowCrossTenantReplication": false,
"minimumTlsVersion": "TLS1_2",
"allowBlobPublicAccess": true,
"allowSharedKeyAccess": true,
"largeFileSharesState": "Enabled",
"networkAcls": {
"bypass": "AzureServices",
"virtualNetworkRules": [],
"ipRules": [],
"defaultAction": "Allow"
},
"supportsHttpsTrafficOnly": true,
"encryption": {
"requireInfrastructureEncryption": false,
"services": {
"file": {
"keyType": "Account",
"enabled": true
},
"blob": {
"keyType": "Account",
"enabled": true
}
},
"keySource": "Microsoft.Storage"
},
"accessTier": "Hot"
}
},
{
"type": "Microsoft.Storage/storageAccounts/blobServices",
"apiVersion": "2024-01-01",
"name": "[concat(parameters('storageAccounts_storageautoinfra001_name'), '/default')]",
"dependsOn": [
"[resourceId('Microsoft.Storage/storageAccounts', parameters('storageAccounts_storageautoinfra001_name'))]"
],
"sku": {
"name": "Standard_RAGRS",
"tier": "Standard"
},
"properties": {
"containerDeleteRetentionPolicy": {
"allowPermanentDelete": false,
"enabled": true,
"days": 7
},
"cors": {
"corsRules": []
},
"deleteRetentionPolicy": {
"allowPermanentDelete": false,
"enabled": true,
"days": 7
}
}
},
{
"type": "Microsoft.Storage/storageAccounts/fileServices",
"apiVersion": "2024-01-01",
"name": "[concat(parameters('storageAccounts_storageautoinfra001_name'), '/default')]",
"dependsOn": [
"[resourceId('Microsoft.Storage/storageAccounts', parameters('storageAccounts_storageautoinfra001_name'))]"
],
"sku": {
"name": "Standard_RAGRS",
"tier": "Standard"
},
"properties": {
"protocolSettings": {
"smb": {}
},
"cors": {
"corsRules": []
},
"shareDeleteRetentionPolicy": {
"enabled": true,
"days": 14
}
}
}
master
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
Json
Go Live
Open
Desktop > AutoInfra
Search AutoInfra
Organize
New folder
Home
Gallery
SHEGIN - Personal
Attachments
Documents
Pictures
Videos
Name
Date modified
Type
Size
StorageAccount-AutoInfra.json
14-05-2025 18:37
JSON Source File
File name: StorageAccount-AutoInfra.json
All files (*.*)
Upload from mobile
Open
Cancel
Home - Microsoft Azure
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Load balancers
Public IP addresses
Recovery Services vaults
Alerts
More services
Last Viewed
e group
3 minutes ago
lancer
51 minutes ago
WinAutoInfraVM2
Virtual machine
59 minutes ago
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Welcome to Azure Cloud Shell
Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Welcome to Azure Cloud Shell
Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
StorageAccount-AutoInfra.json
shegin [~]$ az deployment group create --resource-group AutoInfra-RG --template-file StorageAccount-AutoInfra.json
{
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/Microsoft.Resources/deployments/StorageAccount-AutoInfra",
"location": null,
"name": "StorageAccount-AutoInfra",
"properties": {
"correlationId": "7095690d-7a1f-4a05-9157-07103fb27985",
"debugSetting": null,
"dependencies": [
{
"dependsOn": [
{
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/Microsoft.Storage/storageAccounts/storageautoinfra001",
"resourceGroup": "AutoInfra-RG",
"resourceName": "storageautoinfra001",
"resourceType": "Microsoft.Storage/storageAccounts"
}
],
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/Microsoft.Storage/storageAccounts/storageautoinfra001/blobServices/default",
"resourceGroup": "AutoInfra-RG",
"resourceName": "storageautoinfra001/default",
"resourceType": "Microsoft.Storage/storageAccounts/blobServices"
},
{
"dependsOn": [
{

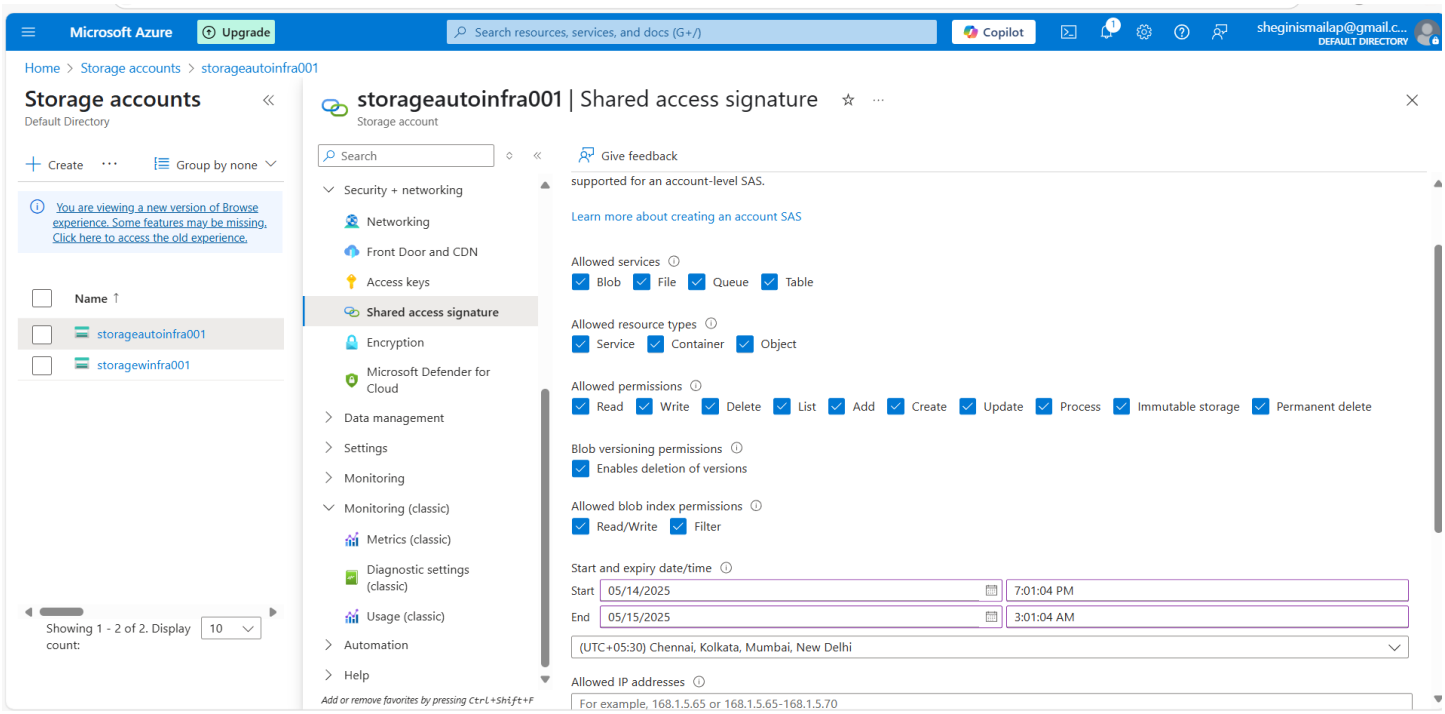
RESULTS

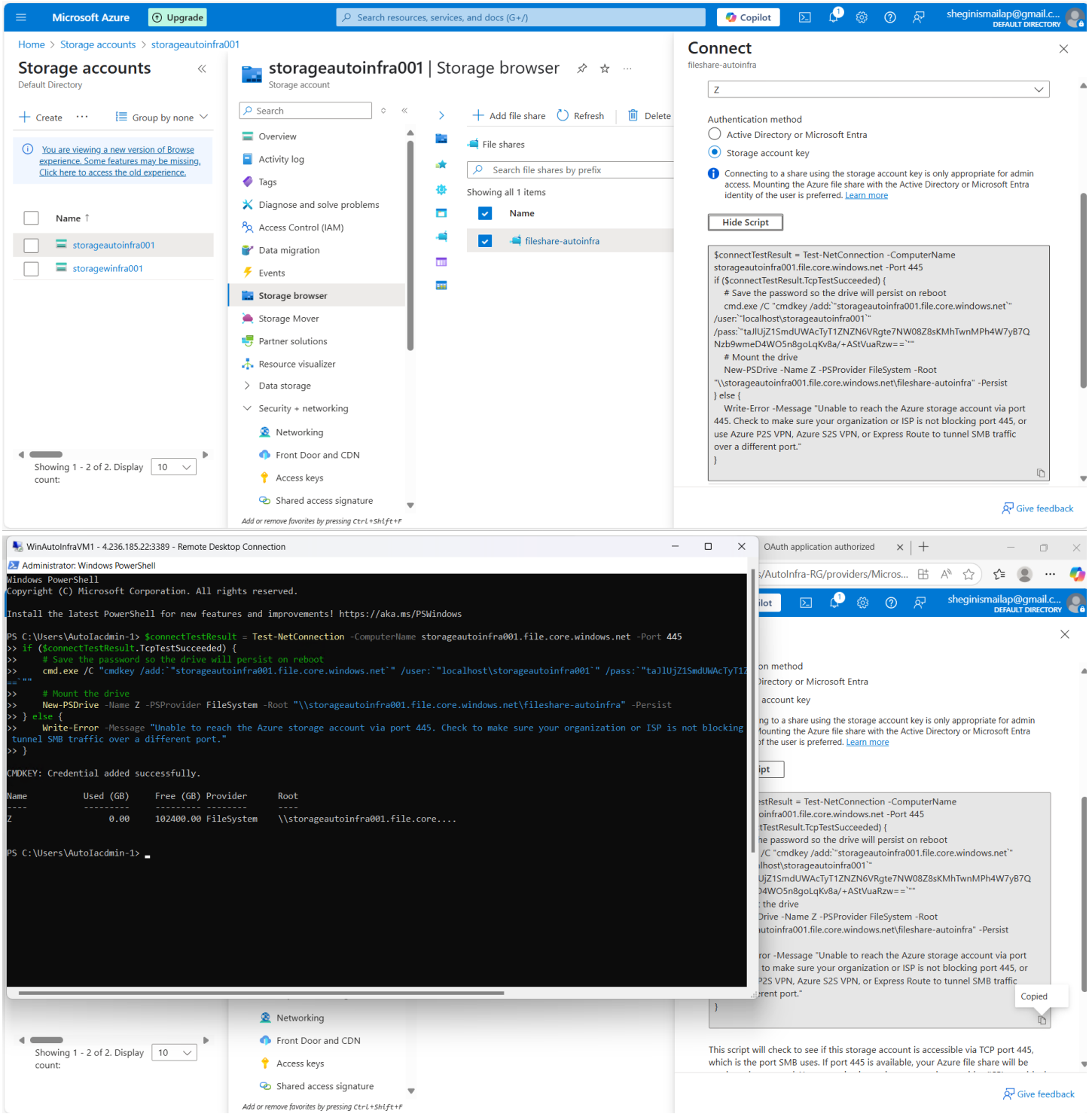

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Storage accounts > storageautoinfra001
Storage accounts
Default Directory
Create
Group by none
You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.
Name
storageautoinfra001
storagewinfra001
Showing 1 - 2 of 2. Display count: 10
storageautoinfra001 | Storage browser
Storage account
Search
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Security + networking
Networking
Front Door and CDN
Access keys
Shared access signature
Add or remove favorites by pressing Ctrl+Shift+F
storageautoinfra001
Favorites
Recently viewed
Blob containers
$logs
blob-autoinfra
View all
File shares
fileshare-autoinfra
View all
Queues
Tables
Add container
Upload
Refresh
Delete
Change access level
Blob containers
Search containers by prefix
Only show active containers
Showing all 2 items
Name
Last modified
Anonymous access level
Lease sta
$logs
14/5/2025, 6:50:26 pm
Private
Available
blob-autoinfra
14/5/2025, 6:50:29 pm
Private
Available



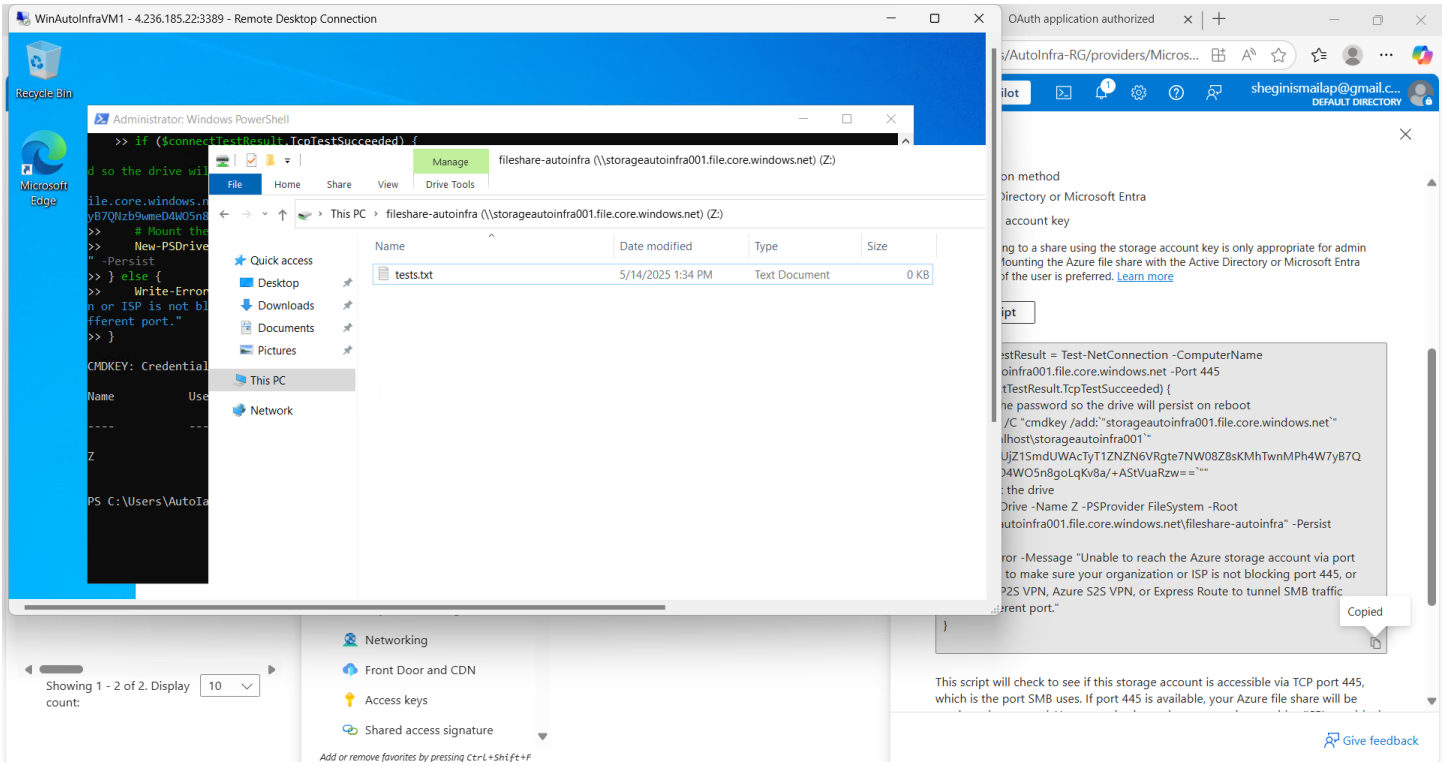
6. Map File Share on VMs

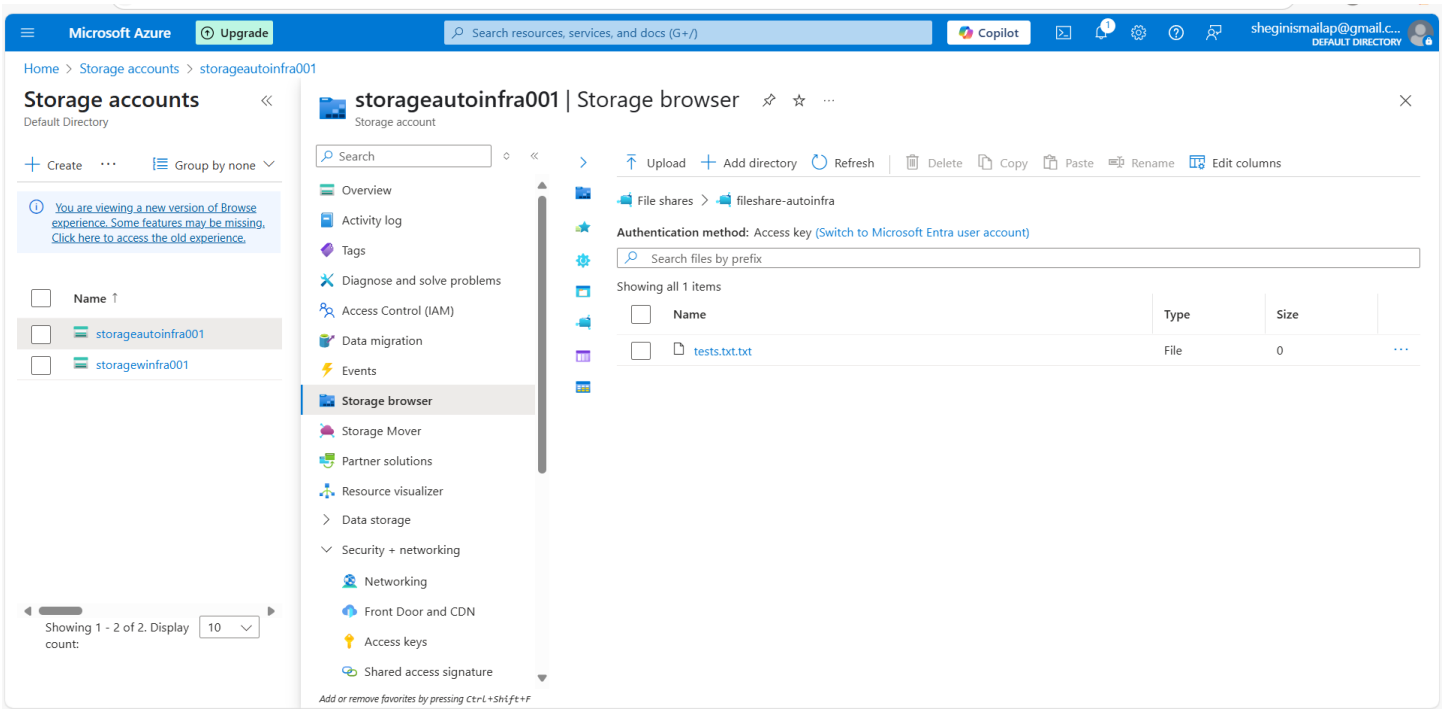
   Demonstrated below is via custom script mapping to Z Drive



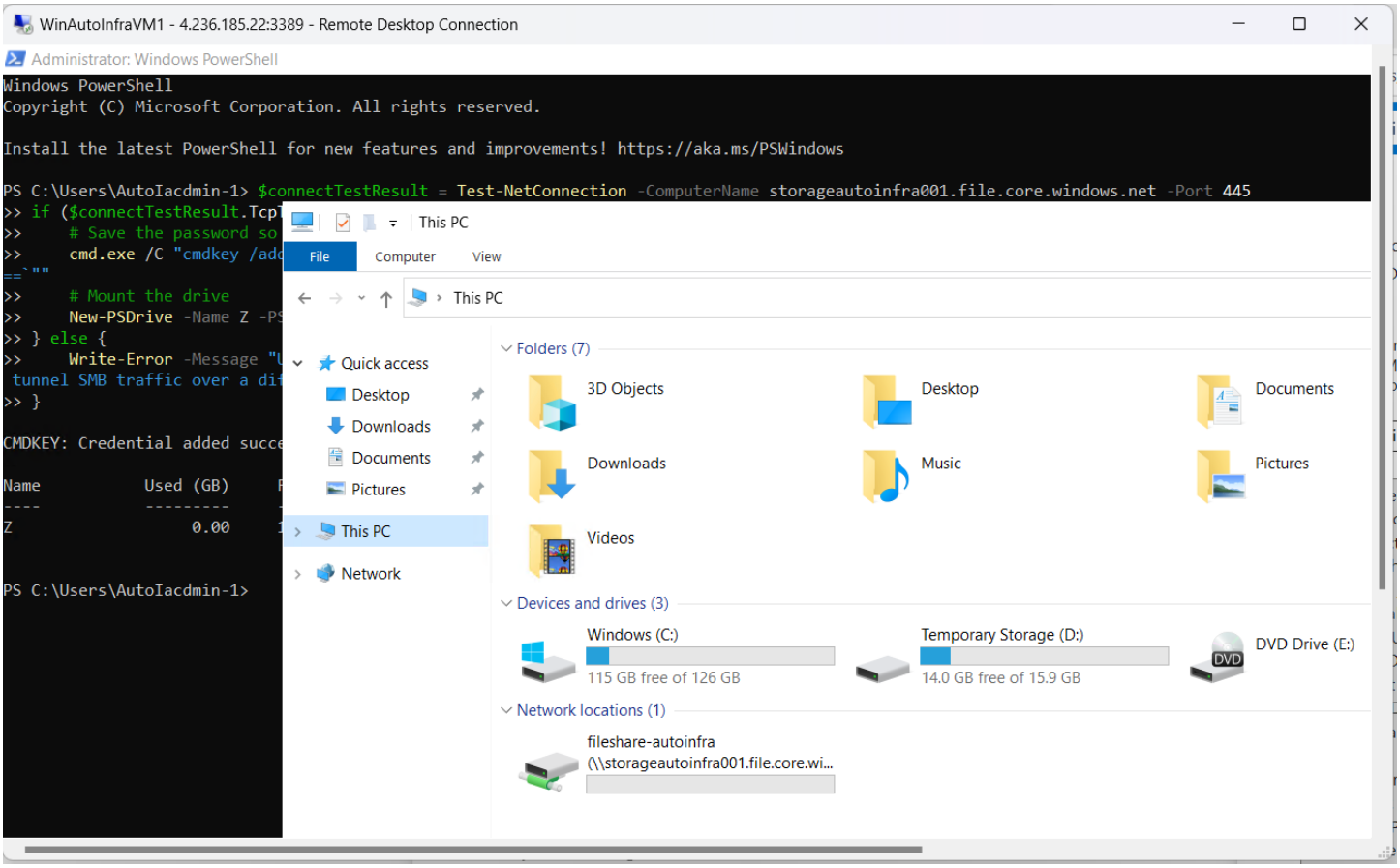


Added test.txt file

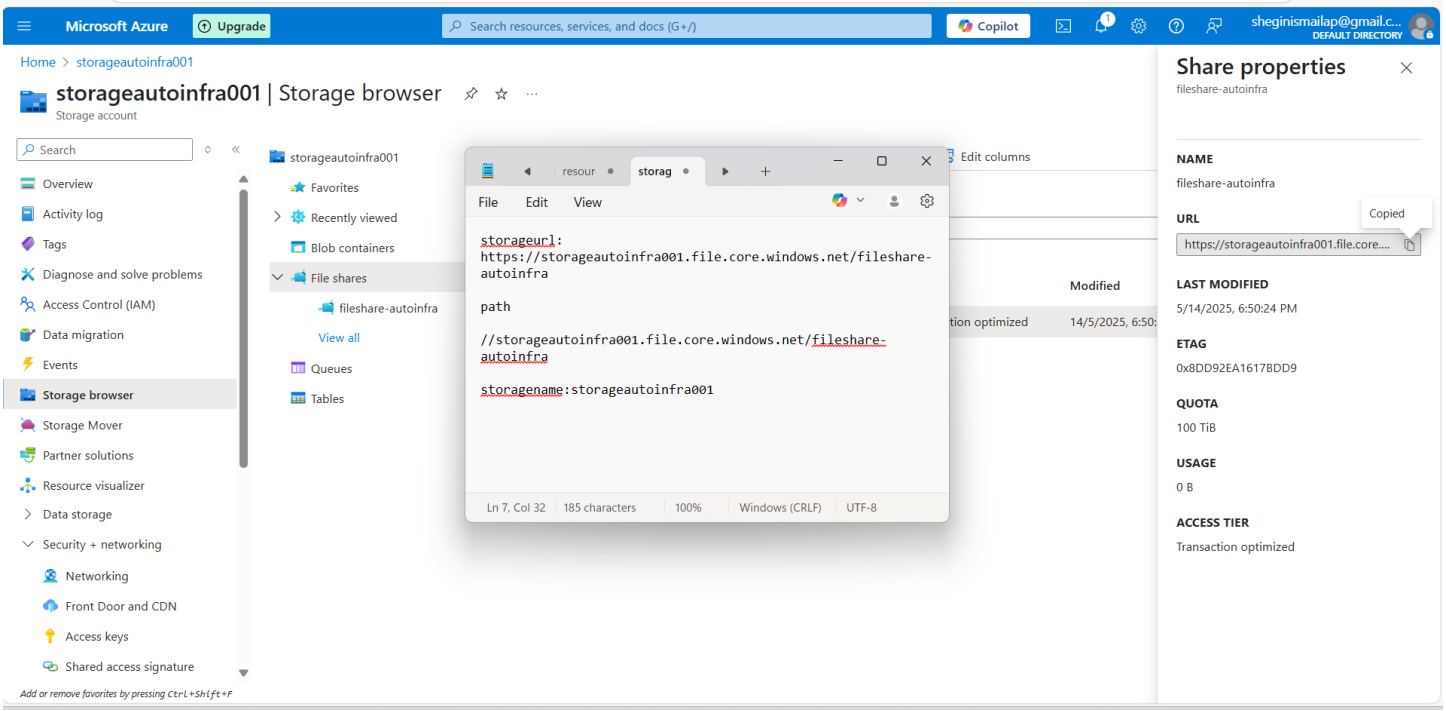

WinAutoInfraVM1 - 4.236.185.22:3389 - Remote Desktop Connection
Recycle Bin
Microsoft Edge
Administrator: Windows PowerShell
fileshare-autoinfra (\\storageautoinfra001.file.core.windows.net) (Z:)
This PC › fileshare-autoinfra (\\storageautoinfra001.file.core.windows.net) (Z:)
Quick access
Desktop
Downloads
Documents
Pictures
This PC
Network
Name
Date modified
Type
Size
tests.txt
5/14/2025 1:34 PM
Text Document
0 KB
OAuth application authorized
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Learn more
Copied
This script will check to see if this storage account is accessible via TCP port 445, which is the port SMB uses. If port 445 is available, your Azure file share will be
Give feedback
Networking
Front Door and CDN
Access keys
Shared access signature
Showing 1 - 2 of 2. Display count:
10

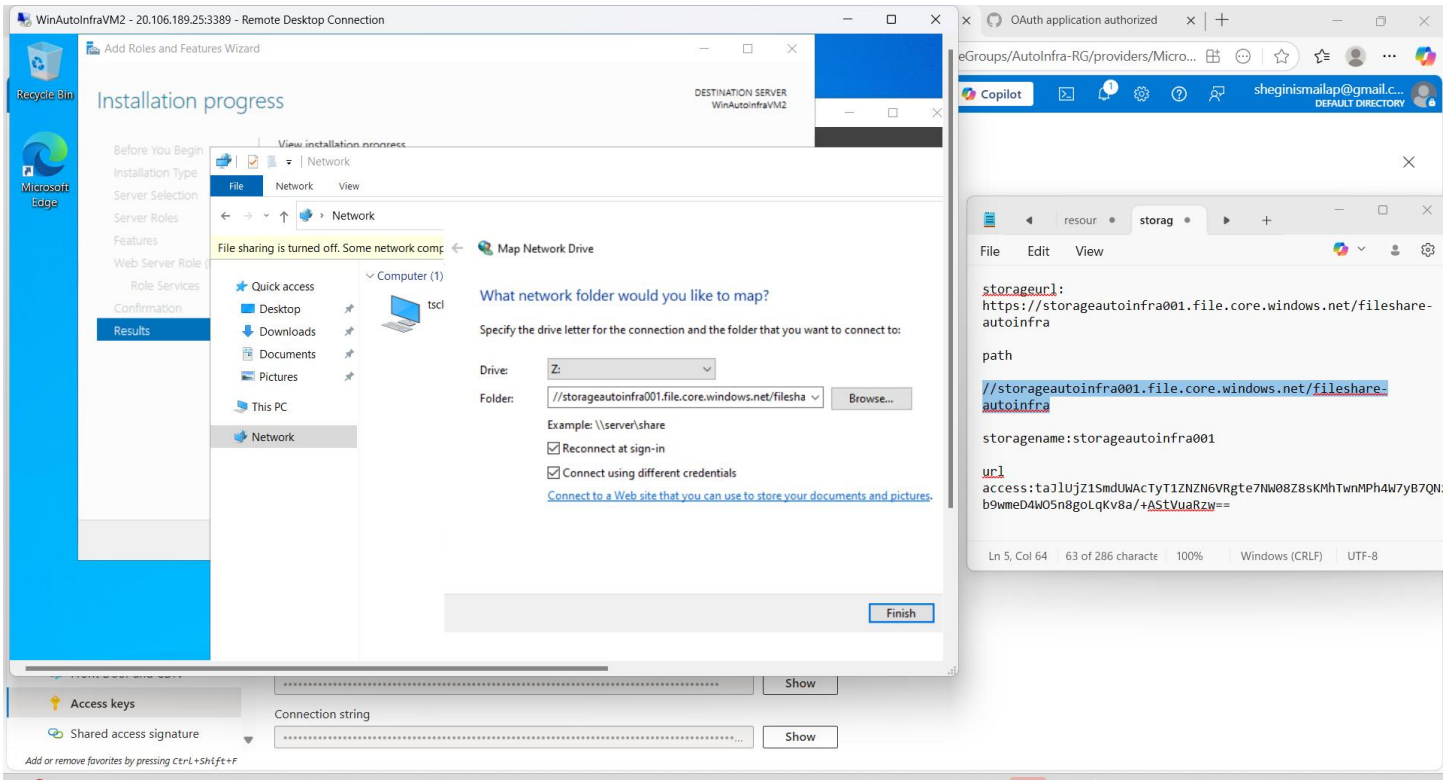

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > Storage accounts > storageautoinfra001
Storage accounts
Default Directory
Create
Group by none
You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.
Name
storageautoinfra001
storagewinfra001
Showing 1 - 2 of 2. Display count:
10
storageautoinfra001 | Storage browser
Storage account
Search
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Security + networking
Networking
Front Door and CDN
Access keys
Shared access signature
Upload
Add directory
Refresh
Delete
Copy
Paste
Rename
Edit columns
File shares > fileshare-autoinfra
Authentication method: Access key (Switch to Microsoft Entra user account)
Search files by prefix
Showing all 1 items
Name
Type
Size
tests.txt.txt
File
0

RESULT

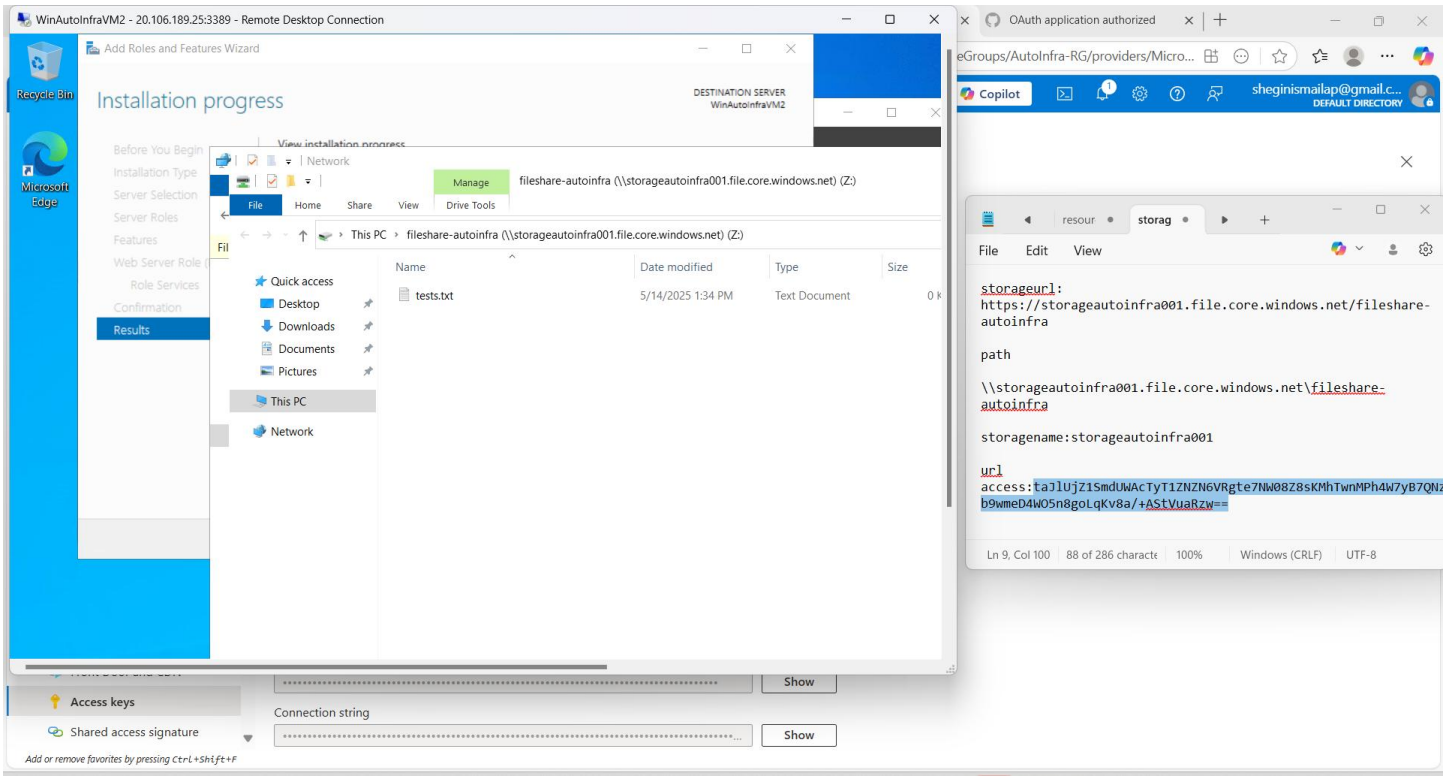


Shown below is key mapping through network drive mapping

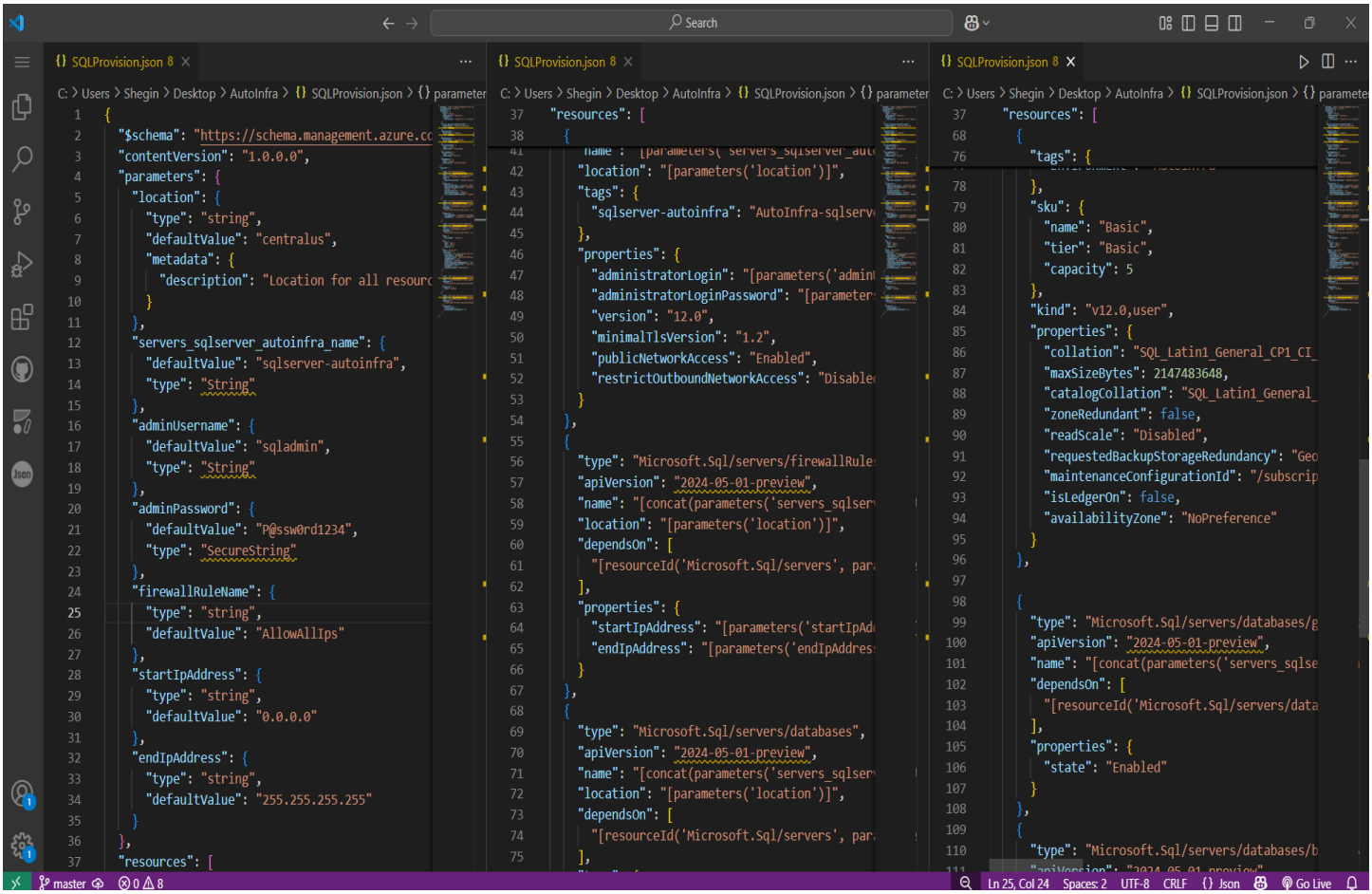


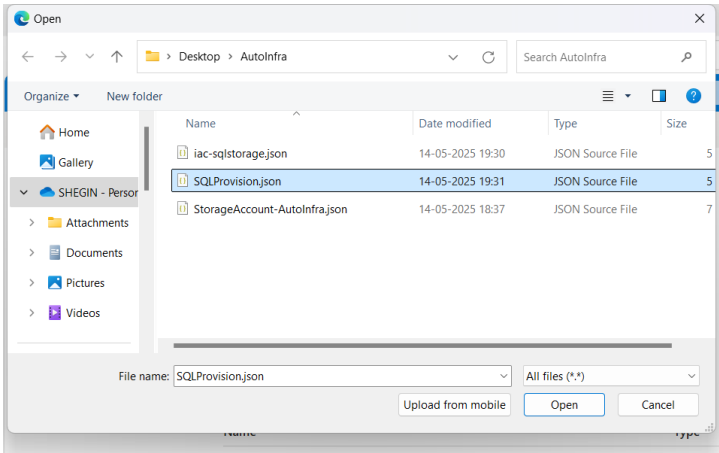


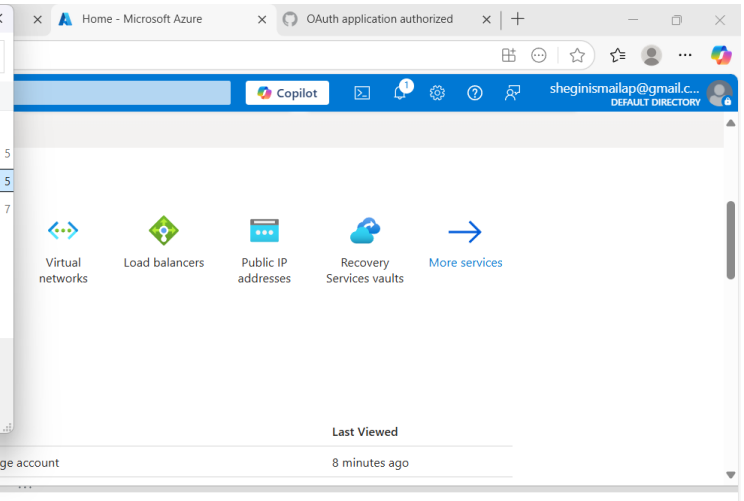
RESULT

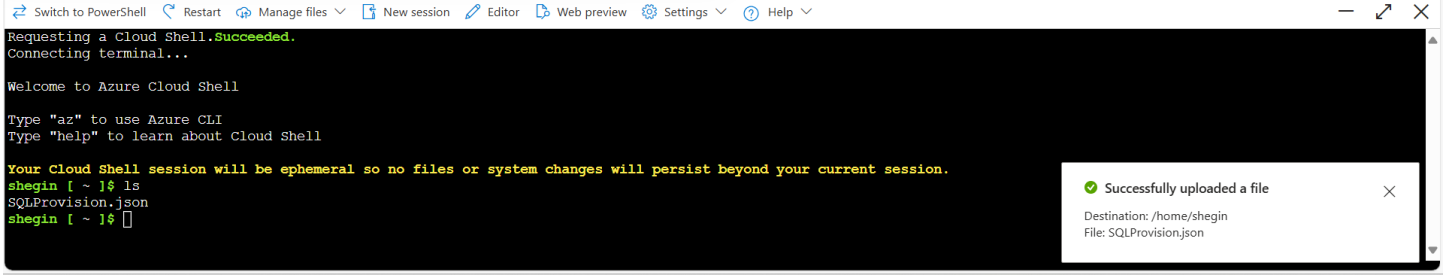


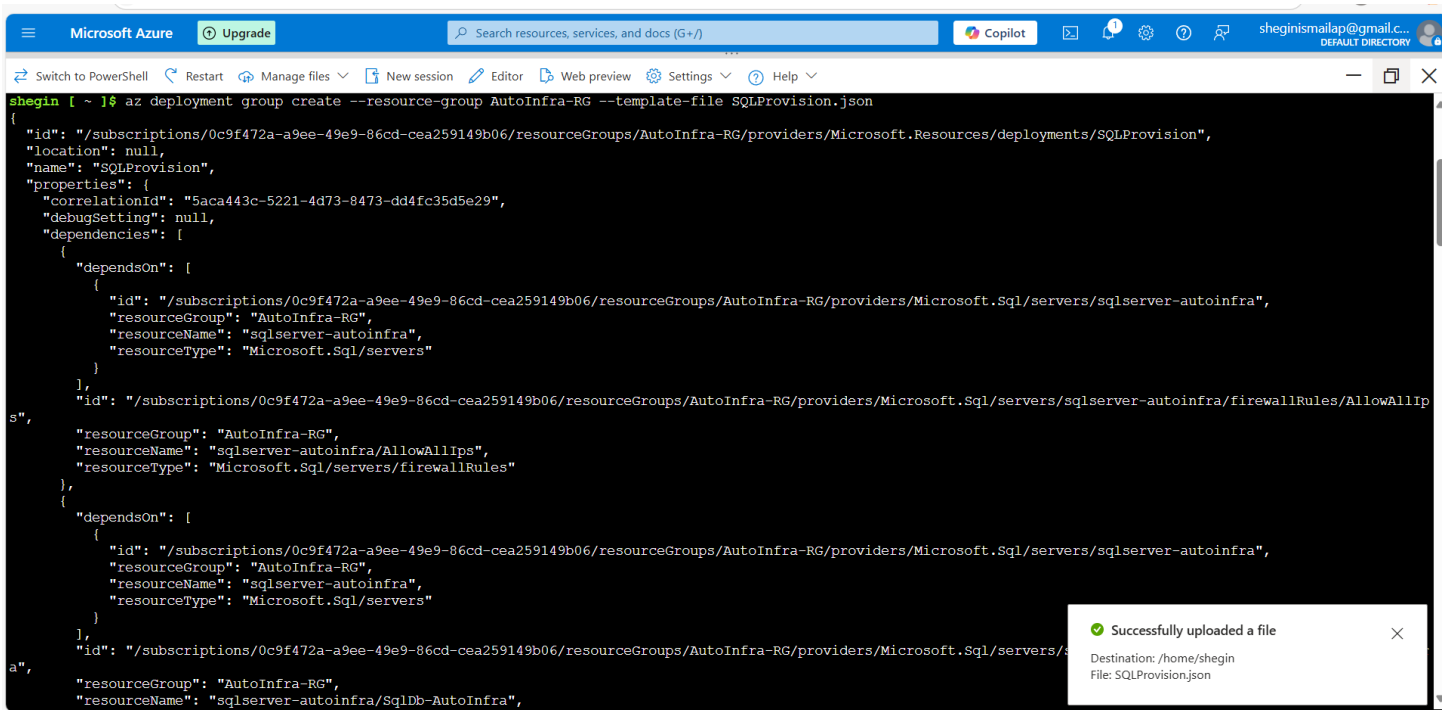
7. The ARM template for Provision Azure SQL Database (basic tier) at SQLProvision.json,the snippets from template,azure cli and outputs are shown


SQLProvision.json 8
C: > Users > Shegin > Desktop > AutoInfra > {} SQLProvision.json > {} parameter
{
"$schema": "https://schema.management.azure.co
"contentVersion": "1.0.0.0",
"parameters": {
"location": {
"type": "string",
"defaultValue": "centralus",
"metadata": {
"description": "Location for all resourc
}
},
"servers_sqlserver_autoinfra_name": {
"defaultValue": "sqlserver-autoinfra",
"type": "String"
},
"adminUsername": {
"defaultValue": "sqladmin",
"type": "String"
},
"adminPassword": {
"defaultValue": "P@ssw0rd1234",
"type": "SecureString"
},
"firewallRuleName": {
"type": "string",
"defaultValue": "AllowAllIps"
},
"startIpAddress": {
"type": "string",
"defaultValue": "0.0.0.0"
},
"endIpAddress": {
"type": "string",
"defaultValue": "255.255.255.255"
}
},
"resources": [
"resources": [
"location": "[parameters('location')]",
"tags": {
"sqlserver-autoinfra": "AutoInfra-sqlserv
},
"properties": {
"administratorLogin": "[parameters('admin
"administratorLoginPassword": "[parameter
"version": "12.0",
"minimalTlsVersion": "1.2",
"publicNetworkAccess": "Enabled",
"restrictOutboundNetworkAccess": "Disable
}
},
{
"type": "Microsoft.Sql/servers/firewallRule
"apiVersion": "2024-05-01-preview",
"name": "[concat(parameters('servers_sqlser
"location": "[parameters('location')]",
"dependsOn": [
"[resourceId('Microsoft.Sql/servers', par
],
"properties": {
"startIpAddress": "[parameters('startIpAd
"endIpAddress": "[parameters('endIpAddres
}
},
{
"type": "Microsoft.Sql/servers/databases",
"apiVersion": "2024-05-01-preview",
"name": "[concat(parameters('servers_sqlser
"location": "[parameters('location')]",
"dependsOn": [
"[resourceId('Microsoft.Sql/servers', par
],
"resources": [
"tags": {
},
"sku": {
"name": "Basic",
"tier": "Basic",
"capacity": 5
},
"kind": "v12.0,user",
"properties": {
"collation": "SQL_Latin1_General_CP1_CI_
"maxSizeBytes": 2147483648,
"catalogCollation": "SQL_Latin1_General_
"zoneRedundant": false,
"readScale": "Disabled",
"requestedBackupStorageRedundancy": "Geo
"maintenanceConfigurationId": "/subscrip
"isLedgerOn": false,
"availabilityZone": "NoPreference"
}
},
{
"type": "Microsoft.Sql/servers/databases/g
"apiVersion": "2024-05-01-preview",
"name": "[concat(parameters('servers_sqlse
"dependsOn": [
"[resourceId('Microsoft.Sql/servers/data
],
"properties": {
"state": "Enabled"
}
},
{
"type": "Microsoft.Sql/servers/databases/b
master
Ln 25, Col 24
Spaces: 2
UTF-8
CRLF
Json
Go Live

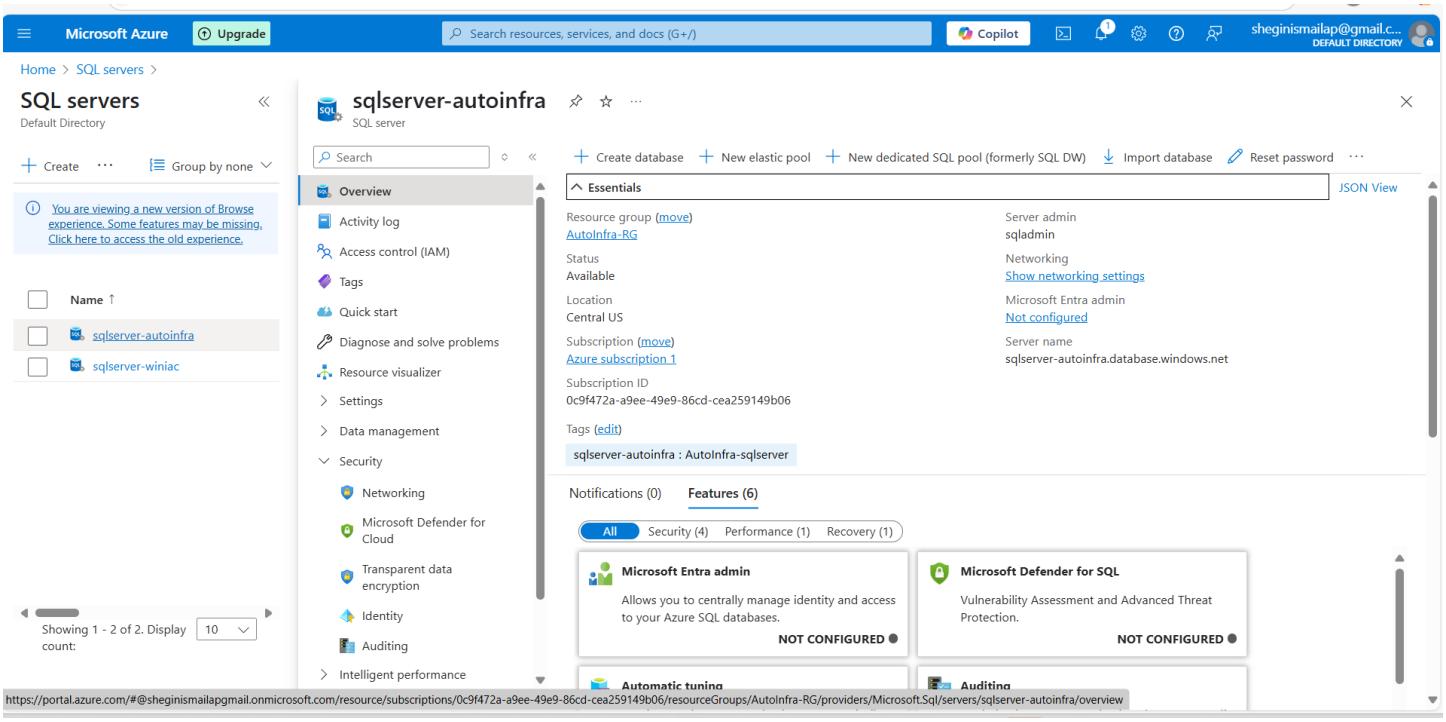

Open
Desktop > AutoInfra
Search AutoInfra
Organize
New folder
Home
Gallery
SHEGIN - Persor
Attachments
Documents
Pictures
Videos
Name
Date modified
Type
Size
iac-sqlstorage.json
14-05-2025 19:30
JSON Source File
5
SQLProvision.json
14-05-2025 19:31
JSON Source File
5
StorageAccount-AutoInfra.json
14-05-2025 18:37
JSON Source File
7
File name: SQLProvision.json
All files (*.*)
Upload from mobile
Open
Cancel


Home - Microsoft Azure
OAuth application authorized
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Virtual networks
Load balancers
Public IP addresses
Recovery Services vaults
More services
Last Viewed
storageautoinfra001
Storage account
8 minutes ago

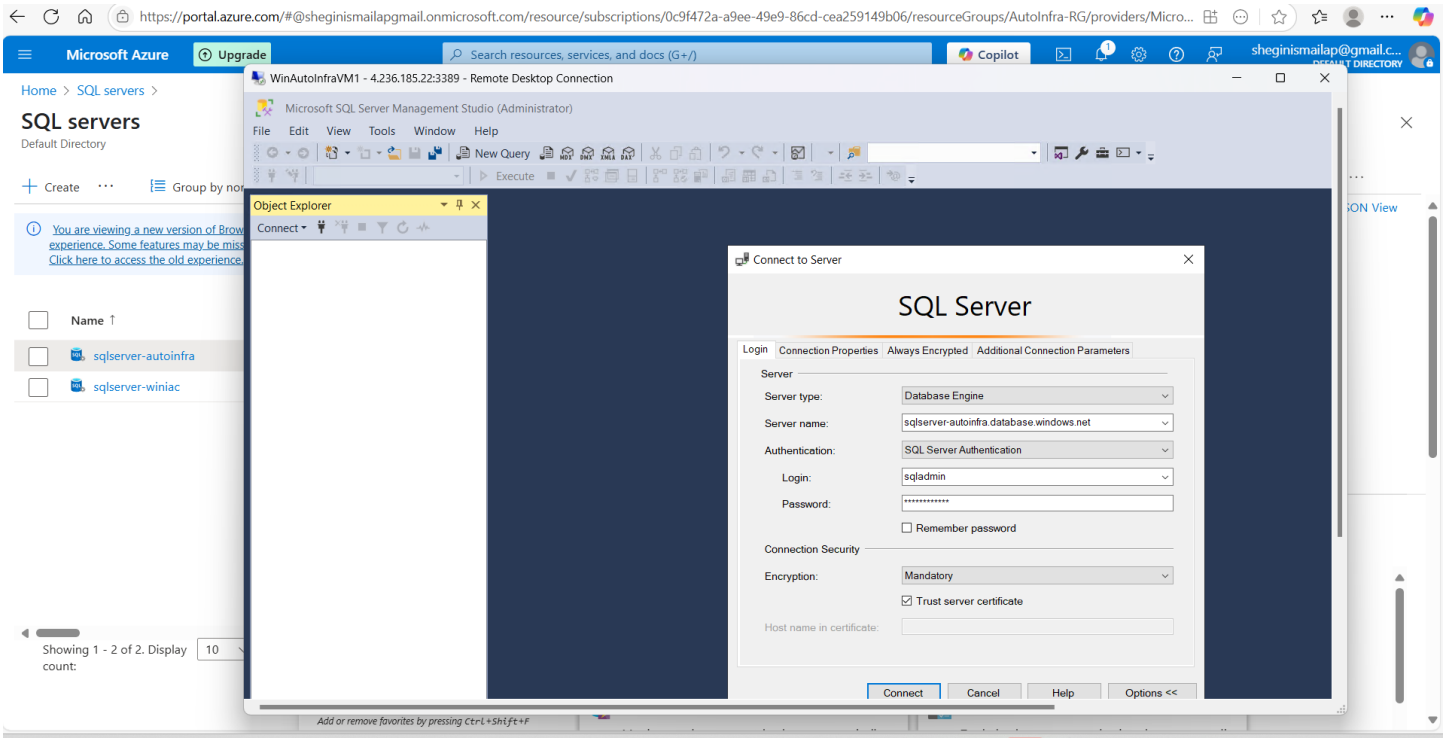

Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Welcome to Azure Cloud Shell
Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
SQLProvision.json
shegin [~]$
Successfully uploaded a file
Destination: /home/shegin
File: SQLProvision.json

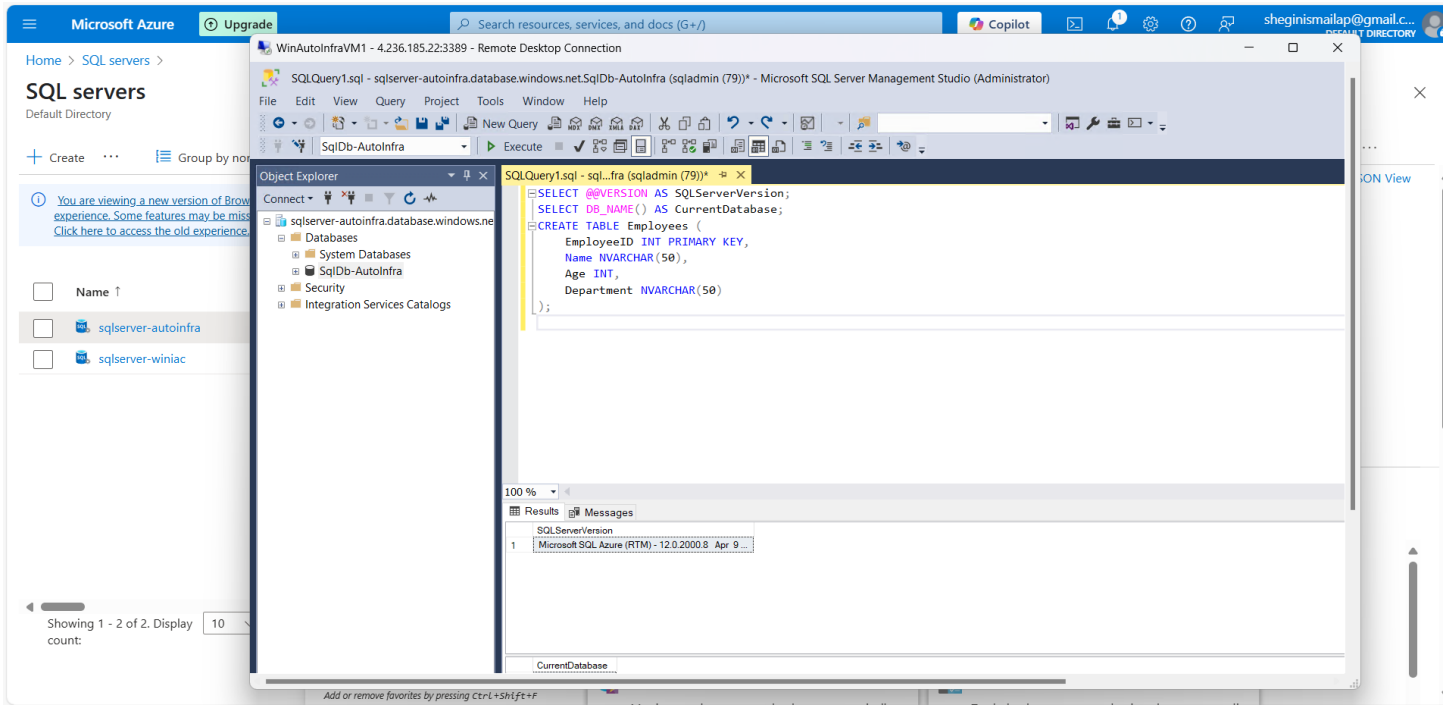


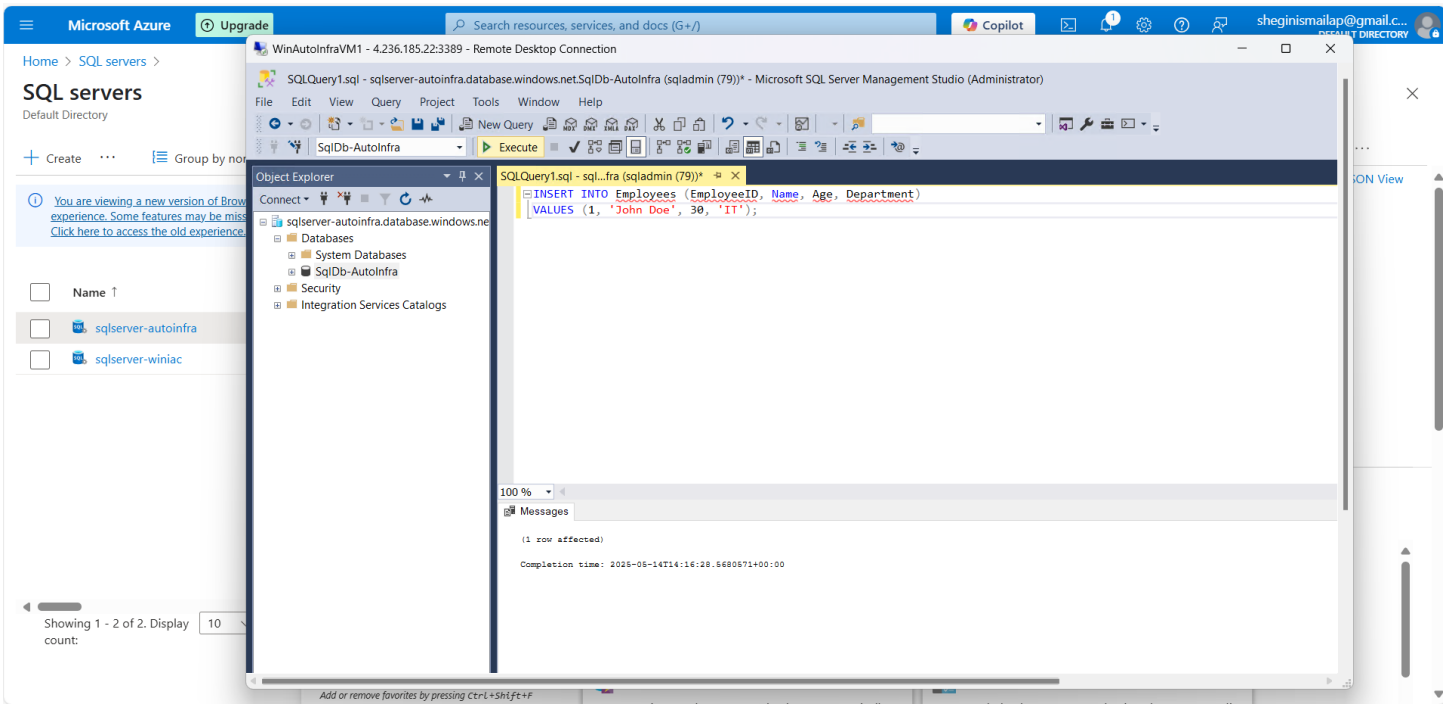
RESULT



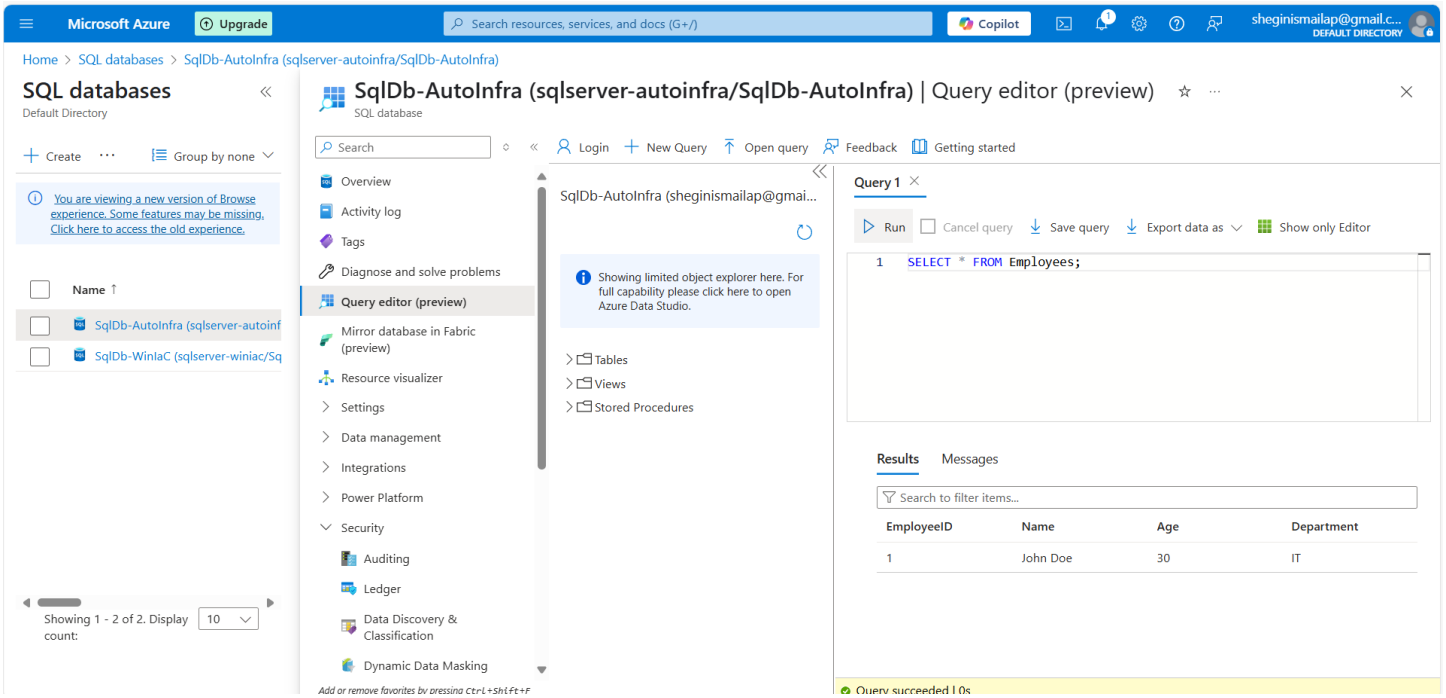
TEST AND VALIDATION


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > SQL servers >
SQL servers
Default Directory
Create
Group by no
You are viewing a new version of Brow
experience. Some features may be miss
Click here to access the old experience.
Name
sqlserver-autoinfra
sqlserver-winiac
Showing 1 - 2 of 2. Display count:
10
ON View
WinAutoInfraVM1 - 4.236.185.22:3389 - Remote Desktop Connection
Microsoft SQL Server Management Studio (Administrator)
File Edit View Tools Window Help
New Query
Execute
Object Explorer
Connect
Connect to Server
SQL Server
Login
Connection Properties
Always Encrypted
Additional Connection Parameters
Server
Server type:
Database Engine
Server name:
sqlserver-autoinfra.database.windows.net
Authentication:
SQL Server Authentication
Login:
sqladmin
Password:
Remember password
Connection Security
Encryption:
Mandatory
Trust server certificate
Host name in certificate:
Connect
Cancel
Help
Options <<
Add or remove favorites by pressing Ctrl+Shift+F


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > SQL servers >
SQL servers
Default Directory
Create
Group by no
You are viewing a new version of Brow
experience. Some features may be miss
Click here to access the old experience.
Name
sqlserver-autoinfra
sqlserver-winiac
Showing 1 - 2 of 2. Display count:
10
ON View
WinAutoInfraVM1 - 4.236.185.22:3389 - Remote Desktop Connection
SQLQuery1.sql - sqlserver-autoinfra.database.windows.net.SqlDb-AutoInfra (sqladmin (79))* - Microsoft SQL Server Management Studio (Administrator)
File Edit View Query Project Tools Window Help
New Query
SqlDb-AutoInfra
Execute
Object Explorer
Connect
sqlserver-autoinfra.database.windows.ne
Databases
System Databases
SqlDb-AutoInfra
Security
Integration Services Catalogs
SQLQuery1.sql - sql...fra (sqladmin (79))*
SELECT @@VERSION AS SQLServerVersion;
SELECT DB_NAME() AS CurrentDatabase;
CREATE TABLE Employees (
EmployeeID INT PRIMARY KEY,
Name NVARCHAR(50),
Age INT,
Department NVARCHAR(50)
);
100 %
Results
Messages
SQLServerVersion
1
Microsoft SQL Azure (RTM) - 12.0.2000.8 Apr 9 ...
CurrentDatabase
Add or remove favorites by pressing Ctrl+Shift+F

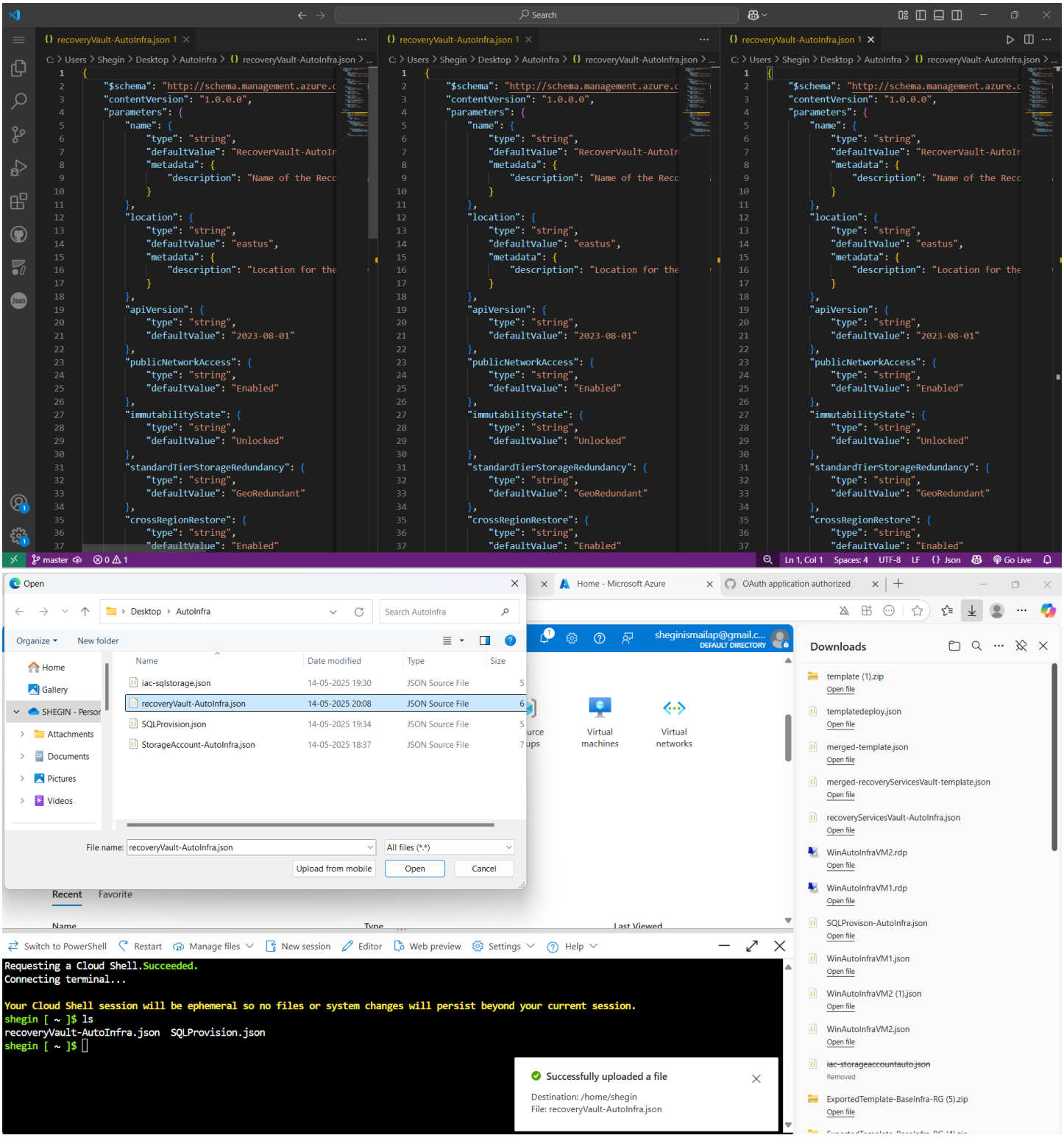


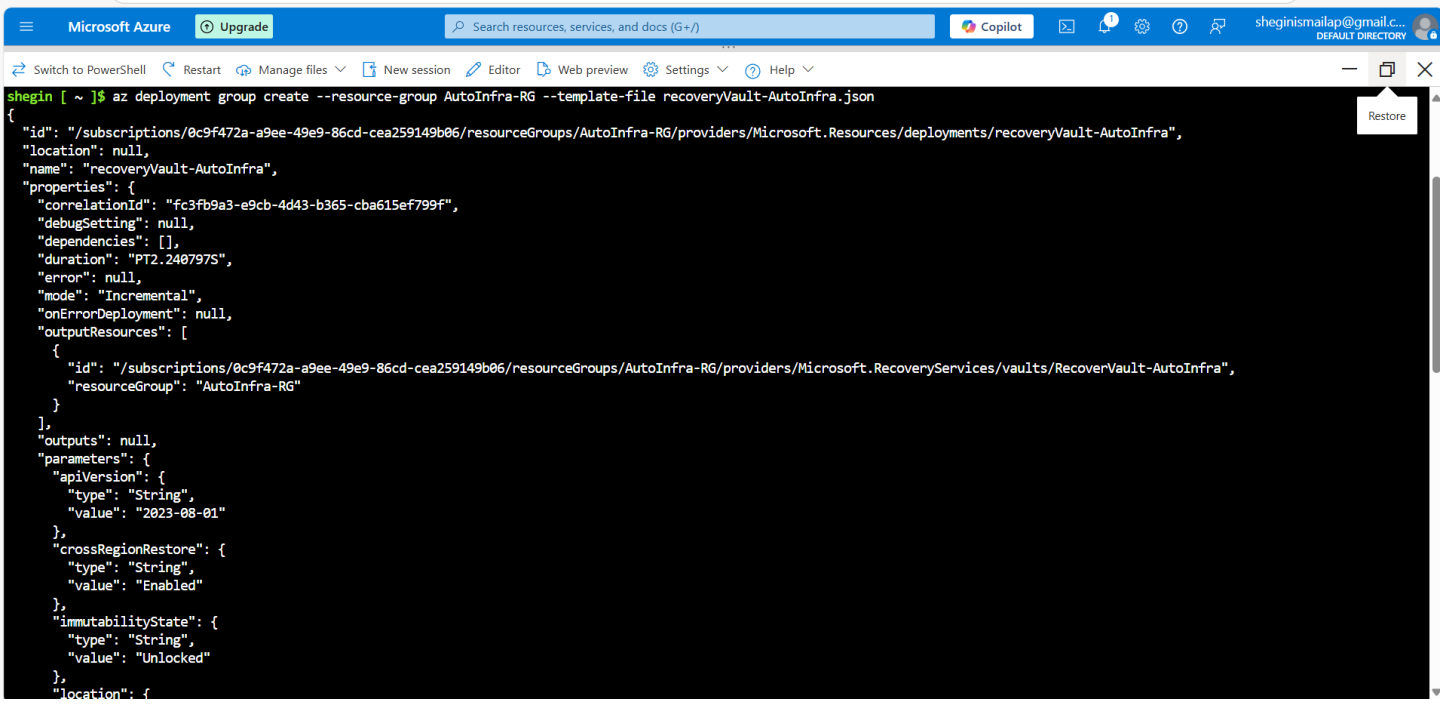
RESULT



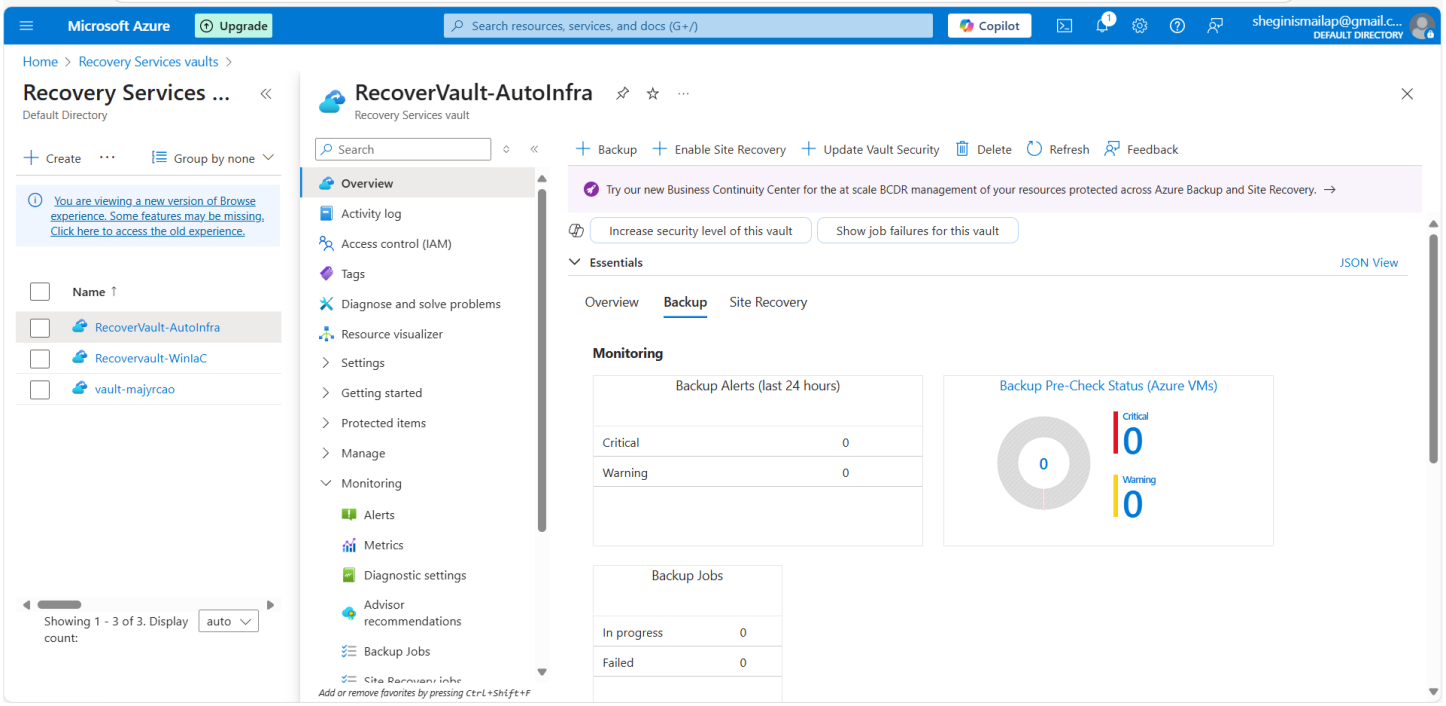
8The ARM template for Set up Recovery Services Vault for VM backups at recoveryVault-AutoInfra.json,the snippets from template,azure cli and outputs are shown

8.

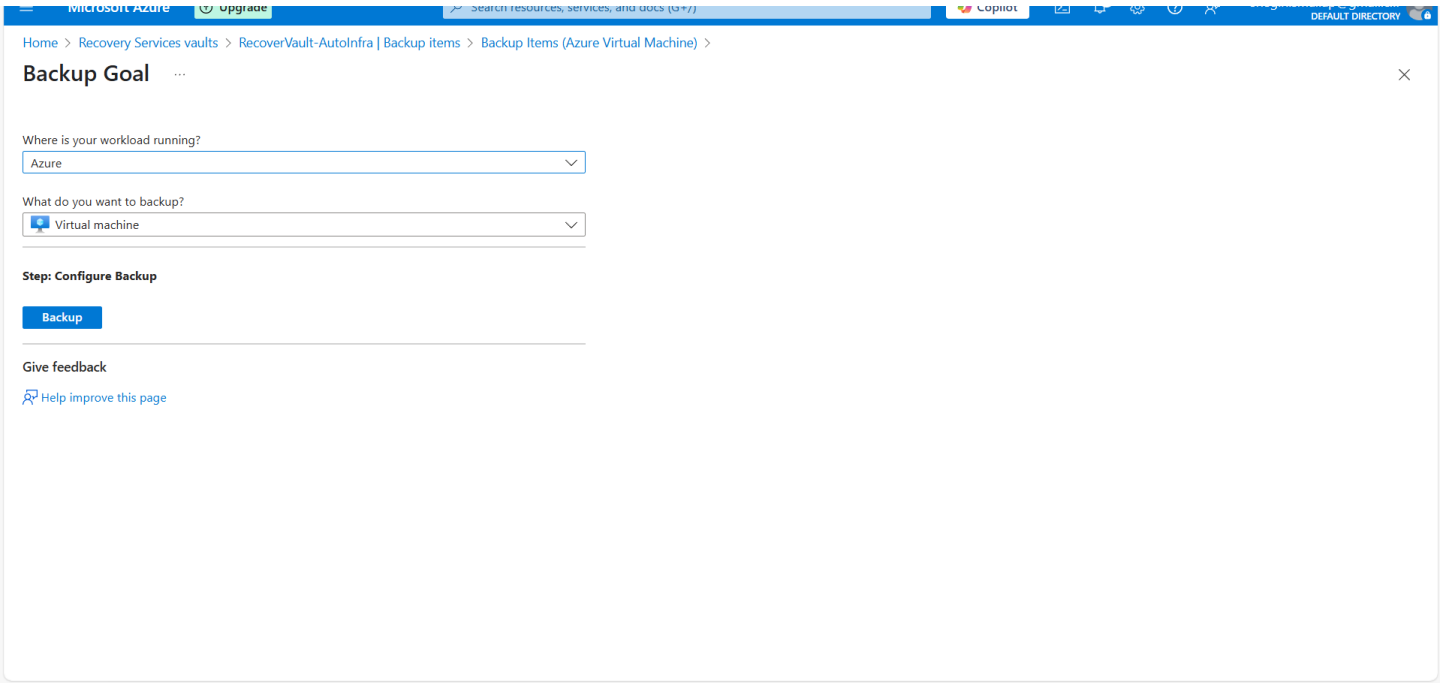

recoveryVault-AutoInfra.json 1
C: > Users > Shegin > Desktop > AutoInfra > recoveryVault-AutoInfra.json
{
"$schema": "http://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
"contentVersion": "1.0.0.0",
"parameters": {
"name": {
"type": "string",
"defaultValue": "RecoverVault-AutoInfra",
"metadata": {
"description": "Name of the Recovery Services Vault"
}
},
"location": {
"type": "string",
"defaultValue": "eastus",
"metadata": {
"description": "Location for the Recovery Services Vault"
}
},
"apiVersion": {
"type": "string",
"defaultValue": "2023-08-01"
},
"publicNetworkAccess": {
"type": "string",
"defaultValue": "Enabled"
},
"immutabilityState": {
"type": "string",
"defaultValue": "Unlocked"
},
"standardTierStorageRedundancy": {
"type": "string",
"defaultValue": "GeoRedundant"
},
"crossRegionRestore": {
"type": "string",
"defaultValue": "Enabled"
master
Ln 1, Col 1
Spaces: 4
UTF-8
LF
Json
Go Live
Open
Desktop > AutoInfra
Search AutoInfra
Organize
New folder
Home
Gallery
SHEGIN - Personal
Attachments
Documents
Pictures
Videos
Name
Date modified
Type
Size
iac-sqlstorage.json 14-05-2025 19:30 JSON Source File 5
recoveryVault-AutoInfra.json 14-05-2025 20:08 JSON Source File 6
SQLProvision.json 14-05-2025 19:34 JSON Source File 5
StorageAccount-AutoInfra.json 14-05-2025 18:37 JSON Source File 7
File name: recoveryVault-AutoInfra.json
All files (*.*)
Upload from mobile
Open
Cancel
Recent
Favorite
Home - Microsoft Azure
OAuth application authorized
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Virtual machines
Virtual networks
Downloads
template (1).zip
Open file
templatedeploy.json
merged-template.json
merged-recoveryServicesVault-template.json
recoveryServicesVault-AutoInfra.json
WinAutoInfraVM2.rdp
WinAutoInfraVM1.rdp
SQLProvison-AutoInfra.json
WinAutoInfraVM1.json
WinAutoInfraVM2 (1).json
WinAutoInfraVM2.json
iac-storageaccountauto.json
Removed
ExportedTemplate-BaseInfra-RG (5).zip
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
recoveryVault-AutoInfra.json SQLProvision.json
shegin [~]$
Successfully uploaded a file
Destination: /home/shegin
File: recoveryVault-AutoInfra.json

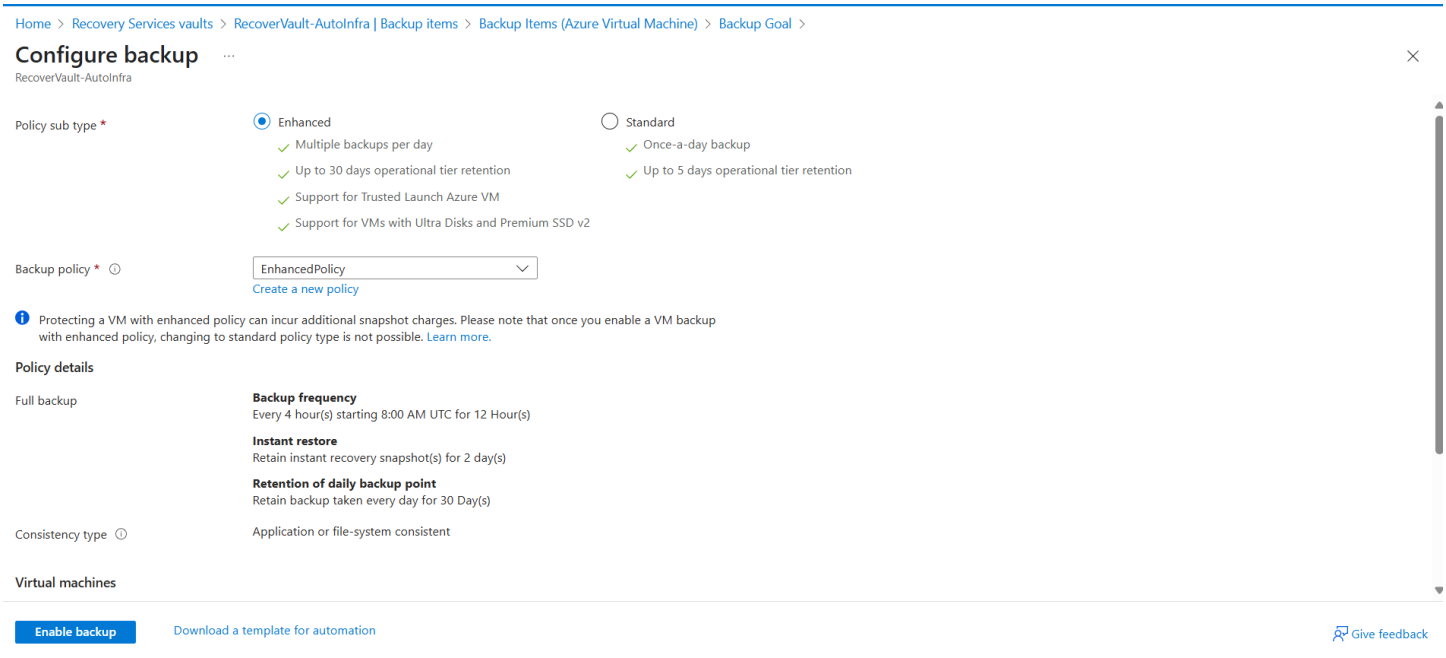


RESULT

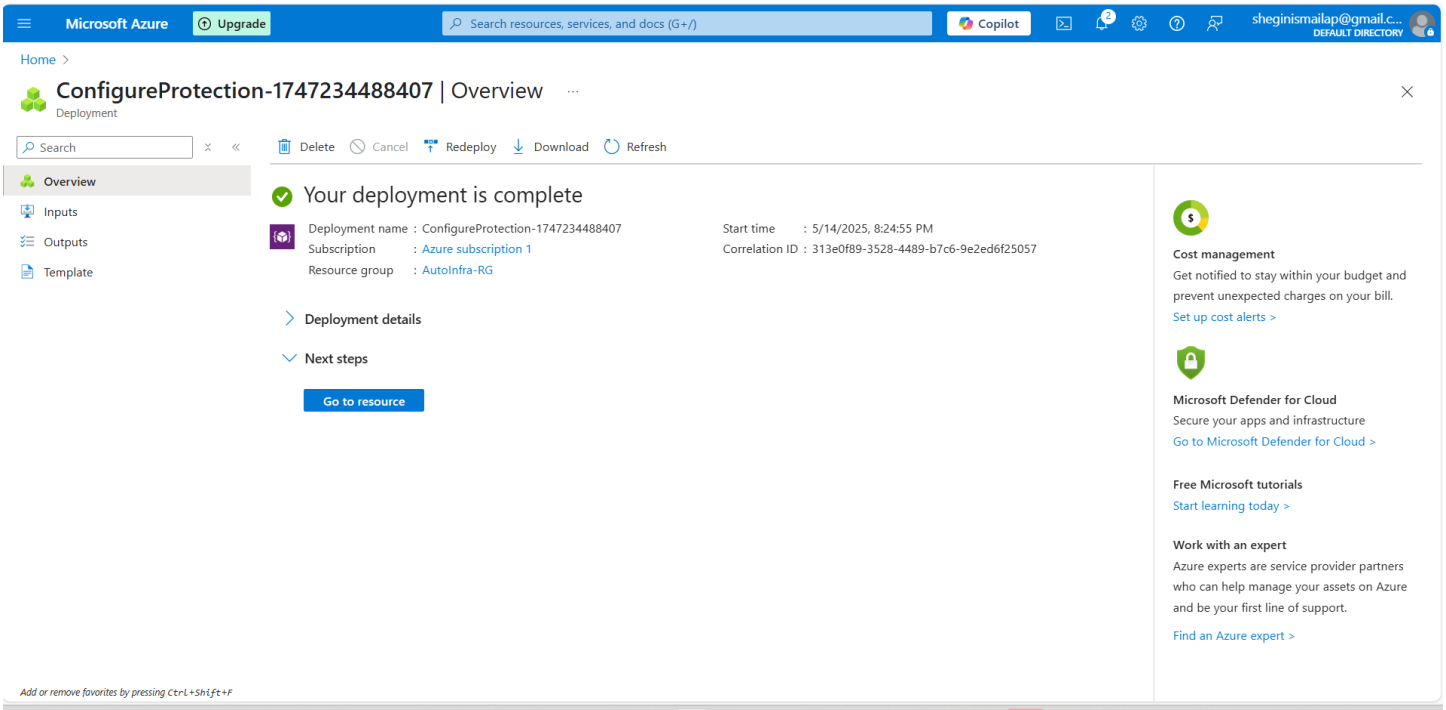


The backup configuration for VMS can only be done via GUI or Azure CLI


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
DEFAULT DIRECTORY
Home > Recovery Services vaults > RecoverVault-AutoInfra | Backup items > Backup Items (Azure Virtual Machine) >
Backup Goal
Where is your workload running?
Azure
What do you want to backup?
Virtual machine
Step: Configure Backup
Backup
Give feedback
Help improve this page

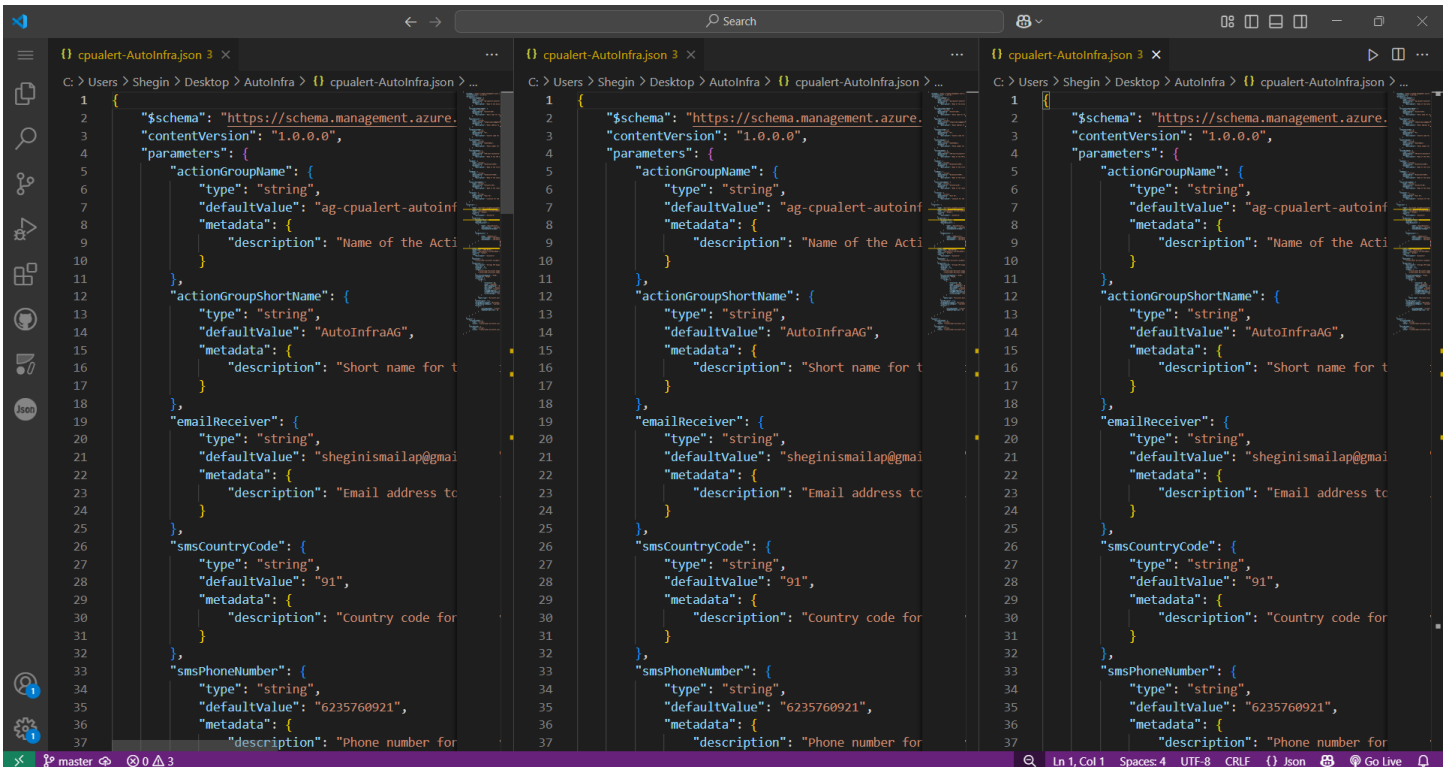

Home > Recovery Services vaults > RecoverVault-AutoInfra | Backup items > Backup Items (Azure Virtual Machine) > Backup Goal >
Configure backup
RecoverVault-AutoInfra
Policy sub type *
Enhanced
Multiple backups per day
Up to 30 days operational tier retention
Support for Trusted Launch Azure VM
Support for VMs with Ultra Disks and Premium SSD v2
Standard
Once-a-day backup
Up to 5 days operational tier retention
Backup policy *
EnhancedPolicy
Create a new policy
Protecting a VM with enhanced policy can incur additional snapshot charges. Please note that once you enable a VM backup with enhanced policy, changing to standard policy type is not possible. Learn more.
Policy details
Full backup
Backup frequency
Every 4 hour(s) starting 8:00 AM UTC for 12 Hour(s)
Instant restore
Retain instant recovery snapshot(s) for 2 day(s)
Retention of daily backup point
Retain backup taken every day for 30 Day(s)
Consistency type
Application or file-system consistent
Virtual machines
Enable backup
Download a template for automation
Give feedback

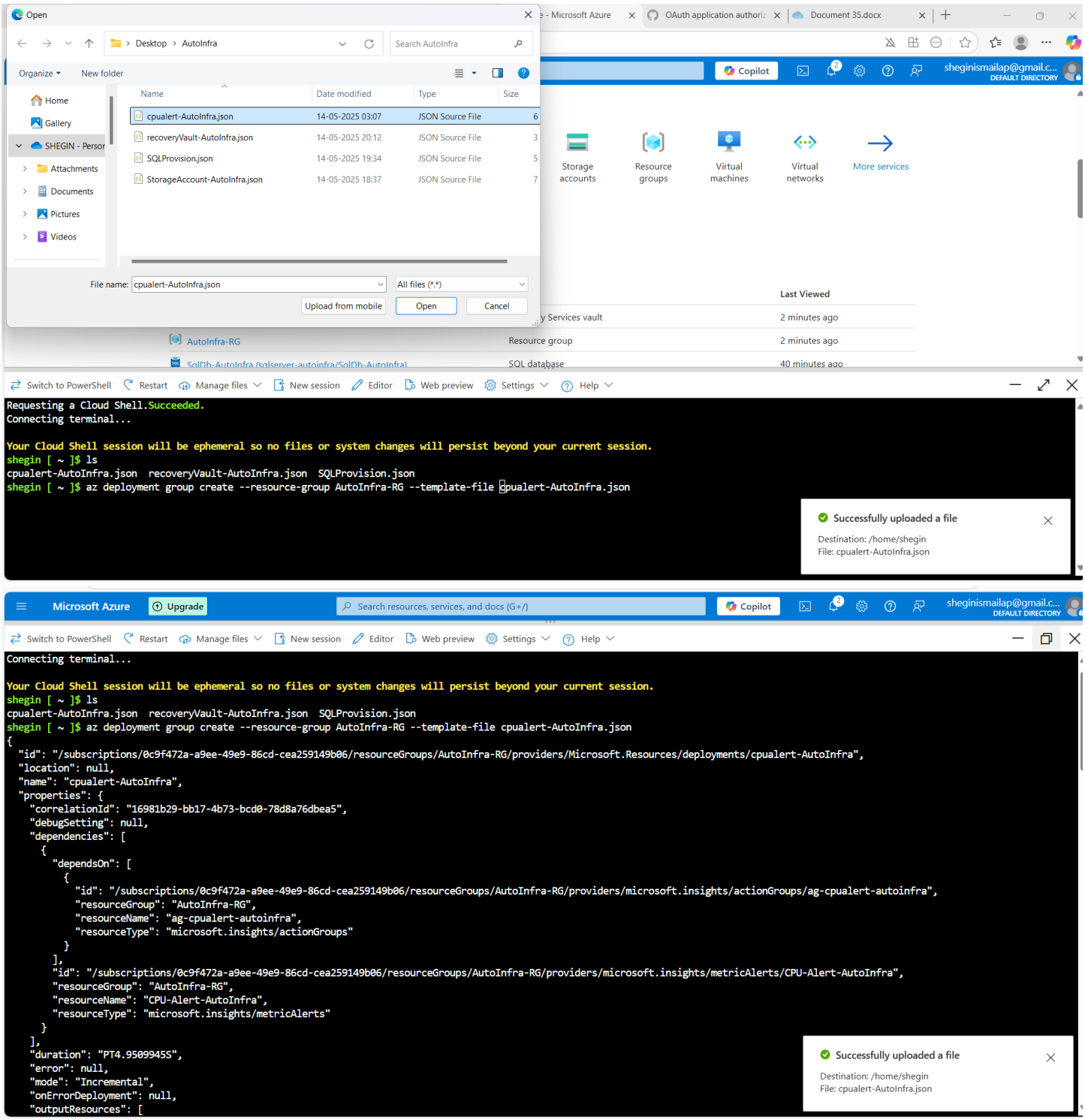
RESULT



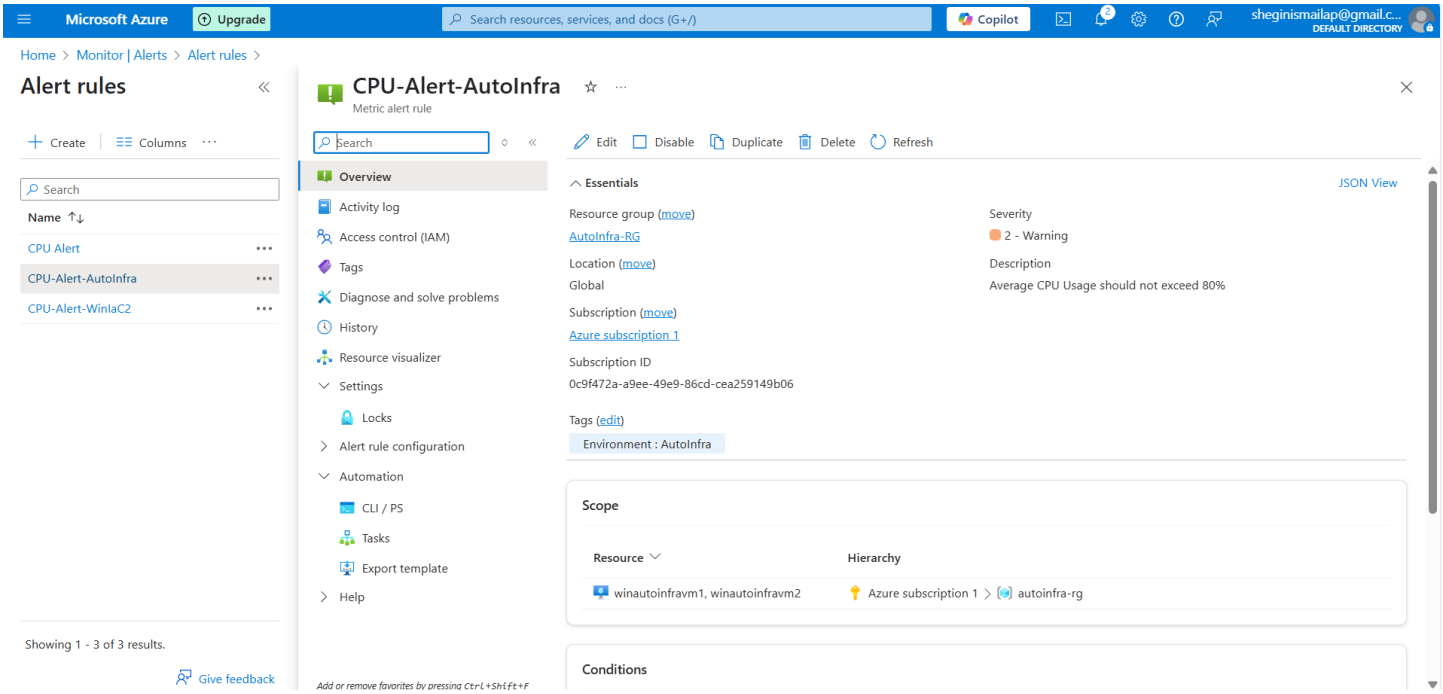
9.The ARM template for Configure CPU alert (80%) via Azure Monitor

at cpuAlert-AutoInfra.json,the snippets from template,azure cli and outputs are shown

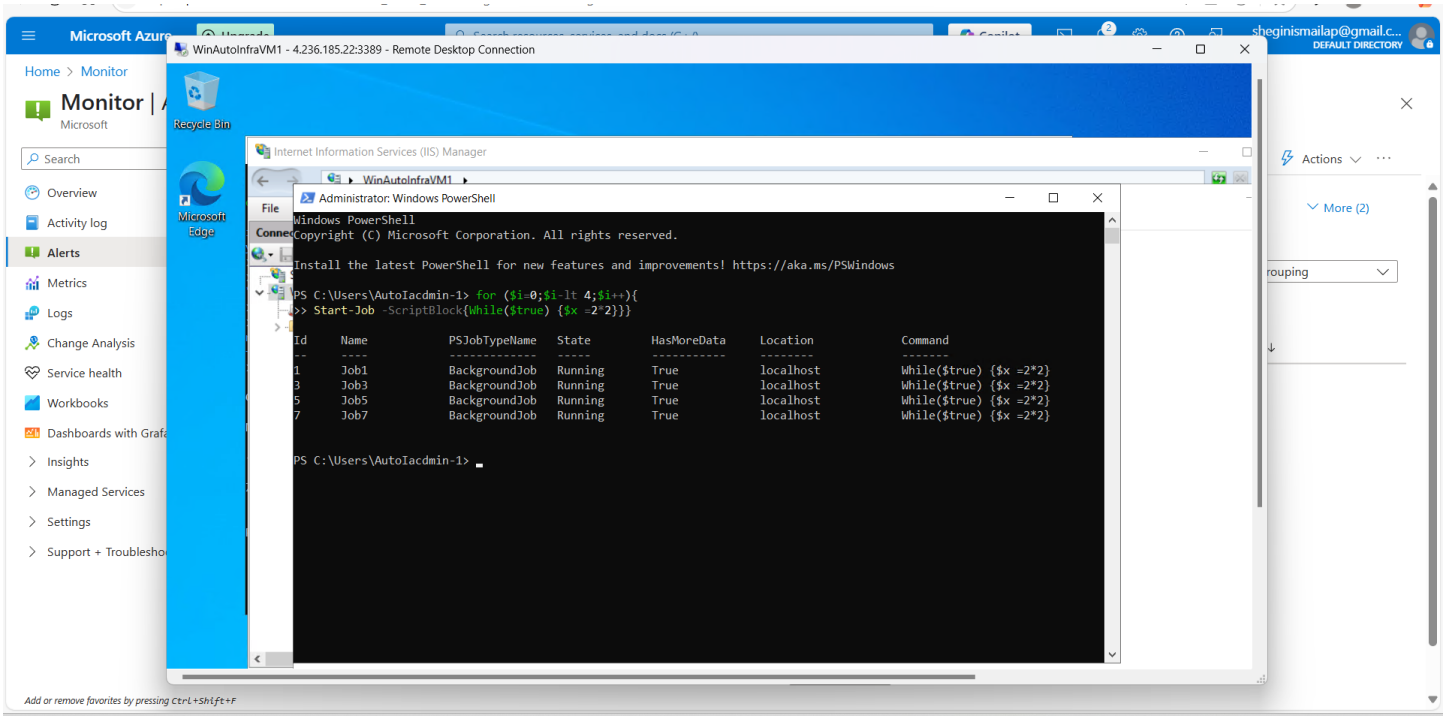



Open
Desktop › AutoInfra
Search AutoInfra
Organize
New folder
Home
Gallery
SHEGIN - Perso
Attachments
Documents
Pictures
Videos
Name
Date modified
Type
Size
cpualert-AutoInfra.json 14-05-2025 03:07 JSON Source File 6
recoveryVault-AutoInfra.json 14-05-2025 20:12 JSON Source File 3
SQLProvision.json 14-05-2025 19:34 JSON Source File 5
StorageAccount-AutoInfra.json 14-05-2025 18:37 JSON Source File 7
File name: cpualert-AutoInfra.json
All files (*.*)
Upload from mobile
Open
Cancel
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Storage accounts
Resource groups
Virtual machines
Virtual networks
More services
Last Viewed
Services vault 2 minutes ago
AutoInfra-RG Resource group 2 minutes ago
SQL database 40 minutes ago
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
cpualert-AutoInfra.json recoveryVault-AutoInfra.json SQLProvision.json
shegin [~]$ az deployment group create --resource-group AutoInfra-RG --template-file cpualert-AutoInfra.json
Successfully uploaded a file
Destination: /home/shegin
File: cpualert-AutoInfra.json

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Switch to PowerShell
Restart
Manage files
New session
Editor
Web preview
Settings
Help
Connecting terminal...
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
shegin [~]$ ls
cpualert-AutoInfra.json recoveryVault-AutoInfra.json SQLProvision.json
shegin [~]$ az deployment group create --resource-group AutoInfra-RG --template-file cpualert-AutoInfra.json
{
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/Microsoft.Resources/deployments/cpualert-AutoInfra",
"location": null,
"name": "cpualert-AutoInfra",
"properties": {
"correlationId": "16981b29-bb17-4b73-bcd0-78d8a76dbea5",
"debugSetting": null,
"dependencies": [
{
"dependsOn": [
{
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/microsoft.insights/actionGroups/ag-cpualert-autoinfra",
"resourceGroup": "AutoInfra-RG",
"resourceName": "ag-cpualert-autoinfra",
"resourceType": "microsoft.insights/actionGroups"
}
],
"id": "/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/microsoft.insights/metricAlerts/CPU-Alert-AutoInfra",
"resourceGroup": "AutoInfra-RG",
"resourceName": "CPU-Alert-AutoInfra",
"resourceType": "microsoft.insights/metricAlerts"
}
],
"duration": "PT4.9509945S",
"error": null,
"mode": "Incremental",
"onErrorDeployment": null,
"outputResources": [
Successfully uploaded a file
Destination: /home/shegin
File: cpualert-AutoInfra.json



TEST AND VALIDATION



← 56767457

5–14 2:47 AM

All done. You're in AutoInfraAG group. Reply 'STOP' to stop all, 'Disable AutoInfraAG' to stop grp, 'HELP' for info. Msg&data rates apply.

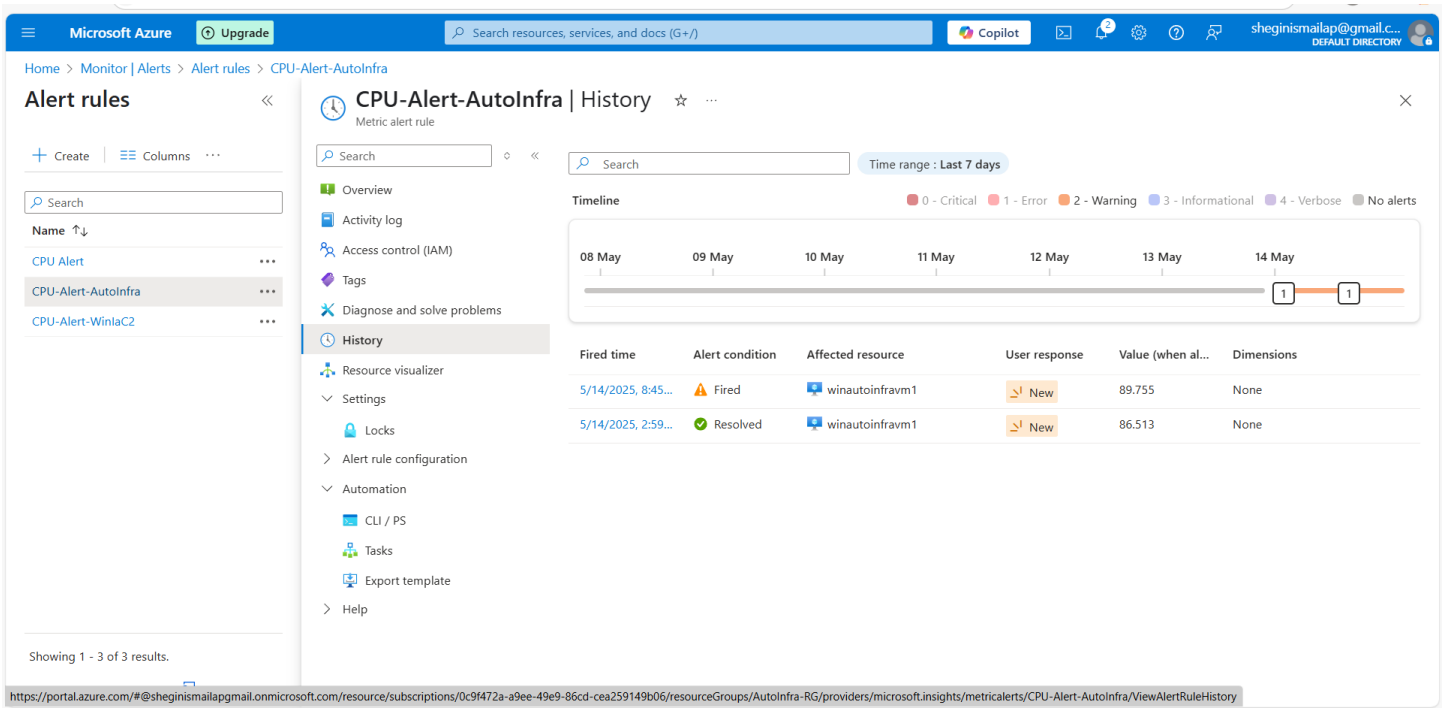
5–14 8:28 PM

All done. You're in AutoInfraAG group. Reply 'STOP' to stop all, 'Disable AutoInfraAG' to stop grp, 'HELP' for info. Msg&data rates apply.

5–14 8:45 PM

AutoInfraAG:Fired:Sev2 Azure Monitor Alert CPU–Alert–AutoInfra on winautoinfravm1

+ Text message ↑



Azure: Activated Severity: 2 CPU-Alert-AutoInfra Inbox ×

Microsoft Azure <azure-noreply@microsoft.com> ... Unsubscribe Wed 14 May, 03:00 (1 day ago)

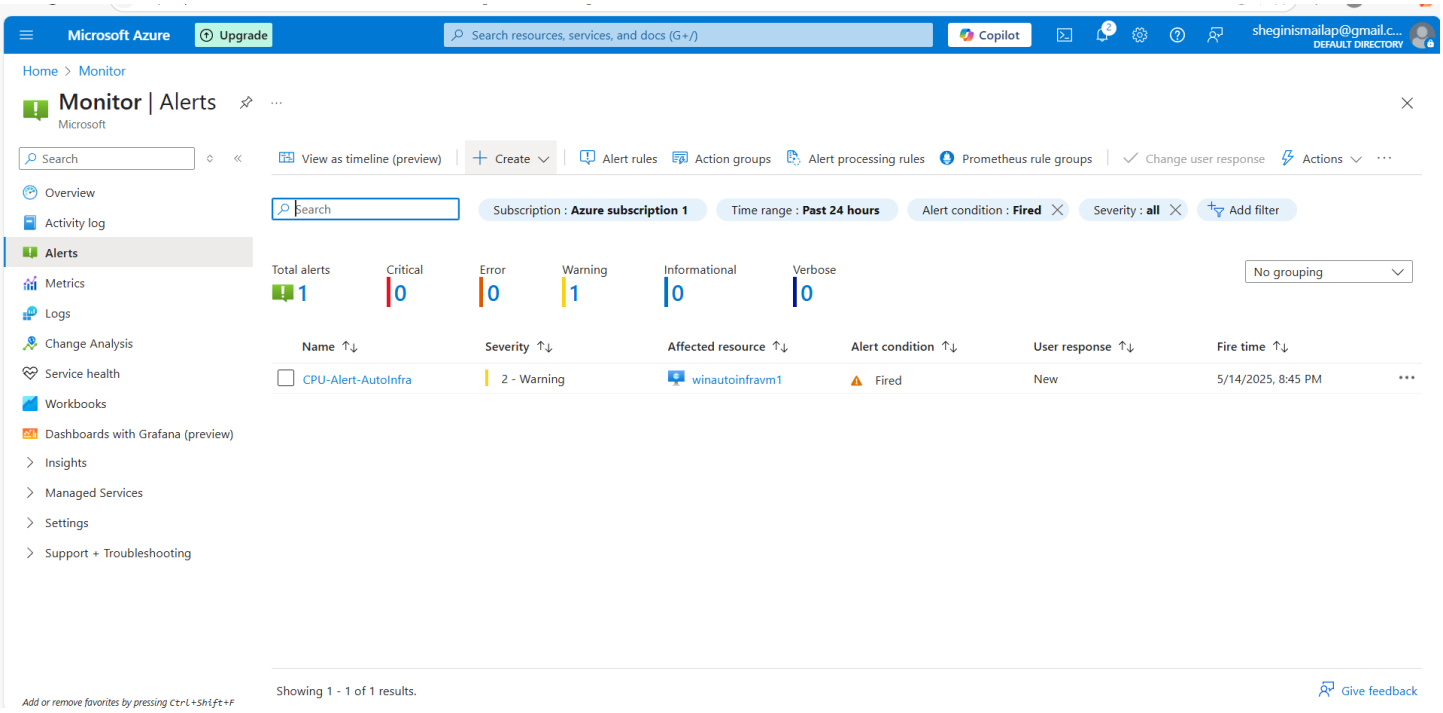
to me

Microsoft Azure

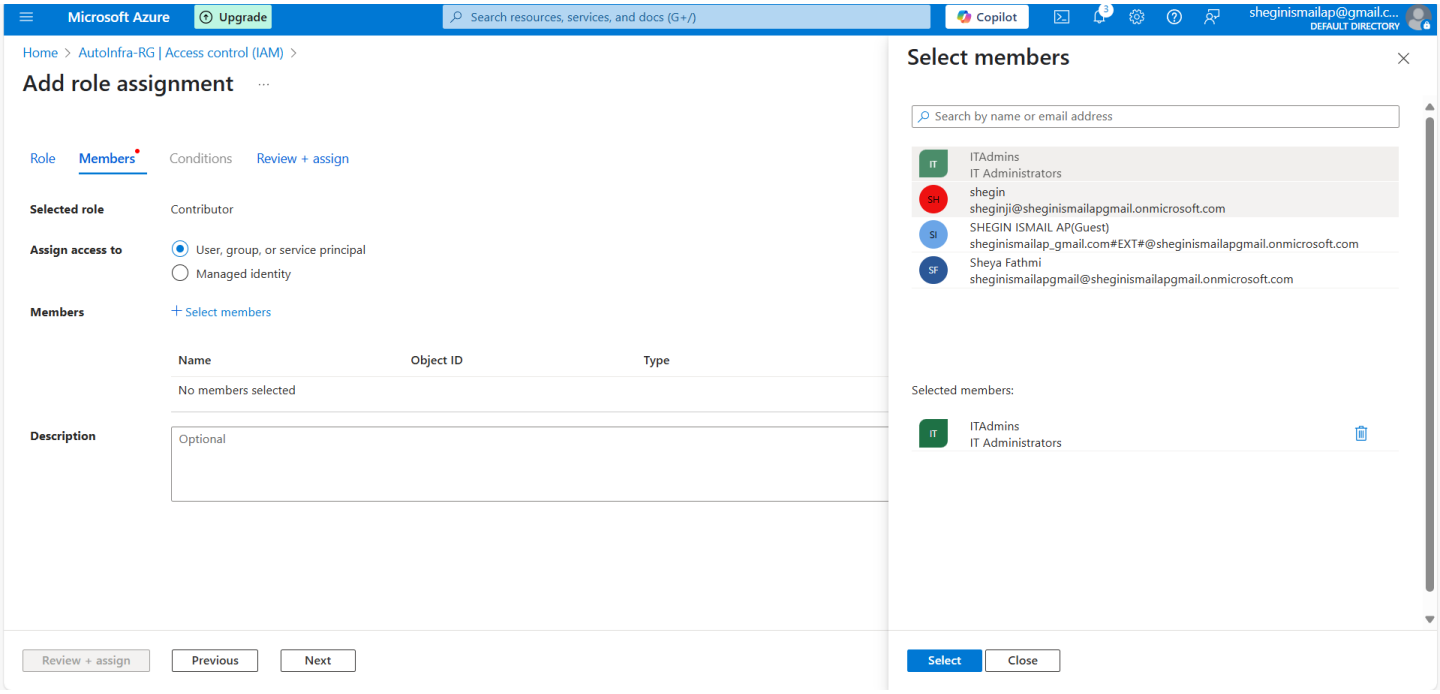
⚠ Your Azure Monitor alert was triggered

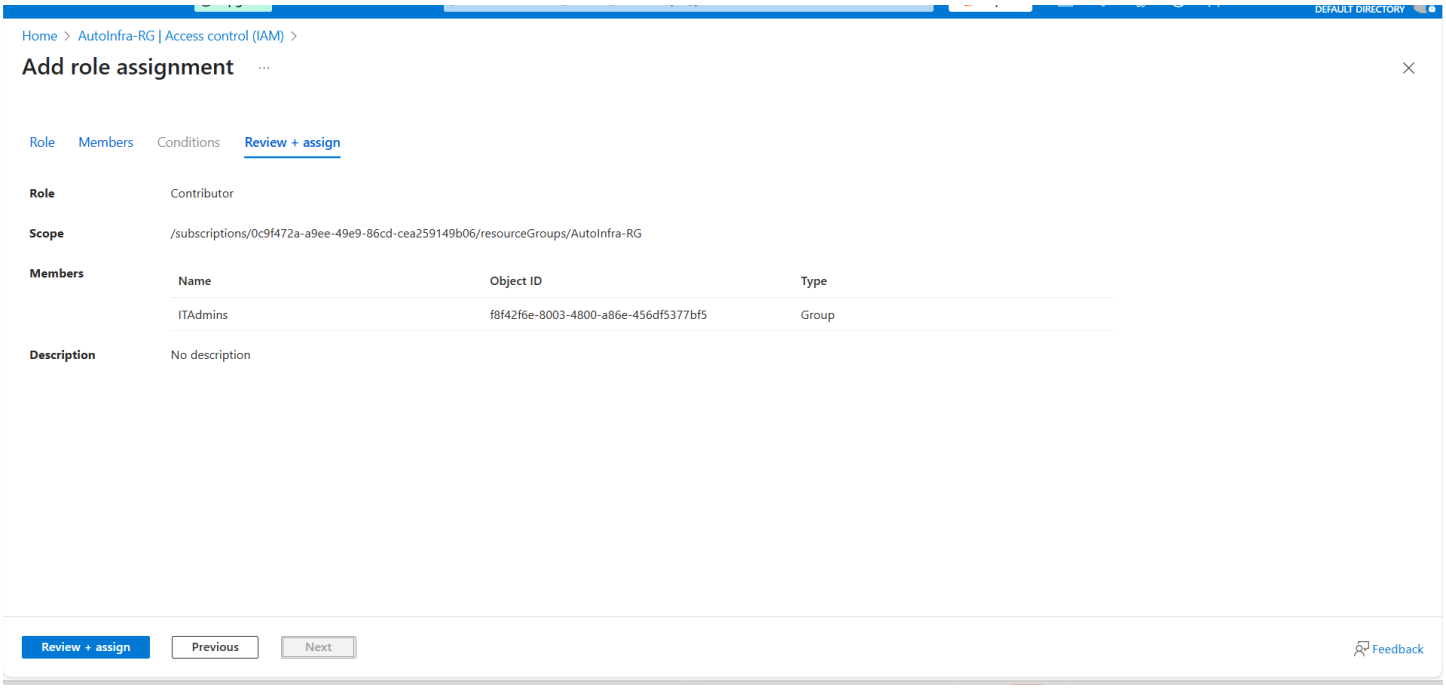
Azure monitor alert rule CPU-Alert-AutoInfra was triggered for WinAutoInfraVM1 at May 13, 2025 21:29 UTC.

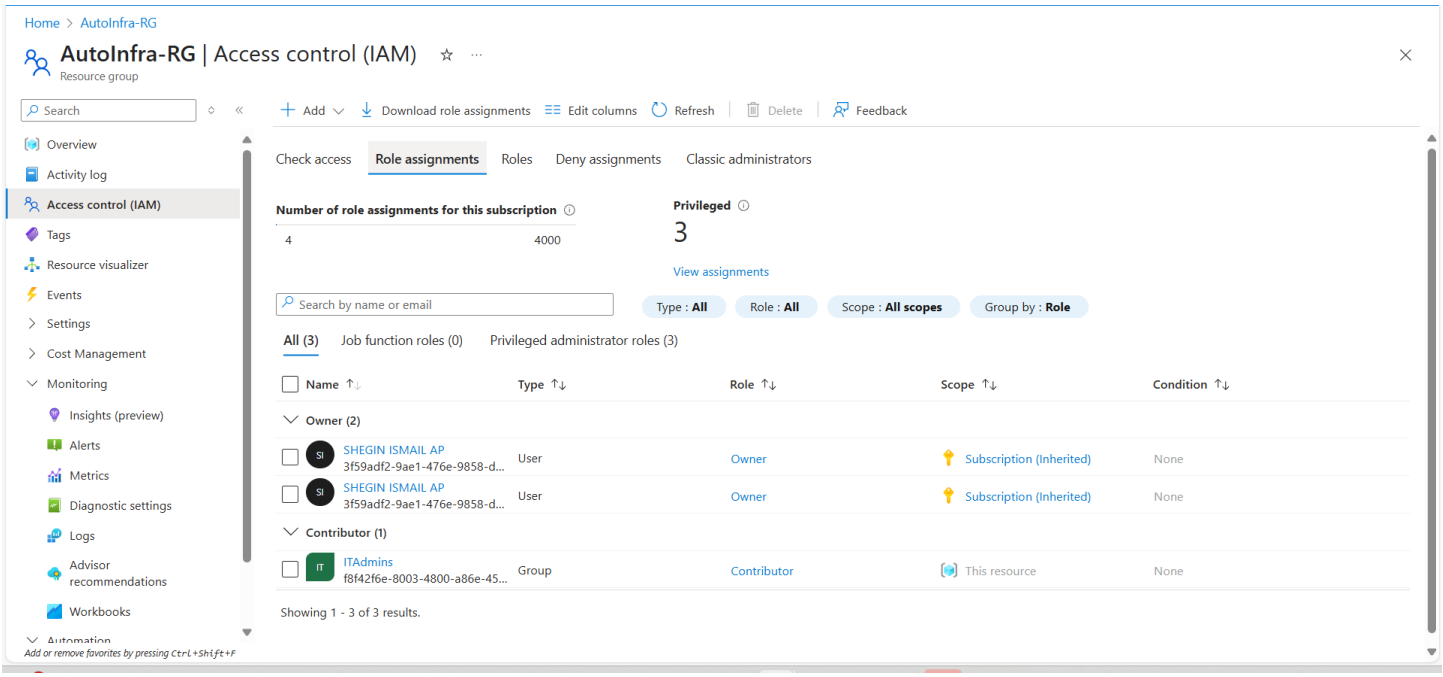
Alert rule description	Average CPU Usage should not exceed 80%
Rule ID	/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourcegroups/AutoInfra-RG/providers/microsoft.insights/metricAlerts/CPU-Alert-AutoInfra View Rule >
Resource ID	/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG/providers/Microsoft.Compute/virtualMachines/WinAutoInfraVM1 View Resource >



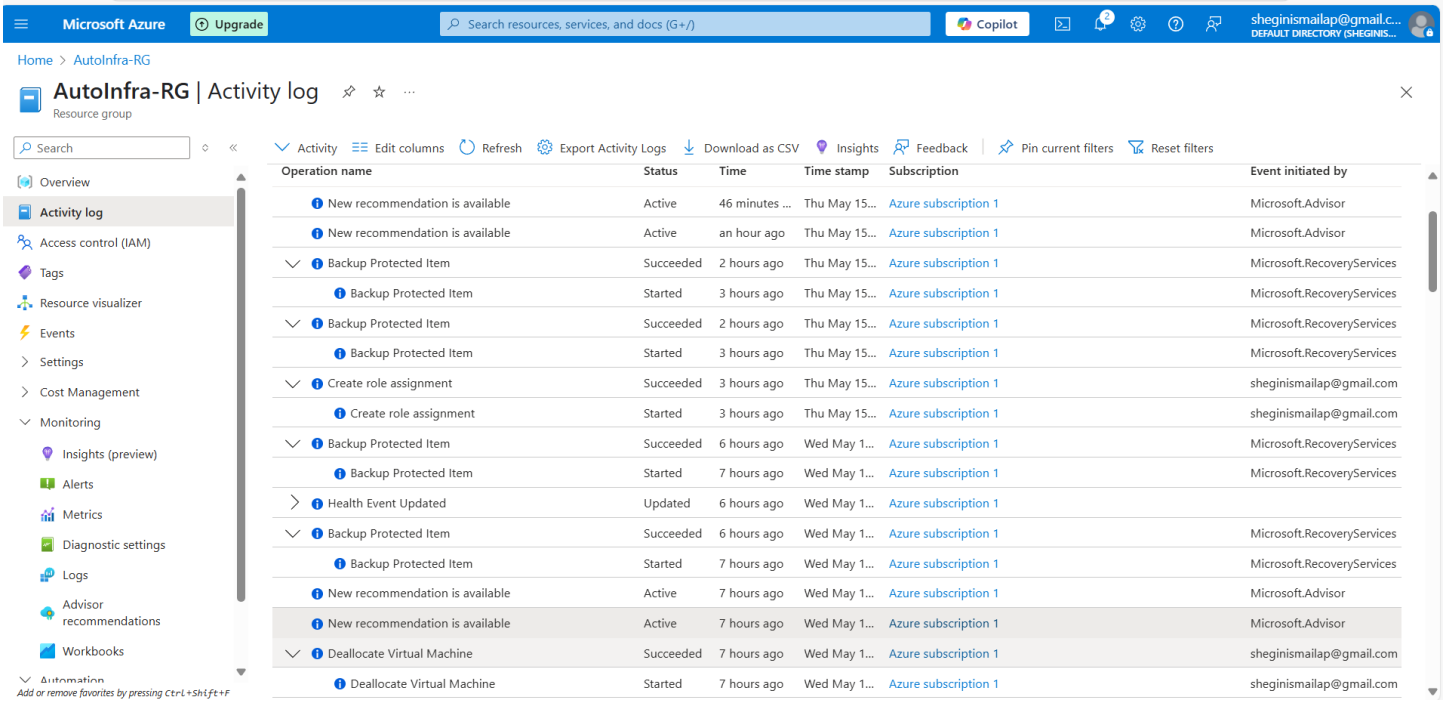
10Create Azure AD Group (e.g., ITAdmins), assign RBAC

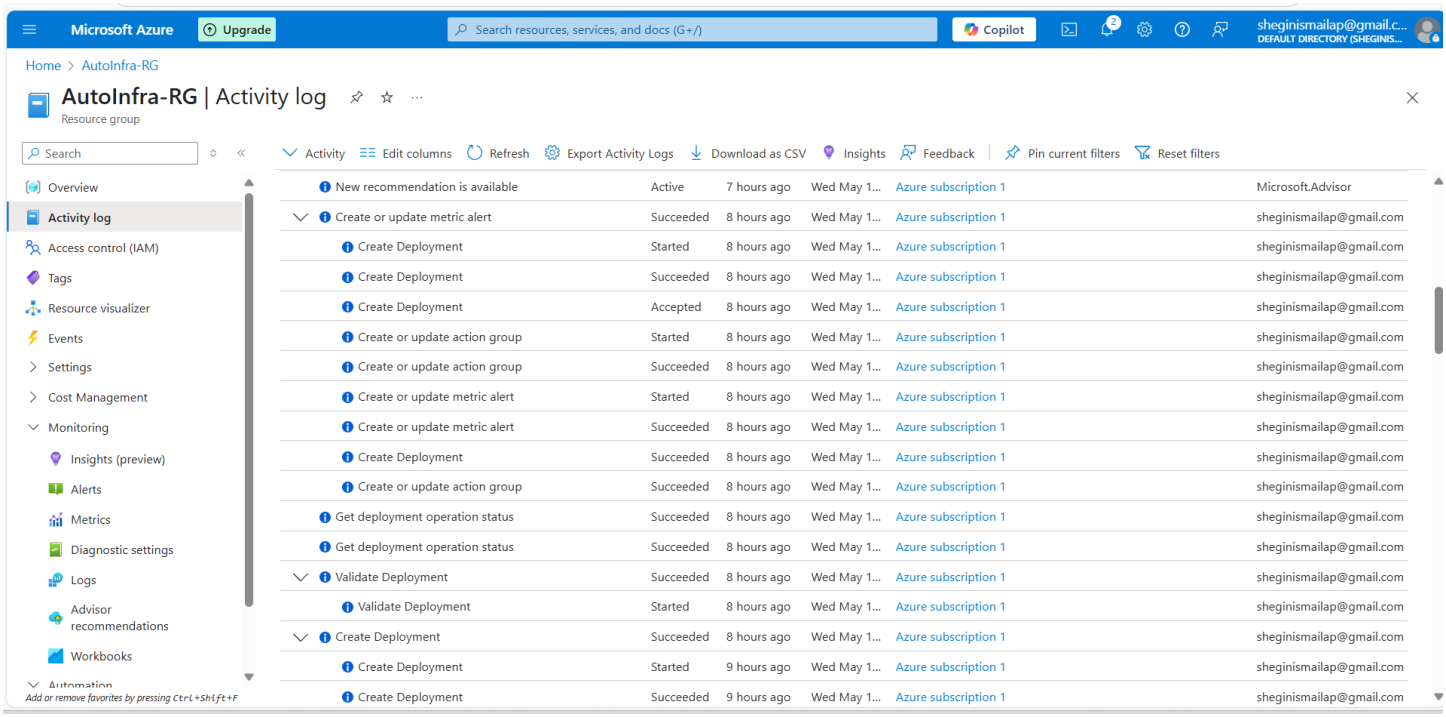

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
sheginismailap@gmail.c...
DEFAULT DIRECTORY
Home > AutoInfra-RG | Access control (IAM) >
Add role assignment
Role
Members
Conditions
Review + assign
Selected role
Contributor
Assign access to
User, group, or service principal
Managed identity
Members
+ Select members
Name
Object ID
Type
No members selected
Description
Optional
Review + assign
Previous
Next
Select members
Search by name or email address
ITAdmins
IT Administrators
shegin
sheginji@sheginismailapgmail.onmicrosoft.com
SHEGIN ISMAIL AP(Guest)
sheginismailap_gmail.com#EXT#@sheginismailapgmail.onmicrosoft.com
Sheya Fathmi
sheginismailapgmail@sheginismailapgmail.onmicrosoft.com
Selected members:
ITAdmins
IT Administrators
Select
Close


DEFAULT DIRECTORY
Home > AutoInfra-RG | Access control (IAM) >
Add role assignment
Role
Members
Conditions
Review + assign
Role
Contributor
Scope
/subscriptions/0c9f472a-a9ee-49e9-86cd-cea259149b06/resourceGroups/AutoInfra-RG
Members
Name
Object ID
Type
ITAdmins
f8f42f6e-8003-4800-a86e-456df5377bf5
Group
Description
No description
Review + assign
Previous
Next
Feedback



4.Validate using Azure Resource Manager logs.





5. Store templates in GitHub for reuse.
   Access the templates here: [Github-link]